Method Handbook

Praxeological policy analysis

Phase 1 deliverable of the Praxeology AI project

draft 1.1

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Introduction

How to analyze the consequences of a policy praxeologically, written so that a newcomer can follow it and a machine can later encode it. This is the Phase 1 deliverable of the project plan (`../Praxeology_AI_Project_Plan_v3.md`, Section 4).

Status: draft 1.1 (12 core rules; catalog 1.1, 105 derived patterns; core rules tested by blind re-runs of all 65 examples), with 65 worked examples (WE-026 to WE-035 probe parts of the method the first 25 did not test; WE-036 to WE-065 are held-out tests of catalogs 0.8, 0.9 and 1.0), written before any expert review. Gaps found in the worked examples have been closed or turned into questions for the experts; see `06_gaps.md`. Every rule, tier decision and source pointer is a proposal. Citations have been checked once against the texts; `07_sources.md` records how, and which still need a direct check.

How to use it

To analyze a policy, work through `03_workflow.md` step by step, using `templates/analysis_template.md` as the form to fill in. Use the patterns in `04_rules.md` where they fit, and cite them by ID. When no pattern fits, derive the step from the core rules in `04a_core_rules.md` and cite those. Mark a step `GAP` only if no core rule applies, and add it to `06_gaps.md`. The worked examples in `05_examples/` show what a finished analysis looks like.

Contents

File	What it holds
`01_foundations.md`	The action axiom and the axiom-level laws that rules rest on, with a proposed tier for each.
`02_ontology.md`	The vocabulary: actor classes, variables, interventions, consequences, assumptions, signs, horizons, visibility. Includes the controlled variable list for version 1.
`03_workflow.md`	The eleven-step procedure, with the questions to ask at each step, common moves, pitfalls and the output each step produces.
`04a_core_rules.md`	The core rules (CR-1 to CR-12): the universal calculus from which every pattern is derived, the tests a rule must pass to count as universal, and a map from workflow steps to core rules.
`04_rules.md`	Rule catalog 1.0, 103 patterns: each with its derivation from the core rules. Patterns cover generic responses, opportunity-cost and time rules, and rules for the three version 1 families (price controls; taxes, tariffs and subsidies; licensing, quotas and prohibition), plus dominance rules.
`05_examples/`	65 fully worked analyses: WE-001 to WE-025 (the core set, from the broken window to a multi-part housing bill, with three repeals), WE-026 to WE-035 (gap-probing cases) and WE-036 to WE-065 (held-out tests of catalogs 0.8, 0.9 and 1.0). Listed in `06_gaps.md`, 6.4.
`06_gaps.md`	Gaps the examples found and how each was resolved, questions that need an expert decision, and next steps.
`05_examples_core/`	WE-001 to WE-065 re-derived blind from the core rules alone, each with a comparison against the pattern-based version.
`08_core_rerun_report.md`	Report of the five rounds of that test: method, results (the core rules derived every valid claim in every round), the patterns and recurring errors each exposed, and the changes made.
`07_sources.md`	Editions, short forms and the verification status of every citation.
`templates/analysis_template.md`	Blank analysis form following the workflow.
`../tools/check_handbook.py`	Consistency checker for the worked examples (sections, rule IDs, variables, signs, assumptions).

Conventions

- **Rule IDs** have a family prefix and a number: `RS` generic responses, `OC` opportunity cost, `TS` time structure, `PC` price controls, `TX` taxes and tariffs, `SB` subsidies, `EN` entry restrictions (licensing, quotas, prohibition), `DM` dominance rules. The IDs used for illustration in the project plan (PR-017 and so on) are superseded by these.
- **Foundations** are cited as `AX-n` (axiom) and `LW-n` (axiom-level law). **Core rules** are cited as `CR-n`, with a letter for a clause (`CR-7b`).
- **Tiers** follow the plan: A (axiom-level only), B (named empirical assumptions, each with an estimate), J (typed-decision trail), C (suggestion, not derived). In this Handbook a rule's tier is the tier of its conclusion when its preconditions are met; a consequence derived through several rules takes the lowest tier in its chain.
- **Signs** are `increase`, `decrease` or `ambiguous`. Always relative to what would have happened without the policy, never relative to the past.
- **Horizons** are `short` (under a year), `medium` (one to five years) and `long` (more than five years).
- **Worked-example probabilities** are illustrative, because the examples are stylized policies in a fictional city. In a real analysis, every estimate must cite its evidence.
- **Historical evaluation cases** (for example the 1994 San Francisco rent control expansion) are reserved for Phase 4 and are deliberately not used as worked examples here.

1. Foundations

Every rule in the catalog rests on the action axiom and a small set of laws derived from it. This file lists them, says how each is used in policy analysis, and proposes a tier. The tier decisions are for the experts to confirm (see `06_gaps.md`, question Q1).

Citations use the short forms defined in `07_sources.md` (HA, MES, P&M, EIOL), which also records how each was checked.

The guiding question, from Bastiat and Hazlitt: **look at the effects of the policy on all groups, not just one, and over the long run, not just the short run.**

AX-1 The action axiom

Statement. Human beings act: they use means to try to replace a less satisfactory state of affairs with a more satisfactory one.

What follows for analysis. A policy does not act; people do. Every consequence must be traced to choices made by members of some actor class who are pursuing their own ends with the means they have. A claim that cannot be stated as "these people, facing this changed situation, will choose differently in this way" is not yet an analysis.

Tier: A. **Source:** HA ch. I §1 "Purposeful Action and Animal Reaction".

LW-1 Scarcity and opportunity cost

Statement. Means are scarce. Using a means for one end means giving up the next-best end it could have served. That forgone end is the cost of the action.

Use. Every resource a policy directs to a use (spending, compliance, enforcement, queuing, a forced purchase) was taken from some other use. The other use is the "unseen" consequence. Rules OC-1 to OC-3 apply this law.

Tier: A. **Source:** HA ch. IV §4 "Action as an Exchange"; Bastiat 1850, §1 "The Broken Window".

LW-2 Marginal valuation

Statement. People value units of a good by the least important use those units would serve (the marginal use). The more units of a good someone has, the less important the use served by one more unit (diminishing marginal utility).

Use. Responses to policy happen at the margin: those actors for whom the policy changes the ranking of their best and next-best options change their behavior; others do not. Ask who is "at the margin" for each change.

Tier: A. **Source:** HA ch. VII §1 "The Law of Marginal Utility"; MES ch. 1 §5.

LW-3 Law of demand

Statement. Other things equal, the quantity of a good that buyers wish to buy at a lower price is not smaller, and is usually larger, than at a higher price.

Use. Price ceilings, subsidies to buyers and taxes on sellers all work through this law.

Tier: A by proposed decision, with a recorded exception: goods whose higher price signals quality, and the theoretical Giffen case, can break it. Rules that rely on LW-3 must state in their exceptions when a good's price is used as a quality signal. **Source:** MES ch. 2 §5 and ch. 4.

LW-4 Resources move toward higher expected returns

Statement. Owners of resources (labor, capital, land, entrepreneurial attention) use them where they expect the highest return, as judged by their own ends. If a policy lowers the expected return of one use relative to others, some resources move to the others, provided other uses are open to them.

Use. The basis of every supply response: withdrawal of rental units, exit from a taxed industry, entry into a subsidized one, choice of occupation.

Tier: the direction is A (from AX-1 and LW-2); whether alternative uses exist, and how fast resources can move, are empirical. Rules that use LW-4 are therefore Tier B and carry an assumption such as `alternative_uses_exist`. **Source:** HA ch. XV §4 "The Sovereignty of the Consumers"; HA ch. XVIII §5 "The Convertibility of Capital Goods" and §8 "The Mobility of the Investor".

LW-5 Voluntary exchange is expected to benefit both parties

Statement. A person enters a voluntary exchange only if they expect to value what they receive more than what they give. So every voluntary exchange is expected, at the moment of choice, to benefit both parties.

Use. When a policy prohibits an exchange that would otherwise have happened (triangular intervention), at least one party, and usually both, loses an expected gain. When a policy forces an exchange that would not otherwise have happened (binary intervention), at least one party loses compared with what they would have chosen. This is how the analysis identifies losers who are never counted.

Tier: A (ex ante, in terms of each party's own ranking). It says nothing about whether the parties are right afterwards. **Source:** MES ch. 2 §4 "Terms of Exchange"; P&M ch. 2 §2 "Direct Effects of Intervention on Utility"; Rothbard, "Toward a Reconstruction of Utility and Welfare Economics" (1956).

LW-6 Non-price rationing

Statement. When a good is scarce and the price is prevented from rising to the level at which quantity demanded equals quantity supplied, the units available are still allocated somehow: by queuing, first come first served, favoritism, personal connections, side payments, discrimination, tie-in requirements or lower quality.

Use. Price ceilings do not abolish rationing; they change who gets the good and what it costs them in non-money terms. The form rationing takes is empirical; the fact that some form appears is not.

Tier: A, given a shortage. **Source:** EIOL 1946 ch. 17 "Government Price-Fixing"; HA ch. XXX §2.

LW-7 Time preference

Statement. Other things equal, people prefer a satisfaction sooner to the same satisfaction later.

Use. Policies that change the return on saving or investment change how much future-oriented production people undertake. Policies with delayed costs and immediate benefits are attractive to actors with short horizons (including politicians facing elections).

Tier: A. **Source:** HA ch. XVIII §2; MES ch. 6 §3.

LW-8 Knowledge and calculation

Statement. Market prices summarize information about scarcities that no single mind possesses. When a price is fixed by law, it stops carrying that information, and actors make choices that no longer reflect the actual scarcity of the good.

Use. A controlled price misallocates in two ways: between buyers (the good may not go to those who value it most, as ranked by their own willingness to give up other goods) and over time (signals to invest or conserve are suppressed). Apply as a qualitative warning attached to PC rules; it does not by itself predict a specific variable.

Tier: A as a statement about information; specific consequences derived from it are B. **Source:** Hayek, "The Use of Knowledge in Society", *American Economic Review* 35(4), 1945; Mises, "Economic Calculation in the Socialist Commonwealth" (1920).

LW-9 The law of returns

Statement. Means are combined to produce an end. For any combination of complementary means there is a most productive proportion; adding more of one means while the others stay fixed eventually adds less and less to the result.

Use. Why the value of one factor depends on how much of its complements is available: labor works with capital, a car with fuel and repairs. Used by CR-9 and by patterns on wages and capital (TS-9).

Tier: A. **Source:** HA ch. VII §2 "The Law of Returns"; MES ch. 1 §6.

LW-10 Imputation

Statement. The value of a means derives from the value of the ends it serves. A means used to produce something else (an input, a factor of production, a complement) is valued for its contribution to that end, so a change in the value or use of the end changes the value of the means in the same direction.

Use. Why changes travel from one market to the markets for its inputs and complements (CR-9): demand for labor follows demand for what labor produces; demand for repairs follows the number of cars in use.

Tier: A. **Source:** Menger, *Principles of Economics*, ch. I and III; HA ch. XVI §3 "The Prices of the Goods of Higher Orders"; MES ch. 7 §1.

Laws that are results of the core rules

LW-3, LW-4 and LW-6 are not independent premises: they follow from the core rules in `04a_core_rules.md` (LW-3 from CR-4c, with CR-4's exception; LW-4 from CR-7; LW-6 from CR-6). They are kept here, with their IDs, because patterns and worked examples cite them.

The intervention typology

Rothbard classifies intervention by the relationship it coerces. The type tells the analyst where to look first.

Type	What it does	Where to look first	Version 1 examples
Autistic	Restricts a person's own action, with no exchange involved.	The restricted person's next-best use of their time and resources.	Not in version 1 scope except as a component (for example, a ban on home production).
Binary	Forces an exchange between the state and a person: the state takes (tax) or gives (subsidy).	Both sides of the budget: who pays and what they would have done with the money (LW-1), and who receives and how they change their behavior.	Taxes, tariffs (tax side), subsidies, conscription.
Triangular	Forces or prohibits an exchange between two private parties.	The exchanges that no longer happen (LW-5), the parties on each side, and how scarcity is now allocated (LW-6).	Price ceilings and floors, licensing, quotas, prohibition of a good, tariffs (as a grant of privilege to domestic producers).

Many real policies combine types. Classify each component separately (workflow step 2).

Source: P&M ch. 2 §1 "Types of Intervention".

2. Ontology

The vocabulary every analysis uses. The plan (Section 4.2) defines the concepts; this file fixes their fields and the controlled lists for version 1. The outcome cards in Phase 4 use the same lists, which is what makes predictions and observations comparable.

2.1 Policy record

The structured statement of what the policy does (workflow step 1).

Field	Content
`policy_id`	Short identifier.
`jurisdiction`	Where it applies. Note the nearest places where it does not apply.
`announced`, `enacted`, `effective`, `expires`	Dates. `announced` is when the policy became publicly expected (a bill's introduction, a credible campaign promise); it drives announcement effects (RS-8). `expires` may be none.
`components`	One entry per distinct instrument (a policy with a cap and an exemption has two components).
`component.instrument`	One of the instrument types in 2.3.
`component.target`	The good, service, activity or factor affected.
`component.bound`	Who must comply.
`component.parameters`	The cap, rate, requirement or quantity, as stated.
`component.exemptions`	Who or what is excluded, and the conditions.
`component.enforcement`	Who enforces, with what penalties and powers.
`component.revenue_use`	For taxes: where the money goes, if the policy says.
`component.funding`	For subsidies and spending: where the money comes from.
`existing_rules`	Rules already in force that the policy interacts with or makes binding (a council-set meter rate that a license cap makes bind). Same sub-fields as a component. They stay in place in the counterfactual.
`source_text`	For each field, the sentence of the policy it comes from. Fields that the text leaves unclear are marked `unclear`, never guessed.

2.2 Actor class

A group defined by its role in the situation, not by identity. Name actor classes by role plus market, for example `tenants.controlled_units`, `landlords.controlled_units`, `prospective_tenants`, `braiders.unlicensed`. Always consider these generic classes, even when the policy does not mention them:

Class	Who
`buyers.current`	People who buy the target good now.
`buyers.prospective`	People who would buy it in future, or who would enter the market.
`sellers.current`	People and firms who supply the target good now.
`sellers.prospective`	Would-be entrants.
`factor_owners.specific`	Owners of resources that are hard to move to other uses (land, licenses, specialized equipment, specialized skills).
`buyers.substitutes`, `sellers.substitutes`	Participants in markets for substitutes.
`sellers.complements`	Suppliers of goods used together with the target good.
`input_suppliers`	Suppliers of inputs to the target good.
`taxpayers`	Those who fund the policy, directly or through the budget.
`officials.enforcement`	The agency that administers and enforces the policy.
`intermediaries`	Brokers, agents, black-market sellers, compliance consultants.
`neighboring_jurisdictions`	Buyers and sellers just across the boundary.
`neighbors`	Those whose situation is changed through proximity, not exchange (CR-9).
`buyers.foreign`, `sellers.foreign`	Foreign participants in a market.
`residual_claimant`	Whoever keeps what a fund collects and does not pay out.

2.3 Intervention instrument types (version 1)

Instrument	Rothbard type	Rule family
`price_ceiling`	triangular	PC
`price_floor` (including minimum wage)	triangular	PC
`per_unit_tax`, `ad_valorem_tax`	binary	TX
`lump_sum_tax` (a fixed sum per household or person)	binary	TX: TX-1 and OC-1 apply; TX-2, TX-9 and DM-1 do not, because no one can reduce it by changing behavior
`income_tax`, `profit_tax`, `capital_tax`	binary	TX
`tariff`	binary (tax) and triangular (privilege)	TX
`per_unit_subsidy`, `lump_sum_subsidy`, `purchase_rebate`	binary	SB
`licensing_requirement`	triangular	EN
`quota`	triangular	EN

Instrument	Rothbard type	Rule family
`prohibition` (of a good or service)	triangular	EN
`own_production_ban` (banning production for one's own use)	autistic	CR-1, CR-4 directly; candidate pattern
`product_standard` (a minimum specification for goods sold)	triangular	EN-14, TS-3
`mandated_contract_term`	triangular	PC-17
`term_prohibition` (a contract term made unenforceable)	triangular	PC-18
`disclosure_mandate`	triangular	CR-11c, EN-15
`deposit_refund` (a charge refunded on a later action)	binary	CR-1, CR-8; candidate pattern
`export_subsidy`	binary	SB-10
`public_supply` (a good supplied by the state at a set price, quantity set by a budget)	binary	PC-19
`government_borrowing`	binary	OC-8
`rights_allocation` (how a created right is first assigned: by history, auction or lottery)	binary	SB-4, CR-7c, EN-12
`mandate_with_charge` (a required term or action that may be bought out by paying a charge, "play or pay")	triangular / binary	PC-17, TX-2

2.4 Variables (controlled list, version 1)

Every consequence names one variable from this list. Add a variable only by editing this file, so that predictions and outcome cards stay comparable.

Prices and quantities

Variable	Meaning
`price_paid`	Money price paid by buyers, including taxes.
`net_price_received`	Money price received by sellers, after taxes and including subsidies.
`quality_adjusted_price`	Price per unit of quality, where quality is variable.
`quantity_demanded`	Amount buyers wish to buy at the prevailing price.
`quantity_supplied`	Amount sellers offer at the prevailing price.
`quantity_traded`	Amount actually exchanged in the legal market.
`excess_demand`	Shortage: quantity demanded minus quantity supplied, at a controlled price.
`excess_supply`	Surplus: quantity supplied minus quantity demanded, at a controlled price.
`price_substitute`, `quantity_substitute`	Price and quantity in the market for a named substitute.
`price_complement`, `quantity_complement`	Same, for a named complement.
`asset_value`	Market value of a durable asset tied to the target (land, building, license, quota right, shares in a firm).
`full_cost_per_unit`	Money price plus the value the buyer places on the wait, search and side payments needed to obtain a unit. Usually `ambiguous` under a ceiling.
`inventory_held`	Stocks of a good held by an actor class at a named time (sellers' stocks before a season; an agency's surplus stocks).
`transaction_size`	Typical amount per exchange (loan size, order size).
`specification_level`	A product's measured specification under a standard, with no presumption that buyers value it (unlike `quality_per_unit`).
`product_attribute`	A named attribute or composition of a product (calories, recycled content).
`discards`	Output caught or produced and then thrown away.
`unit_size`	Physical size of each unit of a durable good (floor area per dwelling, capacity per vehicle).

Allocation, quality and non-price terms

Variable	Meaning
`wait_time`	Time spent queuing or on waiting lists.
`travel_time`	Time spent traveling to obtain a good or reach work.
`search_cost`	Time and money spent finding a seller or buyer.
`side_payments`	Payments outside the controlled price (key money, bribes, tie-in purchases).
`quality_per_unit`	Quality of each unit sold (including maintenance and service).
`quality_variability`	Spread of quality between units bought in a market (for a drug, potency and purity).
`quality_information`	How easily buyers can judge a seller's quality before buying: reviews, brands, certification, legal recourse.
`share_concentrated_forms`	Share of sales in concentrated, high-potency forms of a good.
`allocation_by_non_price_criteria`	Use of connections, discrimination or favoritism in deciding who gets the good.
`turnover`	Rate at which holders of the good change (tenant moves, job changes).
`variety`	Number of distinct versions of the good offered.
`share_of_sellers_licensed`	Share of active legal sellers who hold the required license or permit.
`term_prevalence`	Share of contracts in a market that contain a named term.
`home_production`	Goods or services produced by a household for its own use or to give away.
`consumption`	Spending by a household on present goods (as distinct from `saving`).

Variable	Meaning
`job_changes`	Rate at which members of an actor class change jobs, including moves that require changing home.

Production, capital and entry

Variable	Meaning
`investment_target`	New investment in producing the target good.
`maintenance_spending`	Spending on keeping existing capital in use.
`capital_stock_target`	Existing productive capacity for the target good.
`capital_stock_economy`	Capital across the economy, not only in the target activity.
`saving`	Income not consumed, by actor class.
`interest_rate`	The rate at which a named class of borrowers can borrow.
`right_lease_price`	The price for using a tradable right (a quota, a license) for one period.
`return_on_capital`	Net money return per unit of capital invested, for a named actor class.
`units_held_vacant`	Units of a durable good kept out of use by their owners: neither occupied, rented nor offered.
`number_of_sellers`	Count of active sellers.
`entry_rate`	Rate at which new sellers enter.
`employment`	Number of people employed in the named activity.
`hours_worked`	Hours per worker.
`non_wage_compensation`	Benefits, training, flexibility and other non-money terms of employment.
`wage_rate`	Money wage per hour.
`earnings`	Total money earnings of a group (wage times hours times employment).

Avoidance and the state

Variable	Meaning
`informal_activity`	Trade in the target outside the legal market (black or grey market).
`cross_border_purchases`	Purchases made just outside the jurisdiction.
`reclassification`	Reshaping of goods or contracts to fall outside the policy's definitions.
`compliance_cost`	Resources spent by those bound to comply.
`enforcement_cost`	Resources spent by the state to enforce.
`tax_revenue`	Revenue collected.
`non_tax_receipts`	Fines, fees and unredeemed deposits received by the state or a fund.
`fund_balance`	Balance of a fund that collects and pays out under a policy.
`subsidy_outlay`	Total paid out under a subsidy.
`exchange_rate`	Price of foreign currency in domestic currency.
`legal_breach`	Breach of a rule within the legal market (unpaid mandated leave, unlabeled items), as distinct from `informal_activity`.
`tax_revenue_vs_static_projection`	Revenue collected compared with a projection that multiplies the rate by the pre-tax base (ignoring responses).
`subsidy_share_received`	Share of a subsidy's benefit received by a subgroup, signed relative to the nominal recipients (sellers or asset owners who capture part of it: increase).
`disputes_settled_outside_courts`	Share of disputes in a market settled without legal recourse. Any violence used in such settlement is a safety effect and out of scope for version 1.
`tax_share_borne`	Share of a tax's (or a mandate's) burden borne by a subgroup, signed relative to the other subgroups in the same market (increase means it bears more than they do).
`lobbying_effort`	Resources spent seeking to change, extend or keep the policy.
`resources_available_other_uses`	Resources available to actors for uses outside the target (the "unseen" bucket).
`resource_stock`	The state of a shared natural stock (an aquifer's level, a fish stock). Chains through it are Tier C (CR-9, 0.7).
`enrollment`	Number enrolled in a course of training or study.
`stock_age`	Average age, or age mix, of a durable stock (a car fleet, a housing stock).
`applications`	Number of applications for a permit, license, lottery or program.

Out of scope for version 1. Health, safety, environmental quality and nutrient intake have no variables. Reports say explicitly that these are not analyzed, rather than leaving them out silently.

2.5 Consequence (normal form)

```
consequence:
  id:            C-<n>
  actor_class:   from 2.2
  variable:      from 2.4 (with a named market where needed)
  sign:          increase | decrease | ambiguous
  horizon:       short | medium | long
  visibility:    visible | invisible
  tier:          A | B | J | C
  derivation:    [rule IDs in order]
  assumptions:   [assumption IDs]
  reversal:      what would make the sign flip or the effect vanish
  chains:        for ambiguous consequences, the opposing derivations
```

```
  components:    for multi-component policies, which components the chain runs through
  reading:       optional; where a sign depends on how an unclear provision is read,
                 one entry per reading, such as "U1-a: increase; U1-b: ambiguous"
  persistence:   for repeals only: none | partial | full, compared with the path on
                 which the original policy was never imposed (TS-6); may be given per
                 horizon, such as "partial (medium); none (long)"; `n/a` for effects that
                 exist only during the transition
```

No change. Where a chain gives no change for a class, or a provision has no effect under one reading, the sign is `no_change` (for example "U1-a: increase; U1-b: no_change"). Use it only where the derivation establishes no change; where it establishes nothing, record nothing.

Mixed groups. Where a rule licenses only that at least one of several variables moves, and the variables move in different directions (quality down or waiting up; price up or quantity down), list each variable with its direction, such as "employment (−), hours_worked (−), term_prevalence (+)", and give the sign `at_least_one_as_stated`. A group over actor classes ("taxpayers or recipients of other spending") is recorded the same way in the actor-class column.

Group consequences. Some rules conclude that at least one of several variables moves, without saying which (PC-10's weaker form: employment, hours or non-wage compensation; EN-3; RS-2). Record these as one consequence with the variables listed together and the sign `at_least_one_decrease` or `at_least_one_increase`. It is scored as correct if at least one member moves as predicted and none moves the other way. Writing each variable as its own "decrease" would overclaim.

Timing shifts. A response that moves an action in time (RS-8, SB-6) is written as signed consequences per time window, each against the counterfactual. This applies whenever the action also takes place without the policy, at another time: interest credited before a tax starts, bonuses paid before a contribution rises, purchases brought forward into a subsidy window. Each window gets its own sign ("purchases in the window: increase; in the year after: decrease"). Where there is no counterfactual comparison to make, because without the policy the action would not happen at all, the shift is a note on the path in section 8, not a consequence.

Readings or assumptions. A provision is a reading only if its meaning is genuinely undecided by the text and the record. Where the record supplies evidence about how it will be applied (an agency's practice, a court's earlier decisions), treat it as an assumption with a probability instead.

Readings. Readings of an unclear provision are not given probabilities (workflow step 10); each reading's sign is reported, and the policy text, not the analyst, decides between them.

Every consequence, including an opportunity cost, is a signed change in a variable from 2.4. There is no separate "loss" entry: a forgone purchase is `resources_available_other_uses` decrease, prevented exchanges are `quantity_traded` decrease, time spent queuing is `wait_time` increase. This keeps every consequence scorable in the same way.

The sign is always relative to the counterfactual (the same place and time without the policy), not to the situation before the policy. "Rents rise more slowly than they would have" is a decrease in `price_paid`, even if rents still rise.

2.6 Assumption

```
assumption:
  id:        A-<n>
  text:      plain statement, such as "the cap is enforced against most landlords"
  type:      enforcement | binding | substitutability | mobility | alternative_uses
             | durability | expectations | revenue_use | reversal_cost | counterfactual_trend
             | other
  p:         probability it holds in this case, 0 to 1, stated before any outcome is known
  evidence:  what the estimate rests on
  if_false:  which consequences fail or reverse
```

2.7 Visibility

A consequence is **visible** if it is intended, immediate and concentrated on an identifiable group, so that it is the kind of thing a supporter of the policy would point to. It is **invisible** if it is indirect, delayed, spread thinly over many people, or a forgone alternative. When unsure, ask: would the consequence appear in a news story about the policy's passage? If so, it is visible.

2.8 Market relations

Second-order effects travel only along recorded relations between markets.

```
relation:
  from:      good or market
  to:        good or market
  type:      substitute | complement | input_to | same_market_other_segment | across_border
             | contract | currency | shared_facility | proximity | gift
  strength:  close | partial | weak
  source:    evidence for the relation
```

Two relation types need care:

- **Trip complements.** Goods bought on the same shopping trip are complements in where they are bought, even if they are unrelated in use. Record them as `complement` with a note `trip`. RS-9 uses them.
- **Secondary use.** A good that came free with something else and was reused for another end (carryout bags reused as bin liners) is tied to the goods it substituted for in that end. Record as `substitute` with a note `secondary_use`, so that RS-1 applies and the reason is visible.
- **Contracts.** Parties bound by a continuing contract (a lease, an insurance policy, an employment contract) are linked even when no new exchange takes place: a policy that changes one party's obligations changes the other's situation. Record as type `contract`, naming the contract.

- **Shared facilities, proximity and gifts.** `shared_facility`: users of a road, a vehicle or a common resource, whose use changes what is available to the others. `proximity`: neighbors, whose situation is changed by activity next door. `gift`: a giver and a recipient. Chains through the first two are Tier C pending Q20 (CR-9).
- **Currency.** Payments for imports are the means with which foreigners buy the country's exports or assets; a change in one changes the means available for the other. Record as type `currency`. TX-8 uses it.
- **Location ties to other markets.** Holding a good tied to a place (a controlled tenancy) links the holder's housing market to their labor market. Record as `complement` with a note `location`. RS-11 uses them.

Example: `controlled_rental_units` and `uncontrolled_rental_units` are `same_market_other_segment`, `close`. `sweetened_drinks_in_city` and `sweetened_drinks_outside_city` are `across_border`, with strength depending on distance.

3. The analysis workflow

This is the procedure for analyzing a policy. It has eleven steps. Do them in order on the first pass; later steps often send you back to earlier ones, which is expected. Record everything in `templates/analysis_template.md`.

Before starting, keep four habits in mind:

1. **People act, policies do not.** Every claimed effect must be a choice by some actor class (AX-1).
2. **Compare with the counterfactual, not the past.** The question is always "compared with the same place and time without the policy".
3. **Look for the unseen.** For each effect you find, ask who else is affected and what was given up (LW-1).
4. **Say what you assume.** If a step needs a fact about the world (that the law is enforced, that a substitute exists), write it down as an assumption and estimate how likely it is.

Core rules and patterns. Every step below applies the core rules in `04a_core_rules.md` (the map at the end of that file shows which). The patterns in `04_rules.md` are derived shortcuts for situations that recur. Use a pattern when its preconditions hold, and cite it.

When no pattern fits, do not invent one. Derive the step from the core rules and cite them. For example, "CR-4 (x: comply, y: violate)" or "CR-6, pressure displaced to delivery times". Then state the premises the derivation needs as assumptions. Mark a step `GAP` only if no core rule applies. That would mean the calculus itself is incomplete, which is a finding for the experts. A derivation used in more than one analysis is a candidate for a new pattern, and it enters `04_rules.md` with its derivation written out.

Step 1. State the policy precisely

Goal. A policy record (see `02_ontology.md`, 2.1) that says exactly what is required, prohibited, taxed, subsidized or transferred, by whom, for whom, from when, and with what enforcement.

Questions to ask

- What exactly does each provision require or forbid? Split the policy into components, one per instrument.
- When was it announced or first expected, and when does it take effect? The gap between the two is when announcement effects happen (RS-8).
- Who is bound? Who is exempt, and on what conditions?
- What are the parameters (cap, rate, number of hours, quantity)? How do they change over time?
- Who enforces it, with what penalties, and with what resources?
- For taxes: where does the revenue go? For subsidies and spending: where does the money come from?
- Where is the jurisdiction's boundary, and what lies just outside it?

Pitfalls

- Analyzing the policy's stated purpose instead of its provisions. The title says "affordable housing"; the provisions say "no rent increase above 2% a year".
- Missing exemptions. Exemptions create boundaries, and boundaries create incentives to move across them (RS-5).
- Guessing unclear provisions. Mark them `unclear` and, if they matter, analyze both readings.

Output. Policy record with components.

Step 2. Classify each component

Goal. For each component, its instrument type and Rothbard type (`02_ontology.md`, 2.3), and the rule families that apply.

Questions to ask

- Does it force or prohibit an exchange between private parties (triangular), force an exchange with the state (binary), or restrict a person's own action (autistic)?
- Is it binding? A price ceiling above the market-clearing price, or a licensing requirement that everyone already meets, changes nothing now. Record bindingness as an assumption, with its estimate and how it may change over time (a cap fixed in money terms becomes binding as other prices rise).

Pitfalls

- Classifying the whole policy as one type when components differ. A tariff is a tax on importers (binary) and a grant of privilege to domestic producers (triangular); analyze both sides.

Output. A component table: instrument, Rothbard type, binding (with assumption ID), rule families.

Step 3. Identify the actor classes

Goal. A list of every actor class whose options change, using the generic list in `02_ontology.md`, 2.2 as a checklist.

Questions to ask

- Who is directly bound? Who directly benefits? Who pays?
- Who is a current participant and who is a prospective one? (Prospective buyers and sellers are the most often forgotten.)
- Who owns resources specific to this market, which cannot easily move elsewhere?
- Who participates in markets for substitutes and complements, or just across the border?
- Who administers and enforces the policy? Who would gain from changing it?

Pitfall. Treating a group as homogeneous when the policy splits it. Rent control splits tenants into those who have a controlled unit and those looking for one. Licensing splits providers into those who qualify and those who do not.

Output. Actor-class table: class, role, and how its options change.

Step 4. Identify the direct effect

Goal. The immediate, intended change: the visible consequence the policy's supporters expect.

Questions to ask

- What does the policy change first, for whom, at once? (Rents paid by sitting tenants; prices paid at the shelf; who may legally offer a service.)
- Is the direct effect what the supporters claim, given the provisions? It sometimes is not.

Pitfall. Stopping here. The direct effect is usually real; it is just not the whole story.

Output. One or more consequences in normal form, marked `visible`.

Step 5. Trace the responses

Goal. For each actor class, how its constraints and incentives change and what its members will do differently at the margin.

Questions to ask, for each actor class:

- Which of its options became more costly, cheaper, forbidden or newly available?
- Who in the class is at the margin (LW-2): for whom does the ranking of the best and next-best options change?
- Which generic responses are open to them? Work through the list:

Response	Rule	Ask
Substitute	RS-1	What else serves the same end? Is it covered by the policy?
Adjust quality or terms	RS-2	Can the non-price terms of the exchange change (quality, maintenance, service, contract length)?
Evade	RS-3	Is breaking the rule cheaper than complying, given the enforcement?
Relocate	RS-4	Can the activity move outside the jurisdiction?
Reclassify	RS-5	Can the good or contract be reshaped to fall outside the definitions or into an exemption?
Exit or stop producing	RS-6	Do resources have better uses elsewhere (LW-4)? How quickly can they move?
Lobby	RS-7	Does anyone now have a concentrated stake in extending, changing or repealing the policy?
Change timing	RS-8	Does the policy reward acting sooner or later (deadlines, phase-ins, expected changes)?

Pitfalls

- Assuming everyone responds. Only those at the margin do; the others carry on, which is why the direct effect is often visible while the responses are not.
- Assuming responses are instant. Note how fast each can happen; that becomes step 8.

Output. Responses as consequences in normal form, each with its rule and assumptions.

Step 6. Trace second-order effects

Goal. How the responses affect other actor classes and other markets.

Questions to ask

- When an actor class changes its behavior, whose options change as a result? (Landlords cut maintenance; tenants receive worse quality. Buyers move to substitutes; prices rise there.)
- Which recorded market relations (`02_ontology.md`, 2.8) does the effect travel along? If you need a relation that is not recorded, record it with its evidence first.
- What happens to the people who would have made the exchanges that no longer happen (LW-5)?
- Three second-order patterns to check every time:
    - **Bundled trips (RS-9).** If buyers now shop somewhere else for the target good, what else do they buy on the same trip?
    - **Informal trade (RS-10).** If some trade goes underground, what happens to search costs, quality and dispute resolution?
    - **Lock-in (RS-11).** If people now hold a benefit they lose by moving or by changing status, which other choices (jobs, breaks, schools) do they now turn down?
- Four more links to check:
    - **Complements in use (RS-12).** If the stock or use of a good changes, what is used with it?
    - **Added supply (RS-13).** Does the policy push extra supply into a linked market?
    - **Inputs (RS-14, PC-12).** If the policy changes the cost of an input, what happens to the price of the output? If it caps the output, what happens to the input?
    - **Asset values (SB-4).** Does the policy change the lasting return to an asset in fixed supply? Its value moves at announcement.
- **Policies with several components.** For each pair of components, ask:

    - Does one close or open an exit route from the other?
    - Does one change who is bound by the other?
    - Does one fund the other (SB-7)?
    - Does one tax or restrict the other's intended output?

  Record the components each interaction chain runs through.

Stopping rule. Stop each chain at depth 4 (four links from the policy), and stop any branch earlier if it has reached a diffuse effect on bystanders at depth 3. Note where you stopped. Chains are not stopped because an effect "seems small"; this method cannot measure size.

Pitfall. Following a chain through an unrecorded market relation, such as "this will hurt restaurants" without saying why restaurants are linked to the target.

Output. Second-order consequences in normal form.

Step 7. Combine effects

Goal. One sign for each actor class and variable, where several chains reach the same place.

Procedure (plan, Section 4.5):

1. List every consequence by (actor class, variable, horizon).
2. If all chains for that key have the same sign, that is the combined sign.
3. If signs conflict and a dominance rule (DM family) applies, apply it, and add its assumption.
4. If signs conflict and no dominance rule applies, the sign is `ambiguous`. Keep all chains, and state what would decide it.
5. A one-time cost in one chain against a later stream of gains in another is ambiguous by design (DM-3): say so, and do not try to net them.
6. Never merge different variables. "Wage rate rises" and "hours fall" are two consequences.

Output. Combined consequence list.

Step 8. Add time structure

Goal. A horizon for each consequence, and a note on effects that grow, fade or reverse over time.

Questions to ask

- How durable are the goods and capital involved? Supply responses for durable goods (buildings, equipment, trained skills) arrive as the existing stock wears out and is not replaced (TS-1).
- Does the policy bind more tightly over time (a cap fixed in money terms while other prices rise)?
- What do actors expect: that the policy will be extended, tightened, repealed? Expectations bring responses forward (TS-2).
- Does the policy shift activity in time (a temporary subsidy that pulls purchases forward, followed by a dip; see SB-6)?

Output. Horizon field filled in for every consequence; notes on growth and reversal.

Step 9. Identify opportunity costs

Goal. The unseen consequences: what the resources used by or because of the policy would otherwise have done.

Questions to ask

- What resources does the policy take (taxes, borrowing) or direct (mandated spending, compliance, enforcement)? What would their owners have done with them (OC-1)?
- What time and effort do people now spend that does not appear in any price: queues, searches, paperwork, training hours that serve only to qualify (OC-2)?
- What exchanges no longer happen, and what did each party expect to gain from them (LW-5)?
- What production does not happen because resources moved elsewhere?
- Whose work depended on the spending that was diverted (OC-5, Hazlitt's tailor)?

Pitfall. Counting what the money is spent on as a pure gain. The spending is visible; what the taxpayers would have bought is not. Both must appear.

Output. Opportunity-cost consequences, almost always `invisible`, each written as a signed change in a variable (for example `resources_available_other_uses` decrease, `quantity_traded` decrease for prevented exchanges), never as a bare "loss".

Step 10. State and estimate conditions

Goal. For every Tier B consequence, its assumptions, an estimate of how likely each is to hold in this case, the evidence for the estimate, and what would reverse the conclusion.

Questions to ask

- Is the policy binding, now and later?
- Will it be enforced, and how strictly?
- Do substitutes exist, inside and outside the policy's reach?
- Can resources move to other uses, and how quickly?
- What are actors likely to expect about the policy's future?

Rules for estimates

- Estimates are made now, before any outcome is known, and are not revised after.
- Each estimate cites its evidence: the policy text, enforcement budgets, the existence of a border nearby, conversion rules, past behavior in comparable places.
- A consequence is reported if each of its assumptions has p of at least 0.5. Consequences below the cut-off are listed separately as conditional possibilities. (Whether the cut-off should apply to the joint probability instead is open: `06_gaps.md`, Q9.)
- Readings of an unclear provision are not assumptions and get no probability. Where a sign depends on the reading, report one sign per reading (`02_ontology.md`, 2.5).

Output. Assumption table; each consequence linked to its assumptions.

Step 11. Report

Goal. A report a non-expert can follow, that commits to clear claims.

Structure

1. **The policy in one paragraph:** what it does, not what it is for.
2. **Visible consequences:** the direct effects, with who gains and who pays.
3. **Invisible consequences:** grouped by actor class, each with its chain in plain words, its horizon and its key assumptions.
4. **Ambiguous effects:** what pushes each way, and what would decide it.
5. **Conditions:** the assumptions that matter most, their estimates, and what would change the picture.
6. **What the method cannot say:** sizes, net welfare, whether the policy is good, and (in version 1) health, safety and environmental effects. These are left to the reader or to other methods.

Pitfalls

- Advocacy. Describe consequences by group and horizon; do not weigh them.
- Unsupported claims. Every sentence in the report must trace to a consequence in the list; anything else is labelled as a Tier C suggestion.

Output. The report, and the full consequence list in normal form.

Final check before the report

The core-only re-runs (`08_core_rerun_report.md`) found that most errors, in analyses with and without patterns, are of a few recurring kinds. They also found that an analyst's own verdict that a rule "worked" is not reliable; the blind re-run is the check. Before writing the report, check each consequence for them:

1. **Uncombined chains.** For each actor class and variable, list every chain in the analysis that reaches it, including chains written down in other sections (time structure, announcement effects, other components). If any two have opposite signs, the consequence is `ambiguous` unless a dominance premise is stated (CR-12b, CR-12e). This was the most common sign error.
2. **Single signs for a group.** Where a rule licenses only "at least one of" (employment or hours; waiting, search or side payments; hiring, hours or non-wage terms), record the group with `at_least_one_increase` or `at_least_one_decrease`, not one member alone.
3. **The wrong baseline.** Every sign is against the same place and time without the policy: not against the situation before it, not against a controlled price, and not against a different version of the policy.
4. **Worse off is not poorer.** That a class is worse off by its own ranking (CR-3a) does not mean it has less to spend on other things (CR-3, note). Sign `resources_available_other_uses` only where the class keeps paying, or pays a sum it cannot avoid.
5. **Timing shifts.** Where an action happens with and without the policy, but at different times, sign it per window (`02_ontology.md`, 2.5). Use a path note only where the action would not happen at all without the policy.
6. **Both sides of every transfer.** For every payment, subsidy, grant or revaluation, check that the payer's side and the recipient's side are both recorded (CR-8b).
7. **Every actor class, every core rule.** For each actor class, ask whether each of CR-4 (every pair of options), CR-6, CR-7 (including capitalization, CR-7c) and CR-9 applies. Patterns prompt for some of these and not others, so analyses that rely on patterns tend to miss capitalization, enforcement costs and the payer's side.
8. **Unstated premises.** Every premise a derivation uses is in the assumption table with a probability. That includes "the group can organize" for lobbying, "some actors are at the margin" for any response, and "the alternative is worse" for any quality claim.
9. **The baseline of every pattern (1.0).** For each pattern cited, check that its conclusion is against the no-policy counterfactual. Several patterns were found to compare with something else (another policy, the allocation before trading, uniform enforcement). Re-sign where they do (CR-12, 1.0).

Analyzing a repeal

0.9. Where no earlier analysis of the original rule exists, derive its effects first, from the core rules, as a short table, and run the metamorphic check against that table.

A repeal is analyzed with the same eleven steps, with three differences.

1. **The counterfactual is that the policy stays in force.** Signs are relative to that path, not to the world before the policy.
2. **Check what does not reverse.** For each consequence of the original policy, ask whether the repeal reverses it, partly reverses it or leaves it in place (TS-6). Converted buildings, closed firms, lost skills and displaced people do not come back just because the rule is gone. Record `persistence` on each consequence, and include a table that checks the repeal against the original analysis, as in WE-022 to WE-024.
3. **Ask whether the repeal is believed (TS-5).** If actors think the policy may return, quick-payback responses go ahead and long-payback ones are held back. Ask also who must make up any lost revenue (OC-6), and who now has a stake in restoring the policy (RS-7).

4a. Core rules (the calculus)

Citations use the short forms in `07_sources.md`, which records how each was checked.

The rule catalog has two layers.

- **Core rules** (this file, `CR-1` to `CR-12`) are universal. They are stated over any actor, any option, any means and any time, with no reference to a particular instrument, good or market. Each is derived from the action axiom and the laws in `01_foundations.md`, and each states exactly which part of its conclusion is a priori and which part needs an empirical premise.
- **Patterns** (`04_rules.md`) are derived results for recurring situations: a binding price ceiling, a per-unit tax, a license requirement. Each pattern carries its derivation from the core rules. A pattern is a shortcut, never an independent source of truth.

The relation is that of the limit definition of the derivative to the power rule. Any consequence that a pattern yields can also be derived step by step from the core rules; the pattern saves the work and must give the same answer. When no pattern fits a situation, the analyst derives from the core rules directly and cites them (workflow, "When no pattern fits"). A new pattern enters the catalog only with a written derivation.

What makes a rule universal

A rule is admitted to this file only if it meets all four tests. Patterns must meet the second and fourth.

1. **Generality.** It is stated over arbitrary actors, options, means and times. It names no instrument (tax, ceiling, license), good or market.
2. **Grounding.** It follows from AX-1 and the laws in `01_foundations.md` by a derivation written out below. Whatever empirical content it needs is isolated in named premises, never hidden in its wording.
3. **Symmetry.** It applies in the same way to imposing and to removing a policy, to raising and to lowering a cost, to buyers and to sellers, and to private and to political actors. A rule that works only in one direction is a pattern, not a core rule.
4. **Answerability.** It says what would have to be true for its conclusion to fail, so that a miss in evaluation can be traced to a failed premise rather than explained away.

The objects the rules work on

- An **actor** is an individual. An actor class is a set of actors in the same situation; a class "responds" when some of its members do.
- An actor's **situation** is the set of options they believe are open to them, what they expect each option to bring, and the means they command.
- An actor's **ranking** of ends is their own, ordinal, and revealed only in choice. The rules never assume a particular ranking. They reason only from the fact that there is one.
- A **policy** is a change in the rules that others will enforce. Its effect on any actor is described entirely by CR-1.

CR-1 Representation: what a policy can do to an actor

Statement. A policy affects an actor's action only by doing one or more of four things to their situation:

1. removing an option (a prohibition, a license requirement for those who lack the license, a price above a ceiling);
2. adding an option (a permission, a free service, an exemption);
3. changing what an option brings: attaching a payment, penalty, delay, paperwork or risk to it, or removing one;
4. transferring means to or from the actor (a tax, a grant, a confiscation).

A policy does not change anyone's ranking of ends, except through what it tells them about the world (CR-11).

Derivation. Action is the choice of means for ends within a situation (AX-1). The rule-maker cannot choose for the actor; it can only alter the situation within which the actor chooses: the options, their consequences, and the means available.

Use. Workflow steps 1 and 2: for each component and each actor class, write down which of the four operations it performs. Every later step works on that list.

Premises. Whether a stated rule is actually enforced is empirical (`enforcement`). An unenforced prohibition does not remove an option; it attaches a risk to it (operation 3).

Recording. A provision often does more than one thing: a free tradable right adds an option and transfers means; a void-contract rule removes, for both parties, the option of agreeing to the term, and may add an option for one of them (the worker who may now join a rival); a prohibition removes an option where enforced and attaches a risk where not. Record every operation that applies, for each actor class, and split an actor class where the operation differs within it. Which operation a provision performs depends on what the law would be without it, so state that counterfactual law. Where an operation holds only under one reading of an unclear provision, record the operation per reading (U1-a, U1-b), as consequences are (`02_ontology.md`, 2.5).

Source. HA ch. I §1; HA ch. XXVII; P&M ch. 2 §1.

0.7 clarifications.

- A charge paid only when an option is taken (a deposit on a purchase, a fee on a transaction) is operation 3: it changes what the option brings. Operation 4 is a transfer that no choice avoids (a lump-sum tax, a grant).
- *0.8.* A refund attached to a thing rather than to a person (a deposit refunded to whoever returns the container) is operation 3 on the option of returning it, for anyone who holds the thing; that is what makes collectors and charities recipients. A charge per unit paid by a seller to a fund (distributors' deposits) is also operation 3, on the option of selling.
- Operations may be dated: a provision that removes an option from a future date, or tightens each year, performs its operation on that date. Record the date. Before it, the operation acts only through expectations (CR-10a): an option nobody currently chooses can be removed with no present effect on choices, while assets whose yields depend on the future option change value at once (CR-7c).
- Where every member of a class faces the same operation but responds differently (some return containers and some do not), split the class by response, not by operation.

- Information that changes how much an actor values an option (a calorie count) is not one of the four operations; it works through CR-11c.

0.9 clarifications.

- A single provision can perform different operations for one actor on either side of a threshold: a charge per worker capped per firm is operation 3 below the cap and operation 4 above it. Record the operation per range.

1.0 clarifications.

- *Notches.* A threshold at which a burden jumps on the whole base (a levy on all payroll once payroll passes a line) is a notch, not a range: record the operation on each side, and the jump itself as a cost of crossing (CR-4). A dated operation that repeats (each quota year) is recorded with its period.

1.1 clarifications.

- A line on a ratio (recycled share, land share) can be crossed by moving either term; record both routes. A transfer avoided only by leaving the jurisdiction is operation 4 for those who stay and operation 3 on the option of leaving. An option removed but restored at the other party's election (a buy-out the worker may choose) is recorded as removed, with the restoring option as operation 2 for the party who holds the election.

CR-2 Choice

Statement. Each actor chooses, from the options they believe open, the one they expect to serve their most highly ranked end. What an actor chooses reveals that they ranked it above the alternatives open at the time (demonstrated preference).

Derivation. AX-1, with LW-2 (ordinal valuation).

Premises. None for the statement. What actors believe about their options is a fact about them, and may be wrong (CR-11).

Source. HA ch. I §1 and ch. VII §1 ("action sorts and grades"); Rothbard, "Toward a Reconstruction of Utility and Welfare Economics" (1956), on demonstrated preference.

0.7 clarification: absence of an exchange. That a term or an exchange was absent before a policy demonstrates only that, in the situation as it was, no pair of parties found it worth agreeing on (CR-5). Whether that reflects the parties' own rankings or an obstacle to agreeing (one side knowing more than the other, costs of writing the term) is an empirical question and needs a premise; the rules do not presume either.

CR-3 Cost and loss

Statement. The cost of an action is the most highly ranked alternative given up by taking it. Hence:

- (a) removing an actor's chosen option, or attaching a burden to it, makes the actor worse off by their own ranking, unless their next-best was ranked equal;
- (b) only adding options, or removing burdens, cannot make an actor worse off by their own ranking at the time of choice;
- (c) means used for one end are not available for any other (LW-1).

A loss by the actor's own ranking (a) is not the same as a fall in their spending on other things. An actor who must pay more for something may respond by buying less of it, and so spend more, not less, elsewhere. `resources_available_other_uses` falls for an actor class only if its members keep spending on the burdened option, or are made to pay a sum they cannot avoid. A charge that is later refunded burdens only those who do not claim the refund, and those who claim it bear only the cost of claiming and the time the money is held. An actor who also sold part of what the policy stops them producing loses those receipts too (CR-5a); combine the lost receipts with the saved outlays under CR-12b.

Derivation. CR-2 and LW-1. Under (a) the actor must fall back to an alternative they ranked lower. Under (b) the previously chosen option is still open.

Premises. None. (Statement (b) is about the actor's ranking at the time of choice; it says nothing about how things turn out.)

Source. HA ch. IV §4 (cost as the value of the satisfaction forgone); P&M ch. 2 §2.

0.9 clarifications.

- The note on refunded charges covers partial refunds and refunds computed on a standard rather than on what was used (yield tables, deemed amounts). Those who claim on the standard and use less than it receive more than they paid; those who use more receive less. Split the class by its position against the standard.

1.0 clarifications.

- A refund capped at the charge paid can exceed the cost of the act refunded per unit but never in total. Where the standard is paid to a provider, the position against it that matters is the provider's.

1.1 clarifications.

- A burden attached to an option that exists only under the policy (obligations that come with a newly sold license) is no loss against the counterfactual; the counterfactual is not having the option at all. The note on lost receipts and saved outlays applies to every seller who loses sales, including to a better offer. Where a saving comes back through another term of the same contract (a deductible, a premium share), net it before signing means.

CR-4 Response at the margin

Statement. Suppose a change makes option x worse relative to option y for an actor (x costs more, brings less, or y improves). Then:

- (a) no actor who was choosing y switches to x on account of the change;
- (b) among actors who were choosing x, those for whom x was ranked only narrowly above y (or above some other alternative) switch away from it; the rest keep choosing x and bear the change (CR-3a);

- (c) the same holds for each unit when an actor chooses how many units of x to take: the last unit, which was valued least (LW-2), is the first given up.

The mirror case (x made better) reverses every direction.

What is a priori and what is not. Statement (a) and the direction in (b) and (c) are a priori. That some actors actually are at the margin, so that someone does switch, is empirical: it is the premise behind `substitutability`, `alternative_uses` and `mobility`. **Exception:** when the change also alters the actor's total means (a price rise on something they spend much of their means on; a lower return on something they save toward a fixed sum), the change in means can shift their ranking among other options in the opposite direction. Where such an effect matters, the direction for that actor class is `ambiguous` unless a premise says which effect dominates (this is the exception recorded under LW-3 and TX-6). The exception can bear on the burdened option itself (a heavy buyer of a good whose price rises) and on what an actor offers as well as what they buy (hours of work offered when the wage falls).

A change in means alone. When a policy changes an actor's means without changing the relative cost of any option (a lump-sum tax or grant), CR-3c and CR-8 establish that spending on some uses falls (or rises) by the amount taken (or given). Which uses, and by how much each, CR-4 does not say: it depends on the actor's ranking, which the rules never assume. A claim about a named use ("sales at local shops fall") therefore needs a premise. Whether anything more can be said a priori is `06_gaps.md`, Q19.

Units and lumpy costs. In (c), the unit is the unit of the end served (a dose, not a gram; a trip, not a mile), since that is what the actor ranks. Where a change attaches a large one-time cost to an option (training hours, a license fee), "narrowly ranked" in (b) covers every actor whose ranking of x over y is reversed by that cost, which may be many.

Derivation. CR-2 and CR-3. Under (a), y was ranked above x before, and x is now worse, so y still ranks above it. Under (b) and (c), LW-2.

Use. All the generic responses are this one rule applied to different pairs of options: substitution (x: the good, y: a substitute), evasion (x: comply, y: violate), relocation (x: here, y: there), reclassification (x: the covered form, y: the uncovered form), exit (x: this use, y: another use), lobbying (x: accept, y: lobby), timing (x: now, y: later), entry (x: stay out, y: enter).

Source. HA ch. VII §1; MES ch. 1 §5, ch. 2 §5 and ch. 4.

0.7 clarifications.

- *Means over time.* A transfer financed by borrowing that the recipient expects to repay through later taxes is, for that recipient, partly a loan at the government's rate: it changes the cost of present against future goods for those who could not borrow on such terms, and so falls under CR-4 proper, not only under "a change in means alone".
- *Saving is a use.* Saving and repaying debt are among the uses on which spending rises or falls when means change. By LW-2, the use added is the next in the actor's own ranking and the use given up is the last ranked; this is a statement about rank and names no good. With lumpy goods it may be a new kind of item rather than more of an old one.
- *Fixed costs per item offered.* A cost incurred once for each item a seller offers (testing and labeling a menu item, certifying a model) bears on which items are offered, not on the price of each unit sold: it is a cost of the option of offering the item (x: offer it, y: drop it). *0.8:* the same holds for a cost fixed per outlet or per firm (a menu board, a registration): it bears on whether the outlet or firm operates, and on the size at which it pays to do so, not on the price of each unit. The unit price is signed only with a premise about entry and exit.

0.9 clarifications.

- *Lobbying in repeals.* Applying the rule to political action (x: accept, y: lobby), remember that in a repeal the counterfactual is the old rule still in force, which may itself have been contested. Lobbying is signed against that path, not against no lobbying; where the old rule was contested, the sign needs a premise.

1.0 clarifications.

- *Lobbying, generally.* The 0.9 note on lobbying in repeals applies to any policy: whenever the question the policy settles would have been contested on the counterfactual path (residents campaigning for a curfew, incumbents for a license), lobbying by each side is signed against that contest, and needs a premise.

1.1 clarifications.

- *Several changes in means.* Where a policy changes an actor's means in more than one way (a payment and a consumption tax), net them first, by arithmetic or with a premise, then apply "a change in means alone" to the net. The exception's means effect can run either way, with the sign of the net.
- *The same question.* For lobbying (1.0), "the same question" is the question of the policy field that was contested on the counterfactual path (support for farmers, Sunday opening), not the exact provision. Lobbying over provisions that exist only under the policy (the reading of a new clause) is an increase.

CR-5 Exchange

Statement. A voluntary exchange takes place only if each party ranks what they receive above what they give. Hence:

- (a) a policy that prevents an exchange that would otherwise have taken place deprives both parties of a gain each expected;
- (b) a policy that compels a transfer that would not have taken place voluntarily makes at least one party worse off by their own ranking;
- (c) the terms of any exchange that does take place lie within the range both parties accept, and any change in what one party must give or may receive moves that range. The terms include everything either party gives or receives, money or not: hours that count toward a qualification, training, flexibility, security of tenure.

Derivation. CR-2 applied to both parties; CR-3.

Premises. None for (a) to (c). Whether a given exchange would have taken place without the policy is the empirical question.

Source. MES ch. 2 §4 ("a double inequality of values"); HA ch. XVI §1; P&M ch. 2 §2.

0.7 clarifications.

- A gift is a transfer the giver ranks above keeping the thing given; a policy that removes the means of a gift (a ban on the production whose output was given away) removes a gain for the giver and a receipt for the recipient (CR-8b).
- (a) applies to an exchange that would otherwise have taken place. Where a policy only makes another offer more attractive, so that a buyer takes the better offer instead, the party who loses the sale loses; the buyer does not.

CR-6 Price formation and displaced pressure

Statement. At any price at which the quantity buyers wish to buy differs from the quantity sellers wish to sell (whatever the reason the price is where it is), some unsatisfied party can gain by offering better terms, and some party on the other side can gain by accepting them (CR-5). Prices therefore tend toward the level at which planned purchases and sales are compatible.

If a rule prevents the price from moving, the gain from closing the gap does not disappear. The rule may be the policy, a rule already in force (a regulated fare), or a buyer of last resort that holds the price. The same holds when a rule fixes the quantity (a quota, a cap on the number of sellers) and leaves the price free: the price then rises to ration the fixed quantity, and the gap between that price and what sellers would accept is the pressure that seeks other margins (entry outside the rule, the value of the permission). Unsatisfied parties pursue it along every margin that is still open:

- the terms that are not regulated (quality, service, conditions, timing);
- the criteria by which the scarce side chooses among the other side (queues, connections, preference for some buyers or sellers);
- exchanges outside the rule, where the expected gain exceeds the expected penalty (CR-4 with x: comply, y: violate).

The full price, meaning the money price plus everything else given up, moves toward the clearing level as far as the open margins allow. CR-6 establishes that at least one open margin is used; which ones, and whether each is used, needs a premise per margin. Without such premises, record the margins as a group with sign `at_least_one_increase` (02_ontology.md, 2.5).

What is a priori and what is not. That the pressure exists and moves to open margins is a priori. Which margins are open, and how wide, is a fact about the case: which terms are regulated, and how well the rule is enforced (`enforcement`, `binding`).

Symmetry. The same rule covers a price held too low (pressure from buyers) and too high (pressure from sellers), in any market: goods, labor, loans, housing, permits.

Derivation. CR-2, CR-4, CR-5.

Source. HA ch. XVI §1 and ch. XXX §2; MES ch. 2 §5; P&M ch. 3 §1; EIOL 1946 ch. 15 and 17.

0.7 clarifications.

- *Public supply.* Where the supplier is the rule-maker itself and supplies a fixed quantity at a set price (zero, for a free service), the unsatisfied buyers' pressure goes to the margins open to buyers sharing that quantity: waiting, crowding, time of travel. The supplier does not respond as a seller would, unless the budget rule changes. *0.8:* the mirror case is a public buyer at a fixed price with no fixed quantity (a fund paying a set fee per container redeemed): the quantity supplied to it is set by sellers' response to the fee (CR-4), and any pressure goes to the margins open to sellers (who gets the business, waiting to be paid). A private supplier that chooses not to charge (an animal rescue) is not bound by a rule; its rationing is its own choice (CR-2), and the margins (waiting, criteria) are signed as a group as under the main clause.
- *A fixed quantity of a good priced in a wider market.* Where a cap fixes the quantity of a good whose price is set in a wider market (a catch quota, a local quota on a traded good), the price of the good does not ration the capped quantity. The price (or lease price) of the right to produce or sell does. *0.8:* the wider-market price still moves in the CR-6 direction, by as much as the capped quantity matters to that market; record its sign, with a premise if the effect may be negligible. Where only part of the capped good is sold in the wider market, the part sold at home is rationed by its own price as under the main clause.
- *Margins that move in opposite directions.* Where the open margins differ in direction (quality falls while waiting rises; pay falls while a substitute restraint becomes more common), record them as a group with each member's direction stated and the sign `at_least_one_as_stated` (02_ontology.md, 2.5).
- *Illegal margins.* The price of a legal right can satisfy the "at least one margin" conclusion on its own. Illegal margins (misreporting, discarding) each need their own premise.
- *Existing contracts.* Terms fixed by existing contracts count as regulated while those contracts last. *0.8:* for at-will employment and other contracts without a fixed term, terms are regulated only for as long as changing them costs more than it gains (custom, pay scales, the cost of renegotiating): treat their stickiness as a premise with a probability, not as a rule.
- *0.8. A free price's rise is enough.* Where a margin is a free price (a rater's fee, a lease price), its rise satisfies the "at least one margin" conclusion by itself; waiting or other non-price margins then need their own premise.

0.9 clarifications.

- *Splitting a shift.* A shift in demand (or supply) gives the direction of the pressure; how it divides between price and quantity needs a premise about the other side's response (supply that can or cannot expand in the period). Sign the price and the quantity separately only with that premise; otherwise record them as a group.
- *A public seller with an open quantity.* Where the state sells at a set price and supplies whatever is demanded (a utility, a network with spare capacity), there is no excess demand to ration: the main clause applies in the markets it competes in. Buyers move to it by CR-4 and rival sellers face lower demand (CR-9), with the price and quantity split as above. Where capacity can be added from the fees it earns, but only after a lag, the public-supply clause applies until the capacity arrives.
- *A capped price with a free fee.* Where a capped price can be paid together with a fee that is not capped (a platform's charge on a capped resale), the buyer's price is the total paid. Record the capped part and the fee as two items of `price_paid` (CR-12, two items) where the policy moves them differently.

1.0 clarifications.

- *Scope of the public-supply clause.* The clause applies only where demand at the set price exceeds a fixed quantity. It then gives a group sign over the margins (waiting, crowding, pass-ups), not any one of them alone. Crowding among users of a road or vehicle below its capacity is not rationing: it is the `shared_facility` relation (CR-9, 0.7), at Tier C.
- *One public price, many costs.* Where a public seller charges one price to buyers whose cost to serve differs and rivals price by each buyer's cost, buyers sort: those whose cost exceeds the price move to the public seller, the others stay (CR-4 per buyer). Sign per class of buyer.
- *A quota on a rate.* Where a quota fixes the quantity that pays a low rate and further units pay a higher one (a tariff-rate quota), the price is set at the margin by the higher rate, as long as demand at the old price exceeds the quota.

- *Small effects are signed.* Where a chain moves a price in a wider market by an amount that may be small, record its sign; do not leave it unsigned as negligible. Its size goes in the evidence column.

1.1 clarifications.

- *A third-party payer.* A cap on what the buyer pays, with a third party paying the rest, is not a ceiling on the price the seller receives: there is no shortage from it, and the seller's receipt is set by the payer's terms. Where the payer pays the seller's own posted price up to a limit, the posted price is drawn toward the limit.
- *Margins the policy creates.* A legal exit the policy itself creates (opting out, moving to an exempt form) is one of the margins to which displaced pressure can go.
- *Labor demand.* Where demand for a seller's output changes, the labor it hires moves as a group (`employment`, `hours_worked`), not `employment` alone, unless a premise or a new activity (OC-5, 0.8) says otherwise.

CR-7 Equalization across uses

Statement. Wherever a means (labor, capital, money, land, a purchase, an activity, a moment in time) can be moved between uses that serve the same end, its owners move it from the use that yields less to the use that yields more, until the difference in yield at the margin is no greater than the cost of moving. Hence:

- (a) after any change in yields, yields on the means that can move are brought back into line, net of moving costs;
- (b) the lasting gain or loss from the change settles on the means that cannot move: specific, immobile or fixed resources, and actors without alternatives (this decides who bears a tax and who keeps a subsidy or a privilege). This is a matter of degree: where both sides have some alternatives, each bears part, and the side whose alternatives are clearly worse bears more than the other. Ranking the shares needs a premise comparing the alternatives; with three or more parties (a distributor, a retailer and a buyer), rank each pair. Without such a premise, only the direction of each side's price is signed;
- (c) the price of a durable asset adjusts until a buyer expects to earn on it what they could earn elsewhere, so a change in its expected yield is borne or captured by whoever owns it when the change becomes expected, not by later buyers. A change of limited duration is capitalized for the yields it affects. An asset the policy itself creates (a quota right, a medallion, a transferable license) is valued the same way, and its value is captured by whoever receives it at creation. "Becomes expected" is a matter of probability: as a measure becomes more likely to pass, or more likely to be renewed, the asset's price moves part of the way, so the loss or gain is shared among those who hold the asset as the probability rises. CR-7c states what happens in the market for the asset; CR-10c states why its value reflects expected yields. Cite both.

What is a priori and what is not. The direction of every movement is a priori (CR-4 for each owner). How mobile each means is, and how fast it moves, is empirical (`mobility`, `alternative_uses`, `durability`).

Symmetry. It applies to yields raised and lowered, to buyers and sellers, to the present and to future uses.

Derivation. CR-2 and CR-4 applied to every owner of a movable means; CR-10 for (c).

Source. HA ch. XIV §5 (the evenly rotating economy) and ch. XV §4; MES ch. 6 §1 and ch. 8 §1 (equalization of returns); MES ch. 7 §5 "Capitalization and Rent" (clause c); EIOL 1946 ch. 15. On incidence (clause b), see the dispute recorded under DM-1: the formulation here is the standard one, not Rothbard's.

0.7 clarifications.

- *A policy that makes a means mobile.* Where a policy removes a restraint on moving a means between uses (a ban on non-compete clauses, an end to a residence requirement), the gain from its mobility goes to the owner of the means (the worker), and the loss falls on those who held the restraint. This is (b) applied to the change in mobility itself.
- *Gross and net burdens.* Where a mandated term is worth something to the side that receives it, the burden "borne" by the paying side is the cost net of anything it recovers through the other terms. Say whether a share is gross or net.
- *A tax whose revenue buys something the payers value.* Where the payers of a tax receive part of what it funds (employers whose staff ride the transit their payroll tax pays for), their net burden is smaller than the tax; sign each side separately and combine only with a premise (CR-12e).
- *Claims before creation.* Where the claim to a right the policy will create travels with a tradable asset before the right exists (quota shares allocated to vessels), the claim's value is capitalized into that asset when the allocation rule becomes expected, and captured by its holder then. *0.8:* where the qualifying history was fixed before the measure became expected, the value is a windfall to its holders. Where actors compete to qualify for the claim (a race to build catch history), part of the value is used up in the competition (CR-8c).

0.9 clarifications.

- *Assets without a market price.* CR-7c applies to durable assets whether or not they can be sold. Inventories that take time to finish (cattle on feed, stock in process) are durable assets in this sense. For an asset that cannot be sold (a license, training), the change in its expected yield is a loss or gain by the holder's own ranking (CR-3a); a change in money earnings needs a premise about the holder's next-best use.
- *A contract term that removes an option.* Where an agreement binds an asset's owner (a team's commitment not to relocate), the asset's value falls by the value of the option given up, against whatever the agreement pays for it. Sign the net only with a dominance premise.
- *Persistence of an asset's value.* After a repeal, an asset price persists only as far as the stock its value depends on persists (homes built under the rule, a scarce license count). Give its persistence from that stock, not from who held the asset when the change was capitalized.

1.0 clarifications.

- *Which stock.* Where a policy-created permission and the capital it allowed come apart (a license and the store using it), the permission's value follows the legal count, and changes on the date the count changes; the capital's value follows the stock of capital. A permission that can no longer be sold after a repeal keeps only a value by its holder's own ranking (CR-7, 0.9).

1.1 clarifications.

- *A right sold at creation.* Where the state sells a right it creates (an auction), the price goes to the state and only the value above the price to the winner (CR-5c, CR-8b); "captured by whoever receives it at creation" applies to rights given away.
- *Alternatives taxed too.* Where a tax or subsidy reaches the buyers' alternatives as well as the asset (a wealth tax on homes, shares and cash alike), the asset's value moves with its return relative to those alternatives as the policy leaves them, not with its own return alone.

CR-8 Conservation of means

Statement. A policy does not create means; it redirects or uses them up. Hence:

- (a) every resource the policy directs to one use is withdrawn from another use its owner would have chosen (the seen and the unseen);
- (b) every payment is a receipt to someone, and every transfer adds to the recipient exactly what it takes from the payer;
- (c) resources spent on complying with, administering, enforcing or evading a policy, or on queuing, searching or lobbying because of it, are used up and serve no other end of those who spend them;
- (d) money is fungible: a sum earmarked for one purpose frees other money for other purposes, unless the other spending is also fixed;
- (e) a confiscation (seized stock, a forfeited asset) is a transfer under (b), recorded on both sides: the holder's means fall, and the state's rise by what it can realize from what it seized. Where the seized goods are destroyed, (a) applies instead: the means are used up.

Derivation. LW-1 and CR-3c.

Premises. None for (a) to (d). Which use the means would otherwise have gone to is the empirical question (`revenue_use`).

Symmetry. Applies to taxes and subsidies, imposition and repeal, public and private budgets. A rule that counts a gain to the recipient of a transfer, or to the seller of an asset that has risen in value, must count the matching loss to the payer or buyer.

Source. Bastiat 1850, §1; EIOL 1946 ch. 2 and 4; P&M ch. 5.

0.7 clarifications.

- *Residual claimants.* Where a fund pays out of what it collects (a deposit fund that refunds deposits and keeps the rest), the residual claimant's receipt can be negative; record its sign per reading or with a premise, not as a gain by default.
- *Payers from the general budget.* Where a cost is funded from the general budget, CR-8b establishes that someone pays but not who: record the payers as a group of actor classes (taxpayers, recipients of other spending, future taxpayers through borrowing) with sign `at_least_one_decrease`.
- *Borrowing.* Borrowing now implies taxes later only if the debt is expected to be repaid; that is a premise. The lenders' side of a public borrowing is recorded too: the means they lend are not available to other borrowers (OC-8).
- *Voiding a contract right.* A rule that voids a right one party held under an existing contract (an enforceable non-compete) is a transfer under (b): the holder loses the value of the right and the other party gains it. That includes deferred pay that a voided clause would have let the holder withhold.
- *0.8. Contingent payers and capped budgets.* Where a budget pays only if something happens (a general fund that covers a shortfall in a dedicated tax), record its payers per reading or with a premise on the contingency. Where a law promises more than its budget can pay, record who bears the shortfall per reading (claimants paid less or later, or a group of other payers if it is made up).
- *0.8. Value created by ending a race.* Where a policy ends a competition that used up means (a race to fish, to queue, to qualify), part of the value of the rights it creates is means no longer used up (a), which no one pays. Only the rest is a transfer under (b), with a payer to name.
- *Non-rival means.* Knowledge, techniques and know-how can be used by several actors at once without being used up. (a) applies to the time and materials used with them, not to the knowledge itself. Whether a policy that spreads knowledge raises total output therefore cannot be signed by conservation of means; it needs a premise. Whether this is a gap in the calculus is `06_gaps.md`, Q21.

0.9 clarifications.

- *The borrowing baseline.* Public borrowing is compared with the borrowing that would have taken place without the policy, including private borrowing for the same purpose (a team that would have renovated with its own loan). Where the counterfactual also borrows, the effect on other borrowers' interest rate is `ambiguous` without a premise on the difference. Borrowing repaid from the project's own fees (revenue bonds) implies later taxes only for any shortfall. Interest exempt from tax reduces tax receipts, and those who make up the receipts are payers too. The lenders' side gives no sign by itself: they would have lent elsewhere.
- *A reading that permits.* Where one reading of an unclear provision allows a source of funds without requiring it, record the payers under that reading as a group that includes the permitted source.
- *Mutual rights.* Where every member of a class both holds a right and is bound by it (owners under a covenant that each can enforce), voiding it is not a transfer between classes: each member loses the right to enforce and gains freedom from enforcement. Sign the net per member only with a premise about which they value more.

1.0 clarifications.

- *Receipts into the same budget.* Where fees or fines go into the budget that funds the policy, record them and the outlay as two items of the budget (CR-12, two items), and the net as a separate consequence, `ambiguous` unless arithmetic or a premise settles it.
- *Statutory rights.* The voiding note applies to rights held under statute (a license, a privilege) as well as under contract: the holder loses the right's value; who gains it depends on who now obtains what the right excluded, and may be spread over time.

1.1 clarifications.

- *A dedicated fund.* Where a dedicated fund is fed from the general budget, say whether the two are treated as one budget for netting; record each fund's items and the net under that choice.

CR-9 Propagation

Statement. When an actor changes what they do, the situation of everyone who would have dealt with them changes too: those who would have sold to them or bought from them, those who compete with them for the same means or customers, and those linked to them by any other relation recorded under `02_ontology.md`, 2.8: goods used together, goods bought together on one trip, goods put to a secondary use, contracts (a lease, an insurance policy, an employment contract), and the currency link (payments for imports are the means with which foreigners buy exports or assets). Apply CR-1 to CR-8 to each of these actors in turn, following those relations.

The value to anyone of a means depends on the ends it serves and on what it is used with (LW-10). So when the use of a good changes, the demand for the means that produce it and for the goods used with it changes in the same direction, and the demand for goods that compete with it in the opposite direction.

Derivation. CR-1 (one actor's choice changes the options open to those who deal with them), CR-5, LW-10.

Limits. Propagation follows exchange relations only, and stops at the depth limit in workflow step 6. Every link must name the relation it uses.

Source. Menger, *Principles of Economics*, ch. I and III; HA ch. XVI §3; MES ch. 7 §1.

0.7 clarifications.

- *The currency link runs both ways.* More imports provide foreigners with means to buy more exports or assets; more exports provide the country's residents with means to buy more imports. Exporters of different goods also compete for the same foreign buyers' means. Follow the link in whichever direction the policy moves it.
- *Shared facilities and proximity.* An actor's use of a facility others share (a road, a vehicle, a common fishing ground) changes what is available to them; an actor's activity on their own property can change their neighbors' situation (noise, smell, gifts of produce). These are relations, not exchanges, and are recorded as `shared_facility` and `proximity` (02_ontology.md, 2.8). Propagation along them follows the same rules. Whether such effects belong to praxeological analysis at all, given that the affected party does not choose them, is `06_gaps.md`, Q20; until the experts decide, chains through them are reported at Tier C.

1.0 clarifications.

- *A charge on users of a shared facility.* Where a facility's cost is recovered by a charge on its users (an airport's landing fees), a change in the number of users changes the charge per user. This is an exchange relation (the charge is a price), at Tier B, not a `shared_facility` chain.

CR-10 Time

Statement. Action takes place in time, on expectations, and present satisfaction is preferred to the same satisfaction later (LW-7). Hence:

- (a) a change expected in the future alters choices now, as far as present actions' payoffs depend on it: announcement effects, rushes before deadlines, holding back when a policy may return;
- (b) a stock (buildings, equipment, skills, inventories, a client base) changes only through flows (production, investment, maintenance, wear, conversion), so its response to a change in returns is gradual, and changes that cost something to undo persist after the cause is removed. Two cases need care in repeals. A cost already incurred under the policy (training paid for, a conversion completed) is the same with and without the repeal, so it is not a consequence of the repeal; it enters only through what it makes cheaper or dearer next. An effect that exists only during the transition (units held empty before decontrol, buyouts before a deadline) has no lasting stock to persist: record its `persistence` as `n/a`;
- (c) the value of a durable means reflects the yields it is expected to bring, discounted by time preference (with CR-7c).

Premises. What actors expect (`expectations`); how durable the stock is (`durability`); what reversal costs (`reversal_cost`).

Derivation. AX-1 (action aims at a future state), LW-7, CR-4.

Source. HA ch. XVIII §2, §5 and §8; MES ch. 6 §3 and ch. 7 §5.

CR-11 Knowledge

Statement. Actors act on what they know and believe. Hence:

- (a) a price that is prevented from moving stops conveying the scarcity it would have revealed, and those who rely on it act as if conditions were other than they are;
- (b) where an exchange is moved outside the legal market, the means by which buyers and sellers ordinarily learn about each other (advertising, reputation, brands, certification, courts) are not available to it; where it moves back into the legal market, they become available again;
- (c) a policy can change actions by conveying information (a license as a signal, a record of conviction), and only to that extent can it change what actors believe.

Derivation. AX-1 (action rests on the actor's understanding of the situation), CR-1, LW-8.

Source. Hayek, "The Use of Knowledge in Society" (1945); Mises, "Economic Calculation in the Socialist Commonwealth" (1920); HA ch. XXVI.

0.7 clarifications.

- *What a sign needs.* A consequence of information conveyed by a policy needs two premises: what the actors believed before (otherwise nothing changes), and which way those who learn something move (otherwise the direction is open).
- *Silence.* Where a policy requires some sellers to disclose and exempts others, buyers may infer something from an exempt seller's silence. That inference is information conveyed by the policy and falls under (c), with its own premise.

0.9 clarifications.

- *Sellers' response to silence.* Where buyers infer something from an exempt seller's silence, exempt sellers whose goods would compare well choose to disclose voluntarily (CR-4). Record it.
- *When a policy stops conveying something.* After a repeal, or when a signal becomes noisier, buyers' beliefs adjust as they learn; how fast is empirical. Record such consequences with `persistence: partial` and no long-run sign unless a premise gives one.

CR-12 Composition

Statement. When several chains of reasoning reach the same actor class, variable and horizon:

- (a) if all push in the same direction, that is the combined direction;
- (b) if they push in opposite directions, the combined direction is `ambiguous`, unless a premise establishes that one chain dominates (stated under (e); a dominance pattern may be cited for it where one fits);
- (c) different variables are never netted against each other;
- (d) a consequence takes the lowest tier of any link in its chain, and holds only if all its premises hold. Where the same consequence is reached by independent chains with the same sign, it holds if any one chain's premises hold: list the premises of each chain separately, and report the consequence if at least one chain clears the cut-off.
- (e) a premise that establishes dominance under (b) may be an empirical premise with a probability, an accounting identity, arithmetic on the case's figures (a cost far smaller than a revenue), or the argument that a response cannot reverse the change that causes it (sellers moving sales abroad cannot push the domestic price back below its old level, since the move stops once the yields are equal). Say which.

0.8: the response argument binds the response as a whole, not each member of a class (riders already aboard can be worse off from crowding by new riders, though new riders as a whole cannot reverse the fare cut that drew them); and it does not cover a response driven partly by a second change the policy also causes, which needs its own premise. A count over a distribution (how many homes cross a threshold) is arithmetic only if the distribution is in the case's figures; otherwise it is a premise.

- (f) where chains cancel exactly by construction (one provision undoes what another creates), the consequence is not reported, and section 7 says so.
- (g) chains at different horizons are not combined: a one-time effect in the short run and a lasting effect in the long run are separate consequences, one per horizon.
- (h) where a policy raises an activity in every window against the counterfactual, but also moves some of it from one window to another (bunching before each step-down of a subsidy), sign each window against the counterfactual and describe the bunching as a path note in section 8, with its effect on waiting or prices recorded as consequences where it has one.

Derivation. None needed: this is the logic of combining qualitative signs. The method states directions, not magnitudes, so opposing directions cannot be netted without a further premise.

0.7 clarifications.

- (g) does not forbid a separate consequence for a whole period ("over the ten years") where a dominance premise under (e) establishes its sign, such as a repayment plainly smaller than the receipt. That consequence is recorded with the horizon `whole period`, alongside the per-horizon ones.
- An opposing chain that cannot be derived from the rules is not a chain. Where a derived chain meets a suspected but underivable opposing effect, report the derived sign and record the suspected effect as a Tier C note, not as an ambiguity.
- *0.8. Three kinds of opposing chain.* (1) A chain no rule derives: a Tier C note, as above. (2) A chain the rules derive but that needs a premise the case lacks: it is a chain. State the premise with an estimated probability; if p is at least 0.5 the result is `ambiguous` without a dominance premise; if below 0.5, report the derived sign and record the chain in section 10 with its premise. (3) A chain whose own sign is `ambiguous`: it can push either way, so the combined sign is `ambiguous` unless a dominance premise covers the other chain against both of its directions.
- *0.8. Two items in one variable.* Where a policy moves two named items of the same variable in opposite directions (mandated sick time up, other benefits down, both `non_wage_compensation`), record each item as its own consequence with a qualifier, and record the total separately, combined under (b).
- The same action at different times is "the same" for timing shifts only if it is paid and performed in the same way; where the policy changes how it is paid (paid absence where it was unpaid), record the change in kind separately from any change in timing.

0.9 clarifications.

- *0.9. Tier C against Tier B.* Where a Tier C chain (through a `proximity` or `shared_facility` relation) opposes a Tier B chain on the same key, record two consequences: the Tier B chain's sign at Tier B, qualified "not counting effects through [relation]", and the combined sign at Tier C (`ambiguous` unless a dominance premise covers the Tier C chain). The Tier C chain never lowers the tier of the Tier B consequence by itself, and never raises its own tier by being combined.
- *0.9. What the response argument gives.* CR-12e's response argument shows that the result is not reversed: the price stays at or above its old level. To report `increase` rather than an increase or no change, the response must be partial: state that supply cannot fully expand at the old cost in the period, as a premise or from the case's figures.
- *0.9. Kind 3.* A chain whose own parts push in opposite directions is a kind (3) chain under the 0.8 note, unless its parts cancel by construction (CR-12f).
- *0.9. Activities that exist only under the policy.* An activity the policy creates (filings to hold a place in a queue, refund claims) is an `increase` in its window against a counterfactual of none. Say whether a timing consequence is signed on a flow (actions in the window) or on a stock (employment at a date).

1.0 clarifications.

- *1.0. What the response argument needs.* The response argument shows that yields are equalized, not that the old price returns. It gives `increase` (or `decrease`) for a price only with a premise that cost per unit does not fall as the activity contracts (constant or rising cost): with fixed costs per firm, contraction can restore equal yields at a higher price. State which total the argument bounds (new and imported cars together, not new cars alone). Where the response is a contraction by rivals, "partial" means that their inputs can be shifted to other uses only gradually. The argument covers one response; a second response to the same change, a new competitor attracted by the policy, or a stock built in advance that holds the price for a while, is a second chain to combine.
- *1.0. Tiers and dominance.* A dominance premise settles a sign; it does not raise a tier. A Tier C chain that is itself kind 3 makes the combined Tier C sign `ambiguous` unless the dominance premise covers both of its directions.
- *1.0. The baseline of patterns.* Every pattern's conclusion is against the no-policy counterfactual. Where a pattern compares with anything else (another policy, the allocation before trading, uniform enforcement, a license on one requirement), re-sign its conclusion against the counterfactual before using it.

1.1 clarifications.

- *1.1. The response argument in every direction.* The 1.0 note is written for a contraction raising a price. The mirror cases hold: an expansion that lowers a price is bounded by the old price from above, with a premise that cost per unit does not fall as the activity expands; a price fall is bounded the same way. The argument may bound a counterfactual total (all installations, new and imported cars). Responses within the named total (imports answering the same price) are part of the response; responses through other markets are second chains.
- *1.1. Items that cancel.* Where two items of a variable cancel by construction and nothing else moves the variable, CR-12f applies and no total is recorded. Where a third item remains, record the items and the total. Per-window signs (h) may be summed into a whole-period total under the same rule.
- *1.1. Group members are not chains.* A member of an `at_least_one` group cannot be used as a signed chain in another combination; only the group as a whole carries a sign.
- *1.1. Ratios.* Arithmetic under (e) covers ratios as well as levels where the figures are in the case.

How the earlier laws relate to the core rules

Three of the laws in `01_foundations.md` are results of the core rules rather than independent premises: LW-3 (law of demand) is CR-4c applied to buyers, with CR-4's exception; LW-4 (resources move toward higher returns) is CR-7; LW-6 (non-price rationing) is CR-6. They keep their IDs

because patterns and worked examples cite them.

Map from workflow steps to core rules

Workflow step	Core rules applied
1, 2. State and classify the policy	CR-1
3. Actor classes	CR-1 (whose situation changes), CR-9 (whose situation changes in turn)
4. Direct effect	CR-1, CR-2
5. Responses	CR-4 for every pair of options each actor class has; CR-6 where a price is fixed; CR-5 for prevented or compelled exchanges
6. Second-order effects	CR-9, then CR-5 to CR-7 in each market reached
7. Combine	CR-12
8. Time structure	CR-10
9. Opportunity costs	CR-3, CR-8
10. Conditions	The premises each core rule names

4. Rule catalog 1.1: patterns

This file holds the **patterns**: derived results for recurring situations. They rest on the **core rules** in `04a_core_rules.md`, which are the universal calculus. Every pattern below carries a **Derivation** line that shows which core rules it instantiates and how; a pattern adds no reasoning of its own beyond that derivation and the premises it names. If a pattern and a direct derivation from the core rules ever disagree, the derivation wins and the pattern is corrected.

Each pattern is one reusable inference. Format:

- **If:** the preconditions, stated as facts about the policy record or earlier consequences.
- **Then:** the conclusion, as a consequence in normal form (actor class, variable, sign).
- **Assumes:** empirical premises, each of which needs an estimate in the case at hand.
- **Unless:** known exceptions that block the rule.
- **Tier / visibility / horizon:** proposed defaults. A rule's tier is the tier of its conclusion when its preconditions hold; it is A only if it has no empirical assumptions.
- **Builds on:** foundations from `01_foundations.md`.
- **Source:** where the reasoning comes from, in the short forms of `07_sources.md`, which records how each citation was checked.
- **Derivation:** how the pattern follows from the core rules (`04a_core_rules.md`).

Every conclusion below is stated against the no-policy counterfactual (CR-12, 1.0). Where a pattern says "relative to" anything else, re-sign it before use.

Patterns are deliberately small. A full analysis chains several of them, and each link is reviewed on its own.

Changes in 1.1. From the third held-out test (WE-056 to WE-065) and its core-only re-run (`08_core_rerun_report.md`, fifth round). Re-based on the no-policy counterfactual: RS-17, RS-23, PC-17, EN-13. Corrected: SB-4 (taxes that reach the alternatives), PC-1 and PC-16 (a third-party payer), TX-7 (a matched domestic tax), TX-9 (revenue returned to the payers), PC-17 (buy-outs paid to the counterparty; product sales), EN-14 (a standard buyers value), SB-8, TX-1. Widened: SB-2, TX-2, TX-3, RS-6, TS-3, TS-10, RS-4, RS-7, RS-16, RS-21, RS-22, PC-11. New: PC-20 (a cap on the buyer's share). Promoted after a valid second use: RS-24 (a fixed cost per item reduces variety). Candidates recorded: allocation of a right by sale, a mandated add-on to an unregulated price, a tax on the value of a fixed factor.

Changes in 1.0. From the second held-out test (WE-046 to WE-055) and its core-only re-run (`08_core_rerun_report.md`, fourth round). Re-based on the no-policy counterfactual: RS-20, EN-11, EN-13. Corrected: PC-17 (a charge refunded on the act), RS-16 (prices per unit of quality; fixed costs on the substitute), TX-7 (two-rate tariffs), EN-14 (thinning complements), EN-6 (caps on sellers; tariff-rate quotas), OC-8 (small effects are signed; segments; interest deductions). Widened: EN-1, EN-2, PC-19, RS-7, RS-8, RS-13, RS-17, RS-18, TS-5, OC-1. Promoted after a valid second use: RS-21 (high-grading under a cap in physical units), RS-22 (compensating differential), RS-23 (selection under a uniform price), TS-10 (scheduled tightening faster than the trend), TX-12 (deposit-refund). Not promoted: "tariff-like burden without a tax" (its second use was a different move).

Changes in 0.9. From the held-out test (WE-036 to WE-045) and its core-only re-run. Corrected where a pattern gave a wrong conclusion on a new case: RS-17 (value of firms near a threshold), RS-18 (caps per firm), SB-10 (domestic price; drawback; checks on the return crossing), EN-11 (licenses issued on different requirements), OC-8 (baseline borrowing, revenue bonds, tax-exempt interest), PC-8 (demand withdrawn from the uncontrolled segment). Widened: PC-17 (a mandate with a buy-out charge), PC-19 (a public seller with an open quantity), RS-4, RS-7, RS-8, RS-9, RS-16, OC-3, OC-5, EN-6, TX-8. Promoted from candidates after a second use: RS-19 (round trips), RS-20 (complaint-only enforcement), OC-9 (debt-funded transfer in two windows), EN-12 (allocation of a right by history), EN-13 (consolidation under tradable rights), EN-14 (standard on new units, stock exempt), EN-15 (disclosure mandate). "Fixed cost per item reduces variety" was not promoted: its second use was invalid (`08_core_rerun_report.md`, third round).

Changes in 0.8. From the core-only re-runs of WE-026 to WE-035 (`08_core_rerun_report.md`, second round). Revised: OC-3 (enforcement the policy makes unnecessary), OC-1 and TX-1 (payers from a general budget are a group), RS-1 (the split between price and quantity of a substitute needs a supply premise), OC-5 (new activities), TS-3 (product standards), EN-6 (the wider-market price still moves), SB-10 (checks on the return crossing), RS-8 (opposing timing chains).

Changes in 0.7. From the ten gap-probing examples (WE-026 to WE-035). New patterns for moves that are general or recurred: PC-17 (mandated contract term), PC-18 (prohibited contract term), PC-19 (public supply at a set price), RS-17 (size thresholds), RS-18 (a per-head cost with a cap), OC-8 (crowding out by public borrowing), SB-10 (export subsidy). Widened: TX-8 (both directions of the currency link), RS-4, RS-8, RS-16, OC-1, OC-4, SB-8, EN-6. Moves seen once are listed as candidates in `06_gaps.md`, 6.2e.

Changes in 0.6. New patterns for moves that recurred in the core-only re-run without a pattern to prompt them: PC-16 (a ceiling as a transfer), RS-16 (a substitute's price after a demand shift), EN-11 (a license as a signal). RS-8 and SB-6 now state both sides of a timing shift.

Changes in 0.5. The core-only re-run of the 25 worked examples (`08_core_rerun_report.md`) found no claim the core rules could not derive, but found patterns that overclaim or lack a clause. Revised: EN-3, EN-4, DM-1, DM-2, RS-6, TX-3, RS-7, RS-10, PC-2, PC-8, OC-3, SB-1, SB-3, SB-4.

Changes in 0.4. The core rules (CR-1 to CR-12) are introduced and every pattern is given a derivation from them. Writing the derivations exposed four problems, now fixed: TX-6 stated as a priori a direction that depends on an empirical premise (CR-4's exception); OC-7 counted the gain to an asset's seller without the matching outlay of its buyer (CR-8 symmetry); RS-13 covered added supply but not withdrawn supply (symmetry); TS-4 cannot be derived at all and is quarantined.

Changes in 0.3. New: RS-12 to RS-15, OC-5 to OC-7, TS-5 to TS-9, PC-12 to PC-15, TX-10, TX-11, SB-8, SB-9, EN-9, EN-10, DM-4, DM-5. Widened: RS-4, RS-10, RS-11, TS-1, TS-3, PC-8, TX-6, TX-8, TX-9, SB-4, EN-6, EN-8, DM-1. Each change traces to a gap found in the worked examples (`06_gaps.md`, 6.2).

RS: Generic responses

These apply to any intervention. Workflow step 5 runs through them for every actor class.

RS-1 Substitution

- **If:** a policy raises the cost (money or otherwise) to an actor class of using a good or means, relative to an alternative that serves the same end.
- **Then:** `actor_class.quantity_substitute` increase; use of the target decreases at the margin.
- **Assumes:** a substitute exists and is not covered by the policy (`substitutability`). CR-4 gives only that more buyers turn to the substitute; `quantity_substitute` increase as a traded quantity also needs a premise that the substitute's supply can expand, and its price is then RS-16's question.
- **Unless:** the actor class has no margin of choice (a fixed contract during its term).
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** LW-2, LW-3.
- **Derivation:** CR-4 (x: the target, y: a substitute serving the same end).

RS-2 Adjustment of quality and terms

- **If:** a policy fixes the money price or one term of an exchange below (or above) what the parties would agree.
- **Then:** the party disadvantaged adjusts unregulated terms in its own favor: `quality_per_unit`, `maintenance_spending`, service, contract length, or added conditions move against the other party.
- **Assumes:** some terms of the exchange are unregulated and adjustable.
- **Unless:** every relevant term is fixed and enforced.
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** AX-1, LW-5.
- **Source:** HA ch. XXX §2; EIOL 1979 ch. 18 "What Rent Control Does" (maintenance; not in the 1946 edition).
- **Derivation:** CR-6 (pressure displaced to unregulated terms) with CR-5c.

RS-3 Evasion

- **If:** complying with a rule costs an actor more than the expected cost of breaking it (probability of detection times penalty, plus moral and reputational cost).
- **Then:** `informal_activity` increase; `side_payments` increase where the rule fixes a price.
- **Assumes:** enforcement is imperfect (`enforcement`).
- **Tier** B. **Visibility** invisible. **Horizon** short.
- **Builds on:** AX-1, LW-1.
- **Derivation:** CR-4 (x: comply, y: violate; the expected penalty is a burden attached to y, CR-1).

RS-4 Relocation across a boundary

- **If:** the policy applies within a boundary, and the activity can be carried out or bought on the other side of it.
- **Then:** activity and purchases move to whichever side the policy favors. Where the policy burdens activity inside: `cross_border_purchases` increase (buyers), or sellers and production relocate outside; activity inside decreases. Where it favors activity inside (a floor, subsidy or exemption that applies only there): `entry_rate` increase inside and `quantity_supplied` decrease just outside.
- Where a payment inside the boundary depends on an item crossing it (a refund, a subsidy on exports), items are drawn across it, including round trips made only to claim the payment.
- **Assumes:** the boundary is near enough for crossing to be worth its cost to some actors (`mobility`).
- **Reach of enforcement (0.9).** Where the law claims to reach acts beyond the boundary (resales of a local event wherever made) but can be enforced only within it, the boundary that matters is the limit of enforcement.
- **Two chains (1.1).** Where the policy favors one side on some terms and the other side on others, sign the net under CR-12b.
- **Tier** B. **Visibility** invisible. **Horizon** short (purchases) to long (firms).
- **Builds on:** RS-1.
- **Derivation:** CR-4 (x: act on this side, y: act on the other side), then CR-7 for the market on each side.

RS-5 Reclassification

- **If:** the policy's definitions or exemptions draw a line between covered and uncovered goods, contracts or actors.
- **Then:** `reclassification` increase: goods, contracts and organizations are reshaped to fall on the favorable side (for example, conversion of rental units to owner-occupied units; reformulating a product below a tax threshold).
- **Assumes:** reshaping costs less than the policy's burden for some actors.
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** AX-1.
- **Derivation:** CR-4 (x: the covered form, y: an uncovered form that serves nearly the same end).

RS-6 Exit and reallocation

- **If:** the policy lowers the expected return of using resources in the target activity, relative to other uses.
- **Then:** `investment_target` decrease, then `quantity_supplied` and `capital_stock_target` decrease; `number_of_sellers` decrease or `entry_rate` decrease; labor in the activity: `employment` and `hours_worked` as a group, `at_least_one_decrease` (sellers may cut hours rather than staff). Resources go to other uses, where quantities rise.
- **Assumes:** resources have other uses (`alternative_uses`); speed depends on how specific and durable the resources are (`durability`; see TS-1).
- **Mirror (1.1).** A tax removed or cut reverses these moves.
- **Tier** B. **Visibility** invisible. **Horizon** medium to long.
- **Builds on:** LW-4.
- **Derivation:** CR-4 for each owner (x: this use, y: another use), aggregated by CR-7; timing by CR-10b.

RS-7 Lobbying and rent-seeking

- **If:** the policy creates a gain that is concentrated on an identifiable group and depends on political decisions (renewal, extension, exemption, enforcement).

- **Then:** `lobbying_effort` by that group increase. Those concentrated groups harmed by the policy also lobby, for repeal or exemption.
- **Assumes:** the group can organize at a cost lower than its stake (`other`). This premise must be stated and estimated in each case, like any other.
- **Unless:** the counterfactual also has lobbying on the same question (a repeal, where the policy's beneficiaries would lobby to keep it in force). Then compare with that: the side that gains from the change lobbies more, the side that loses may lobby more or less, and the combined sign for a class that lobbied in both worlds is `ambiguous` without a premise.
- **Repeals (0.9).** In a repeal, whether the old rule would have been contested on the counterfactual path is a premise, not a given (CR-4, 0.9).
- **1.0.** The Unless clause covers impositions as well as repeals: wherever the question was contested on the counterfactual path, lobbying is signed against that contest (CR-4, 1.0).
- **The same question (1.1).** See CR-4, 1.1.
- **Tier** B. **Visibility** invisible. **Horizon** medium.
- **Builds on:** AX-1, LW-1.
- **Derivation:** CR-4 applied to political action (x: accept the decision, y: spend means to change it). The payoff to y rises with the concentrated gain or loss at stake.

RS-8 Change of timing

- **If:** the policy treats the same action differently depending on when it is taken (announcement before effective date, phase-in, deadline, expected tightening or repeal).
- **Then:** actions are moved toward the favorable time: rush before a tax or cap takes effect, delay until after a repeal is expected, purchases pulled forward into a temporary subsidy. Every such move has two sides: the action rises in the favored window and falls in the window it was taken from. Record both, each signed against the counterfactual (`02_ontology.md`, 2.5).
- **Unless:** where a stock must be run down to a common deadline with no grandfathering (a phase-out), there is no favorable window to move toward, and the sign before the deadline is `ambiguous` rather than a rush. Where a claim on a future payment can be acquired before the charge that funds it starts (containers refundable before deposits are charged), holders acquire it in the gap. Where the policy also improves the earlier window (a clause applying the new rule to cases already pending, so that acting before the effective date is already partly protected), the two timing chains oppose each other. Check every provision for this before signing: the sign in that window is `ambiguous` without a dominance premise.
- **Assumes:** actors know the timing and can shift their actions.
- **More timing chains (0.9).** The Unless clause also covers an opposing chain from a second incentive the policy creates or meets (a leasing season), and cases where sellers and buyers move in opposite directions in time. Sign each window by combining all the timing chains under CR-12b.
- **Races and rivals (1.0).** A race to qualify adds activity in the window without taking it from another window; a rush whose other side is rivals in the same window (near-line sellers taking jobs from others) is recorded on both classes.
- **Tier** B. **Visibility** invisible. **Horizon** short.
- **Builds on:** LW-7.
- **Derivation:** CR-10a with CR-4 (x: act at one time, y: act at another).

RS-9 Bundled trips carry other purchases with them

- **If:** buyers of a good change where they buy it (RS-4 cross-border buying, or a switch to another store or channel), and they usually buy other goods on the same trip (a `complement` relation noted `trip`).
- **Then:** the other goods bought on those trips move with it: `quantity_traded` of untaxed or unregulated goods decrease at the sellers left behind and increase at the sellers gained.
- **Assumes:** buyers bundle purchases on trips (`other`); the destination sells the other goods too.
- **Location complements (0.9).** The same move applies to purchases tied to a location rather than a trip (visitors who shop where they stay). When a policy changes who occupies the location, record the new occupants' purchases as the opposing chain.
- **Tier** B. **Visibility** invisible. **Horizon** short.
- **Builds on:** LW-1 (a trip has a cost, so buyers spread it over several purchases), RS-4.
- **Derivation:** CR-3c (a trip is a joint cost spread over the purchases made on it), so CR-4 moves the other purchases with the trip; CR-9 carries the change to sellers.

RS-10 Driving trade into the informal market

- **If:** some trade in a good or service moves outside the legal market (RS-3, PC-6, EN-3, EN-7).
- **Then:** for that trade: `search_cost` increase (sellers cannot advertise openly); `quality_information` decrease and `quality_variability` increase (no brands, licensing, reviews or recourse to rely on); `disputes_settled_outside_courts` increase; sellers charge for the risk of punishment. In reverse (trade returning to the legal market after a repeal), `search_cost` and `disputes_settled_outside_courts` decrease and `quality_information` increases; `quality_variability` is `ambiguous` if the repeal also admits sellers who were previously excluded by a training or quality requirement.
- **Assumes:** penalties for informal trade are enforced at least sometimes (`enforcement`).
- **Tier** B. **Visibility** invisible. **Horizon** short.
- **Builds on:** AX-1, LW-1.
- **Derivation:** CR-1 (the options of advertising, open reputation and court enforcement are removed for illegal trade), CR-11b (less information), CR-7 (sellers' net yield is equalized, so the price includes compensation for penalty risk).

RS-11 Lock-in spills into linked markets

- **If:** an actor class holds a non-transferable benefit that is lost by a change of location or status: moving (PC-7: a controlled tenancy, a place on a list), leaving an occupation, or letting a registration lapse (a grandfathered status).
- **Then:** members of the class turn down opportunities that would cost them the benefit: `job_changes` decrease, longer commutes, forgone breaks or better work elsewhere.
- **Assumes:** some valued opportunities would require the change (`mobility`); the benefit is not transferable.
- **Tier** B. **Visibility** invisible. **Horizon** medium.
- **Builds on:** LW-1 (losing the benefit is part of the cost of moving), PC-7.
- **Derivation:** CR-3 (moving or changing status now also costs the forfeited benefit), then CR-4 (x: take the opportunity, y: keep the benefit).

RS-12 Complement spillover

- **If:** the quantity traded, or the stock in use, of a good changes, and another good is recorded as its `complement` in use (not `trip`): cars and repairs, cars and fuel.
- **Then:** `quantity_complement` changes in the same direction; where the complement's supply is slow to adjust, `price_complement` moves the same way too.
- **Assumes:** the complement relation is close, and the complement has few other uses (`substitutability`).
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** LW-2.
- **Derivation:** CR-9 with LW-10 (complements: the value of a means depends on what it is used with).

RS-13 Supply spillover

- **If:** a policy adds quantity supplied to a linked market (exempt new building next to a capped segment, subsidized output fed into a wholesale market, released public stocks), or withdraws it (units removed from a segment, production stopped).
- **Then:** the price in that market decreases where supply is added, and increases where it is withdrawn (`price_substitute`, or that market's `price_paid`), compared with the counterfactual.
- **Assumes:** the added supply is not offset by withdrawal elsewhere; buyers there do not absorb it at an unchanged price (`substitutability`).
- **1.0.** Where the added supply draws inputs from other uses, record the crowding out of other output and combine; the response argument (CR-12, 1.0) bounds the price only with its premises.
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** LW-3. This is the general form of the opposing chain noted under PC-8.
- **Derivation:** CR-9 then CR-6 in the linked market: added supply at an unchanged price leaves sellers unsatisfied, so the price falls; withdrawn supply is the mirror case.

RS-14 Input cost pass-through

- **If:** a policy raises the cost of an input (a tax or floor on labor, a tariff on a material, a regulated price) that is a material part of the cost of some output.
- **Then:** for the output: `price_paid` by its buyers increase and/or `quantity_supplied` decrease (at least one). Where the sellers' competitors face the same increase, the price rise dominates; where competitors are exempt (imports, untaxed or uncovered sellers), RS-6 applies and the quantity effect dominates.
- **Assumes:** the input cannot be fully replaced at the old cost (`substitutability`); the competitors' exposure (`other`).
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** LW-4, TX-2. The upstream counterpart, for a cap on the output, is PC-12.
- **Derivation:** CR-9 with LW-10 (an input's price is part of the cost of the output), then CR-7: where competitors face the same cost, no seller can avoid it and the price of the output moves; where they do not, the exposed sellers are the ones that must move (CR-4, exit).

RS-15 Discretionary relief is rationed by non-price criteria

- **If:** relief from a policy's burden (an exclusion, waiver, permit or exemption) is granted case by case at officials' discretion, and is worth more to applicants than its fee.
- **Then:** `allocation_by_non_price_criteria` increase among applicants; `compliance_cost` of applying increase; `lobbying_effort` increase, both by applicants and by those who gain when relief is denied.
- **Assumes:** the grant is discretionary and valuable (`other`).
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** LW-6 (relief is scarce and not priced), RS-7.
- **Derivation:** CR-6 (relief is scarce and not priced, so it is allocated by other criteria) and CR-4 applied to applicants and to political action.

RS-16 A substitute's price after a demand shift

- **If:** a policy moves buyers from a good to a substitute (by taxing, banning, capping or licensing the good), outside any price control on the substitute.
- **Then:** `quantity_substitute` increases, and `price_substitute` increases where the substitute's supply cannot expand at an unchanged cost in the period considered.
- **Assumes:** `substitutability`; the substitute's supply is not perfectly elastic in the period (`durability`, `alternative_uses`).
- **Unless:** the policy also adds supply to the substitute's market (RS-13), in which case the price is `ambiguous` without a premise.
- **Mirror case:** a policy that makes a good cheaper or more available (a free public service) pulls buyers from its substitutes: `quantity_substitute` decreases and `price_substitute` decreases where their supply cannot contract at once.
- **Costs on the substitute (0.9).** Where the policy also attaches a cost to the substitute itself (a record-keeping charge per domestic animal), that cost is an opposing chain on the substitute's net price to its sellers; combine under CR-12b.
- **Per unit of quality (1.0).** Sign the substitute's price per unit of a given quality or risk class: a change in the mix of buyers or goods is not a price change. Only a cost per unit sold opposes sellers' net price; a cost fixed per facility bears on the number of facilities (CR-4, 0.8); a cost on owners (a road charge) combines with quantity bought, not with sellers' net price.
- **Waiting (1.1).** Where the substitute's market is rationed partly by waiting (dealer queues), its price and wait are a group.
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** LW-3.
- **Derivation:** CR-9 (buyers turn to the substitute) then CR-6 in the substitute's market (at the old price, buyers wish to buy more than sellers wish to sell, so the price rises as far as supply cannot expand). PC-8 is the same move for a controlled market.

RS-17 Size thresholds

- **If:** a policy applies only above a size threshold (outlets, employees, revenue), measured at a date or over a past period.
- **Then:** firms near the threshold stay below it (`number_of_sellers` just below the threshold increases relative to just above), split activity across separately counted units (second brands, separate entities: `reclassification` increase), and the value of growing past the threshold falls (CR-7c: `asset_value` of firms near it decreases). Where the threshold is measured over a past period, the response comes in that period, before the policy takes effect.
- **Assumes:** the cost of the policy is material relative to the gain from growing (`binding`); splitting is legal and cheap enough (`other`).

- **Value of firms near the line (0.9).** Against the no-policy counterfactual, firms just below a threshold also gain from the burden on their rivals above it, so their `asset_value` is `ambiguous` unless a premise says which dominates. The same clause applies to a line on hours or any other measure ("job" for "firm").
- **Notches, mirrors, eligibility (1.0).** Distinguish a notch (the burden jumps on the whole base) from a kink (it applies only above the line). The mirror case, a burden below the line (a training requirement grandfathered above an experience line), is signed the same way. Losing eligibility for a program at the line is a further cost of crossing it.
- **Baseline (1.1).** Where this pattern says "relative to just above the line", re-sign against the counterfactual: firms below the line bear no burden from it, so their response is signed against no policy, and firms above against no policy too. A line above which holding more is forbidden (a cap per bidder) is a prohibition, not a burden: it removes the option of holding more (CR-1).
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** LW-2.
- **Derivation:** CR-1 (the operation applies only above the threshold), CR-4 (x: grow past it, y: stay below or split), CR-7c, CR-10a.

RS-18 A per-head cost with a cap

- **If:** a policy imposes a cost per worker (or per unit) that stops at a cap (a contribution up to a ceiling, a leave entitlement up to a maximum).
- **Then:** below the cap the cost raises the price of every hour and every head; above it, an extra hour from an existing worker adds nothing while an extra worker adds the full cost. Employers shift hours toward fewer heads: `hours_worked` per worker increases and `employment` decreases where workers are above the cap, as a mixed group.
- **Assumes:** hours and heads are substitutes in production (`substitutability`).
- **Caps per firm (0.9).** Where the cap is per firm rather than per head (a penalty capped at a sum per firm), once a firm is above the cap its cost no longer varies with heads: it is a fixed cost (CR-4, 0.8), and there is no shift of hours toward fewer heads above the cap. One switch can move both hours and heads; sign each where CR-4 gives it, rather than as a group.
- **Any cost capped per firm (1.0).** The per-firm cap clause applies to any cost capped per firm, per crown of payroll as well as per head.
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** LW-2.
- **Derivation:** CR-4 (x: another worker, y: more hours from existing workers) with the cost attached only to x above the cap.

RS-19 Round trips under a payment for crossing a boundary

- **If:** a payment is made (or a charge avoided) when a good crosses a boundary in one direction, and the cost of bringing it back is less than the payment.
- **Then:** goods are sent across and brought back to claim the payment: `quantity_traded` across the boundary increases with no change in final use; resources are used up in the trips (CR-8c, `compliance_cost`); the payer's outlay rises (SB-1). A check on the return crossing reduces round trips from what they would be (CR-4), never below none.
- **Assumes:** the return crossing is not checked, or checked imperfectly (`enforcement`).
- **Tier** B. **Visibility** invisible. **Horizon** short.
- **Derivation:** CR-4 (gain per round trip against its cost), CR-8c. Seen in WE-030 and WE-042.

RS-20 Enforcement only on complaint

- **If:** a rule is enforced only when a protected or affected party complains.
- **Then:** the expected penalty for breach differs by counterparty: breach is concentrated where complaint is least likely (short tenure, ignorance of the law, fear of retaliation, absent neighbors), and `legal_breach` increases there relative to uniform enforcement. Parties who can complain gain a bargaining threat. Where complaint is costless and can be strategic, complaints also rise for reasons unrelated to breach.
- **Assumes:** the likelihood of complaint differs across counterparties (`enforcement`).
- **Baseline (1.0).** Against no policy, breach of a new rule increases from none wherever enforcement is less than certain. "Concentrated where complaint is least likely" describes where the breach that occurs falls, not a comparison with uniform enforcement.
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Derivation:** CR-4 (the expected cost of breach is the penalty times the chance of a complaint), CR-5c. Seen in WE-029 and WE-044.

RS-21 High-grading under a cap counted in physical units

- **If:** a cap, quota or right counts units (tonnes, movements, licences per tonne) whose value per unit varies, and the actor chooses the mix or the size of each unit.
- **Then:** the value per counted unit rises: high-value fish are landed and low-value ones discarded, larger aircraft fly each slot, white sugar fills tonnage licences rather than raw. `quality_per_unit` (or `unit_size`) increases; the counted good's lower-value varieties fall.
- **Assumes:** the mix or size can be changed at a cost below the gain (`alternative_uses`).
- **Charges counted in units (1.1).** The same move applies to a tax or payment counted in physical units; under a count by weight, items become lighter, and under a payment per tonne, output shifts toward lower-value tonnes.
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Derivation:** CR-4 with the unit of the constraint (x: fill the unit with a higher-value variety; y: a lower-value one). Seen in WE-028, WE-047, WE-054.

RS-22 Compensating differential from a non-wage term

- **If:** a policy changes a term of a job other than the wage (the cost of reaching it, its schedule, its security) for some jobs and not others.
- **Then:** for jobs made worse, `wage_rate` or other terms move in workers' favor, or those jobs are harder to fill, as a group (`at_least_one_as_stated`); for jobs made better, the reverse.
- **Assumes:** workers can move between jobs (`mobility`).
- **1.1.** Not for terms the employer changes as a response to the policy (those are PC-17's adjusting terms).
- **Tier** B. **Visibility** invisible. **Horizon** medium.
- **Derivation:** CR-5c, CR-7a. Seen in WE-032, WE-046.

RS-23 Selection under a uniform price

- **If:** a policy fixes one price or term for counterparties whose cost to serve differs, while their alternative is priced by each counterparty's cost.
- **Then:** those whose cost exceeds the uniform price take it; the others take the alternative. The provider's cost per counterparty rises; under the alternative, the remaining pool is cheaper to serve. Sign per class.
- **Assumes:** the alternative is priced by each counterparty's cost (`other`).
- **Baseline (1.1).** "The remaining pool is cheaper to serve" compares two pools under the policy. Against no policy, sign the alternative provider's cost per counterparty only with a premise on what the pool would have been.
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Derivation:** CR-4 per counterparty, CR-6 (1.0). Seen in WE-029 and, on the buyers' side, WE-049.

RS-24 A fixed cost per item reduces variety

- **If:** a policy attaches a cost fixed per item, model or version offered (testing, certification, labeling, supplying parts for each model), and some items sell in small numbers.
- **Then:** sellers drop the items whose sales do not cover the fixed cost: `variety` decreases, against a counterfactual in which those items are offered.
- **Assumes:** some items' sales are small relative to the fixed cost (`other`).
- **Tier** B. **Visibility** invisible. **Horizon** medium.
- **Derivation:** CR-4 (0.7, fixed cost per item: x offer, y drop). Seen in WE-035 and WE-059.

OC: Opportunity cost

Opportunity costs are recorded as signed changes in variables (`02_ontology.md`, 2.5), never as a separate "loss" entry.

OC-1 Diverted resources

- **If:** the policy takes resources from an actor class (tax, mandated purchase, compliance) or directs resources to a use.
- **Then:** `resources_available_other_uses` decrease for that actor class; the goods they would have bought or produced with those resources are not produced, so `quantity_traded` decrease in the markets where they would have spent it. The spending or use is visible; the forgone alternative is not.
- **Naming the forgone uses.** Where means are taken directly (a tax, a levy), which uses are given up needs a premise (CR-4, a change in means alone). Where they are taken through a market (public borrowing draws on the loan market), the forgone uses are those of that market's marginal participants (the borrowers crowded out), which can be named with a premise about who they are.
- **Who pays.** Where the means come from a general budget and the record names no tax, the payers are a group (taxpayers, recipients of other spending, future taxpayers) with `at_least_one_decrease`; do not name taxpayers alone. Where the means are borrowed, the lenders' side is recorded too (OC-8).
- **Saving is a use.** For payers who may cut saving rather than purchases, `quantity_traded` falls in named markets only with a premise (CR-4, 0.7).
- **Receipts into the same budget (1.0).** Where fees or fines flow into the budget that pays, record both and the net (CR-8, 1.0).
- **Tier** A, given that resources are taken. **Visibility** invisible. **Horizon** short.
- **Builds on:** LW-1.
- **Source:** Bastiat 1850, §1 "The Broken Window"; EIOL 1946 ch. 2 to 4.
- **Derivation:** CR-8a.

OC-2 Unpriced time and effort

- **If:** the policy causes actors to spend time or effort that buys nothing they value for its own sake: queuing (PC-2), searching, paperwork, qualification hours beyond what buyers would demand (EN-1).
- **Then:** `wait_time`, `search_cost` or `compliance_cost` increase; the forgone uses of that time are an invisible cost.
- **Tier** A, given the time is spent. **Visibility** invisible. **Horizon** short.
- **Builds on:** LW-1.
- **Derivation:** CR-3c and CR-8c.

OC-3 Administration and enforcement

- **If:** the policy must be administered or enforced.
- **Then:** `enforcement_cost` increase for `officials.enforcement`. Who bears it depends on the funding, which must be stated:
  - from a budget that rises with the new task: `taxpayers` (OC-1 applies);
  - from a fixed budget: enforcement of other rules falls instead (CR-8a), and the cost falls on those whom that enforcement protected;
  - from the policy's own revenue: it reduces what the revenue does elsewhere (CR-8d), not general taxpayers' means.
  - from a general budget where the record names no tax: the payers are a group (taxpayers, recipients of other spending, future taxpayers through borrowing) with `at_least_one_decrease` (CR-8, 0.7).
- **Against the right baseline.** The comparison is enforcement without the policy, not zero. Where the policy shrinks an activity that was itself enforced (complaints about backyard flocks under a nuisance law), enforcement of the old rule falls; the net sign is `ambiguous` without a dominance premise (arithmetic on the case's figures may supply one).
- **Fixed budget, more precisely (0.9).** From a fixed budget, the resources come from some other use of the agency's time: slower refunds, slower clearance, fewer other checks. Record it as a group with `at_least_one_as_stated`, not as less enforcement of other rules by default.
- **Tier** A, given positive enforcement activity and no enforcement the policy makes unnecessary; otherwise B. **Visibility** invisible. **Horizon** short.
- **Builds on:** LW-1.
- **Derivation:** CR-8c, with CR-8a for the funding.

OC-4 Prevented exchanges

- **If:** the policy prevents exchanges that buyers and sellers would otherwise have made (a ceiling, a floor, a license requirement, a prohibition, a tax that prices some buyers out).
- **Then:** `quantity_traded` decrease for the prevented exchanges; both parties to each lose the gain they expected from it. Name both sides.
- **Unless:** the exchange is not prevented but lost to a better offer the policy creates elsewhere (a subsidy draws a buyer to another seller). Then only the party who loses the sale loses (CR-5).
- **Tier** A (ex ante, by each party's own ranking). **Visibility** invisible. **Horizon** short.
- **Builds on:** LW-5.
- **Source:** P&M ch. 2 §2 and ch. 3 §1.
- **Derivation:** CR-5a.

OC-5 Diverted spending diverts work (Hazlitt's tailor)

- **If:** a policy changes where an actor class spends (OC-1: money taken and spent by others; SB-8: money freed), so that `quantity_traded` falls in some markets and rises in others.
- **Then:** `employment` or `hours_worked` decrease in the markets that lose the spending and increase in those that gain it. The jobs the policy creates are visible; the jobs it prevents elsewhere are not.
- **Assumes:** sellers adjust their labor to their sales (`alternative_uses`).
- **New activities.** Where the activity that gains is new (a redemption center, a new service), it starts from zero, so hours cannot substitute for jobs: `employment` increase alone is derivable there. The group form applies to existing activities.
- **New to the place (0.9).** "New" means new to the place considered: activity that moves from one place to another is new at the destination and lost at the origin. Apply the clause per location.
- **Tier** B. **Visibility**: the gain is visible, the loss invisible. **Horizon** short to medium.
- **Builds on:** OC-1, LW-4.
- **Source:** EIOL 1946 ch. 2 (the tailor who does not get the order for a suit) and ch. 4 ("for every public job created ... a private job has been destroyed somewhere else"); Bastiat 1850, §1, where the forgone purchase is shoes or a book.
- **Derivation:** CR-8b (spending diverted from some sellers is received by others), then CR-9 with LW-10 (demand for labor is derived from demand for what it produces).

OC-6 Lost revenue is made up elsewhere

- **If:** a repeal or cut reduces `tax_revenue` that the budget was spending.
- **Then:** other taxes rise, borrowing rises (implying later taxes), or spending falls. Apply TX-1 and OC-1 to whichever group bears the replacement, named from the budget response. If the record does not say, state the replacement chain with a `revenue_use` assumption; where it frees resources for the same group that the repeal relieves, combine the chains (workflow step 7).
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** LW-1, SB-1 (its mirror case).
- **Derivation:** CR-8b (the budget's spending must be paid for by someone), with CR-8a for whoever pays.

OC-7 Revaluation changes holders' means

- **If:** `asset_value` of an asset held by an actor class rises or falls (SB-4).
- **Then:** when they sell or borrow against it, their `resources_available_other_uses` change in the same direction, and so do their purchases in linked markets (homeowners whose homes gain value bid for larger homes). The buyers of the asset have correspondingly fewer means for other purchases; record that chain too.
- **Assumes:** the change can be realized by sale or borrowing (`other`), and is not offset by an equal change in the price of what they intend to buy.
- **Tier** B. **Visibility** invisible. **Horizon** medium.
- **Builds on:** LW-1.
- **Derivation:** CR-8b: when an asset that has risen in value is sold, the seller's gain in means is the buyer's outlay. Symmetry requires the matching chain: buyers of the asset have fewer means for other purchases. Combine the two with CR-12.

OC-8 Crowding out by public borrowing

- **If:** `government_borrowing` finances a policy.
- **Then:** the means lent to the state are not available to other borrowers (CR-8a): `interest_rate` increases for them and their borrowing (for investment, house purchase) decreases, unless the lending comes from abroad; repayment implies taxes later (CR-8b), recorded as a group of later payers. Lenders receive the interest (CR-8b).
- **Assumes:** the loans come from domestic saving that would otherwise have been lent at home (`other`); the debt is expected to be repaid, not rolled over (`expectations`).
- **Baseline and kind of debt (0.9).** Compare with the borrowing that would have taken place without the policy (CR-8, 0.9): where private actors would have borrowed for the same purpose, the effect on `interest_rate` is `ambiguous` without a premise on the difference. Revenue bonds imply later taxes only for a shortfall. Tax-exempt interest lowers tax receipts; record those who make them up as payers. A borrower too small to move its loan market: the rate effect is recorded as negligible, as under EN-6. A legal debt limit: borrowing for this project displaces the government's own other borrowing first. Lenders: no sign against the counterfactual.
- **1.0.** The small-borrower case is signed, not left as negligible (CR-6, 1.0). Where an exemption creates a segment (bonds exempt only from one state's tax), the borrowing can be large in that segment: yields rise there and holders of existing bonds in it lose. Tax-exempt borrowing for private projects also reduces the projects' interest deductions; net receipts need arithmetic.
- **Tier** B. **Visibility** invisible. **Horizon** short to long.
- **Builds on:** LW-1, LW-7.
- **Derivation:** CR-8a and CR-8b; CR-6 in the loan market (more demand for loans at the old rate); CR-7a (lenders move to the higher rate); CR-10 for the later taxes.

OC-9 Debt-funded transfer in two windows

- **If:** a transfer (a payment to households, a subsidy, a lease below market) is funded by borrowing that is repaid later from taxes or a budget.

- **Then:** two windows, signed separately (CR-12g): in the first, recipients gain and the loan market supplies the funds (OC-8); in the repayment window, payers lose (OC-1, group). A whole-period sign for a class needs a dominance premise or arithmetic on the case's figures. Recipients who expect to be among the later payers treat the transfer partly as a loan (CR-4, 0.7).
- **Assumes:** the debt is expected to be repaid (`other`).
- **Tier** B. **Visibility**: the transfer is visible; the repayment is invisible. **Horizon** short to long.
- **Derivation:** CR-8 (borrowing, payers), CR-10, CR-12g. Seen in WE-033 and WE-041.

TS: Time structure

TS-1 Durable capital lag

- **If:** a consequence works through reduced investment or maintenance in durable capital (buildings, equipment, trained skills).
- **Then:** the effect on `capital_stock_target` and `quantity_supplied` is small at first and grows as the existing stock wears out and is not replaced. Set horizon to medium or long; `quality_per_unit` often falls before quantity does.
- **In the other direction:** when a policy (or a repeal) raises the return on durable capital, the stock grows at the speed of new investment (planning, permits, finance, construction, training), which is not the mirror of wear-out. State that lag as a `durability` assumption; TS-5 may slow it further.
- **Assumes:** the durability of the capital involved (`durability`).
- **Tier** B. **Horizon** medium to long.
- **Builds on:** LW-4, LW-7.
- **Derivation:** CR-10b.

TS-2 Expectations

- **If:** actors expect the policy to be extended, tightened, made permanent or applied to new categories.
- **Then:** responses that the policy would cause in future are brought forward: investment falls now in categories expected to be covered; actors rush to get into exempt categories before the rules change.
- **Assumes:** what actors expect (`expectations`).
- **Tier** B. **Horizon** short to medium.
- **Builds on:** LW-7.
- **Derivation:** CR-10a.

TS-3 Erosion of a fixed parameter

- **If:** a cap, floor, rate, threshold or quantity is fixed in money terms or in physical units (a number of licenses, a tonnage), or indexed at a rate different from the change in the relevant market price or demand.
- **Then:** the component becomes more binding (or less) over time as prices or demand move. A ceiling or quota that is not binding at enactment can become binding later.
- **Assumes:** the expected path of market prices or demand.
- **1.1.** State the expected path of prices or demand as a premise.
- **Tier** B. **Horizon** medium to long.
- **Product standards.** A standard on a product's specification has no clearing price to move past. It binds on the share of designs actors would choose below it (CR-1, CR-4 over the spread of designs), and that share changes as costs and the counterfactual trend move (CR-10a).
- **Derivation:** CR-6 (a fixed parameter binds when the clearing price or demand moves past it) with CR-10a (expected path); for a product standard, CR-1 and CR-4 with CR-10a.

TS-4 The intervention spiral (proposed, contested)

- **If:** a policy produces consequences that its supporters regard as problems (shortages, black markets, evasion).
- **Then:** pressure for further intervention to address them (extending controls to substitutes, rationing schemes, penalties), or for repeal.
- **Tier** C. **Status:** not derivable (see Derivation); kept out of Tier A and B results until the experts decide what premises would make it a pattern. See `06_gaps.md`, Q4.
- **Source:** Mises, *A Critique of Interventionism* (1929), Part I; HA ch. XXX §2 (a price ceiling must be extended to other goods) and ch. XXXVI.
- **Derivation:** Not derivable from the core rules. It needs a premise about what a policy's supporters believe about its effects and how political decisions respond, which is neither a core rule nor a fact about the policy record.

TS-5 Credibility of a repeal

- **If:** a policy is repealed (or not renewed), and actors give a meaningful probability to its return within the payback period of their investments.
- **Then:** responses that pay back quickly (repairs, pricing, lettings, short contracts) go ahead; responses that pay back only over long periods (major renovation, new construction, entry with sunk costs) are smaller and later than the mirror of the original policy's effects. Where the expected return would exempt existing practitioners (grandfathering), it brings entry forward instead. The losing side responds too: those who gained from the policy and expect it back delay scrapping, selling or retraining, and keep capacity on standby.
- **Assumes:** the probability of re-imposition over a named period (`expectations`).
- **Lines fixed before announcement (1.0).** Where grandfathering is decided by a date or history already past when the measure is announced, there is no rush to qualify (EN-12).
- **Tier** B. **Horizon** medium to long.
- **Builds on:** LW-7, TS-2.
- **Derivation:** CR-10a with CR-4: the expected yield of a long-lived investment includes the chance that the old policy returns; quick-payback actions depend on it less.

TS-6 Stock changes persist after repeal or erosion

- **If:** a policy is repealed, or its incentive erodes (TS-3), and it had caused reclassification (RS-5), exit (RS-6, EN-3), or the loss of specific capital, skills or client relationships, and reversing these has a cost (reassembling converted units, rebuilding capacity, retraining, re-

establishing a client base).
- **Then:** a repeal (or erosion) reverses the flow but not the stock already changed, except where the restored return exceeds the cost of reversal. Responses that required sunk investment (a smuggling network, converted buildings) shrink by less than the incentive does. Former sellers return only partly and slowly; firms that closed re-enter as the sunk costs of entry are paid; people displaced stay displaced. Record the expected `persistence` (none, partial, full) on each repeal consequence.
- **Assumes:** the cost of reversal (`reversal_cost`), including how long people and capital have been away.
- **Tier** B. **Horizon** medium to long.
- **Builds on:** LW-1, LW-4, TS-1.
- **Derivation:** CR-10b with CR-3: undoing a stock change has a cost, so it is undone only where the restored yield exceeds it (CR-4).

TS-7 Expected caps reduce stocks held

- **If:** sellers expect a `price_ceiling` to apply in future periods of scarcity (an anti-gouging rule, a standing threat of price controls).
- **Then:** `inventory_held` before those periods decreases, because the expected return to storing goods for the scarce period falls. Less is available when scarcity arrives.
- **Assumes:** storage is costly and the cap is expected to bind (`expectations`, `binding`).
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** LW-4, LW-7.
- **Derivation:** CR-7 across time (storage moves a good from a time of plenty to a time of scarcity) and CR-10a: a cap expected in the scarce period lowers the yield of storing.

TS-8 Habit adjustment

- **If:** a policy changes the cost of a routine, often-repeated action (carrying bags, a commute, a shopping pattern).
- **Then:** responses are smaller at first and grow over weeks or months as routines change; some short-run effects (fee payments, use of the costlier option) shrink over the same period without any change in prices or capital. Set the horizon of the full response to short-to-medium and describe the path.
- **Assumes:** the action is repeated often (`other`).
- **Tier** B. **Horizon** short to medium.
- **Builds on:** AX-1, LW-1 (learning a new routine has a cost).
- **Derivation:** CR-3 and CR-10b: changing a routine has a cost (attention, learning) paid over time, so CR-4's switches happen gradually.

TS-9 Capital per worker and wages

- **If:** a policy reduces saving or investment across the economy (TX-6), so that `capital_stock_economy` is smaller over time than it would have been.
- **Then:** `wage_rate` across the economy is lower in the long run than it would have been, because labor works with less capital.
- **Assumes:** no offsetting inflow of foreign capital (`other`).
- **Tier** B. **Visibility** invisible. **Horizon** long.
- **Builds on:** LW-7, LW-4.
- **Source:** HA ch. XXI §6 "Wages and Subsistence".
- **Derivation:** CR-7 (labor's price tends to its discounted marginal value product) with LW-9 and LW-10 (labor's product depends on the capital it works with).

TS-10 Scheduled tightening faster than the trend

- **If:** a standard, cap or ban tightens on an announced schedule faster than the counterfactual trend.
- **Then:** in each window, the share of actors it binds on is the gap from the trend (TS-3, 0.8); asset values capitalize the whole schedule when it becomes expected (CR-7c); activity bunches before each step (RS-8, CR-12h), signed per window against the trend.
- **Assumes:** the counterfactual trend (`counterfactual_trend`).
- **Taxes (1.1).** The same holds for a tax or charge rising on a schedule.
- **Tier** B. **Visibility** invisible. **Horizon** medium to long.
- **Derivation:** CR-10c, CR-7c, CR-12h. Seen in WE-031, WE-050.

PC: Price controls

PC-1 Binding ceiling: excess demand

- **If:** `price_ceiling` on a good, below the price at which quantity demanded would equal quantity supplied.
- **Then:** `excess_demand` increase: at the controlled price, quantity demanded exceeds quantity supplied.
- **Assumes:** the ceiling is binding (`binding`) and enforced (`enforcement`).
- **Unless:** the ceiling is above the market price (not binding); price is a quality signal (LW-3 exception).
- **A third-party payer (1.1).** The If clause requires that the cap bind on the price the seller receives. Where it caps only what the buyer pays and a third party pays the rest, use PC-20.
- **Tier** B. **Visibility** invisible (the shortage is often noticed, but its cause is not attributed to the policy). **Horizon** short.
- **Builds on:** LW-3, LW-4.
- **Source:** HA ch. XXX §2; P&M ch. 3 §1; EIOL 1946 ch. 17.
- **Derivation:** CR-6 (a price held below the clearing level).

PC-2 Shortage is rationed by other means

- **If:** `excess_demand` increase (PC-1).
- **Then:** for `buyers.prospective`, the group `wait_time`, `search_cost`, `side_payments`, `allocation_by_non_price_criteria`: `at_least_one_increase`. Any one member is signed on its own only with a premise that that margin is open and used in this market. The good does not go to whoever is willing to give up the most for it, but by these other criteria.
- **Tier** A, given the shortage; the particular form is B. **Visibility** invisible. **Horizon** short.
- **Builds on:** LW-6.

- **Derivation:** CR-6 (the scarce side chooses by other criteria).

PC-3 Buyers split into insiders and outsiders

- **If:** `excess_demand` increase (PC-1).
- **Then:** `buyers.current` who obtain the good at the controlled price gain (`price_paid` decrease for them); `buyers.prospective` who cannot obtain it lose: they pay more elsewhere, wait, or go without.
- **Tier** A, given the shortage. **Visibility**: the insiders' gain is visible; the outsiders' loss is invisible. **Horizon** short.
- **Builds on:** LW-5, LW-6.
- **Derivation:** CR-6 and CR-5: those who obtain the good gain the gap between the controlled and the clearing price; those who do not, lose the exchange (CR-5a) or pay more elsewhere.

PC-4 Quality falls under a ceiling

- **If:** binding `price_ceiling` (PC-1), and sellers control unregulated aspects of quality.
- **Then:** `quality_per_unit` and `maintenance_spending` decrease; the quality-adjusted price moves back toward the market level.
- **Assumes:** quality and maintenance are not fully regulated or enforced (`enforcement`).
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** RS-2.
- **Derivation:** CR-6 (pressure displaced to quality) with CR-5c.

PC-5 Supply shrinks under a ceiling

- **If:** binding `price_ceiling` (PC-1).
- **Then:** `investment_target` decrease, then `quantity_supplied` decrease, as resources move to uncontrolled uses.
- **Assumes:** `alternative_uses` exist for the resources (for housing: conversion to owner-occupation or other uses, demolition, keeping units off the market); speed governed by TS-1.
- **Unless:** the resources are completely specific and have no other use, in which case quantity supplied from the existing stock holds until it wears out (TS-1 still applies to new investment).
- **Tier** B. **Visibility** invisible. **Horizon** medium to long.
- **Builds on:** LW-4, RS-6, TS-1.
- **Source:** EIOL 1979 ch. 18 "What Rent Control Does" (not in the 1946 edition).
- **Derivation:** CR-7 (owners move resources to uses whose yield is not capped), timed by CR-10b.

PC-6 Black and grey markets under a ceiling

- **If:** binding `price_ceiling` (PC-1).
- **Then:** `informal_activity` and `side_payments` increase: some exchanges happen at prices above the ceiling, disguised or illegal (key money, tie-in sales, sublets at market rent). RS-10 applies to that trade.
- **Assumes:** imperfect `enforcement`.
- **Tier** B. **Visibility** invisible. **Horizon** short.
- **Builds on:** RS-3.
- **Derivation:** CR-6 (exchanges outside the rule) via CR-4 (x: comply, y: violate).

PC-7 Reduced turnover where the benefit is tied to holding

- **If:** binding `price_ceiling`, and the right to buy at the controlled price is tied to keeping the good (a tenancy, a place on a list), and cannot be sold.
- **Then:** `turnover` decrease among insiders: they move less, even when another unit, job or place would otherwise suit them better.
- **Assumes:** the benefit is not transferable (`other`).
- **Tier** B. **Visibility** invisible. **Horizon** medium.
- **Builds on:** LW-2, LW-5.
- **Derivation:** CR-3 (giving up the good now forfeits the gap between controlled and market price) and CR-4.

PC-8 Spillover to uncontrolled substitutes

- **If:** a binding `price_ceiling` or `price_floor` covers part of a market, and an uncontrolled substitute or segment exists (recorded market relation).
- **Then:** demand moves to the uncontrolled segment (under a ceiling, from outsiders who cannot obtain the good, PC-3; under a floor, from buyers avoiding the higher price): `price_substitute` increase, which harms `buyers.substitutes`.
- **Assumes:** `substitutability` between the segments.
- **Unless:** the policy also pushes sellers into the uncontrolled segment (sellers shut out of the controlled one by a cap on their number, owners moving units out of the controlled segment, exempt new building). Then RS-13 adds supply there, and the combined sign of `price_substitute` is `ambiguous` unless a premise says which chain dominates.
- **Withdrawn demand (0.9).** The same policy may also withdraw demand from the uncontrolled segment (intermediaries who bought there to resell into the controlled one). Combine that chain with the inflow under CR-12b.
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** RS-1, PC-3.
- **Note:** another chain can push `price_substitute` the other way: resources withdrawn from the controlled segment (PC-5) may reappear in the uncontrolled one (for example, new building that is exempt from the cap), adding supply there (RS-13). Combine the chains using the procedure in workflow step 7.
- **Derivation:** CR-9 (unsatisfied buyers, or buyers avoiding a floor, turn to the uncontrolled segment) and CR-6 in that segment.

PC-9 Binding floor: excess supply

- **If:** `price_floor` on a good, above the price at which quantity demanded would equal quantity supplied.
- **Then:** `excess_supply` increase; `quantity_traded` decrease (buyers buy less at the higher price).
- **Assumes:** `binding`, `enforcement`.

- **Unless:** the state buys the surplus (then see SB rules for the purchase).
- **Tier** B. **Visibility** invisible. **Horizon** short.
- **Builds on:** LW-3, LW-6.
- **Derivation:** CR-6 (a price held above the clearing level).

PC-10 Minimum wage: fewer hires of the least productive

- **If:** `price_floor` on labor (minimum wage) above the wage some workers can command.
- **Then:** for workers whose labor employers value below the floor: `employment` decrease or `hours_worked` decrease, or `non_wage_compensation` decrease (RS-2). Workers who keep their jobs at the floor: `wage_rate` increase.
- **Assumes:** `binding` for an identifiable group; `enforcement`; employers can adjust hours, staffing, automation or non-wage terms.
- **Tier** B. **Visibility**: the wage rise is visible; lost hires and cut hours are invisible. **Horizon** short to medium.
- **Builds on:** PC-9, LW-4.
- **Source:** HA ch. XXX §3 "Minimum Wage Rates"; EIOL 1946 ch. 18 "Minimum Wage Laws" (ch. 19 in 1979).
- **Note:** the empirical size of the employment effect is contested (Card and Krueger, *American Economic Review* 84(4), 1994, and the literature since). Earnings of the low-wage group as a whole combine opposite chains and are `ambiguous` unless a dominance rule applies. This is a designated contested case for evaluation.
- **Derivation:** CR-6 applied to labor (a price held above the level some workers can command), with CR-4 for employers' margins: hiring, hours, non-wage terms.

PC-11 Non-price selection under a floor

- **If:** binding `price_floor` (PC-9).
- **Then:** buyers choose among excess sellers by non-price criteria: `allocation_by_non_price_criteria` increase. Under a minimum wage, employers become more selective (credentials, experience), and the cost to them of discriminating falls.
- **1.1.** Where the counterfactual already rations by non-price criteria, sign only the change in that rationing.
- **Tier** A, given the surplus; its form is B. **Visibility** invisible. **Horizon** short.
- **Builds on:** LW-6 (applied to the selling side).
- **Derivation:** CR-6 (the scarce side, here buyers, chooses by other criteria).

PC-12 A ceiling passes up to the inputs

- **If:** a binding `price_ceiling` on a good, and the capped sellers buy most of a named input (relation `input_to`).
- **Then:** `net_price_received` by input suppliers for sales into the capped market decreases, and/or the capped sellers buy less of the input (`quantity_demanded` of the input decreases).
- **Assumes:** the input has few uncapped outlets in the period considered (`alternative_uses`).
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** LW-4, PC-5. The downstream counterpart is RS-14.
- **Derivation:** CR-9 with LW-10 (the input's demand derives from the capped output), then CR-7 (the input supplier with few other outlets bears it).

PC-13 Ceilings that allow documented costs

- **If:** a binding `price_ceiling` allows increases only for documented costs (invoiced freight, wholesale prices), not for costs that leave no paper trail (the owner's labor, risk, the value of the goods elsewhere).
- **Then:** `number_of_sellers` decreases most among sellers whose main costs are undocumented (occasional and informal suppliers, people bringing goods from outside); `compliance_cost` increases for sellers who use the defense.
- **Assumes:** the enforcing agency's definition of a documentable cost (`enforcement`).
- **Tier** B. **Visibility** invisible. **Horizon** short.
- **Builds on:** LW-4, RS-6.
- **Derivation:** CR-1 (the cap with a documented-cost exception burdens sellers differently by the kind of cost they bear), then CR-4 for each kind of seller.

PC-14 Rate caps bind hardest on small and short transactions

- **If:** a binding `price_ceiling` is stated as a rate per unit of value and time (an annual interest rate, a percentage markup), and a material part of the seller's cost is fixed per transaction (processing, collection).
- **Then:** sellers raise the minimum size and length of the transactions they offer: `transaction_size` increase; `quantity_supplied` of small, short transactions decreases most; `variety` decreases at the small end.
- **Assumes:** the share of fixed per-transaction costs (`other`).
- **Tier** B. **Visibility** invisible. **Horizon** short.
- **Builds on:** LW-4, RS-2.
- **Derivation:** CR-6 with CR-4: where part of the cost is fixed per transaction, a cap stated as a rate leaves small transactions unprofitable first.

PC-15 Regulated cost recovery over a shrinking base

- **If:** a regulator sets a price to recover largely fixed costs per unit sold (utility rates), and a policy reduces the units sold through the regulated channel.
- **Then:** at the next rate review, `price_paid` per unit increases for the remaining buyers.
- **Assumes:** the regulator's rate method (`other`); fixed costs do not fall with sales.
- **Tier** B. **Visibility** invisible. **Horizon** medium.
- **Builds on:** LW-1.
- **Derivation:** CR-1 (the regulator's pricing rule is part of the situation), applied with CR-2 to the regulator.

PC-16 A binding ceiling is a transfer

- **If:** a binding `price_ceiling` (PC-1), and some buyers obtain the good at the controlled price.

- **Then:** for each unit traded at the controlled price, the buyer pays less and the seller receives less than at the clearing price, by the same amount: `price_paid` decrease for buyers who obtain the good and `net_price_received` decrease for sellers. Record both sides. Side payments (PC-6) and lower quality (PC-4) take back part of the transfer.
- **1.1.** Not where a third party pays the difference (PC-20).
- **Tier** A, given the ceiling binds and is enforced for those units. **Visibility**: the buyers' gain is visible; the sellers' loss is invisible. **Horizon** short to long.
- **Builds on:** LW-5.
- **Derivation:** CR-8b (every payment is a receipt: what the buyer does not pay, the seller does not receive) with CR-6 (the clearing price is the comparison).

PC-17 Mandated contract term

- **If:** a `mandated_contract_term` requires every contract of a kind to include a term (paid sick leave, a notice period), and the term is absent from some contracts that would otherwise be made.
- **Then:** for those contracts, the cost to the party who must provide the term rises (CR-1, operation 3); the bargaining range moves (CR-5c), so the other terms adjust: for an employment term, a mixed group of `wage_rate` (−), `non_wage_compensation` (−) and `employment/hours_worked` (−), sign `at_least_one_as_stated`. Parties who already included the term are not bound (not binding for them). Where the term is worth something to the side that receives it, part of the cost is recovered through the other terms (CR-7, gross and net).
- **Assumes:** the term was absent for some contracts (`binding`); `enforcement`. Why it was absent (the parties' own rankings, or an obstacle to agreeing) is a premise (CR-2), and decides whether the receiving side is better off by its own ranking.
- **A mandate with a buy-out charge (0.9).** Where the employer may pay a charge instead of providing the term ("play or pay"), split the bound class by response (CR-1, 0.7): PC-17 applies to those who provide the term, TX-2 to those who pay the charge. Where the charge is capped per firm, RS-18 (0.9) applies.
- **A charge refunded on the act (1.0).** Where the charge is paid first and refunded when the mandated act is bought (a training levy repaid as vouchers), the act is free at the margin up to the refund, so it rises; the wedge (CR-5c) falls on the whole payroll of those who do not recover the charge, not on the contracts that carry the term.
- **1.1.** The terms that adjust are those of the exchange that carries the term: for a product, its price and other features (warranty, design); for a term owed to third parties (parts sold to independent repairers), the term moves the bargaining range of that third exchange. Compare with no term, not with the same contract minus the term. A buy-out at the worker's option paid to the worker is not a TX-2 wedge but a transfer within the exchange, and where it is added to a price the payer can still set freely, it can be undone through that price (CR-6).
- **Tier** B. **Visibility**: the term is visible; the adjustment of other terms is invisible. **Horizon** short to medium.
- **Builds on:** LW-5.
- **Derivation:** CR-1 (operation 3 on the option of contracting without the term), CR-5c (the range moves), CR-6 (pressure displaced to the other terms), CR-7b (who bears the net cost), CR-12 (the group).

PC-18 Prohibited contract term

- **If:** a `term_prohibition` makes a contract term unenforceable (a non-compete clause, an exclusivity clause), in new contracts and possibly in existing ones.
- **Then:** the option of buying the term's protection is removed for one party and the option of breaking it is added for the other (CR-1). The bargaining range moves (CR-5c): the party who valued the term offers less on other terms or shifts to substitute terms (a mixed group: `wage_rate` (−), training it funds (−), `term_prevalence` of substitute terms (+), sign `at_least_one_as_stated`). Where existing contracts are voided, the right voided is transferred from its holder to the other party (CR-8, voiding). Where the term restrained the mobility of a means, the gain from mobility goes to its owner (CR-7, a policy that makes a means mobile).
- **Assumes:** the term was in use (`binding`); substitute terms are available or not (`substitutability`).
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** LW-5.
- **Derivation:** CR-1, CR-5c, CR-6 (pressure displaced to the other terms), CR-7 (mobility), CR-8 (voiding), CR-12.

PC-19 Public supply at a set price

- **If:** `public_supply` of a good at a set price below the clearing price (zero, for a free service), with the quantity fixed by a budget.
- **Then:** excess demand at the set price (PC-1); the pressure goes to the margins open to buyers sharing the fixed quantity: `wait_time`, crowding (`quality_per_unit` down), `travel_time`, as a mixed group (CR-6, public supply). Buyers who would have paid anyway gain what they no longer pay (SB-8). Buyers move from substitutes (RS-16, mirror case). The funding is binary and analyzed separately (SB-1).
- **Assumes:** `binding`; the budget does not expand with use (`revenue_use`).
- **Open quantity (0.9).** Where the state sells at a set price and supplies whatever is demanded, there is no rationing. Buyers move to it (CR-4), rival sellers face lower demand (CR-9, RS-13), and the split between their price and quantity needs a premise (CR-6, 0.9). Rivals' plant that cannot be moved keeps operating as long as it covers its running costs (CR-7, CR-10b).
- **Selection (1.0).** Where the public price is uniform and rivals price by each buyer's cost, buyers sort (CR-6, 1.0): high-cost buyers move to the public seller and its outlay per buyer rises. Crowding below capacity is not rationing (CR-6, 1.0) but a Tier C chain.
- **Tier** B. **Visibility**: the free service is visible; crowding and the funding are invisible. **Horizon** short to medium.
- **Builds on:** LW-3, LW-6.
- **Derivation:** CR-6 (public supply), CR-8b, CR-9.

PC-20 A cap on the buyer's share

- **If:** a policy caps what buyers pay out of pocket for a good, and a third party (an insurer, a fund) must pay the seller the rest.
- **Then:** no shortage follows from the cap itself. Buyers whose share falls buy more of the good and its close varieties (CR-4); the third party's outlay rises, and its payers bear it (CR-8b; premiums, a budget). The seller's receipt is set by the payer's terms; where the payer pays the seller's own price up to a limit, that price is drawn toward the limit (CR-6, 1.1). Buyers already paying less than the cap are unaffected.
- **Assumes:** the third party pays as required (`enforcement`).
- **Tier** B. **Visibility**: the cap is visible; who pays for it is invisible. **Horizon** short to medium.
- **Derivation:** CR-1, CR-4, CR-6 (1.1), CR-8b. Seen in WE-056.

TX: Taxes and tariffs

TX-1 Tax takes resources

- **If:** a tax is imposed and collected.
- **Then:** `resources_available_other_uses` decrease for the actor classes that bear it (see TX-2 for who bears it); `tax_revenue` increase. Apply OC-1 to what those actors would have done with the money, and analyze the spending of the revenue as a separate component. Where a shortfall in revenue is made up from a general budget, its payers are a group (OC-1, Who pays).
- **Rate rises (1.1).** For a rise in the rate of an existing tax, `tax_revenue` increase needs a premise that the base does not fall enough to offset it: Tier B.
- **Tier** A, given collection. **Visibility**: the payment is visible; the forgone uses are invisible. **Horizon** short.
- **Builds on:** LW-1.
- **Source:** P&M ch. 4.
- **Derivation:** CR-8a and CR-8b.

TX-2 Per-unit tax: wedge and smaller market

- **If:** `per_unit_tax` or `ad_valorem_tax` on the sale of a good.
- **Then:** `price_paid` by buyers increase and/or `net_price_received` by sellers decrease (at least one must happen); `quantity_traded` decrease.
- **Assumes:** at least one side can reduce its quantity in response (some `substitutability` for buyers, or `alternative_uses` for sellers).
- **A fixed factor (1.1).** Where the base is a factor in fixed supply (land), quantity does not fall; the tax is capitalized into the factor's value (CR-7c) and borne by its owner when the tax becomes expected.
- **Tier** B. **Visibility**: the price rise is visible; the fall in quantity and in sellers' net price is invisible. **Horizon** short.
- **Builds on:** LW-3, LW-4.
- **Note:** how the burden splits between buyers and sellers is not fixed by this rule. Use DM-1, or report the split as ambiguous.
- **Derivation:** CR-5c (the tax is a wedge between what buyers pay and sellers keep) with CR-4 for each side.

TX-3 Taxed activity shrinks; resources move to untaxed uses

- **If:** TX-2 applies.
- **Then:** `investment_target` in the taxed activity decreases over time; `employment` and `hours_worked` there, as a group, `at_least_one_decrease`; `quantity_substitute` increase.
- **Assumes:** `alternative_uses`, `substitutability`.
- **Mirror (1.1).** A tax removed or cut reverses these moves, with persistence following the stock (CR-7, 1.0).
- **Tier** B. **Visibility** invisible. **Horizon** medium.
- **Builds on:** RS-1, RS-6.
- **Derivation:** CR-7.

TX-4 Avoidance of a local tax

- **If:** a tax applies within a jurisdiction with nearby untaxed places, or with definitions that can be avoided.
- **Then:** RS-4 (cross-border purchases), RS-5 (reformulation, reclassification) and RS-3 (evasion) apply. Taxed sales inside the boundary fall by more than total consumption does.
- **Assumes:** `mobility`; cost of reformulating.
- **Tier** B. **Visibility** invisible. **Horizon** short.
- **Builds on:** RS-3, RS-4, RS-5.
- **Derivation:** CR-4 (relocate, reclassify, evade), as RS-3 to RS-5.

TX-5 Compliance cost

- **If:** a tax requires record-keeping, filing, labeling or collection by private parties.
- **Then:** `compliance_cost` increase for those parties; OC-1 applies to those resources.
- **Tier** A, given positive compliance work. **Visibility** invisible. **Horizon** short.
- **Builds on:** LW-1.
- **Derivation:** CR-8c.

TX-6 Taxes on the return to saving and investment

- **If:** a tax falls on interest, dividends, capital gains or profits.
- **Then:** the net return on providing future goods falls; at the margin, saving and `investment_target` decrease, and the capital stock is smaller over time than it would have been.
- **Assumes:** for the group as a whole, the substitution response (save less as the future becomes dearer) outweighs the response of savers aiming at a fixed sum (`other`). Without this premise the direction is `ambiguous` (see Derivation).
- **Unless:** some savers aim at a fixed future sum, for whom a lower net return calls for more saving. Where such savers are a material part of the group, total saving by the group is `ambiguous`; the sign holds firmly only for saving in the taxed forms by savers who have untaxed alternatives. (Pending expert decision, `06_gaps.md` Q14.)
- **Tier** B. **Visibility** invisible. **Horizon** long. Variables: `saving`, `investment_target`, `capital_stock_economy`; TS-9 carries the effect on wages.
- **Builds on:** LW-7, LW-4.
- **Derivation:** CR-4 with x: future goods, y: present goods; the tax raises the cost of the future relative to the present. CR-4's exception applies: savers aiming at a fixed future sum must save more. So the a priori direction for total saving is `ambiguous`; the stated decrease holds only with the premise that the substitution response dominates for the group (`other`), or for saving in the taxed forms by savers with untaxed alternatives.

TX-7 Tariff: domestic buyers pay more, protected sellers gain

- **If:** `tariff` on an imported good.

- **Then:** `price_paid` for the imported good and for domestic substitutes increase for `buyers.current; sellers.substitutes` (domestic producers of competing goods) see `net_price_received` and `quantity_supplied` increase. Domestic firms that use the good as an input face higher costs.
- **Assumes:** domestic substitutes exist; the tariff is not absorbed entirely by foreign sellers.
- **Two-rate tariffs (1.0).** Where a quota sets the quantity at a low rate and further imports pay a higher rate (a tariff-rate quota), apply this pattern at the marginal rate, not at the in-quota rate; the in-quota rate only divides the gap between the domestic and world price between the state and the holders of in-quota rights (DM-4).
- **Unless (1.1).** Where domestic output pays the same tax as imports, there are no protected sellers; this pattern does not apply.
- **Tier** B. **Visibility**: gains to protected producers are visible; higher prices to buyers and input users are invisible. **Horizon** short to medium.
- **Builds on:** TX-2, RS-1.
- **Source:** EIOL 1946 ch. 11 "Who's 'Protected' by Tariffs?"; Bastiat, *Economic Sophisms*, First Series (1845), ch. VII (the candlemakers' petition).
- **Derivation:** CR-5c and CR-7 for the importing side; CR-9 for domestic sellers of the same good (their buyers' alternative has become dearer) and for users of the good as an input.

TX-8 Import restrictions: export industries lose

- **If:** a `tariff` (TX-7) or an import `quota` (EN-6) reduces imports.
- **Then:** foreigners earn less of the domestic currency from selling to the country, and so have less to spend on its exports: `quantity_supplied` and `employment` in export industries decrease over time.
- **Assumes:** trade is not offset by capital flows or other adjustments in the period considered (`other`).
- **Mirror case:** a policy that raises exports (an export subsidy, SB-10) gives the country's residents more foreign means, so imports rise and `quantity_supplied` and `employment` (as a group with `hours_worked`) in import-competing industries decrease over time; exporters of other goods also lose foreign buyers' means to the subsidized exporters (CR-9, currency).
- **A change in the form of imports (0.9).** Where a policy reduces imports of one good but raises imports of goods made from it, the currency chain is itself `ambiguous` (CR-12, kind 3) unless a premise says which dominates.
- **Tier** B. **Visibility** invisible. **Horizon** medium.
- **Builds on:** LW-5, OC-4.
- **Source:** EIOL 1946 ch. 11 and ch. 12 "The Drive for Exports" ("Without imports we can have no exports, for foreigners will have no funds with which to buy our goods").
- **Derivation:** CR-8b (foreigners' receipts from selling to the country are the means with which they buy its exports or assets) and CR-9.

TX-9 Revenue falls short of a static projection

- **If:** TX-2 applies (the taxed quantity falls), or TX-4 applies (avoidance shrinks the base).
- **Then:** `tax_revenue_vs_static_projection` decrease: revenue is less than the rate times the pre-tax base.
- **Unless:** the tax is ad valorem on a price that the tax itself pushes up (a tax on interest that raises the gross interest rate). Then the remaining base is worth more per unit and the sign is open: combine the chains (workflow step 7).
- **Unless (1.1).** Where the revenue is returned to the same buyers (a payment funded by the tax), the base need not fall; the sign is `ambiguous` without a premise.
- **Tier** B (inherits the assumptions of TX-2 or TX-4). **Visibility** invisible until revenue is reported. **Horizon** short.
- **Builds on:** TX-2, TX-4.
- **Derivation:** Arithmetic with CR-4 (the base falls).

TX-10 Base erosion and base shift across taxes

- **If:** a policy reduces `quantity_traded`, wages, employment or sales that bear other taxes in the jurisdiction (directly, or through RS-4 and RS-9), or a prohibition removes the base of an existing tax.
- **Then:** `tax_revenue` from those other taxes decreases. Where part of the spending moves to other goods taxed in the jurisdiction, revenue there increases, and the combined sign is `ambiguous` unless the substitutes are untaxed (informal or cross-border purchases), in which case it is a decrease.
- **Tier** B, inheriting the assumptions of the rules that shrink the base. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** TX-1, RS-1.
- **Derivation:** CR-8b and CR-9 (spending moved elsewhere carries other taxes' bases with it).

TX-11 Tax on nominal returns

- **If:** a tax falls on nominal interest or gains, and prices are rising.
- **Then:** the share of the real return taken exceeds the statutory rate, and rises with the rate of inflation; the TX-6 responses are correspondingly stronger.
- **Tier** A for the arithmetic, B for the responses. **Visibility** invisible. **Horizon** medium to long.
- **Builds on:** TX-6.
- **Derivation:** Arithmetic (a tax on the nominal return takes a larger share of the real return) with TX-6.

TX-12 Deposit-refund

- **If:** a charge is paid on an act and refunded, up to the charge, to whoever performs a later act (returns a container, buys approved training) within a period; what is not refunded goes to a fund.
- **Then:** the later act rises (it is paid up to the refund); those who do not perform it bear the charge; the fund receives the unclaimed part. Refunds attach to the thing or the act, not to the payer, where the law says so (CR-1, 0.8). Those who claim bear the cost of claiming and the time the money is held (CR-3, 0.9).
- **Assumes:** the later act is feasible for some (`alternative_uses`).
- **Tier** B. **Visibility**: the charge is visible; who bears it is invisible. **Horizon** short to medium.
- **Derivation:** CR-1, CR-3, CR-4, CR-8b. Seen in WE-027, WE-048.

SB: Subsidies

SB-1 Subsidy is funded by someone

- **If:** a subsidy is paid.
- **Then:** apply TX-1 and OC-1 to its funding source (taxes now, borrowing that implies taxes later, or monetary expansion). Name the payers. By CR-8b, the recipients' means increase by what the payers give up, before any response: record both sides.
- **Tier** A, given payment. **Visibility** invisible. **Horizon** short to long, depending on funding.
- **Builds on:** LW-1.
- **Derivation:** CR-8a and CR-8b.

SB-2 Subsidized activity expands

- **If:** `per_unit_subsidy` or `purchase_rebate` on a good.
- **Then:** `price_paid` by buyers decrease and/or `net_price_received` by sellers increase; `quantity_traded` increase.
- **Assumes:** at least one side can expand its quantity.
- **1.1.** Where the good is priced in a wider market, the domestic price is set there (CR-6, 0.7); sign the small effect on the wider price (CR-6, 1.0). A payment indexed to the seller's own posted price draws that price up to the limit (CR-6, 1.1).
- **Tier** B. **Visibility** visible. **Horizon** short.
- **Builds on:** LW-3, LW-4.
- **Derivation:** CR-5c (the subsidy is a wedge in the other direction) with CR-4 for each side.

SB-3 Resources drawn from unsubsidized uses

- **If:** SB-2 applies.
- **Then:** resources move into the subsidized activity from other uses, where quantities fall: `quantity_substitute` decrease and output elsewhere decreases.
- **Assumes:** resources are scarce and in use elsewhere.
- **Unless:** the policy only moves an activity in time (a temporary program that brings work forward and leaves it undone later). Then the effect on other uses is a timing shift (RS-8, SB-6), signed per window, not a lasting reduction.
- **Tier** B. **Visibility** invisible. **Horizon** medium.
- **Builds on:** LW-1, LW-4.
- **Derivation:** CR-7 and CR-8a.

SB-4 Capitalization of lasting changes in returns

- **If:** a policy expected to last for some period raises or lowers the expected net return from an asset in fixed or slowly expandable supply (land, buildings, licenses, quota rights, shares in firms with specific capital): a subsidy or a restriction that raises returns (EN-2, EN-6), or a tax, cap or competing entry that lowers them.
- **Then:** `asset_value` changes in the same direction from the time the change becomes expected. Owners at that time capture the gain or bear the loss; later buyers pay the new price, so they neither gain nor lose from the policy as it stands. When an expected gain later erodes (the subsidy ends, exempt competitors enter, repeal becomes likely), the value falls on whoever holds the asset then, including recent buyers who paid the capitalized price.
- **Assumes:** fixed or slowly expandable supply of the resource; how long the change is expected to last (`expectations`). A change of limited duration is capitalized only for the yields it affects, so its effect on `asset_value` is smaller, but its sign is the same. Assets the policy itself creates (quota rights, medallions, transferable licenses) are covered too: their value is the expected yield of the privilege, captured by whoever receives them at creation.
- **Alternatives reached too (1.1).** Where the policy also changes the return on buyers' alternatives (a tax on all forms of wealth), sign the asset's value by its return relative to those alternatives (CR-7, 1.1); it may be `ambiguous` even for a taxed asset. For a right sold at creation, only the value above the price is captured (CR-7, 1.1).
- **Tier** B. **Visibility** invisible. **Horizon** short (announcement) to medium.
- **Builds on:** LW-4, LW-7.
- **Derivation:** CR-7c with CR-10c.

SB-5 Seeking to qualify

- **If:** a subsidy is conditional on criteria.
- **Then:** RS-5 (reshaping to qualify) and RS-7 (lobbying to widen or keep the criteria) apply.
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Derivation:** CR-4 (x: as before, y: qualify), as RS-5 and RS-7.

SB-6 Temporary purchase subsidy: pull-forward and dip

- **If:** a purchase subsidy available only during a limited window.
- **Then:** purchases are moved into the window (RS-8): `quantity_traded` increase during it and decrease after it, compared with the counterfactual. If the subsidy requires destroying a used good (scrappage), the stock of used goods falls: `price_substitute` increase for buyers of used goods.
- **Assumes:** buyers can shift purchase timing; for scrappage, the scrapped goods would otherwise have stayed in use.
- **Price side:** in the dip after the window, sellers face fewer buyers than they would have, so `net_price_received` decreases for them in that window where their supply cannot adjust at once (CR-6).
- **Tier** B. **Visibility**: the surge is visible; the dip and the used-goods effect are invisible. **Horizon** short.
- **Builds on:** RS-8, OC-1.
- **Derivation:** CR-10a with CR-4 (x: buy at another time, y: buy in the window).

SB-7 Earmarked revenue is fungible

- **If:** a tax's revenue is earmarked for a purpose that the budget already funds, or would have funded, from general revenue.

- **Then:** general funding for the purpose decreases in part, so spending on the purpose rises by less than the earmarked amount, and the rest of the revenue in effect funds other spending or lower borrowing.
- **Assumes:** the purpose already receives, or would have received, general funding; budget-makers can reduce it (`revenue_use`).
- **Unless:** the law also fixes the general funding level (a maintenance-of-effort clause) and it is enforced.
- **Tier** B. **Visibility** invisible. **Horizon** medium.
- **Builds on:** LW-1, AX-1 (budget-makers act on the whole budget).
- **Derivation:** CR-8d with CR-2 applied to budget-makers.

SB-8 In-kind provision of a good the recipient would have bought anyway

- **If:** the state provides a good, free or below cost, to recipients some of whom would have bought it themselves (or had it paid for, for example by an insurer).
- **Then:** for those recipients (and their payers), the provision works like a cash transfer: `resources_available_other_uses` increase, and they spend the freed money elsewhere (OC-5 applies there). The quantity of the provided good rises only for recipients who would not have bought it. Providers of the good who would have been paid privately are paid by the state instead.
- Where the good is supplied with fixed capacity, those who would have bought anyway share it with new users: `wait_time` or crowding (`quality_per_unit`) worsens for them (CR-6, public supply).
- **Assumes:** the share of recipients who would have bought anyway (`other`).
- **1.1.** Where the saving comes back through another term of the same contract (a deductible) or recipients pay part of the funding, net those first (CR-3, 1.1).
- **Tier** B. **Visibility** invisible. **Horizon** short.
- **Builds on:** LW-1, SB-1.
- **Derivation:** CR-8b (for those who would have bought anyway, the good provided frees the money they would have spent) and CR-4.

SB-9 Public stocks and their disposal

- **If:** the state buys a surplus at a support price (PC-9 Unless clause).
- **Then:** `inventory_held` by the agency increases while the floor binds. When the stock is released, `quantity_traded` decreases for the private sellers it displaces (donation, school meals) or `price_paid` decreases in the market it is sold into (resale, including abroad, RS-13); if it is destroyed, OC-1 applies to the resources used to produce and store it.
- **Assumes:** the product's shelf life and the disposal routes allowed (`durability`, `other`).
- **Tier** B. **Visibility** invisible. **Horizon** medium.
- **Builds on:** LW-1, RS-13.
- **Derivation:** CR-8, CR-10b (the stock accumulates) and CR-6 in the market the stock is released into.

SB-10 Export subsidy

- **If:** an `export_subsidy` pays a fixed sum per unit of a good exported.
- **Then:** sellers move sales abroad until the domestic price rises toward the export price plus the subsidy (CR-7a): domestic buyers' `price_paid` increases and their `quantity_traded` decreases; producers' `net_price_received` and `quantity_supplied` increase; `asset_value` of the land or capital specific to the good increases (SB-4); foreign buyers gain (`price_paid` for `buyers.foreign` decreases); TX-8's mirror case applies; round trips to claim the subsidy appear (RS-4), unless the policy checks the return crossing (rules of origin, re-import bans); where it does, combine the round-trip chain with the check under CR-12b. The funding is analyzed under SB-1.
- **Assumes:** the domestic market is open to the world price (`mobility`); producers can expand (`alternative_uses`).
- **Limits (0.9).** The domestic price rises toward the export price plus the subsidy only where buyers cannot import the good, or goods made from it, free of the policy; otherwise imports cap it. A refund of a tariff on inputs used in exports (drawback) is an export subsidy only to the extent that the refund exceeds the tariff paid on what was used (refunds on a standard, CR-3 0.9), net of the cost of holding the tariff until refunded. A check on the return crossing is a burden on the round-trip option (CR-4), not an opposing chain: it reduces round trips from what they would be, and never below the counterfactual of none.
- **Tier** B. **Visibility**: the subsidy is visible; the higher domestic price is invisible. **Horizon** short to medium.
- **Builds on:** LW-4.
- **Derivation:** CR-7a (sales move to the higher-yielding market), CR-12e (the response cannot push the domestic price below its old level), CR-7c, CR-8b, CR-9 (currency).

EN: Licensing, quotas and prohibition

EN-1 Licensing requirement restricts supply

- **If:** `licensing_requirement` whose conditions (training hours, exams, fees, waiting periods) cost would-be sellers more than some of them would otherwise spend.
- **Then:** `entry_rate` and `number_of_sellers` decrease; `quantity_supplied` decrease and `price_paid` increase compared with no licensing.
- **Assumes:** `binding` (the requirement goes beyond what some current or prospective sellers already meet); `enforcement`.
- **Unless:** existing sellers are exempt (grandfathered), which delays the effect on supply (TS: horizon medium).
- **Refill (1.0).** Where the requirement removes part of the stock of sellers at one date, the higher fee draws new entrants who meet it: entry in the medium run is `ambiguous`; the long-run decrease needs a premise on exit rates.
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** LW-1, LW-4, LW-5.
- **Source:** P&M ch. 3 §3 "Product Control: Grant of Monopolistic Privilege".
- **Derivation:** CR-1 (the option of selling is removed for those without a license, and a cost is attached to obtaining one), CR-4 for would-be sellers, CR-6 in the market.

EN-2 Licensed incumbents gain

- **If:** EN-1 applies.
- **Then:** `sellers.current` who hold licenses: `net_price_received` increase. If licenses are transferable, `asset_value` of a license increases (SB-4).

- **Cost recovery (1.0).** Where the authority recovers its cost by a charge spread over the capped quantity, a smaller quantity raises the charge per unit, which opposes incumbents' net price; combine. Where the license cannot be sold, its value is by the holder's own ranking only (CR-7, 1.0).
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** EN-1, SB-4.
- **Derivation:** CR-6 and CR-7b: with entry restricted, the gain from the higher price settles on those who hold the scarce permission.

EN-3 Excluded sellers and priced-out buyers

- **If:** EN-1 applies.
- **Then:** would-be sellers who do not qualify move to their next-best occupation (OC-4); by their own ranking they are worse off (CR-3a). Their money `earnings` decrease only with the premise that the next-best occupation pays less in money (`other`): an occupation chosen second may pay more and be valued less for other reasons. Buyers priced out of the licensed service substitute toward doing it themselves, unlicensed providers (RS-3) or going without (RS-1).
- **Assumes:** `substitutability` for buyers.
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** OC-4, RS-1, RS-3, RS-10.
- **Derivation:** CR-3a and CR-4 for excluded sellers and priced-out buyers.

EN-4 Quality received by buyers: ambiguous by default

- **If:** EN-1 applies and the requirement is related to the quality of the service.
- **Then:** two chains push `quality_per_unit` received by buyers in opposite directions: licensed providers meet a minimum standard (up), while buyers priced out move to self-provision or unlicensed providers (EN-3, down). The "down" chain needs its own premise: that what buyers move to is of lower quality than the paid provision it replaces (`other`). Without it, that chain gives no sign. The combined sign is `ambiguous` unless DM-2 applies.
- **Tier** B for each chain. **Visibility** the first chain is visible; the second is invisible. **Horizon** medium.
- **Note:** the canonical example for workflow step 7.
- **Derivation:** CR-12 (two chains with opposite signs).

EN-5 Incumbents push for tighter requirements

- **If:** EN-2 applies (incumbents gain from restriction).
- **Then:** RS-7: `lobbying_effort` by incumbents for stricter or broader requirements.
- **Tier** B. **Visibility** invisible. **Horizon** medium to long.
- **Derivation:** CR-4 applied to political action, as RS-7.

EN-6 Quota

- **If:** `quota` limiting the quantity produced, sold or imported, below the quantity that would otherwise be traded.
- **Then:** `quantity_traded` decrease; `price_paid` increase; holders of quota rights gain, and their `asset_value` increases where rights are tradable (SB-4). Domestic sellers of the same good (for an import quota): `net_price_received` and `quantity_supplied` increase, as under TX-7. Buyers lose (OC-4). For an import quota, TX-8 applies to export industries; DM-4 governs how the gap between the domestic and world price is split.
- **Assumes:** `binding`, `enforcement`.
- **Unless:** the capped good's price is set in a wider market (fish sold into a regional market, a traded commodity). Then the lease price of the right (`right_lease_price`) rations the capped quantity (CR-6), not `price_paid`. `price_paid` still moves up in the wider market, by as much as the capped quantity matters there: record it as `increase` with a premise that the effect is not negligible, or note it as negligible. Where part of the good is sold only at home, that part's price rises as in the main case.
- **Inputs (0.9).** Where the cap is on an input whose output is sold in a wider market (groundwater for crops), the lease price of the right rations the input (CR-6), and the wider-market clause applies to the output through CR-9.
- **Caps on sellers; small effects (1.0).** A cap on the number of sellers is not a cap on quantity: its removal lowers price only with a premise that the cap held quantity down. The wider-market price is signed even when the effect may be small (CR-6, 1.0). For a tariff-rate quota, see TX-7 (1.0).
- **Tier** B. **Visibility** invisible. **Horizon** short.
- **Builds on:** LW-3, LW-5.
- **Derivation:** CR-6 (quantity held below the clearing level), CR-7b (the gain settles on the holders of quota rights, the scarce side) and CR-9 (domestic sellers of the same good).

EN-7 Prohibition of a good moves trade underground

- **If:** `prohibition` of a good that some buyers continue to value highly.
- **Then:** `quantity_traded` legally falls to near zero; `informal_activity` increase; `price_paid` increase (sellers require compensation for the risk of punishment). RS-10 applies to the informal trade: higher search costs, less predictable quality, disputes settled outside the courts.
- **Assumes:** demand persists for some buyers; imperfect `enforcement`.
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** RS-3, RS-10, LW-3.
- **Derivation:** CR-1 (legal trade removed), CR-6 and CR-4 (trade outside the rule where the expected gain exceeds the expected penalty), CR-7 (the price includes compensation for risk).

EN-8 Prohibition favors concentrated forms

- **If:** EN-7 applies, and penalties or detection risk depend on bulk or volume rather than on potency.
- **Then:** sellers shift toward more concentrated, potent forms of the good, which are cheaper to hide per unit of effect: `share_concentrated_forms` increase.
- **Assumes:** penalties scale with volume (`enforcement`).
- **Tier** B. **Visibility** invisible. **Horizon** medium.

- **Source:** the "iron law of prohibition" (Richard Cowan, "How the Narcs Created Crack", *National Review*, 1986; Mark Thornton, *The Economics of Prohibition*, 1991). Not a classical praxeological text; flagged for expert decision (Q7).
- **Derivation:** CR-7 with CR-4: where penalties scale with volume, sellers minimize volume per unit of effect, so the concentrated forms yield more net of expected penalty.

EN-9 Supervised-practice requirement

- **If:** a license requires a period of practice supervised by existing license holders.
- **Then:** incumbents' willingness to supervise becomes a bottleneck on entry: `entry_rate` is limited by the number of supervised places, and where places are scarce relative to candidates, `wage_rate` of those in supervised practice decreases, as incumbents capture part of the value of the license through their trainees' pay.
- **Assumes:** supervision is costly or limited per supervisor (`other`).
- **Tier** B. **Visibility** invisible. **Horizon** short to medium.
- **Builds on:** EN-1, LW-6.
- **Derivation:** CR-1 (entry requires a supervised place), then CR-6: supervised places are scarce and the gap is closed through trainees' pay.

EN-10 Penalty records

- **If:** a prohibition (EN-7) or licensing rule is enforced by criminal convictions that are recorded and checked by employers.
- **Then:** for those convicted, `employment` and `earnings` decrease over the long run, compared with the counterfactual.
- **Assumes:** record-keeping and checking practice (`other`).
- **Tier** B. **Visibility** invisible. **Horizon** long.
- **Builds on:** LW-1.
- **Derivation:** CR-11c (a record conveys information employers act on) with CR-4 for employers.

EN-11 A license as a signal

- **If:** a license requirement tests something buyers care about but cannot easily judge for themselves (competence, safety, honesty), and the license is visible to buyers.
- **Then:** for licensed sellers, `quality_information` increases for their buyers. Where the requirement tests something buyers do not care about, the license conveys nothing about the quality they want, and this chain gives no sign.
- **Assumes:** the requirement's content bears on the quality buyers value (`other`); buyers can see who holds a license.
- **Unless (0.9).** Where the same license is issued on different requirements (recognized from other jurisdictions, or with standards that vary by issuer), it conveys less about the quality buyers care about than a license issued on one requirement: `quality_information` decreases for buyers of the sellers whose license now covers more variation (CR-11c).
- **Baseline of the Unless (1.0).** The 0.9 Unless compares with a license issued on one requirement. Where the counterfactual has no license, the main clause applies: `quality_information` increases, by less where requirements vary.
- **Tier** B. **Visibility** visible. **Horizon** short to medium.
- **Builds on:** AX-1.
- **Derivation:** CR-11c (a policy changes actions by conveying information) with CR-2 for buyers.

EN-12 Allocation of a right by history

- **If:** a policy creates a valuable right and allocates it in proportion to an activity measured over a base period.
- **Then:** where the base period ends before the measure became expected, the allocation is a windfall to those with history (CR-7c, 0.8), and their `asset_value` increases. Where the base period runs after the announcement, actors race to build history: the activity, investment in it and resources used up rise in the window (CR-10a, CR-8c), and part of the right's value is spent obtaining it. A share reserved for newcomers by lottery or open eligibility draws entry to qualify.
- **Assumes:** the allocation rule is known (`other`).
- **Tier** B. **Visibility** invisible. **Horizon** short.
- **Derivation:** CR-7c, CR-10a, CR-8c. Seen in WE-028 and WE-039.

EN-13 Consolidation under tradable rights

- **If:** a policy creates tradable rights to a capped quantity, and holders' costs or the value they get from the right differ.
- **Then:** rights move to the holders who value them most: `number_of_sellers` using the right decreases, the quantity per remaining user increases, and lessors receive `right_lease_price` instead of producing. The workers and suppliers of those who sell or lease out lose (CR-9).
- **Assumes:** rights can be traded or leased at low cost (`mobility`); values differ (`other`).
- **Baseline (1.0).** Against no policy, the number of users of the capped resource falls through blocked entry and holders who sell or lease their whole holding, not by comparison with the initial allocation. Where rights can go to anyone, a premise is needed that the highest-value users are among the holders.
- **No users without the policy (1.1).** Where the resource had no users at all without the policy (an unused band), the number of users rises from none; consolidation only lowers it relative to the first allocation.
- **Tier** B. **Visibility** invisible. **Horizon** medium.
- **Derivation:** CR-5, CR-7a. Seen in WE-028 and WE-039.

EN-14 A standard on new units, with the stock exempt

- **If:** a policy forbids new units of a durable good below a specification (a home, a lot, a vehicle), and existing units are exempt.
- **Then:** new units cost more or are not built where the standard binds (TS-3, 0.8); existing units, which now face less competition from cheaper new ones, rise in value (CR-7c) and are kept in use longer (CR-10b). A repeal reverses these effects as the stock turns over: asset values persist only as far as the stock does (CR-7, 0.9).
- **Assumes:** the standard binds on some designs (`binding`).
- **Thinning complements (1.0).** As the exempt stock shrinks, the goods and services used with it (fuel stations, repair) thin out and cost more (CR-9, CR-4 0.8), which opposes the rise in the stock's value; in the long run that value is `ambiguous` without a dominance premise.
- **Unless (1.1).** Where buyers value the standard, covered units draw demand from the exempt stock, which opposes the rise in its value.
- **Tier** B. **Visibility** invisible. **Horizon** medium to long.
- **Derivation:** CR-1, CR-4, CR-7c, CR-10b. Seen in WE-031 and WE-043.

EN-15 Disclosure mandate

- **If:** a policy requires sellers to disclose something about their goods (origin, calories, energy use).
- **Then:** buyers' choices change only where they did not already know and care (CR-11, 0.7): a sign needs a premise on prior belief and a premise on direction. Covered sellers bear a compliance cost, fixed per item, outlet or firm (CR-4, 0.8). Where some sellers are exempt, buyers may read their silence, and exempt sellers whose goods compare well disclose voluntarily (CR-11, 0.9). Whether buyers who choose differently are better off is `06_gaps.md`, Q22.
- **Assumes:** see above (`other`).
- **Tier** B. **Visibility**: the label is visible; the changed choices are invisible. **Horizon** short to medium.
- **Derivation:** CR-11c, CR-4, CR-2. Seen in WE-035 and WE-040.

DM: Dominance rules

Dominance rules decide the combined sign when chains conflict (workflow step 7). Each is itself an assumption and needs an estimate.

DM-1 Incidence follows the alternatives

- **If:** TX-2 (a tax) or SB-2 (a subsidy) applies, and one side of the market has clearly fewer or worse alternatives than the other (buyers with no substitutes; sellers with highly specific, immobile resources).
- **Then:** that side bears more of a tax, or receives more of a subsidy, than the other side does (CR-7b, with the premise that the difference in alternatives is clear). This compares the two sides' shares; it says nothing about how large either is. For a tax: if buyers have fewer alternatives, the rise in `price_paid` exceeds the fall in `net_price_received`; if sellers do, the reverse. For a subsidy, the same with the signs reversed. Without a clear difference in alternatives, the split is not signed: each side's price moves in the direction TX-2 or SB-2 gives, and `tax_share_borne` or `subsidy_share_received` is `ambiguous`. Record the split with `tax_share_borne` or `subsidy_share_received`.
- **Assumes:** the comparison of alternatives (`substitutability`, `mobility`), with evidence.
- **Tier** B. **Horizon**: over time, the side that owns redeployable capital gains alternatives as it redeploys (TS-1), so the burden shifts away from that side. In product markets the sellers own the capital, so incidence shifts toward buyers; in factor markets such as labor the buyers (employers) own it, so a payroll tax's incidence shifts toward workers.
- **Source and dispute:** the rule as stated is the standard treatment of incidence, not Rothbard's. Rothbard's "first law of incidence" (P&M ch. 4 §3A) holds that no tax can be shifted forward: a general sales tax is in effect a tax on the incomes of original factors, and where a price later rises, that comes about through reduced supply, which he classes as an effect of the tax rather than shifting. The two accounts agree that the burden falls first on those who cannot escape it, and that prices rise only as supply contracts; they differ in what they call shifting, and in Rothbard's case for general taxes. Whether to encode Rothbard's account as an option is Q3 in `06_gaps.md`.
- **Derivation:** CR-7b: the side whose means can move away bears less; the burden or benefit settles on the side with fewer alternatives.

DM-2 Quality under licensing

- **No default rule.** Both chains of EN-4 need their premises: the "up" chain that the requirement bears on the quality buyers care about, and the "down" chain that the alternatives buyers move to are of lower quality. EN-4 stays ambiguous unless the case supplies evidence that one chain dominates: for example, the requirement is unrelated to the quality buyers care about (then only the "down" chain is active), or close unlicensed substitutes are unavailable (then the "down" chain is weak). Record such evidence as an assumption.
- **Derivation:** CR-12 (no dominance premise by default).

DM-3 One-time cost against later gains (ambiguous by design)

- **No default rule.** When one chain imposes a one-time cost (training, conversion, relocation) and another produces a later stream of gains (a higher price, a lower cost), the combined effect on the actor class's earnings or welfare is `ambiguous`. Comparing them needs magnitudes and a discount rate, which the method does not supply. Report both chains, and state that the actors who chose to bear the cost expected the gains to be worth it (LW-5), while those who did not, did not.
- **Derivation:** CR-12 (the method has no magnitudes or discount rates with which to net the two).

DM-4 Quota rent follows the scarce side

- **If:** EN-6 (an import quota) raises the domestic price above the world price.
- **Then:** if foreign supply comes from many competing sellers, the holders of quota rights capture most of the gap; if it comes from a few sellers who can coordinate, or if exporting countries administer their own shares of the quota, foreign sellers capture part of it.
- **Assumes:** the structure of foreign supply and of quota administration (`other`).
- **Tier** B.
- **Derivation:** CR-7b: the gap between domestic and world price settles on whichever side is scarce and cannot be competed away.

DM-5 Netting a timing shift

- **If:** SB-6 (or RS-8) produces a surge in one period and a dip in another, and the question is the total over a horizon that includes both.
- **Then:** the total rises if some purchases in the surge are additional, provided the dip consists only of purchases moved into the surge. If the policy is expected to recur, buyers wait for it, and the dip can include purchases that are merely delayed; then the total is `ambiguous`.
- **Assumes:** what the dip contains (`expectations`).
- **Tier** B.
- **Derivation:** CR-12 with CR-10a: the total over the horizon is the sum of the surge and the dip, whose net sign depends on what the dip contains.

6. Gaps and open questions

The to-do list for the catalog and the agenda for the first expert review. Gaps come from the worked examples: each is a move an analysis needed that the catalog does not yet cover. Questions are decisions the Handbook cannot make on its own.

6.1 Gaps found in the worked examples

Status as of catalog 0.2. "Closed" means a rule or variable was added and the worked example re-run; it still needs expert review like everything else.

ID	Found in	Gap	Status
GAP-1	WE-001	Announcement effects: the policy record had no announcement date, so RS-8 could not be applied.	Closed: `announced` added to the policy record (2.1); WE-001 C-17.
GAP-2	WE-001	No rule linked lower housing turnover to the labor market.	Closed: RS-11 lock-in, variable `job_changes`, relation note `location`; WE-001 C-18. Expert check requested (Q12).
GAP-3	all	Opportunity costs were recorded as losses, not signed variables.	Closed: every consequence is now a signed variable (2.5, workflow step 9, OC rules); duplicate OC entries removed from WE-001 to WE-003.
GAP-4	WE-002	No rule for bundled shopping trips.	Closed: RS-9, relation note `trip`; WE-002 C-11 and C-12 now Tier B.
GAP-5	WE-002	No variables for health or nutrient intake.	Out of scope for version 1; stated in 2.4 and in every report's "cannot say" section.
GAP-6	WE-002	No rule for fungibility of earmarked revenue.	Closed: SB-7; WE-002 C-20.
GAP-7	WE-002	No variable for revenue against a static projection.	Closed: TX-9, variable `tax_revenue_vs_static_projection`; WE-002 C-21.
GAP-8	WE-003	Informal-market effects were written only into EN-7.	Closed: generic RS-10, now used by PC-6, EN-3 and EN-7; WE-003 C-12 now Tier B.
GAP-9	WE-003	The direct effect of licensing had no variable.	Closed: `share_of_sellers_licensed`; WE-003 C-1.
GAP-10	WE-003	No way to compare a one-time cost with later gains.	Resolved as ambiguous by design: DM-3. Expert confirmation requested (Q13).
GAP-11	WE-002	Tax incidence by subgroup had no variable.	Closed: `tax_share_borne`; WE-002 C-13.

Gaps found in WE-004 to WE-025 are listed in 6.2.

6.2 Gaps found in WE-004 to WE-025, resolved in catalog 0.3

The 22 later examples raised 60 gaps, and updating them to catalog 0.3 raised four more (the last four rows). Many were the same gap found independently in different examples. They are resolved below in catalog 0.3 and ontology 0.3. "Closed" means a rule, variable or format was added and the example updated; expert review is still needed for all of it.

Gaps	What was missing	Resolution
004-1	State provision of a good recipients would have bought anyway	SB-8
004-2	Diverted spending diverts work (Hazlitt's tailor)	OC-5
004-3	No lump-sum tax instrument	`lump_sum_tax` in 2.3
005-1	A ceiling passes up to inputs	PC-12
005-2	Full cost per unit to the buyer	`full_cost_per_unit`
006-1, 008-2 (variable)	Stocks held against future scarcity	`inventory_held`; TS-7
006-2	Ceilings that allow documented costs	PC-13
007-1	Rate caps and transaction size	PC-14; `transaction_size`
008-1	Activity moves into the favored jurisdiction	RS-4 widened
008-2	Public stocks and their disposal	SB-9
008-3	Spillover under a floor	PC-8 widened to floors
009-1	Disjunctive ("at least one") conclusions	Group consequences, sign `at_least_one_decrease` (2.5)
009-2, 012-1, 013-3	Input cost pass-through to output buyers	RS-14
010-1, 017-1, 024-3	Quality uncertainty in informal trade	`quality_variability`, `quality_information`; named in RS-10
010-2, 012-2, 017-3	Other taxes' revenue when a base shrinks or shifts	TX-10
011-1	Saving, economy-wide capital and wages	`saving`, `capital_stock_economy`; TS-9
011-2, 013-2, 015-1, 020-1, 025-1	Capitalization of losses, and decapitalization when a gain erodes	SB-4 generalized
011-3	Tax on nominal returns under inflation	TX-11
011-4	Savers aiming at a fixed sum	TX-6 Unless clause, pending expert decision (Q14)
011-5	Revenue sign for an ad valorem tax on a price it raises	TX-9 Unless clause
012-3	DM-1's time note gave the wrong direction in factor markets	DM-1 note restated by who owns the redeployable capital
013-1	Discretionary relief (exclusions, waivers)	RS-15
013-4	Retaliation by foreign governments	Not encoded; added to Q5
013-5	Return on capital	`return_on_capital`
014-1	Export losses under a quota	TX-8 widened to quotas

Gaps	What was missing	Resolution
014-2	Split of the quota rent	DM-4
014-3	Domestic sellers under a quota	EN-6 widened
015-2	Caps fixed in physical units	TS-3 widened
015-3	Existing rules the policy makes binding	`existing_rules` in the policy record (2.1)
016-1	Supervised-practice requirements	EN-9
016-2	Lock-in by status, not only location	RS-11 widened
017-2	Disputes settled outside the courts	`disputes_settled_outside_courts` (violence remains out of scope)
017-4	Criminal records and later employment	EN-10
017-5	Mix of concentrated forms	`share_concentrated_forms`; used in EN-8
018-1	Free by-products reused for another end	Relation note `secondary_use` (2.8)
018-2	Responses that grow as routines change	TS-8
019-1	Complements in use	RS-12
019-2, 021-1	Subsidy incidence	DM-1 extended to SB-2; `subsidy_share_received` (020-3)
019-3	Netting a surge against a dip	DM-5
020-2	Spending out of asset gains	OC-7
020-3	Split of a subsidy	`subsidy_share_received`
021-2	Regulated cost recovery over a shrinking base	PC-15
021-3	Added supply lowers price in a linked market	RS-13
022-1, 023-2, 024-2	Whether a repeal is believed	TS-5
022-2, 023-1, 024-1	Changes that persist after a repeal	TS-6; assumption type `reversal_cost`; `persistence` field (2.5); "Analyzing a repeal" in the workflow
022-3	Speed of rebuilding after a repeal	TS-1 extended
023-3	Lost revenue made up elsewhere	OC-6
025-2	Interactions between components	Pairwise check in workflow step 6; `components` field (2.5)
025-3	Signs that depend on an unclear provision	`reading` field (2.5); readings get no probability (step 10)
025-4	Units held vacant	`units_held_vacant`
010-3	Responses with sunk investment outlast the erosion of an incentive	TS-6 widened to erosion (TS-3)
022-4	Persistence that changes with the horizon	`persistence` may be given per horizon (2.5)
023-4	The losing side's response to an expected re-imposition	TS-5 losing-side clause
024-4	Timing shifts had no signed form	Convention in 2.5: signed per window, or a path note in section 8; WE-003 C-7 moved to section 8

6.2a Audit of the patterns against the core rules (catalog 0.4)

Catalog 0.4 adds a layer of twelve core rules (`04a_core_rules.md`), each universal: stated over any actor, option and means, derived from the action axiom, symmetric, and explicit about its empirical premises. Every pattern was then given a derivation from them. Results:

- **77 of 81 patterns derive cleanly** from the core rules plus the premises they already named.
- **Four problems were exposed and fixed:**
  - **TX-6** asserted as a priori that saving falls when its return is taxed. CR-4's exception (a change in total means, here savers aiming at a fixed sum) means the direction is ambiguous without a premise. The premise is now stated.
  - **OC-7** counted the gain to an asset's seller without the buyer's matching outlay, breaking CR-8's symmetry. The matching chain is now required. WE-020 already recorded it, as C-25.
  - **RS-13** covered added supply but not withdrawn supply. It now covers both.
  - **TS-4** (the intervention spiral) cannot be derived. It needs a premise about what supporters of a policy believe and how politics responds. It stays Tier C and quarantined.
- **Two foundations were missing** and have been added: the law of returns (LW-9) and imputation (LW-10). Patterns on complements, inputs, pass-through and wages relied on them without saying so.
- **Three laws are results, not premises:** LW-3 (law of demand) follows from CR-4, LW-4 from CR-7, and LW-6 from CR-6. They keep their IDs.
- **The workflow changes.** When no pattern fits, derive from the core rules. Mark a GAP only when no core rule applies.

6.2b Source check (catalog 0.4)

Every citation was checked against the texts. The editions, the method and the findings are in `07_sources.md`. The main corrections:

- Cost is defined in HA ch. IV, not ch. VII.
- The mutual benefit of exchange is in P&M ch. 2, not ch. 1.
- The tailor is Hazlitt's; Bastiat's own example is shoes or a book.
- The iron law of prohibition is Richard Cowan's.
- The rent control chapter is only in the 1979 edition of Hazlitt.
- DM-1's incidence rule is the standard treatment, not Rothbard's (Q3).

Most citations could be confirmed only from search extracts of the online texts, not by opening them. Opening them directly is still to be done.

6.2c Core-only re-run (catalog 0.5)

All 25 worked examples were re-derived blind from the core rules alone and compared with the pattern-based versions; see `08_core_rerun_report.md`. No valid claim needed anything beyond the core rules (Q16, first half). The test led to:

- revisions of 14 patterns that overclaimed or lacked a clause;
- clearer wording in CR-1, CR-3, CR-6, CR-7 and CR-12;
- a final checklist in the workflow for the errors that recur in both kinds of analysis.

The pattern-based examples are being corrected for the errors the comparison found.

6.2d Gaps closed in catalog 0.6

The core-only re-runs flagged points where a core rule was unclear, and catalog 0.5 did not address all of them. Catalog 0.6 closes the rest.

Raised in	Gap	Resolution
WE-002 core	CR-1 could not record an operation that holds under one reading only	CR-1 "Recording": record operations per reading
WE-004 core	Which named purchases fall when means fall	CR-4 "A change in means alone": a premise is required; Q19 remains open
WE-010, WE-012 core	Whether CR-4's exception applies to the burdened good itself, and to what an actor offers	CR-4: it applies to both
WE-017, WE-003 core	The unit in CR-4c (a gram or a dose); "narrowly ranked" under a large one-time cost	CR-4 "Units and lumpy costs"
WE-016 core	Non-money terms of exchange (hours toward a license)	CR-5c: terms include everything given or received
WE-001, WE-013 core	"Becomes expected" for a measure likely but not certain; overlap of CR-7c and CR-10c	CR-7c: value moves with the probability; cite both rules
WE-017 core	Confiscation: a capital loss or a transfer?	CR-8e: a transfer, recorded on both sides; destruction uses up means
WE-018, WE-013, WE-001, WE-004 core	CR-9 covered only goods used together; leases, insurance and the currency link had no relation type	CR-9 widened; relation types `contract` and `currency` (2.8)
WE-022, WE-024 core	Persistence of transition-only effects; sunk costs in repeals	CR-10b: sunk costs are not consequences of a repeal; `persistence: n/a` for transition effects
WE-024 core	CR-11b covered only trade moving out of the legal market	CR-11b: symmetric
WE-021 core	Bunching before subsidy step-downs	CR-12h
WE-004 core	A reading that the record gives a probability	2.5: that is an assumption, not a reading
WE-025 core	No variable for unit size	`unit_size` (2.4)
Comparison	Recurring moves with no pattern to prompt them	PC-16 (ceiling as transfer), RS-16 (substitute's price), EN-11 (license as signal); RS-8 and SB-6 now state both sides of a timing shift

Still open, and for the experts: GAP-011-4 (Q14), GAP-013-4 (Q5), and Q19.

6.2e Gap-probing examples and catalog 0.7

Ten new examples (WE-026 to WE-035) were written to probe parts of the method the first 25 had not tested: autistic interventions, refundable charges, tradable rights, mandated and prohibited contract terms, export subsidies, product standards with a schedule, public supply at a zero price, debt-funded transfers, non-competes and disclosure mandates. They found two calculus gaps, both now expert questions, and many unclear wordings and missing words, which catalog 0.7 closes.

Calculus gaps (for the experts).

Gap	Step no core rule covers	Status
GAP-026-1 (also WE-032, congestion)	Effects through physical proximity or a shared facility, not exchange	CR-9 0.7: recorded as `proximity` or `shared_facility` relations, chains at Tier C; Q20
GAP-034-1	A non-rival means (know-how used by a rival without loss to the first firm)	CR-8 0.7: no claim on total output; Q21

Closed in 0.7.

Raised in	Gap	Resolution
WE-026	A derived chain against a suspected opposing chain that cannot be derived	CR-12 0.7: report the derived sign; the suspected effect is a Tier C note
WE-026, WE-031	Phased and scheduled operations	CR-1 0.7: operations carry the date they take effect; expected future operations act now through CR-7c and CR-10c
WE-027	A per-unit charge is attached to an option; refunds with residual claimants	CR-1 and CR-8 0.7; instrument `deposit_refund`; actor class `residual_claimant`; variable `non_tax_receipts`
WE-028	Fixed quantity priced in a wider market; who captures a created asset when receipt depends on later actions; the unit of a constraint	CR-6, CR-7 and CR-4 0.7; EN-6 Unless; instrument `rights_allocation`; variables `discards`, `right_lease_price`
WE-029	Direction of terms under a mandated term; burden of a mandate; change of kind against change of timing	CR-5, CR-7 and CR-12 0.7; PC-17; variable `legal_breach`
WE-030, WE-013	The currency link in one direction only; better offers against prevented exchanges; "a response cannot reverse its cause"	CR-9 0.7 (both directions), OC-4 Unless, CR-12(e); TX-8 widened; SB-10; variable `exchange_rate`; `buyers.foreign`, `sellers.foreign`
WE-031	Product standards; counterfactual trend	Instrument `product_standard`; variable `specification_level`; assumption type `counterfactual_trend`; Q11 updated
WE-032	Price set by a public supplier; fixed capacity	CR-6 0.7; PC-19; SB-8 widened; instrument `public_supply`; variable `travel_time`

Raised in	Gap	Resolution
WE-033	Debt-funded transfers; which uses fall	OC-8; instrument `government_borrowing`; variables `interest_rate`, `consumption`; Q19 updated
WE-034	Prohibited contract terms	PC-18; instrument `term_prohibition`; variable `term_prevalence`
WE-035	Disclosure mandates	CR-11 0.7; instrument `disclosure_mandate`
Several	"No change" and mixed-direction groups had no sign	Signs `no_change` and `at_least_one_as_stated` (2.5), also per reading
WE-026	Own-use bans, gifts, neighbors	Instrument `own_production_ban`; variable `home_production`; relation `gift`; actor class `neighbors`
WE-029, WE-012	Size thresholds; per-head costs with a cap	RS-17, RS-18

Candidate patterns. These moves were derived from the core rules once each. They become patterns when a second example needs them; until then the derivation in the example stands.

Candidate	Seen in	Derivation
Ban on own production	WE-026	CR-1 (option removed), CR-3a, CR-9 to the sellers of the bought substitute
Refundable charge (deposit-refund) (promoted to TX-12 in 1.0)	WE-027	CR-1 (charge on an option, refund on a later act), CR-4 per unit, CR-8b with a residual claimant
A claim payable inside a boundary draws items from outside	WE-027	CR-4, CR-7a across the boundary
Circular trade under a payment for crossing a boundary (promoted to RS-19 in 0.9)	WE-030	CR-4 (gain per round trip against its cost), CR-8c
Allocation of a right by history after announcement (promoted to EN-12 in 0.9)	WE-028	CR-10a, CR-7a, CR-8c
High-grading under a physical-unit cap (promoted to RS-21 in 1.0)	WE-028	CR-4c with the unit of the constraint
Consolidation under tradable rights (promoted to EN-13 in 0.9)	WE-028	CR-5, CR-7a (rights move to the lowest-cost users)
Standard on new units, stock exempt (promoted to EN-14 in 0.9)	WE-031	CR-1, CR-7c on the exempt stock, CR-10b
Scheduled tightening faster than the trend (promoted to TS-10 in 1.0)	WE-031	CR-10c, CR-7c, CR-12h
A standard per unit of a chosen attribute shifts the attribute	WE-031	CR-4
Certifiers hired by those they certify	WE-031	CR-2, CR-11
Compensating differential from a non-wage term (promoted to RS-22 in 1.0)	WE-032	CR-5c, CR-7a
Debt-funded transfer in two windows (promoted to OC-9 in 0.9)	WE-033	CR-8, CR-10, CR-12g
Ban on a restraint over carryable training; retroactive voiding	WE-034	CR-5c, CR-7c, CR-10b
Fixed cost per item reduces variety (promoted to RS-24 in 1.1; second use WE-059)	WE-035 (WE-040's use was invalid)	CR-4b with a lumpy cost
Selection under a uniform term (promoted to RS-23 in 1.0)	WE-029	CR-5c, CR-4 per counterparty
Complaint-only enforcement (promoted to RS-20 in 0.9)	WE-029	CR-4 (expected cost of breach differs by counterparty)

6.2f Held-out test of catalog 0.8 (WE-036 to WE-045)

Catalogs 0.7 and 0.8 were tuned on WE-026 to WE-035. Ten new examples were then written with the catalog frozen at 0.8, to see whether the changes work on cases they were not drawn from. Each example ends with a "Test of 0.8" part in its Gaps found, giving a verdict on each change it used.

Calculus gaps. One new one: GAP-041-1, residents who value having a team in their city without buying a ticket. No exchange, shared-facility or proximity relation reaches them. It is referred to Q20. Every other step in the ten examples was derived from CR-1 to CR-12.

Patterns whose conclusion was wrong although their preconditions held.

Pattern	Problem	Found in
RS-18	Predicts hours moving to fewer heads under a cap per firm; the derivation (CR-4 0.8, fixed cost per firm) gives `no_change`.	WE-036
RS-17	Signs the value of firms near a threshold as a decrease; the policy also burdens their rivals above the line, so the sign is `ambiguous`.	WE-040
SB-10	The domestic price need not rise toward the export price plus the subsidy when imports of the finished good cap it. Its If clause fits a refund for exporters (drawback) only in part.	WE-042
EN-11	A license that is issued on different requirements (recognized from other states) conveys less information, not more.	WE-045
PC-8	Its Unless clause misses demand the same policy withdraws from the uncontrolled segment (brokers).	WE-037

Changes that needed interpretation. Most 0.7 and 0.8 changes worked as written. These did not:

- **Tier C against Tier B.** CR-12 0.8 does not say how a Tier C chain (shared facility, proximity) combines with an opposing Tier B chain. Seen in WE-039, WE-041 and WE-044; each read CR-12d as giving `ambiguous` at Tier C. The groundwater plan's central purpose could be reported only at Tier C.
- **OC-8** (WE-038, WE-041): revenue bonds are repaid from fees, not later taxes; no clause for a borrower too small to move its loan market, or for a legal debt limit; the lenders' side gives no sign against the right baseline.
- **Public supply** (WE-038): a public seller at a set price with no fixed quantity is covered by no clause; PC-19 rightly does not apply.
- **PC-17** (WE-036): does not cover a mandate the employer can buy out with a charge.
- **EN-6 0.8 Unless** (WE-039): written for a capped good, needed here for an input (water) whose output is priced in a wider market.
- **CR-8 0.8, value from ending a race** (WE-039): correct, but the race runs through a shared aquifer, so the split is only Tier C.
- **RS-17** for an hours line rather than firm size; **RS-4** where the law reaches resales wherever made; **RS-8 0.8** where the opposing timing chain comes from a second incentive; **TS-3 0.8** where other rules bind first; **CR-11 0.7** on what exempt sellers do, and on buyers who learn

a signal became noisier; **CR-7c** for a license that cannot be sold.
- **Repeals** (WE-043): the workflow assumes an analysis of the original rule exists; here it had to be derived first.

Candidate patterns needed a second time. By the rule in 6.2e these are promoted: consolidation under tradable rights (WE-039); allocation of a right by history, in a wider form where history was fixed before announcement (WE-039); fixed cost per item reduces variety (WE-040); debt-funded transfer in two windows (WE-041); circular trade under a payment for crossing a boundary (WE-042); standard on new units with the stock exempt (WE-043); complaint-only enforcement (WE-044); and disclosure mandate (WE-040). New candidates proposed: a mandate with a buy-out charge, public entry at a set price against sunk incumbents, a tariff-like burden without a tax, public enforcement of a private term, a below-market lease of a public asset.

Vocabulary. A `not_signed` per-reading sign; a way to record a change in the probability of an action (WE-041); variables for the stock of a shared resource, amenity, complaints, and land per dwelling; the line between `legal_breach` and `informal_activity` where a ban has a permitted sub-class.

Re-run and catalog 0.9. The blind core-only re-run of WE-036 to WE-045 found K = 0 again, confirmed the pattern failures above and found more (OC-8's baseline, OC-3's fixed-budget clause, RS-16, RS-7). It also showed that the second use of "fixed cost per item reduces variety" was invalid, so that candidate is not promoted. Catalog 0.9 applies the fixes and promotes the other seven (`08_core_rerun_report.md`, third round). WE-036 to WE-045 are then corrected against their comparisons.

6.2g Held-out test of catalog 0.9 (WE-046 to WE-055)

Ten new examples were written with the catalog frozen at 0.9, on kinds of policy the 0.9 changes were not drawn from. Each ends with a "Test of 0.9" part giving a verdict on each change it used. These verdicts are the analysts' own; the last round showed that a blind re-run contradicts some of them.

Calculus gaps. None. Every step in the ten examples was derived from CR-1 to CR-12. Chains through roads, noise and proximity stay at Tier C under GAP-026-1 (Q20). GAP-039-C2 (the value of a lottery entry) was derived from CR-10c, CR-7c and CR-7a in WE-051.

The new Tier C rule (CR-12 0.9) worked as written on its main test, the congestion charge (WE-046, nine consequences), and in WE-051, WE-052 and WE-054. It needed interpretation where the Tier C chain itself pushes both ways (night noise moved to shoulder hours, WE-054) and on whether to record a Tier C sign identical to the Tier B sign (WE-046: recorded both).

Failures: a pattern or 0.9 clause whose conclusion was wrong although its conditions held.

Item	Problem	Found in
CR-12e 0.9, response argument	Assumes the response restores the old price; with a fixed cost per firm, yields can be equalized at a higher price. Needs a constant-cost premise, or should speak of yields. In WE-051 a partial response required inputs that *can* expand, the reverse of the clause's wording.	WE-049, WE-051
PC-17 0.9, buy-out	Where the charge is paid first and refunded on training, training is free at the margin and demand for the mandated term rises; the terms that adjust are those of employers who do not recover the charge.	WE-048
RS-16 0.9, costs on the substitute	A cost fixed per facility does not oppose net price (CR-4, 0.8); a cost on owners, not sellers, combines with quantity, not price. Prices should be signed per unit of a given quality or risk, so that a change in mix is not a price change.	WE-053, WE-050, WE-049
TX-7	Under a tariff-rate quota its conditions are met by the cut in the in-quota rate and give a lower price; the right sign comes from the marginal rate.	WE-047
RS-20	Its main sign compares with uniform enforcement, another policy, not with no policy.	WE-055
EN-11 0.9 Unless	Its decrease is against a license issued on one requirement, not against no license.	WE-055
EN-6 (in reverse)	A cap on sellers is not a cap on quantity; a price fall needs a premise.	WE-052
EN-14	Omits the chain by which a shrinking stock loses its complements (fuel stations, repair), so long-run used values are `ambiguous`.	WE-050

Needed interpretation. The 0.9 rule that an asset's persistence follows its stock (a license's value follows a legal count that ends on one date; new supply can return at once; WE-052, WE-050); TX-8 0.9 for services (WE-053); EN-6 for a tariff-rate quota (WE-047); OC-8 0.9's small-borrower clause where a state exemption creates a segment (WE-051); CR-8's voiding clause for a license rather than a contract (WE-052); CR-3 0.9 for a refund capped at what was paid (WE-048); RS-17 for a notch against a kink and for a line in time (WE-048, WE-055); CR-8d's earmark, which needs a premise that the council does not cut its own funding by the full amount (WE-046); EN-12, which does not say where the value of a lottery right settles (WE-051).

Worked as written in most uses: OC-8 0.9 (baseline borrowing, revenue bonds, lenders), CR-8 0.9 (a reading that permits, contingent payers), CR-6 0.9 (splitting a shift; public seller with an open quantity), PC-19 0.9, RS-8 0.9, RS-9 0.9, RS-18 0.9, OC-3 0.9, OC-5 0.9, CR-7 0.9 for assets without a market price, CR-12 0.9 on activities that exist only under the policy, the 0.9 workflow note on repeals, EN-12 and EN-13.

Candidate patterns used validly a second time (and so to be promoted, subject to a blind re-run): high-grading under a physical-unit cap (WE-047, WE-054), compensating differential from a non-wage term (WE-046), selection under a uniform term (WE-049), deposit-refund (WE-048), scheduled tightening faster than the trend (WE-050), tariff-like burden without a tax (WE-053). Proposed new clauses: a selection clause for PC-19; a notch/kink distinction for RS-17.

Re-run and catalog 1.0. The blind core-only re-run of WE-046 to WE-055 found K = 0 again and confirmed the failures above. It found one more pattern compared with the wrong baseline (EN-13), showed that "record as negligible" (OC-8, EN-6) conflicts with CR-6, and settled that congestion below capacity is a Tier C `shared_facility` chain, not CR-6 rationing. Of the six claimed second uses, five were valid; "tariff-like burden without a tax" was not (WE-053 is a different move), so it stays a candidate. Catalog 1.0 applies the fixes (`08_core_rerun_report.md`, fourth round).

6.2h Held-out test of catalog 1.0 (WE-056 to WE-065)

Ten examples were written with the catalog frozen at 1.0 and then re-run blind from the core rules (`08_core_rerun_report.md`, fifth round). No calculus gaps were found: every claimed gap was judged a premise, a vocabulary item, a recording limit, or out of scope. The 1.0 baseline rule caught most pattern problems in use, but the pattern texts still needed re-wording (RS-17, RS-23, PC-17, EN-13). New failures: SB-4 when a tax reaches buyers' alternatives, PC-1 and PC-16 with a third-party payer, TX-7 with a matched domestic tax, TX-9 when revenue returns to the payers, EN-14 for a standard buyers value. "Fixed cost per item reduces variety" had a valid second use and is promoted (RS-24). New candidates

(one use each): allocation of a right by sale (WE-064), a mandated add-on to an unregulated price (WE-061), a tax on the value of a fixed factor (WE-058), a payment counted in tonnes shifting output to lower-value tonnes (WE-065). Catalog 1.1 applies the fixes.

6.3 Questions for the experts

Q1. Tier A decisions. Which foundations and rules may produce Tier A conclusions? The draft proposes A for AX-1, LW-1, LW-2, LW-5, LW-6, LW-7, the OC rules, PC-2, PC-3 and PC-11 (given their preconditions), and A for LW-3 by decision, with the price-as-quality-signal and Giffen exceptions recorded. Is that right?

Q2. Contested empirical rules. PC-10 (minimum wage) is a standard praxeological conclusion whose empirical support is disputed. Should it stay as written, with the dispute recorded, or be weakened (for example, to "employment, hours or non-wage compensation decrease, in some combination")? The draft already uses the weaker form.

Q3. Rothbard on incidence. DM-1, and the incidence clause of CR-7b, state the standard treatment: the burden of a tax falls on the side with fewer alternatives. Rothbard's "first law of incidence" (P&M ch. 4 §3A) holds that no tax can be shifted forward. On his account a general sales tax is in effect a tax on the incomes of original factors, and any later price rise comes from reduced supply, which he does not call shifting. The two accounts agree on the sequence of events but differ in what they call shifting, and for general taxes. Should the Handbook adopt Rothbard's terms, keep the standard ones with the dispute noted (the current draft), or encode both as selectable options?

Q4. The intervention spiral. Should TS-4 (Mises: interventions produce problems that lead to further intervention or repeal) be a rule? If so, which variables does it predict, and how could it be checked?

Q5. Political responses. RS-7 (lobbying) is an application of the action axiom to political actors. How far should the method go in predicting political responses, as opposed to market ones? This includes responses by foreign governments, such as retaliation against a tariff (WE-013).

Q6. Depth limit. Is a limit of four links, with pruning of diffuse bystander effects after three, close to how experts actually stop? (WE-001 needed depth 3 at most.)

Q7. Non-classical sources. EN-8 (prohibition favors concentrated forms) comes from Cowan and Thornton rather than Mises or Rothbard. What is the standard for admitting a rule that is consistent with praxeology but not from its canonical texts?

Q8. Visibility test. Is "would this appear in a news story about the policy's passage?" a good enough test for visible versus invisible? WE-002's C-2 (fewer drinks sold) is both intended and visible; C-13 (who bears the tax) is neither.

Q9. Reporting cut-off. The plan reports a consequence if each of its assumptions has p of at least 0.5. But a consequence that rests on three independent assumptions at 0.6 each holds with probability of only about 0.2. Should the cut-off apply to the joint probability of a consequence's assumptions instead? This changes the plan (Section 7.3) and needs deciding before Phase 4.

Q10. Grants of privilege. Rothbard treats licensing, quotas and tariffs as "grants of monopolistic privilege". Should the Handbook adopt that category explicitly, as a shared parent of EN-1, EN-6 and TX-7?

Q11. The counterfactual. Every sign is relative to "the same place and time without the policy". When a market is trending (rents rising 5% a year), is that trend part of the counterfactual by default? The draft assumes yes. WE-031 adds a case: part of the trend is driven by an existing rule (a standard already tightening). The draft keeps existing rules in the counterfactual (`existing_rules`) and records the trend as an assumption of type `counterfactual_trend`.

Q12. Lock-in and jobs. RS-11 says that people who lose a benefit by moving turn down jobs that would require a move. Is this a praxeological conclusion (moving now costs more, LW-1) or an empirical claim that belongs in an assumption?

Q13. One-time costs against later gains. DM-3 leaves these ambiguous rather than netting them. Do experts reason about them differently, for example by appeal to the actors' own revealed choices?

Q14. Savers aiming at a fixed sum. A lower net return makes some savers save more to reach a fixed target. The draft adds this to TX-6 as an exception that makes total saving ambiguous where such savers matter (WE-011). How do Mises's and Rothbard's treatments of time preference bear on this?

Q15. Violence. Disputes in informal markets may be settled by force (WE-017). Version 1 treats this as a safety effect, out of scope. Should a later version add a variable for it, and what assumption should govern it?

Q16. Are the core rules complete and minimal? Every pattern derives from CR-1 to CR-12, but that shows only that the patterns we have are covered. Is there a kind of praxeological reasoning that experts use and none of the twelve captures? Conversely, can any of them be derived from the others? CR-6 and CR-7 might be two faces of one rule about gains not yet taken.

Q17. Tendencies and equilibrium. CR-6 and CR-7 speak of tendencies: prices tend toward clearing, and yields toward equality across uses. Mises grounds these in the imaginary construction of the evenly rotating economy. Is it legitimate to use the direction of a tendency as an a priori conclusion while treating its speed and completeness as empirical, as the core rules do?

Q18. CR-4's exception. The direction of response to a change in cost is a priori except where the change also alters the actor's total means. That exception covers the Giffen case under LW-3 and the fixed-target saver under TX-6. Is this the right single statement of the exception, and is it complete?

Q19. Which purchases fall when means fall? When an actor's means are reduced, the core rules say that some other use is given up but not which (CR-3, CR-8; WE-004). The method treats this as empirical and requires a premise. Is there a praxeological ground (for example, the marginal use is the one given up, LW-2) for saying more? WE-033 tested this. LW-2 says the use given up is the lowest-ranked one served, but names no good, and saving is one of the uses. A named use was reached only through a market, with premises: public borrowing raises the interest rate and crowds out private borrowing (OC-8).

Q20. Shared facilities and proximity. Some effects travel without exchange: a neighbor's hens, a congested road, a shared fishing ground (GAP-026-1, WE-032), a shared aquifer (WE-039), and residents who value a team's presence without buying anything (GAP-041-1). Catalog 0.7 records them as `proximity` and `shared_facility` relations and reports chains through them at Tier C. Do such effects belong to praxeological analysis, given that the affected party does not choose them? If so, under what premises may they be signed at Tier B?

Q21. Non-rival means. CR-8 treats every means as rival. Know-how carried to a rival firm is used there without being lost to the first (GAP-034-1). Should a core rule cover non-rival means, and what could it say a priori about total output?

Q22. Choices made with more knowledge. A disclosure mandate adds no option and removes none, so CR-3 cannot say whether a buyer who changes an order after seeing a calorie count is better or worse off (GAP-035-C1, from the core-only re-run of WE-035). Is a choice made with more knowledge revealed to be better by the actor's own ranking, or must the method stay silent on it? The same question arises for any policy that works through information (CR-11c).

6.4 Next steps for Phase 1

The Handbook now has 25 worked examples, the number the plan set for Phase 1, and catalog 0.3 closes the gaps they found. What remains before the Phase 1 exit test:

1. **Expert review.** Go through Q1 to Q15, then review each rule and worked example step by step. Log findings by severity, as in Phase 3. The rules added in 0.3 need the same scrutiny as the originals, since they came from the examples rather than from the canonical texts.
2. **Finish the source check.** Every citation has been checked once (`07_sources.md`), mostly from search extracts. Open each text directly, confirm the quotations, and add sources for the patterns added in 0.3, most of which cite only the core rules they derive from.
3. **Record expert walkthroughs.** Ask two or three praxeologists to analyze a fresh policy aloud, and compare their moves with the workflow. Every move they make that is not in the workflow or catalog becomes a gap.
4. **Run the Phase 1 exit test.** Give someone new the Handbook and three unseen policies, and have experts judge their analyses.
5. **Start the calibration question bank** from the worked examples: every assumption and every workflow step becomes one or more typed questions with an expert-labelled answer.

The metamorphic checks in the three repeal examples (WE-022 to WE-024) are already a first test of consistency: each records which of the original consequences reverse, partly reverse or persist.

The worked examples

#	Policy	Main rule families
WE-001	Rent freeze with a new-construction exemption	PC, RS, TS
WE-002	Local excise tax on sweetened drinks, earmarked for pre-school	TX, SB, RS
WE-003	Cosmetology license required for hair braiding	EN, RS, OC
WE-004	Broken window: a town pays to replace storm-damaged windows, funded by a levy	OC, TX
WE-005	Price cap on gasoline after a refinery outage	PC
WE-006	Anti-price-gouging law after a hurricane	PC, TS
WE-007	Cap on interest rates for small consumer loans	PC, RS
WE-008	Milk price floor, with the state buying the surplus	PC, SB
WE-009	Minimum wage increase with a lower youth rate (contested case)	PC
WE-010	Cigarette tax increase in one state, neighbors taxed less	TX, RS
WE-011	Tax on interest income	TX, TS
WE-012	Payroll tax increase paid by employers	TX, DM
WE-013	Tariff on imported steel	TX, RS
WE-014	Import quota on sugar	EN, DM
WE-015	Cap on the number of taxi licenses	EN, SB
WE-016	License requirement for interior designers, with grandfathering	EN
WE-017	Ban on a recreational substance, penalties by weight	EN, RS
WE-018	Ban on single-use plastic bags, paper bags for a fee	EN, RS, TS
WE-019	Temporary car scrappage subsidy	SB, RS
WE-020	First-time homebuyer grant	SB
WE-021	Per-watt subsidy for rooftop solar, funded by a surcharge	SB, PC
WE-022	Repeal of the WE-001 rent freeze	PC, TS
WE-023	Repeal of a long-standing lumber tariff	TX, TS
WE-024	Abolition of the WE-003 braiding license	EN, TS
WE-025	Multi-part housing bill: rent cap, vacancy tax, construction subsidy	PC, TX, SB
WE-026	Ban on keeping chickens on residential lots, with a phase-out (autistic)	EN, RS
WE-027	Bottle deposit-refund scheme	TX, SB, RS
WE-028	Tradable fishing quota shares allocated by history	EN, SB
WE-029	Paid sick leave mandate	PC-17, RS
WE-030	Export subsidy on wheat	SB-10, TX-8
WE-031	Building energy code with a tightening schedule	TS, EN
WE-032	Fare-free buses and trams funded by a payroll tax	PC-19, SB-8
WE-033	One-time household cash payment funded by borrowing	OC-8, SB
WE-034	Ban on non-compete clauses	PC-18
WE-035	Calorie labels on restaurant menus	EN, RS
WE-036	Employer health-insurance mandate with a per-worker charge	PC-17, RS-17, RS-18
WE-037	Cap on ticket resale prices; ban on bulk-buying software	PC, RS
WE-038	Municipal broadband at a set price, revenue bonds	OC-8, RS-16
WE-039	Groundwater pumping cap with tradable rights	EN-6, SB

#	Policy	Main rule families
WE-040	Country-of-origin labels on meat	EN, RS-17
WE-041	Stadium funded by city bonds, leased below market	OC-8, OC-5, SB
WE-042	Aluminum tariff with drawback for exporters	TX-7, TX-8, SB-10
WE-043	Repeal of a minimum lot size	TS-3, TS
WE-044	Short-term rental ban	EN, RS-16, RS-8
WE-045	Recognition of other states' occupational licenses	EN, TS
WE-046	Congestion charge on a city center	TX-1, RS-9, RS-16
WE-047	Sugar tariff-rate quota, licenses by history	EN-6, EN-12, EN-13, TX-7
WE-048	Payroll training levy refunded as vouchers	PC-17, RS-17, RS-18
WE-049	Public auto insurer at a set premium	PC-19, RS-16
WE-050	Ban on new combustion-car sales from 2035	EN-14, TS-3, RS-8
WE-051	Tax-exempt revenue bonds for rental housing, by lottery	OC-8, EN-12, SB-4
WE-052	Repeal of a liquor-license cap	EN-6, EN-13, TS
WE-053	Data-localization requirement	RS-17, TX-8, RS-16
WE-054	Airport night curfew with tradable slots	EN-12, EN-13, RS-8
WE-055	Licensing of home inspectors	EN-1 to EN-4, EN-11, RS-20
WE-056	Insulin out-of-pocket cap with a third-party payer	PC-20, CR-6
WE-057	Wealth tax above 10 million with an exit charge	RS-17, TX-6, SB-4
WE-058	Revenue-neutral shift to a land value tax	TX-2, CR-7c, TS-10
WE-059	Right to repair	PC-17, PC-18, RS-24
WE-060	Basic income funded by VAT	OC-1, TX-9, CR-4
WE-061	Gig-worker reclassification with a buy-out	PC-17, PC-10, RS-23
WE-062	Plastic packaging tax with a recycled-content exemption	TX-2, TX-7, RS-21
WE-063	Repeal of a Sunday-trading ban	RS-9, RS-8, TS
WE-064	Spectrum auction with a build-out obligation	EN-13, PC-17, CR-7c
WE-065	Crop deficiency payments	SB-2, CR-8, EN-12

8. Core-only re-run: report

All 25 worked examples were re-derived using only the core rules (`04a_core_rules.md`), with no patterns, and each re-run was compared with the pattern-based version. The re-runs are in `05_examples_core/`. Each ends with a section comparing it with the pattern-based version, and the repeals also have a metamorphic check against the core-only version of the original.

Method

1. **Inputs.** For each case, only the policy record and the case assumptions (with probabilities) were given. The intro paragraph, the "If false" column and every reference to a consequence or a pattern were removed.
2. **Blind derivation.** Five agents each re-derived five cases. They could read only the core rules, the foundations, the ontology, the workflow and the template, not the existing analyses or the pattern catalog. Where a derivation needed a premise that was not among the case assumptions, they added it (marked A-N) and flagged it. `tools/check_handbook.py --core-only` failed any re-run that cited a pattern.
3. **Comparison.** Five further agents, which had not done the re-runs, matched every consequence of each pair. They classified each one:

   - **M:** same claim, same sign.
   - **V:** same claim, recorded differently.
   - **S:** the signs differ.
   - **OP:** only in the core-only version.
   - **OC:** only in the pattern version.

   Every difference (S, OP and OC) was then given a cause:

   - **P:** the pattern version is wrong.
   - **K:** the core rules cannot derive a valid claim.
   - **A:** a difference of premises.
   - **E:** an analyst error or omission in one version.
   - **J:** a judgment call.

Results

	Count
Consequences compared	786
Same claim, same sign (M + V)	499
Matched, signs differ (S)	42
Only in the core-only version (OP)	162
Only in the pattern version (OC)	83

Of the 541 matched pairs, 499 (92%) agree in sign. The 287 differences break down by cause:

Cause	Count
K: the core rules cannot derive a valid claim	**0**
P: the pattern version is wrong	16
A: a difference of premises	27
E: an analyst error or omission, in either version	178
J: a judgment call, both defensible	66

Per-case counts are in each core file's comparison section.

What the test shows

The core rules are sufficient for these 25 cases. No valid claim in any pattern-based analysis needed anything beyond CR-1 to CR-12, the foundations and arithmetic. The hardest steps all came out of the core rules:

- incidence of taxes and subsidies (CR-7b);
- capitalization (CR-7c);
- quota rents;
- the currency channel from import restrictions to exporters (CR-8b, CR-9);
- the time path of repeals (CR-10b);
- the weak, "at least one of" form of the minimum-wage conclusion (CR-6, CR-4).

The one pattern claim the core rules cannot derive is TS-4 (the intervention spiral), which the catalog already marks as not derivable. It had nonetheless been reported as a consequence in WE-010.

The core rules have one limit by design. When an actor's means fall, CR-3 and CR-8 say that some other use is given up. They do not say which. "The levy reduces sales at local shops" (WE-004) therefore needs a stated premise. This is the method working as intended: which purchases fall is an empirical question.

No pattern is wrong in its sign, but five overclaim and several were misapplied.

Pattern	Problem	Found in
EN-3	Says excluded sellers' money `earnings` fall. CR-3a gives only a loss by their own ranking; their next job may pay more.	WE-003

Pattern	Problem	Found in
EN-4, DM-2	The "down" chain for quality assumes, without saying so, that home and informal provision is lower in quality than the paid provision it replaces.	WE-003
DM-1	Says the side with fewer alternatives bears or receives "the larger part". CR-7b gives a direction, not a comparison of shares.	WE-019, WE-020
RS-6, TX-3	Conclude `employment` decrease. The core rules license only a group: employment or hours.	several
RS-10 (reverse)	Asserts that quality variability falls after a repeal, ignoring that a repeal also admits untrained entrants.	WE-024

Pattern	Missing clause, which led to misapplication	Found in
PC-8	The opposing supply chain (sellers shut out of the controlled segment move into the uncontrolled one) was in a note, so it was missed.	WE-015
OC-3	No case for a fixed enforcement budget, or for enforcement funded from the policy's own revenue.	WE-010, WE-012
PC-2	No group form, inviting separate signs for waiting, search and non-price allocation.	WE-006, WE-009
SB-4	Limited to "lasting" policies, so a five-year subsidy's capitalization was missed.	WE-021
SB-1	Records the payers of a subsidy but not its recipients, against CR-8's symmetry.	WE-021
SB-3	Applied to a delay, when the policy only shifted timing.	WE-004
RS-7	Applied without estimating its own premise (that the group can organize). In repeals, it ignores that the other side lobbies in the counterfactual.	all; WE-022 to WE-024

Both versions make the same kinds of analyst errors. Most differences (178 of 287) are omissions or slips that the rules would have caught.

- **Failing to combine opposing chains** (CR-12b) that the analysis had itself derived. This was the most common sign error in both versions. Examples: WE-001 short-run rent, WE-002 private pre-schools, WE-007 under reading U1-b, WE-011 capital stock, WE-025 starting rents.
- **Signing one member of a group** (`employment` alone) where only "employment or hours" is licensed.
- **Recording a timing shift as a note** when the action also takes place in the counterfactual, so it should be signed per window.
- **Inferring `resources_available_other_uses` decrease from "worse off"** (CR-3a). The inference fails when some members of the class respond by buying less.
- **Measuring against the wrong baseline:** against the controlled rent, or against a different policy, rather than against no policy (WE-001 C-10, WE-025 C-19, WE-018).
- **Omissions specific to the pattern versions.** These repeat across cases, because no pattern prompts for them:
  - capitalization (SB-4), and the creation of an asset by the policy;
  - the enforcement-cost half of OC-3;
  - effects on the payer of a transfer as well as the recipient.
- **Omissions specific to the core-only versions:** many effects that were easy to derive but were not followed. Without a pattern to prompt them, the analyst must remember to apply every core rule to every actor class.

Premises that the patterns had left implicit. The blind re-runs added about 40 premises. Some recur:

- that an interest group can organize (behind every use of RS-7);
- that a new license is actually used;
- that sellers sell at world prices (behind the tariff pass-through);
- that an earmark exceeds what would have been spent anyway;
- that an asset's expected life covers the policy's duration.

Changes made in response

The following changes are made in catalog 0.5 and the accompanying files.

- **Patterns:** EN-3, EN-4, DM-1, DM-2, RS-6, TX-3, RS-10, PC-8, OC-3, PC-2, SB-4, SB-1, SB-3 and RS-7 are revised as listed above.
- **Core rules:** wording is clarified where the re-runs found it unclear:
  - CR-1: provisions that fit two operations;
  - CR-3: a loss by the actor's own ranking is not a fall in their other spending;
  - CR-6: fixed quantities, existing rules and buyers of last resort;
  - CR-7b: no comparison of shares, and three or more parties;
  - CR-7c: assets created by the policy, and changes of limited duration;
  - CR-12: what counts as a dominance premise, chains that exactly cancel, and consequences reached by independent chains.
- **Workflow:** a final check, before the report, for each of the recurring errors above.
- **Checker:** warns when TS-4 appears in a consequence's derivation.

The 25 pattern-based worked examples were then corrected against their comparisons and the new final checklist. Each correction was checked against the core rules before it was applied. Most comparison conclusions were accepted. A few were rejected, and the reasons are recorded in the example:

- **WE-006:** a lobbying premise was kept at p 0.6.
- **WE-009:** the core version's "output and consumption fall" was not added, as it would overclaim in the contested case.
- **WE-012:** "employment or hours" was kept for workers below the contribution ceiling, rather than a plain fall in hours.
- **WE-019 and WE-020:** the split of a subsidy was kept as a comparison of shares, with its premise.

The corrections also went beyond the comparison in places, for consistency with the revised patterns. Some premises they added need expert review. For example:

- WE-025's A-29, the dominance premise behind new rental investment;
- WE-006 and WE-009's reading of enforcement budgets as fixed.

The core-only re-runs are kept unchanged, as the record of the test.

Second round: WE-026 to WE-035

The ten gap-probing examples were re-run the same way: blind derivation from the policy record and the case assumptions only, then a comparison by an agent that had written neither version. The derivations used core rules 0.7; the pattern versions were written under catalog 0.6.

This round is not fully independent. Many 0.7 clarifications to the core rules, and much of the 0.7 vocabulary, were drawn from these same ten examples' "Gaps found" sections. So agreement on those points is expected, not a test. The comparisons flag where it matters (WE-029, WE-030, WE-033, WE-035 especially). A clean test of 0.7 needs new cases.

Results

Case	M	V	S	OP	OC	P	K	A	E	J
WE-026 own-production ban	11	12	3	14	0	1	0	2	13	1
WE-027 deposit-refund	16	9	0	17	6	1	0	3	16	3
WE-028 fishing quota shares	19	15	1	19	4	3	0	1	18	2
WE-029 paid sick leave	18	10	0	19	3	0	0	4	14	4
WE-030 export subsidy	27	7	3	17	2	4	0	2	13	3
WE-031 energy code	16	13	3	15	2	0	0	1	16	3
WE-032 fare-free transit	25	15	5	22	5	2	0	1	17	12
WE-033 debt-funded payment	9	0	9	15	4	4	0	1	19	4
WE-034 non-compete ban	13	8	3	16	1	0	0	0	15	6
WE-035 calorie labels	11	10	3	5	1	1	0	0	5	3
Total	**165**	**99**	**30**	**159**	**28**	**16**	**0**	**15**	**146**	**41**

Of 481 consequences compared, 294 were matched; 264 of these (90%) agree in sign. The comparisons assigned 218 causes to the 217 differences (one difference was given two). **K is again 0**: no valid claim in a pattern version needed anything beyond the core rules.

What the round shows

The core rules held on new kinds of policy. Autistic bans, refundable charges, created tradable rights, mandated and prohibited contract terms, an export subsidy, a scheduled standard, public supply at zero price, a debt-funded transfer and a disclosure mandate were all derivable from CR-1 to CR-12. WE-033 confirms the answer to Q19: for a pure change in means the rules give totals and the rank of the use given up, never a named good, and that is correct rather than a gap.

Calculus gaps the core versions raised. Most are wording or premises, not missing rules. Grouped:

Gap	Assessment
GAP-035-C1: whether a buyer who changes an order after a disclosure is better or worse off. CR-3 is stated over options, and a disclosure changes none.	**A real question** for the experts (new Q22): is a choice made with more knowledge revealed to be better by the actor's own ranking, or does the method stay silent?
GAP-028-C1: growth of a shared natural stock (fish) after the cap	Physical relation; belongs with Q20 (shared facilities). Needs an empirical premise, not a rule.
GAP-032-C1: bus dwell time	Technical relation; an empirical premise, not a rule.
GAP-034-C1: non-rival know-how	Same as GAP-034-1 (Q21).
GAP-030-C1: sign of export receipts (price falls, quantity rises)	Correctly ambiguous without a premise; not a gap.
GAP-033-C1: spending rounds after the first recipient	Stops at the monetary boundary, out of scope in version 1.
GAP-034-C2: cumulative location effects	Depth limit (Q6).
GAP-032-C3: the council's choice at review	Political response (Q5).
GAP-027-C2, GAP-032-C2, GAP-033-C2, GAP-027-C1: a public buyer at a fixed fee; a contingent payer; an entitlement larger than its budget; local program budgets	Extensions of the 0.7 clarifications to CR-6 and CR-8. Wording.
GAP-028-C2: who pays for the value of a right created by ending a race	Derivable: the value comes partly from means no longer used up in the race (CR-8c), which is not a transfer. Wording for CR-7c.
GAP-026-C1 (culling), GAP-029-C1, GAP-029-C2, GAP-030-C2, GAP-030-C3, GAP-031-C1, GAP-031-C2, GAP-035-C2	Vocabulary, premises or wording; the comparisons judged GAP-029-C1 and C-2 not real (CR-1 and CR-12e cover them).

Pattern problems. No pattern is wrong in sign. Problems found:

Pattern	Problem	Found in
OC-3	Signs enforcement cost against zero enforcement. When the policy shrinks an activity that was itself enforced (complaints under a nuisance law), there is an opposing chain.	WE-026
OC-3, TX-1, OC-1	Name taxpayers alone as payers from the general budget. Already fixed by the 0.7 CR-8 clarification (a group); the pattern texts should say so.	WE-027, WE-028, WE-029, WE-030, WE-031
RS-1	Signs `quantity_substitute` from CR-4 alone; the split between price and quantity needs a supply premise, as RS-16 asks.	WE-031
OC-5	Gives a group sign (employment or hours) even where the activity is new and starts from zero, where a plain rise in employment is derivable.	WE-027
TS-3	Its derivation through a clearing price does not fit a product standard; binding is CR-1 and CR-4 over the spread of designs.	WE-031
EN-6 (0.7 Unless)	Says `price_paid` is not signed when the good is priced in a wider market. The price does not ration the cap, but it still moves in the CR-6 direction.	WE-028
SB-10 (new in 0.7)	Should note that round trips (RS-4) are opposed by any check on the return crossing.	WE-030
RS-8	Applied one way only where a pending-case clause gives an opposing timing chain.	WE-034

The recurring analyst error is unchanged: failing to combine opposing chains the analysis itself derived (WE-028 C-29, WE-030 C-19 and C-25, WE-034 C-11, WE-035 C-14 in the pattern versions; WE-032 C-5 and C-32 in the core versions). Both versions also again signed one member of a group (a price alone where CR-6 gives price or waiting).

Core-rule wording the 0.7 clarifications left unclear.

- CR-12's 0.7 note ("an opposing chain that cannot be derived is not a chain") was read too widely. It should separate an underivable chain (a Tier C note) from a derivable chain that needs a premise the case lacks (state the premise with p) and from a chain whose own sign is ambiguous (WE-030, WE-031, WE-035).
- CR-1's recording note on a void-contract rule says one side loses an option and the other gains one; both lose the option of agreeing to the term (WE-034).
- CR-6: a private supplier that chooses not to charge (WE-026); a capped good only partly priced in a wider market (WE-028); whether a free price's rise satisfies "at least one margin" by itself (WE-031).
- CR-7c: claims fixed before the announcement (a windfall) against history still to be earned (WE-028).
- CR-12e: whether "a response cannot reverse its cause" binds members of a class who did not respond (WE-032); when counts over a distribution are arithmetic (WE-031).
- CR-4's fixed-cost note: whether it covers costs fixed per outlet as well as per item (WE-035).

Changes made in response

- **Catalog 0.8.** The pattern fixes above (OC-3, OC-1, TX-1, RS-1, OC-5, TS-3, EN-6, SB-10, RS-8) and the core-rule wording above. In CR-12, three kinds of opposing chain: one no rule derives (a Tier C note), one that needs a premise the case lacks (state it with p), and one whose own sign is ambiguous. Also in CR-12: two items moving in opposite directions within one variable. In CR-8: contingent payers, capped budgets and value created by ending a race. In CR-6: public buyers at a fixed fee, and when a free price's rise is enough.
- **Q22** added to `06_gaps.md`.
- **WE-026 to WE-035 corrected** against their comparisons, now at catalog 0.8. Each correction was checked against the core rules before it was applied. Each example has a section "Corrections after the core-only comparison" listing what changed and which comparison conclusions were rejected and why. Some of the rejections:
  - WE-026: the core version's waiting time alone, where CR-6 gives a group.
  - WE-029: a ranking of short-run shares, which needs a premise.
  - WE-030: home flour sales, which are ambiguous once the opposing flour chain is.
  - WE-032: a medium-run ambiguity for employers whose staff do not use transit.
  - WE-034: the core version's single-member sign for funded training.
  - WE-035: a fall in item variety, since the opposing chain (new lighter items) is derivable.
- In all, the corrections changed about 100 consequences, added about 120 and removed 5 (WE-028 C-15, WE-032 C-9 and C-14, WE-033 C-20 and C-21). The removals are claims that needed a premise the case lacks or that duplicated another row.

The core-only re-runs are kept unchanged, as the record of the test.

Third round: the held-out test, WE-036 to WE-045

WE-036 to WE-045 were written with the catalog frozen at 0.8, on cases the 0.7 and 0.8 changes were not drawn from (`06_gaps.md`, 6.2f). They were then re-run blind from the core rules, as before, and compared by agents that had written neither version. Unlike the second round, this one is independent: no catalog change was made between writing the examples and the re-runs.

Two lapses from the blind protocol were reported by the agents themselves: one read past the cut-off in the format reference (WE-021's comparison section, which cites patterns for another case); another ran one `git log` line after finishing its analyses. Neither re-run cites a pattern, and neither lapse concerned its own case.

Results

Case	M	V	S	OP	OC	P	K	A	E	J
WE-036 health cover mandate	20	19	6	11	4	1	0	0	18	2
WE-037 ticket resale cap	24	9	3	11	7	0	0	1	16	4
WE-038 municipal broadband	14	10	4	8	1	0	0	2	7	4
WE-039 groundwater allocation	23	5	2	14	4	0	0	0	20	0
WE-040 origin labeling	25	13	0	10	7	1	0	0	10	6
WE-041 stadium bond	15	10	3	12	10	3	0	0	16	6
WE-042 tariff drawback	25	11	4	12	7	3	0	1	15	4
WE-043 lot size repeal	11	9	2	13	6	0	0	1	17	3
WE-044 short-term rental ban	30	10	6	14	5	4	0	1	14	6
WE-045 license recognition	19	6	3	12	7	2	0	2	15	3
Total	**206**	**102**	**33**	**117**	**58**	**14**	**0**	**8**	**148**	**38**

Of 516 consequences compared, 341 were matched; 308 of these (90%) agree in sign. The 208 differences have 208 causes. **K is 0 for the third time.** Every step in ten cases written against a frozen catalog was derivable from CR-1 to CR-12.

What the round shows

The core rules held; several patterns did not. The comparisons confirmed, from the core rules alone, every pattern failure the pattern versions had reported (RS-18, SB-10, EN-11, PC-8, PC-17 and RS-17's value clause), and found more:

- **OC-8** gives the wrong sign where the counterfactual also borrows (a team that would have renovated with its own loan): other borrowers' rate is `ambiguous`. The CR-8 borrowing note shared the flaw. (WE-041)

- **RS-17's value clause** failed a second time, on the 50-staff line. (WE-036)
- **OC-3's fixed-budget clause** assumes the time comes from enforcing other rules; it can come from any other use of the agency's time. (WE-042)
- **RS-16** misses a cost the policy puts on the substitute itself. (WE-040)
- **RS-7's repeal clause** reads as if the old rule was always contested. (WE-045)

One promotion was wrong. "Fixed cost per item reduces variety" was counted as used a second time in WE-040. The comparison showed the use was invalid: in the counterfactual each cut had one version, so a fixed cost per version cannot reduce the count. It stays a candidate.

Some "worked as written" verdicts were wrong. The comparisons contradicted the pattern versions' own verdicts on CR-12 0.8 kind 2 (WE-039, misapplied), RS-17 (WE-036), the drawback as an export subsidy (WE-042, once holding costs are counted), and RS-8's Unless (WE-036, WE-044). Self-assessment by the analyst is not enough; the blind re-run is needed.

The Tier C problem is real. Both versions, independently, met a Tier C chain opposing a Tier B chain with no rule for combining them (WE-039, WE-041, WE-044).

New calculus gaps raised by the core versions: GAP-039-C1 (the counterfactual race to pump a shared stock: a physical relation, under Q20), GAP-039-C2 (the value of a lottery entry; changed no sign), GAP-045-C1 (how beliefs adjust after a policy stops conveying something: empirical). None produced a difference.

The recurring analyst errors are unchanged: uncombined opposing chains (in both versions), employment signed alone, signs taken against the other reading or against no lobbying instead of the counterfactual.

Changes made in response: catalog 0.9

- **Core rules.** CR-12: a rule for Tier C against Tier B (report the Tier B sign, qualified, and the combined sign at Tier C); what the response argument gives (not reversed; `increase` needs a partial response); kind 3; activities that exist only under the policy; flows and stocks. CR-8: the borrowing baseline, revenue bonds, tax-exempt interest, a permissive reading, mutual rights. CR-6: splitting a shift into price and quantity needs a premise; a public seller with an open quantity; a capped price with a free fee. CR-7: assets without a market price, contract terms that remove an option, persistence following the stock. CR-11: sellers' response to silence; beliefs after a signal ends. CR-1, CR-3, CR-4: operations that switch at a cap, refunds on a standard, lobbying in repeals.
- **Patterns.** Corrected: RS-17, RS-18, SB-10, EN-11, OC-8, PC-8. Widened: PC-17, PC-19, RS-4, RS-7, RS-8, RS-9, RS-16, OC-3, OC-5, EN-6, TX-8. Promoted: RS-19, RS-20, OC-9, EN-12, EN-13, EN-14, EN-15. 98 patterns.
- **Ontology and workflow.** Instrument `mandate_with_charge`; variables `resource_stock`, `enrollment`; for a repeal with no earlier analysis, derive the original rule first.
- **WE-036 to WE-045 corrected** against their comparisons under catalog 0.9. Each correction was checked against the core rules before it was applied; each example has a section "Corrections after the core-only comparison", and its part (e) "Test of 0.8" is kept as the record of the test, with a "Re-run note" where the comparison contradicted a verdict. Five consequences were removed: WE-037 C-20 and C-43, WE-038 C-14, WE-040 C-10 and WE-045 C-15. The new Tier C rule is applied throughout, for example in WE-039 (Tier B signs "not counting effects through the aquifer", with combined signs at Tier C).

Fourth round: the held-out test of catalog 0.9, WE-046 to WE-055

WE-046 to WE-055 were written with the catalog frozen at 0.9 (`06_gaps.md`, 6.2g), then re-run blind from the core rules and compared by agents that had written neither version. The blind protocol was tightened after the third round (a line limit on the format reference; no git history); every re-run agent reported keeping to it.

Results

Case	M	V	S	OP	OC	P	K	A	E	J
WE-046 congestion charge	27	13	4	9	12	1	0	0	23	1
WE-047 sugar tariff-rate quota	27	8	5	7	5	2	0	0	11	4
WE-048 training levy	18	13	4	10	6	1	0	2	14	3
WE-049 public auto insurer	13	14	0	9	3	0	0	2	7	3
WE-050 combustion-car ban	24	7	2	17	4	2	0	0	15	6
WE-051 housing revenue bonds	21	9	2	9	4	1	0	1	10	3
WE-052 liquor license cap repeal	23	8	6	11	3	0	0	2	16	2
WE-053 data localization	16	10	5	9	4	0	0	3	11	4
WE-054 airport night curfew	27	10	0	7	5	0	0	1	9	2
WE-055 home inspector licensing	19	7	2	7	3	0	0	1	6	5
Total	**215**	**99**	**30**	**95**	**49**	**7**	**0**	**12**	**122**	**33**

Of 488 consequences compared, 344 were matched; 314 of these (91%) agree in sign. The 174 differences have 174 causes. **K is 0 for the fourth time.** "Pattern version wrong" fell from 14 to 7, and most of those were analysts leaving a chain uncombined rather than a pattern giving a wrong sign.

What the round shows

Every failure the analysts reported was confirmed by the blind re-runs: the 0.9 response argument (fixed costs per firm), PC-17's buy-out clause, RS-16's 0.9 clause, TX-7 under a tariff-rate quota, EN-14, RS-20 and EN-11's baselines. Unlike the third round, few "worked" verdicts were contradicted (WE-047's response argument in part, WE-052's CR-7 0.9 "not needed").

One systemic fault. Three patterns (RS-20, EN-11's 0.9 clause, EN-13) compared with something other than the no-policy counterfactual. Catalog 1.0 re-bases them and adds a general rule and a final-check item.

"Record as negligible" is wrong. OC-8's small-borrower clause and EN-6's wider-market clause let an effect go unsigned; CR-6 asks for a sign. Both versions left prices unsigned because of it.

Tier C settled. In WE-046 the core version put road and bus crowding at Tier B through CR-6's public-supply clause. The comparison found CR-9 (Tier C) right: the public-supply clause covers only excess demand over a fixed quantity, and gives a group sign. Catalog 1.0 says so.

Promotions. Five candidates had a valid second use and are promoted; "tariff-like burden without a tax" did not (WE-053 is a location requirement with a route around it, a different move).

No new calculus gaps. The core versions raised several (GAP-046-C1 to GAP-055-C2); the comparisons judged each to be a premise, a vocabulary item, or out of scope (safety, productivity, security).

Changes made in response: catalog 1.0

- **Core rules.** CR-12: what the response argument needs (constant cost; which total; a second response is a second chain); dominance does not raise a tier; the baseline of patterns. CR-6: the scope of the public-supply clause; one public price for many costs; a quota on a rate; small effects are signed. CR-8: receipts into the same budget; statutory rights. CR-9: a charge on users is an exchange relation. CR-7: which stock. CR-1, CR-3, CR-4: notches, capped refunds, lobbying generally.
- **Patterns.** Re-based: RS-20, EN-11, EN-13. Corrected: PC-17, RS-16, TX-7, EN-14, EN-6, OC-8. Widened: EN-1, EN-2, PC-19, RS-7, RS-8, RS-13, RS-17, RS-18, TS-5, OC-1. Promoted: RS-21, RS-22, RS-23, TS-10, TX-12. 103 patterns.
- **Workflow and ontology.** Final check item 9 (the baseline of every pattern); variables `stock_age`, `applications`.
- **WE-046 to WE-055 corrected** against their comparisons under catalog 1.0. Each correction was checked against the core rules before it was applied; each example has a section "Corrections after the core-only comparison", and its part (e) "Test of 0.9" is kept as the record of the test, with a "Re-run note" where the comparison contradicted a verdict. One consequence was removed (WE-050 C-6, a change of good recorded as a price change). Examples of the 1.0 rules at work: bus crowding below capacity moved to Tier C (WE-046); the importers' count re-based on no policy under EN-13 (WE-047); small effects now signed (WE-047, WE-051, WE-054); response arguments state their total and their constant-cost premise throughout.

Fifth round: the held-out test of catalog 1.0, WE-056 to WE-065

WE-056 to WE-065 were written with the catalog frozen at 1.0 (`06_gaps.md`, 6.2h), re-run blind from the core rules, and compared by agents that had written neither version. The re-run agents all reported keeping to the blind protocol.

Results

Case	M	V	S	OP	OC	P	K	A	E	J
WE-056 insulin cap	24	9	5	13	4	1	0	0	16	5
WE-057 wealth tax	20	8	3	9	5	0	0	0	13	4
WE-058 land value tax	32	7	3	8	8	1	0	1	11	6
WE-059 right to repair	23	6	6	5	7	1	0	0	13	4
WE-060 basic income	29	12	7	6	14	0	0	1	23	3
WE-061 gig workers	20	12	1	7	6	1	0	0	11	2
WE-062 packaging tax	30	15	2	13	8	2	0	0	21	0
WE-063 Sunday trading repeal	17	13	2	13	9	1	0	0	23	1
WE-064 spectrum auction	23	6	5	8	6	0	0	0	14	5
WE-065 crop payments	18	11	1	11	6	1	0	0	10	7
Total	**236**	**99**	**35**	**93**	**73**	**8**	**0**	**2**	**155**	**37**

Of 536 consequences compared, 370 were matched; 335 of these (91%) agree in sign. **K is 0 for the fifth time.** "Pattern version wrong" stayed low (8), and every one of those was an analyst misapplying a pattern, not a pattern giving a wrong sign as written.

What the round shows

- **The baseline rule (1.0) works in use but the pattern texts lagged.** Analysts re-signed RS-17, RS-23, PC-17 and EN-13 against the counterfactual, as the rule asks; 1.1 rewrites those clauses.
- **Failures confirmed by the core versions:** SB-4 (a tax that reaches buyers' alternatives), PC-1 and PC-16 (a cap on the buyer's share with a third-party payer), TX-7 (a matched domestic tax), TX-9 (revenue returned to the payers), EN-14 (a standard buyers value), PC-17's buy-out clause (a payment to the counterparty).
- **Core-rule wording** that both versions found hard: the response argument for an expansion or a price fall; items that cancel against CR-12f; using a group member as a chain; "the same question" for lobbying; a right sold at creation; a third-party payer.
- **No calculus gaps.** The core versions raised fifteen; the comparisons judged each to be a premise, a recording limit, vocabulary, or out of scope.
- **The analysts' own verdicts were mostly confirmed;** a few "worked" verdicts were contradicted, nearly all for an uncombined opposing chain.

Changes made in response: catalog 1.1

- **Core rules.** CR-12: the response argument in every direction; items that cancel; group members are not chains; ratios. CR-6: a third-party payer; margins the policy creates; labor demand as a group. CR-7: a right sold at creation; alternatives taxed too. CR-4: several changes in means; the same question. CR-1, CR-3, CR-8: lines on ratios, burdens inside a new option, dedicated funds.
- **Patterns.** Re-based: RS-17, RS-23, PC-17, EN-13. Corrected: SB-4, PC-1, PC-16, TX-7, TX-9, EN-14, SB-8, TX-1. Widened: twelve more. New: PC-20. Promoted: RS-24. 105 patterns.

- **WE-056 to WE-065 corrected** against their comparisons under catalog 1.1, each with a section "Corrections after the core-only comparison" and its part (e) "Test of 1.0" kept as the record of the test, with a "Re-run note" where the comparison contradicted a verdict. Four consequences were removed (WE-058 C-23 merged into C-22, WE-060 C-27 merged into C-26, WE-061 C-32 and C-39), each because 1.1 or the comparison showed it rested on a misapplied pattern or a missing premise.

7. Sources

The texts the Handbook cites, the editions used, and how each citation was checked.

How the citations were checked

Every citation in `01_foundations.md`, `04a_core_rules.md` and `04_rules.md` was checked against the texts in October 2026. Each was checked in one of three ways:

- **[d] direct:** the text itself was opened and read.
- **[s] search extract:** the chapter and section were confirmed from the publisher's online edition (chapter and section titles in the page addresses and extracts), but the full page was not opened. Quotations marked [s] are close, and should be checked against the page before they are printed.
- **[u] unverified:** the source could not be checked. The citation is the best available, and must be checked before it is relied on.

The online editions of Mises, Rothbard and Bastiat (mises.org, econlib.org) could not be opened directly from the working environment, so most citations are [s]. Opening them directly is the first task for whoever reviews this file.

Editions and abbreviations

Short form	Work	Edition used
HA	Ludwig von Mises, *Human Action: A Treatise on Economics*	Scholar's Edition (Mises Institute, 1998); chapter and section numbers as in the online edition at mises.org
MES	Murray N. Rothbard, *Man, Economy, and State*	Scholar's Edition with *Power and Market* (Mises Institute, 2004); MES chapters 1 to 12
P&M	Murray N. Rothbard, *Power and Market*	In the same volume, with its own chapters numbered 1 to 7. Cite as "P&M ch. n", not as an MES chapter. MES ch. 12 covers some of the same ground in condensed form.
EIOL 1946	Henry Hazlitt, *Economics in One Lesson*	First edition (Harper, 1946), as reprinted by FEE (1952) [d]
EIOL 1979	Henry Hazlitt, *Economics in One Lesson*	Revised edition (Crown, 1979), which adds ch. 18 "What Rent Control Does" and renumbers the chapters after it
Bastiat 1850	Frédéric Bastiat, "That Which Is Seen, and That Which Is Not Seen"	1850 essay; section 1 is "The Broken Window"

The two Hazlitt editions number their chapters differently from ch. 18 onward:

- **Chapters 1 to 17** are the same in both: "The Lesson", "The Broken Window", ... "Government Price-Fixing".
- **1946:** ch. 18 is "Minimum Wage Laws" and the book ends at ch. 24, "The Lesson Restated".
- **1979:** ch. 18 is "What Rent Control Does", ch. 19 is "Minimum Wage Laws", and ch. 26 is "The Lesson After Thirty Years".

The 1946 edition has no chapter on rent control.

Bibliography, with verification

Work	Status	Used for
HA ch. I §1 "Purposeful Action and Animal Reaction"	[s] "Human action is purposeful behavior."	AX-1, CR-1, CR-2
HA ch. IV §4 "Action as an Exchange"	[s] "Costs are equal to the value attached to the satisfaction which one must forego in order to attain the end aimed at."	LW-1, CR-3
HA ch. VII §1 "The Law of Marginal Utility"	[s]	LW-2, CR-2, CR-4
HA ch. VII §2 "The Law of Returns"	[s] "for the combination of economic goods of the higher orders (factors of production) there exists an optimum"	LW-9
HA ch. XIV §5 "The State of Rest and the Evenly Rotating Economy"	[s]	CR-7
HA ch. XV §4 "The Sovereignty of the Consumers"	[s]	LW-4, CR-7
HA ch. XVI §1 "The Pricing Process"	[s] each party values what he receives above what he gives	CR-5, CR-6
HA ch. XVI §3 "The Prices of the Goods of Higher Orders"	[s]	LW-10, CR-9
HA ch. XVIII §2 "Time Preference as an Essential Requisite of Action"	[s]	LW-7, CR-10
HA ch. XVIII §5 "The Convertibility of Capital Goods"; §8 "The Mobility of the Investor"	[s]	LW-4, CR-7, CR-10
HA ch. XXI §6 "Wages and Subsistence"	[s] for the argument; quotation [u]	TS-9
HA ch. XXVI	[u] (calculation under socialism)	CR-11
HA ch. XXX §2 "The Market's Reaction to Government Interference"	[s] "A different principle for the allocation of the goods ... comes into play"; price ceilings must be extended to other goods	LW-6, CR-6, PC-1, PC-2, TS-4
HA ch. XXX §3 "Minimum Wage Rates"	[s]	PC-10
HA ch. XXXVI "The Crisis of Interventionism"	[s]	TS-4
MES ch. 1 §5 "Further Implications: Ends and Values"	[s]; quotation [u]	LW-2, CR-2
MES ch. 1 §6 "Factors of Production: The Law of Returns"	[s]	LW-9
MES ch. 2 §4 "Terms of Exchange"	[s] exchange involves "a double inequality of values"	LW-5, CR-5
MES ch. 2 §5 "Determination of Price: Equilibrium Price"; ch. 4 "Prices and Consumption"	[s]; an explicit "law of demand" sentence [u]	LW-3, CR-4, CR-6
MES ch. 6 §1 "Many Stages: The Pure Rate of Interest"; §3 "Time Preference and Individual Value Scales"	[s]	LW-7, CR-7, CR-10

Work	Status	Used for
MES ch. 7 §1 "Imputation of the Discounted Marginal Value Product"; §5 "Capitalization and Rent"	[s] "the capital value of a 'whole factor' will be equal to the sum of its future rents"	LW-10, CR-7c, CR-9, SB-4
MES ch. 8 §1 "Entrepreneurial Profit and Loss"	[s]	CR-7
P&M ch. 2 §1 "Types of Intervention"	[s] "the intervener compels or prohibits exchanges between sets of two other individuals"	Intervention typology, CR-1
P&M ch. 2 §2 "Direct Effects of Intervention on Utility"	[s] "Any exchange on the free market ... occurs because it is expected to benefit each party concerned"	LW-5, CR-3, CR-5, OC-4
P&M ch. 3 §1 "Price Control"	[s] a maximum price creates a shortage, a minimum price an unsold surplus	PC-1, PC-9, OC-4
P&M ch. 3 §3 "Product Control: Grant of Monopolistic Privilege"	[s] "Licensing is an example in which those denied a license by the government are barred from pursuing the trade or business."	EN-1, EN-6
P&M ch. 4 "Binary Intervention: Taxation", §3A "The General Sales Tax and the Laws of Incidence"	[s] "No tax can be shifted forward."	TX-1, DM-1 (note)
P&M ch. 5 "Binary Intervention: Government Expenditures"	[s]	CR-8, SB-1
Rothbard, "Toward a Reconstruction of Utility and Welfare Economics" (1956)	[s] demonstrated preference	LW-5, CR-2
EIOL 1946 ch. 2 "The Broken Window"	[d] "The glazier's gain of business, in short, is merely the tailor's loss of business."	OC-1, OC-5, CR-8
EIOL 1946 ch. 4 "Public Works Mean Taxes"	[d] "for every public job created by the bridge project a private job has been destroyed somewhere else"	OC-5, CR-8
EIOL 1946 ch. 11 "Who's 'Protected' by Tariffs?"; ch. 12 "The Drive for Exports"	[d] "Without imports we can have no exports, for foreigners will have no funds with which to buy our goods."	TX-7, TX-8
EIOL 1946 ch. 15 "How the Price System Works"	[d] supply expands "until the profit margin on that article once more falls to the general level of profits"	CR-6, CR-7
EIOL 1946 ch. 17 "Government Price-Fixing"	[d] "fixing a maximum price ... would be to bring about a shortage of that commodity"	LW-6, PC-1, PC-2
EIOL 1979 ch. 18 "What Rent Control Does"	[u] (quoted only at second hand: landlords "will not even keep rented houses or apartments in tolerable repair")	RS-2, PC-4, PC-5
Bastiat 1850, §1 "The Broken Window"	[s] the six francs would have "replaced his old shoes, or added another book to his library"	LW-1, OC-1, CR-8
Bastiat, *Economic Sophisms*, First Series (1845), ch. VII, "A Petition" (the candlemakers' petition)	[s]	TX-7
Menger, *Principles of Economics* (1871), ch. I and ch. III "The Theory of Value"	[s] "the value of goods of higher order is always and without exception determined by the prospective value of the goods of lower order in whose production they serve"	LW-10, CR-9
Mises, *A Critique of Interventionism* (1929; trans. Sennholz, 1977), Part I	[s] for the argument; section heading [u]	TS-4
Hayek, "The Use of Knowledge in Society", *American Economic Review* 35(4), 1945, pp. 519 to 530	[s]	LW-8, CR-11
Mises, "Economic Calculation in the Socialist Commonwealth" (1920; English translation in Hayek, ed., *Collectivist Economic Planning*, 1935)	[s]	LW-8, CR-11
Richard Cowan, "How the Narcs Created Crack", *National Review*, 1986	[s]; exact issue date [u]	EN-8
Mark Thornton, *The Economics of Prohibition* (University of Utah Press, 1991)	[s]; chapter on potency [u]	EN-8
Card and Krueger, "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania", *American Economic Review* 84(4), 1994, pp. 772 to 793	[s]	PC-10 (note)

Findings from the check

- **Cost is defined in HA ch. IV**, not ch. VII. Ch. VII is marginal utility and the law of returns.
- **P&M ch. 1 is "Defense Services on the Free Market"**, which is the wrong place for the mutual benefit of exchange. That is P&M ch. 2 §2.
- **The tailor is Hazlitt's** (EIOL 1946 ch. 2). Bastiat's example of the forgone purchase is shoes or a book. OC-5 is renamed accordingly.
- **The iron law of prohibition is Richard Cowan's** (1986), not "Cowen".
- **The rent control chapter exists only in the 1979 edition** of Hazlitt. Its arguments on maintenance are cited at second hand only.
- **"Incidence follows the alternatives" is not Rothbard's position.** DM-1, and the incidence clause of CR-7b, state the standard treatment. Rothbard's first law of incidence holds that no tax can be shifted forward: the burden falls on the incomes of original factors, and any later rise in price comes about through reduced supply, which he classes as an effect of the tax rather than shifting. The two agree on what happens (the burden first falls on those who cannot escape it, and prices rise only as supply contracts) but describe it differently. The disagreement is recorded under DM-1 and in Q3.

Appendix A. Analysis template

WE-XXX / Analysis:

Analyst: | Date: | Rule catalog version: 0.1

Fill in each section in order, following `../03_workflow.md`. Cite patterns (`../04_rules.md`) and core rules (`../04a_core_rules.md`) by ID. Where no pattern fits, derive the step from the core rules and cite them; mark `GAP` only if no core rule applies, and copy it to `../06_gaps.md`.

1. Policy record

Field	Value	Source sentence
Jurisdiction (and what lies just outside)		
Announced / enacted / effective / expires		
Enforcement (who, penalties, resources)		

Component	Instrument	Target	Bound	Parameters	Exemptions
K1					

Unclear provisions:

2. Classification

Component	Rothbard type	Binding? (assumption ID)	Rule families
K1			

3. Actor classes

Actor class	Role	How its options change

4. Direct effect

ID	Actor class	Variable	Sign	Horizon	Rules	Assumptions
C-1						

5. Responses

For each actor class, check RS-1 to RS-8: substitute, adjust quality or terms, evade, relocate, reclassify, exit, lobby, change timing.

ID	Actor class	Variable	Sign	Horizon	Rules	Assumptions

6. Second-order effects

Depth limit 4; prune diffuse bystander effects after depth 3. Name the market relation each step uses. Check bundled trips (RS-9), informal trade (RS-10) and lock-in (RS-11).

ID	Actor class	Variable	Sign	Horizon	Rules	Market relation	Assumptions

7. Combined effects

Actor class, variable, horizon	Chains (+)	Chains (−)	Dominance rule	Combined sign

8. Time structure

Notes on durability, effects that grow or fade, expectations and timing shifts.

9. Opportunity costs

Write each as a signed change in a variable, not as a bare loss.

ID	Actor class	Variable	Sign	What is given up	Rules

10. Assumptions

ID	Statement	Type	p	Evidence	If false
A-1					

11. Report

The policy:

Visible consequences:

Invisible consequences:

Ambiguous effects:

Key conditions:

What this analysis cannot say:

Consequence list (normal form)

ID	Actor class	Variable	Sign	Horizon	Visibility	Tier	Derivation	Assumptions

Gaps found

-

Appendix B. Sample worked examples

Three of the 65 worked examples, printed in full: a price control (WE-001), its repeal with the metamorphic check (WE-022), and a held-out case corrected under catalog 1.0 (WE-046). The others are in `handbook/05_examples/`, and the blind core-only re-runs in `handbook/05_examples_core/`.

WE-001 Rent freeze with a new-construction exemption

Analyst: draft (Handbook 0.5) | Rule catalog version: 0.5 | Status: awaiting expert review

A stylized policy in a fictional city. It is close to real rent-control ordinances but deliberately differs from the historical cases reserved for evaluation. The probabilities are illustrative; the evidence column says what a real analysis would cite.

1. Policy record

Riverton Rent Stabilization Ordinance. Riverton is a mid-sized city. Its neighbors, Eastfield and Northgate, are 15 to 25 minutes away by car or train and have no rent regulation. Market rents in Riverton have risen about 5% a year for several years. The ordinance was put on the ballot eight months before it took effect, and polls showed it was likely to pass.

Component	Instrument	Target	Bound	Parameters	Exemptions
K1 Rent freeze and cap	`price_ceiling`	Rent on residential rental units existing on the effective date	Landlords of those units	Rent frozen at its level on the effective date for 3 years; increases capped at 2% a year after that. The cap applies to the unit, including when a new tenant moves in.	Units first occupied after the effective date.
K2 Tenancy rules	Restriction on contract terms (part of K1)	Leases of covered units	Landlords and tenants	Eviction only for cause. Tenancies cannot be assigned or sold.	None
K3 Withdrawal route	Permit regime	Conversion of covered buildings to owner-occupied condominiums	Landlords	Permitted after a 12-month permit process; sitting tenants must be given notice and relocation payments.	None
K4 Rent board	Administration and enforcement	All of the above	City	Complaint-driven enforcement; fines for overcharging; funded from the city's general budget.	None

Unclear provisions: whether major renovations allow a rent increase. Assumed not.

2. Classification

Component	Rothbard type	Binding?	Rule families
K1	Triangular: fixes the terms of exchange between landlords and tenants	Not at the moment of enactment (the cap equals the current rent), but becomes binding as market rents rise (TS-3). A-1.	PC, RS, TS
K2	Triangular	Yes, by its text	PC-7
K3	Triangular: limits the exit route	Partly; it delays rather than prevents exit	RS-5, PC-5
K4	Binary: funded from the general budget, which rises with the new task (A-19), so taxpayers bear it (OC-3, first funding case)	Yes	OC-3

3. Actor classes

Actor class	Role	How its options change
`tenants.controlled`	Sitting tenants of covered units	Rent held below the market level; cannot assign the tenancy; lose the benefit if they move.
`tenants.prospective`	People looking for a rental in Riverton, including newcomers and young residents leaving home	Fewer covered units come up for rent; uncontrolled units cost more.
`landlords.controlled`	Owners of covered units	Rental return capped; can cut maintenance, convert, or evade.
`developers` (`sellers.prospective`)	Builders of new rental housing	New units exempt; but must judge whether the exemption will last.
`landlords.uncontrolled`	Owners of new, exempt units	Face more demand.
`homebuyers`	Buyers of owner-occupied housing	More condos come on the market through conversion.
`renters.neighbors`	Renters in Eastfield and Northgate	Some Riverton seekers move to their market.
`officials.enforcement`, `taxpayers`	Rent board and its funders	New administrative task.

4. Direct effect

ID	Actor class	Variable	Sign	Horizon	Rules	Assumptions
C-1	`tenants.controlled`	`price_paid`	decrease	medium to long, growing over time	PC-3, TS-3; dominance over C-17 by A-14 (CR-12b, CR-12e)	A-1, A-2, A-14
C-21	`tenants.controlled`	`price_paid` (covered units, first months after the effective date)	ambiguous	short	PC-3 (−) against RS-8 (+, C-17 lifts the frozen base); CR-12b	A-1, A-2, A-11
C-27	`landlords.controlled`	`net_price_received` (covered units)	decrease	medium to long	PC-3 (the mirror of C-1: the rent sitting tenants no longer pay is rent landlords no longer receive), CR-3a; dominance over C-17 by A-14	A-1, A-2, A-14

The visible effect: sitting tenants pay less than they would have, and the gap widens each year that market rents rise faster than 2% (C-1). In the first months the sign is not determined: landlords whose leases renewed in the announcement window raised rents to lift the frozen base (C-17),

and until the counterfactual market path passes that raised base the two chains conflict (C-21). A pre-emptive raise cannot exceed the market rent on the effective date (CR-5c), and the counterfactual rent passes that level within a lease cycle or two (A-14), so from the medium horizon the cap chain dominates. Landlords receive less by the same amount (C-27).

5. Responses

ID	Actor class	Variable	Sign	Horizon	Rules	Assumptions
C-2	`tenants.prospective` (covered units)	`excess_demand`	increase	short	PC-1	A-1, A-2
C-3	`tenants.prospective`	`search_cost`	increase	short	PC-2 (search margin open and used: A-15)	A-1, A-2, A-15
C-4	`tenants.prospective`	`allocation_by_non_price_criteria` (landlords choose by income, credit, connections, perceived reliability)	increase	short	PC-2 (selection margin open and used: A-16; K2's eviction-for-cause rule makes screening worth more)	A-1, A-2, A-16
C-5	`tenants.controlled`	`turnover`	decrease	medium	PC-7	A-1, A-6
C-6	`landlords.controlled`	`maintenance_spending`, `quality_per_unit`	decrease	short to medium	PC-4, RS-2	A-1, A-4
C-7	`tenants.prospective`	`side_payments` (key money)	increase	short	PC-2, PC-6, RS-3 (side-payment margin open and used: A-3)	A-1, A-3
C-22	`tenants.controlled`	`informal_activity` (illegal sublets at market rent)	increase	medium	PC-6, RS-3 (x: comply with K2's no-assignment rule, y: sublet)	A-1, A-3, A-6
C-24	`landlords.controlled`	`units_held_vacant`	increase	medium	PC-5, RS-5: a vacated unit re-let to a new tenant would need eviction for cause or a relocation payment before conversion (K2, K3), so CR-4 (x: re-let, y: hold vacant pending conversion or sale)	A-1, A-2, A-5
C-25	`landlords.controlled`	`compliance_cost` (permit process, board proceedings)	increase	medium	OC-2, CR-8c	A-5
C-8	`landlords.controlled`	`reclassification` (conversion to condominiums)	increase	medium (after the 12-month permit process)	RS-5, PC-5	A-1, A-5
C-9	`landlords.controlled`	`quantity_supplied` of covered rental units	decrease	medium to long	PC-5, RS-6, TS-1	A-1, A-5, A-10
C-10	`tenants.controlled` (in converted buildings)	`turnover` (they must leave when the building is converted)	increase	medium	PC-5, RS-5, then CR-9 (the landlord's exit ends the lease)	A-1, A-2, A-5
C-11	`tenants.controlled`, `landlords.controlled`, `developers`	`lobbying_effort` (tenants to keep the ordinance, landlords to loosen it, developers to keep the exemption)	increase	medium	RS-7	A-1, A-18
C-17	`tenants.controlled` whose leases renew before the effective date	`price_paid` (in the announcement window)	increase	short	RS-8	A-11

C-10 is worth stating separately: the ordinance protects sitting tenants as a group, but it gives landlords a reason to remove buildings from the rental market, and the tenants in those buildings must move. (Catalog 0.3 recorded this as "`price_paid` increase after relocation". That compared their new market rent with the controlled rent, not with the counterfactual: without the ordinance they would also pay a market rent. No sign for their rent follows; the move itself, and the relocation payment they receive (C-30), do.)

C-3, C-4 and C-7 sign three members of PC-2's group separately; each has its own premise that the margin is open and used (A-15, A-16, A-3). There is no waiting list or allocation rule in the record, so `wait_time` is not signed.

6. Second-order effects

ID	Actor class	Variable	Sign	Horizon	Rules	Market relation	Assumptions
C-12	`tenants.prospective` in uncontrolled units	`price_substitute` (rent of exempt new units)	increase	short to medium	PC-8, through units withdrawn from the covered stock (C-9, C-24) and sitting tenants' larger consumption at the lower rent (CR-4c mirror: fewer moves to smaller units, fewer roommates, A-17)	covered and exempt units: `same_market_other_segment`, close	A-1, A-7, A-17
C-13	`developers`	`investment_target` (new rental construction)	see section 7	medium	LW-4, TS-2		A-7, A-9
C-14	`homebuyers`	`quantity_substitute` (condos for sale)	increase	medium	RS-5	rental and owner-occupied housing: `substitute`, partial	A-5

ID	Actor class	Variable	Sign	Horizon	Rules	Market relation	Assumptions
C-15	`renters.neighbors`	`price_paid` in Eastfield and Northgate	increase	medium	RS-4, PC-3	Riverton and neighbors: `across_border`, partial	A-8
C-20	`landlords.controlled`	`asset_value` of covered buildings	decrease	short (from the ballot announcement)	SB-4	sale of buildings	A-1, A-13
C-23	`tenants.prospective` (subtenants in illegal sublets)	`disputes_settled_outside_courts`	increase	medium	PC-6 → RS-10 (an illegal sublet has no recourse to the courts)	the sublet (tenant to subtenant)	A-1, A-3, A-6

PC-8's Unless clause (0.5) does not apply. It applies when the policy itself pushes sellers into the uncontrolled segment. Here covered units stay covered whoever lets them, and converted units leave rental use altogether (C-8, C-14), so no covered landlord can move a unit into the exempt segment. New exempt building is a response to the higher exempt rent (C-13), not an independent addition: an induced supply response moderates C-12 but cannot reverse its sign. Rationing alone changes who occupies the covered stock, not how many households it houses, so the sign rests on the withdrawn units and the insiders' larger consumption, which push the same way (CR-12a).

C-20: a lasting cap on the income from a building lowers what buyers will pay for it, from the moment the cap is expected. Owners at the time of the ballot bear the loss; anyone who buys a covered building later pays the lower price, so the ordinance does not lower their return further.

C-15 is at depth 3 and reaches a diffuse group of bystanders, so the chain stops there (workflow step 6). The price of condominiums is not signed: converted units add to their supply, while some rationed-out seekers buy instead of renting (CR-12b).

7. Combined effects

Actor class, variable, horizon	Chains (+)	Chains (−)	Dominance rule	Combined sign
`developers`, `investment_target` (new rental construction), medium	Exempt units command higher rents (C-12), so building them pays better (LW-4). Rests on A-7 and A-9 (developers expect the exemption to last, p = 0.6).	Developers fear the exemption will be withdrawn once the new units exist, so expected returns fall (TS-2). Rests on the negation of A-9 (p = 0.4).	None	**increase**. The − chain rests on an assumption below the 0.5 cut-off, so it is listed as a conditional possibility, not reported. If evidence moved A-9 below 0.5, the sign would flip.
`tenants.controlled`, `quality_adjusted_price`, medium	Quality falls (C-6), so they get less for each unit of rent.	Rent is lower (C-1).	None	**ambiguous**. The method cannot say whether lower rent outweighs lower quality for sitting tenants as a group.
`tenants.controlled`, `price_paid`, short	Pre-emptive raises in the announcement window lift the frozen base (C-17, RS-8).	Rent capped below a rising market (PC-3).	None at this horizon: the size of the raises against the counterfactual path is a magnitude.	**ambiguous** (C-21).
`tenants.controlled`, `price_paid`, medium to long	As above.	As above.	A-14 with CR-5c: a pre-emptive raise is bounded by the market rent on the effective date, which the counterfactual path passes within a lease cycle or two (CR-12e, empirical premise).	**decrease** (C-1); likewise `landlords.controlled`, `net_price_received` (C-27).
`landlords.controlled`, `resources_available_other_uses`, medium	Key money received by landlords who evade (C-7).	Rent forgone on every covered unit (C-27); relocation payments and permit costs (C-25, C-30).	A-2: the cap is enforced against most landlords, so key money is confined to a minority and to new lettings, while the cap lowers the rent on every covered unit each month (CR-12e, empirical premise).	**decrease** (C-28).

Catalog 0.3 also recorded a combined "total cost of housing" for seekers across segments. That netted different variables (rents in two segments, search cost, side payments) into one, which CR-12c forbids; each is recorded separately (C-3, C-7, C-12, C-15).

8. Time structure

- The freeze is not binding at first and tightens every year (TS-3). Most effects therefore start small and grow.
- Buildings last decades (A-10). Less maintenance shows up within a year or two (C-6); reduced supply through conversion follows the 12-month permit process (C-8); reduced investment in the existing stock compounds over the long run (C-9, TS-1).
- Announcement effect: the ordinance was expected eight months before it took effect. Landlords whose leases come up for renewal in that window raise rents to lock in a higher base before the freeze (C-17, RS-8). This is a timing shift, signed per window: rents in the announcement window increase (C-17); in the first months after the effective date the sign is ambiguous (C-21); from the medium horizon, a decrease (C-1). Building values fall from the ballot campaign, not from the effective date (C-20).

9. Opportunity costs

ID	Actor class	Variable	Sign	What is given up	Rules
C-16	`tenants.prospective`, `landlords.controlled`	`quantity_traded` (rentals of covered units)	decrease	Rentals that would have been agreed at market rent and now do not happen, because units are converted or kept off the market. Both sides lose the gain they expected.	OC-4
C-18	`tenants.controlled`	`job_changes`	decrease	Moves not made because moving means losing the controlled rent: a job in another part of the region, a smaller unit after children leave.	RS-11, PC-7

ID	Actor class	Variable	Sign	What is given up	Rules
C-19	`taxpayers`	`resources_available_other_uses`	decrease	What the rent board's budget would otherwise have funded (OC-3, first funding case: the general budget rises with the new task, A-19).	OC-3, OC-1
C-26	`officials.enforcement`	`enforcement_cost`	increase	Staff time and budget of the rent board, used up (CR-8c).	OC-3
C-28	`landlords.controlled`	`resources_available_other_uses`	decrease	Rent forgone on covered units (it stays with sitting tenants, C-29), relocation payments and permit costs. See section 7.	CR-8b, CR-8c
C-29	`tenants.controlled`	`resources_available_other_uses`	increase	The matching side of C-28: rent they keep (CR-8b).	CR-8b
C-30	`tenants.controlled` (in converted buildings)	`resources_available_other_uses`	increase	The relocation payment landlords must make under K3 (CR-8b), received when the building is converted.	CR-8b

Time spent searching and waiting is already recorded as C-3, so it is not listed again here.

10. Assumptions

ID	Statement	Type	p	Evidence	If false
A-1	Market rents keep rising above 2% a year, so the cap binds within the first year.	binding	0.9	Five-year rent trend; vacancy rate; pipeline of new supply.	Almost every consequence weakens or disappears until the cap binds.
A-2	The rent board enforces the cap against most landlords.	enforcement	0.8	Board's budget and staffing; complaint-driven model; size of fines relative to the rent gap.	C-1 shrinks; C-7 grows.
A-3	Some landlords and tenants evade (key money, illegal sublets) without being caught.	enforcement	0.8	Complaint-driven enforcement; tenancies cannot be assigned, which creates an incentive for illegal sublets.	C-7, C-22 and C-23 not reported.
A-4	Housing codes leave landlords room to reduce maintenance above the legal minimum.	other	0.85	Content of the city's habitability code and inspection frequency.	C-6 not reported.
A-5	The conversion route is usable for a meaningful number of buildings.	alternative_uses	0.7	Permit approval rates; relocation payment size; demand for condominiums.	C-8, C-9 (via conversion), C-10, C-14, C-24, C-25 and C-30 weaken; supply falls only through wear (TS-1).
A-6	Sitting tenants cannot transfer their tenancy.	other	0.95	Ordinance text (K2).	C-5 weakens if tenancies can be passed on informally.
A-7	Covered and exempt units are close substitutes.	substitutability	0.85	Similar locations and sizes; same tenant pool.	C-12 weakens.
A-8	Commuting from neighboring cities is practical for many seekers.	mobility	0.7	Travel times of 15 to 25 minutes; transit links.	C-15 not reported.
A-9	Developers expect the new-construction exemption to last.	expectations	0.6	Political history of exemptions in the region; whether the exemption is in the city charter or an ordinance.	The sign of C-13 flips to decrease.
A-10	Rental buildings last several decades, so supply adjusts slowly.	durability	0.95	Age of the housing stock.	Supply responses come sooner.
A-11	Enough leases renew during the announcement window, and landlords can raise rents at renewal.	expectations	0.7	Share of leases renewing in any eight-month period; lease terms; whether the ordinance rolls rents back to an earlier date.	C-17 not reported; C-21 becomes a decrease.
A-12	Some job opportunities for sitting tenants would require moving or a much longer commute.	mobility	0.75	Size of the region; where jobs are located relative to covered housing.	C-18 not reported.
A-13	Buyers of buildings expect the ordinance to last for many years.	expectations	0.75	Margin by which the ballot measure passed; history of repeal attempts in the region.	C-20 weakens.
A-14	Without the ordinance, sitting tenants' rents would follow the market path within one or two lease cycles, so the counterfactual rent passes any pre-emptive raise made in the announcement window within the medium horizon.	other	0.7	Typical lease length of 12 months; share of renewals priced at market; a pre-emptive raise is bounded by the market rent on the effective date (CR-5c).	C-1, C-27, C-28 and C-29 become ambiguous at the medium horizon too.
A-15	There is no public waiting list or allocation rule for covered units, so seekers find vacancies by their own search.	other	0.9	Ordinance text (no allocation provision); how vacancies are advertised in Riverton.	C-3 not reported; PC-2's group remains.
A-16	Landlords may choose among applicants (no first-come or lottery rule), and fair-housing law leaves room to screen on income, credit and references.	other	0.85	Ordinance text; state and city fair-housing rules.	C-4 not reported; PC-2's group remains.
A-17	Some sitting tenants would, at market rents, have moved to smaller units or taken in roommates.	substitutability	0.8	Household size and unit size of sitting tenants; share of empty-nest households.	C-12 rests on withdrawn units alone (C-9, C-24, through A-5).
A-18	Landlords, tenant groups and developers can organize and spend to change or defend the ordinance at a cost below their stakes.	other	0.8	The ordinance came by ballot measure, which shows organized interests on both sides; existing landlord and tenant associations.	C-11 not reported.
A-19	The rent board is a new appropriation that raises the general budget, rather than being taken from other enforcement within a fixed budget.	revenue_use	0.75	Ordinance text ("funded from the city's general budget"); the city's budget practice for new agencies.	C-19 is replaced by a fall in the other enforcement the fixed budget would have funded (OC-3, second funding case).

11. Report

The policy. Rents on all existing Riverton rental units are frozen for three years and may then rise by at most 2% a year, whoever the tenant. New buildings are exempt. Owners may convert buildings to condominiums after a year-long permit process.

Visible consequences. Sitting tenants pay less rent than they would have from the medium horizon on, and the gap grows every year that market rents rise faster than the cap (C-1); they keep the difference (C-29), and landlords of covered units receive less (C-27). In the first months after the freeze the sign is not determined, because landlords raised rents at renewals in the announcement window to lift the frozen base (C-17, C-21).

Invisible consequences.

- *People looking for a home in Riverton* find fewer covered units available, search longer, face landlords who choose tenants by income, credit and connections, and sometimes pay key money (C-2 to C-4, C-7). Subtenants in illegal sublets have no recourse to the courts (C-23). Many end up in exempt new buildings, where rents rise because units are withdrawn from the covered stock and sitting tenants occupy more space per household (C-12), or in neighboring cities, where they push rents up (C-15).
- *Sitting tenants* receive less maintenance (C-6) and move less often, even when a different home or a job elsewhere would suit them better (C-5, C-18); some sublet illegally (C-22). Those whose leases renewed in the months before the freeze paid higher rents than they otherwise would have, as landlords locked in a higher base (C-17). Tenants in buildings that are converted must move (C-10), with a relocation payment (C-30).
- *Landlords* receive less on covered units and have less to spend elsewhere (C-27, C-28); the buildings lose value as soon as the ordinance is expected, and owners at that time bear the loss (C-20). They convert some buildings to condominiums (C-8), hold some vacated units empty ahead of conversion (C-24), bear permit and board costs (C-25), and invest less in the rest; over many years the covered rental stock is smaller than it would have been (C-9).
- *Homebuyers* see more condominiums for sale than they otherwise would (C-14).
- *Taxpayers* fund the rent board (C-19), whose staff time is used up on administration and enforcement (C-26).

Ambiguous effects. Sitting tenants' rent in the first months after the freeze (C-21): the cap pushes it down, the pre-emptive raises in the announcement window push it up. Whether sitting tenants are better off overall, taking lower rent and lower quality together, cannot be determined by this method. Condominium prices are not signed.

Key conditions. The most important assumption is that market rents keep rising above 2% (A-1, p = 0.9); if they do not, the ordinance changes little. That sitting tenants pay less from the medium horizon rests on counterfactual rents tracking the market within a lease cycle or two (A-14, p = 0.7). The direction of new construction depends on whether developers trust the exemption (A-9, p = 0.6): if they come to expect it to be removed, new construction falls instead of rising.

What this analysis cannot say. How large any of these effects will be; whether gains to sitting tenants outweigh losses to others; whether the policy is desirable. Health, safety and housing-condition effects beyond the listed variables are not analyzed in version 1.

Consequence list (normal form)

ID	Actor class	Variable	Sign	Horizon	Visibility	Tier	Derivation	Assumptions
C-1	tenants.controlled	price_paid	decrease	medium–long	visible	B	PC-1, PC-3, TS-3; CR-12b with A-14 against C-17	A-1, A-2, A-14
C-2	tenants.prospective	excess_demand	increase	short	invisible	B	PC-1	A-1, A-2
C-3	tenants.prospective	search_cost	increase	short	invisible	B	PC-1, PC-2	A-1, A-2, A-15
C-4	tenants.prospective	allocation_by_non_price_criteria	increase	short	invisible	B	PC-1, PC-2	A-1, A-2, A-16
C-5	tenants.controlled	turnover	decrease	medium	invisible	B	PC-7	A-1, A-6
C-6	tenants.controlled	quality_per_unit	decrease	short–medium	invisible	B	PC-4, RS-2	A-1, A-4
C-7	tenants.prospective	side_payments	increase	short	invisible	B	PC-2, PC-6, RS-3	A-1, A-3
C-8	landlords.controlled	reclassification	increase	medium	invisible	B	RS-5, PC-5	A-1, A-5
C-9	landlords.controlled	quantity_supplied	decrease	medium–long	invisible	B	PC-5, RS-6, TS-1	A-1, A-5, A-10
C-10	tenants.controlled (converted buildings)	turnover	increase	medium	invisible	B	PC-5, RS-5, CR-9	A-1, A-2, A-5
C-11	landlords, tenants, developers	lobbying_effort	increase	medium	invisible	B	RS-7	A-1, A-18
C-12	tenants.prospective	price_substitute (exempt units)	increase	short–medium	invisible	B	PC-3, PC-8, CR-4c	A-1, A-7, A-17
C-13	developers	investment_target (new rental)	increase	medium	invisible	B	PC-8, LW-4	A-7, A-9
C-14	homebuyers	quantity_substitute (condos)	increase	medium	invisible	B	RS-5	A-1, A-5
C-15	renters.neighbors	price_paid	increase	medium	invisible	B	PC-3, RS-4	A-1, A-8
C-16	tenants.prospective, landlords	quantity_traded (covered rentals)	decrease	medium	invisible	B	OC-4	A-1, A-5
C-17	tenants.controlled (renewing before effective date)	price_paid	increase	short	invisible	B	RS-8	A-11
C-18	tenants.controlled	job_changes	decrease	medium	invisible	B	PC-7, RS-11	A-1, A-6, A-12
C-19	taxpayers	resources_available_other_uses	decrease	short	invisible	B	OC-3, OC-1	A-19
C-20	landlords.controlled	asset_value	decrease	short	invisible	B	SB-4	A-1, A-13
C-21	tenants.controlled	price_paid (first months after the effective date)	ambiguous	short	visible	B	PC-3 vs RS-8 (C-17); CR-12b	A-1, A-2, A-11
C-22	tenants.controlled	informal_activity (illegal sublets)	increase	medium	invisible	B	PC-6, RS-3	A-1, A-3, A-6

ID	Actor class	Variable	Sign	Horizon	Visibility	Tier	Derivation	Assumptions
C-23	tenants.prospective (subtenants)	disputes_settled_outside_courts	increase	medium	invisible	B	PC-6, RS-10	A-1, A-3, A-6
C-24	landlords.controlled	units_held_vacant	increase	medium	invisible	B	PC-5, RS-5, CR-4	A-1, A-2, A-5
C-25	landlords.controlled	compliance_cost	increase	medium	invisible	B	OC-2, CR-8c	A-5
C-26	officials.enforcement	enforcement_cost	increase	short	invisible	A	OC-3	none
C-27	landlords.controlled	net_price_received (covered units)	decrease	medium–long	visible	B	PC-3, CR-3a; CR-12b with A-14	A-1, A-2, A-14
C-28	landlords.controlled	resources_available_other_uses	decrease	medium	invisible	B	CR-8b, CR-8c; CR-12b with A-2 against C-7	A-1, A-2, A-14
C-29	tenants.controlled	resources_available_other_uses	increase	medium	visible	B	CR-8b (matching C-28)	A-1, A-2, A-14
C-30	tenants.controlled (converted buildings)	resources_available_other_uses	increase	medium	invisible	B	CR-8b (relocation payment, K3)	A-1, A-5
—	tenants.controlled	quality_adjusted_price	ambiguous	medium	invisible	B	C-1 vs C-6	A-1, A-4

Gaps found

- GAP-1 (closed in catalog 0.2): announcement date added to the policy record; C-17 now reported.
- GAP-2 (closed in catalog 0.2): RS-11 and the variable `job_changes`; C-18 now a signed consequence.
- GAP-3 (closed in catalog 0.2): opportunity costs written as signed variables; the duplicate entry for search time removed.
- Catalog 0.3: SB-4 now covers losses as well as gains; C-20 added.
- Catalog 0.5: corrected after the core-only comparison (see ../05_examples_core/WE-001_rent_freeze_core.md): sitting tenants' short-run rent made ambiguous (C-21, CR-12b with C-17) and C-1 limited to medium–long with new premise A-14; C-10 replaced by a turnover increase (the price claim used the controlled rent as baseline); added landlords' net price, both sides of the rent transfer and the relocation payment, enforcement and compliance cost, units held vacant, illegal sublets and subtenants' disputes (C-22 to C-30); PC-2 members given per-margin premises (A-15, A-16; wait_time dropped); RS-7 premise A-18; OC-3 funding stated (A-19); PC-8 mechanism and Unless clause addressed; the cross-variable "total cost of housing" row removed (CR-12c).

WE-022 Repeal of the Riverton rent ordinance

Analyst: draft (Handbook 0.5) | Rule catalog version: 0.5 | Status: awaiting expert review

A stylized repeal of the ordinance analyzed in WE-001, in the same fictional city, ten years after it took effect. It doubles as a metamorphic check: the analysis of a repeal should reverse the signs of WE-001, except where the ordinance caused lasting changes. The check is at the end of section 7. The counterfactual throughout is **the ordinance stays in force on its current terms**; every sign below is relative to that, not to the situation before the repeal and not to a Riverton that never had the ordinance. Most rules in the catalog are written for imposing a policy. "In reverse" after a rule ID means the rule is applied to the removal of its precondition; where that direction is not obviously valid, the text says so. The catalog states some directions explicitly: SB-4 covers returns that rise or fall, TS-1 has a clause for capital that grows, TS-5 and TS-6 are written for repeals, RS-7 has a clause for repeals (lobbying takes place in the counterfactual too) and RS-10 has a reverse clause (see "Analyzing a repeal" in the workflow). The probabilities are illustrative; the evidence column says what a real analysis would cite.

1. Policy record

Riverton Rent Stabilization Repeal. Ten years after the WE-001 ordinance took effect, voters approve a ballot measure to repeal it. The measure qualified for the ballot six months before its effective date, and polls showed it was likely to pass. The starting point, taken from WE-001's analysis and stated here as the situation at repeal rather than as findings: controlled rents were frozen for three years and have risen at most 2% a year since, while rents for comparable exempt units have kept rising faster; some covered buildings have been converted to condominiums through the K3 permit route; maintenance in covered buildings has been below what it would otherwise have been; Eastfield and Northgate remain unregulated.

Component	Instrument	Target	Bound	Parameters	Exemptions
R1 Decontrol with phase-in	Removal of a `price_ceiling`	Rent on units covered by WE-001's K1	Landlords and tenants of covered units	Vacancy decontrol from the effective date: a covered unit that becomes vacant may be let at any rent. Sitting tenants: during each of the two years after the effective date, rent may rise by at most 15% at renewal. From the second anniversary, no limit.	None
R2 Tenancy rules	Removal of a restriction on contract terms	Leases of covered units (WE-001 K2)	Landlords and tenants	Eviction only for cause, and the ban on assigning a tenancy, stay in force for the two phase-in years and end at the second anniversary.	None
R3 Rent board	End of administration	WE-001 K4	City	The board handles phase-in disputes and overcharge claims for two years, then is dissolved.	None
(unchanged)	—	Condominium-conversion permit process (WE-001 K3)	—	Not repealed: the 12-month permit process and relocation payments continue to apply to all rental buildings.	—

Unclear provisions: whether a unit vacated during the phase-in by a tenant who accepted a payment to leave counts as a vacancy, and is therefore decontrolled at once. Assumed yes; the measure does not regulate such payments.

2. Classification

Component	Rothbard type	Binding?	Rule families
R1	Removal of a triangular intervention: landlords and tenants may again agree the rent	Yes: for most covered units the controlled rent is now well below the rent of a comparable exempt unit (A-1)	PC in reverse, RS-8, SB-4, TS-1, TS-5, TS-6

Component	Rothbard type	Binding?	Rule families
R2	Removal of a triangular intervention	Yes, from the second anniversary	PC-7 and RS-11 in reverse
R3	Removal of a binary intervention (taxpayer-funded board)	Yes, from the second anniversary	OC-3 in reverse

3. Actor classes

Actor class	Role	How its options change
`tenants.controlled`	Sitting tenants of covered units at the effective date	Rent rises toward the market level over two years; lose the reason to stay put; may be offered payments to leave during the phase-in.
`tenants.controlled.displaced`	Sitting tenants who will not or cannot pay the higher rent	Must find other housing in Riverton or a neighboring city.
`tenants.prospective`	People looking for a rental in Riverton	Formerly covered units come up for rent at market rents; no shortage at a controlled price to queue for.
`landlords.controlled`	Current owners of covered buildings, including those who bought during the ten years at prices that reflected the control	Can charge market rents; the return on maintaining and improving the buildings rises; conversion to condominiums becomes less attractive.
`developers` (`sellers.prospective`)	Builders of new rental housing	The rent premium in the exempt segment shrinks; must judge whether controls might return.
`landlords.uncontrolled`	Owners of exempt units built since the ordinance	Face competition from decontrolled units.
`homebuyers`	Buyers of owner-occupied housing	Fewer rental buildings are converted to condominiums.
`buyers.covered_buildings`	People who buy covered buildings after the repeal is expected	Pay a price that reflects decontrolled rents.
`renters.neighbors`	Renters in Eastfield and Northgate	Some Riverton seekers return to Riverton; some displaced sitting tenants arrive.
`officials.enforcement`, `taxpayers`	Rent board and its funders	Board's work ends after two years.

4. Direct effect

ID	Actor class	Variable	Sign	Horizon	Rules	Assumptions
C-1	`tenants.controlled`	`price_paid`	increase	short to medium (phased over two years)	PC-3 in reverse, TS-3	A-1, A-2
C-2	`landlords.controlled`	`net_price_received`	increase	short to medium	PC-3 in reverse	A-1, A-2

The visible effect: sitting tenants pay more than they would have under the ordinance, and their landlords receive more. The size of the step is the gap that built up over ten years of a 2% cap (TS-3), released at once for vacant units and over two years for sitting tenants.

5. Responses

ID	Actor class	Variable	Sign	Horizon	Rules	Assumptions
C-3	`tenants.prospective` (formerly covered units)	`excess_demand`	decrease	short	PC-1 in reverse	A-1, A-2
C-4	`tenants.prospective`	`search_cost`	decrease	short	PC-2 in reverse	A-1, A-2, A-21
C-5	`tenants.prospective`	`allocation_by_non_price_criteria`	decrease	short	PC-2 in reverse	A-1, A-2, A-21
C-6	`tenants.prospective`	`side_payments` (key money, premiums on illegal sublets)	decrease	short to medium (sublets of units still in the phase-in continue until it ends)	PC-6, RS-3 in reverse	A-1, A-2, A-21
C-28	`tenants.prospective` who obtain formerly covered units	`price_paid` (covered units let on vacancy)	increase	short to medium	PC-3 in reverse	A-1, A-2
C-30	`tenants.controlled`	`informal_activity` (illegal sublets)	decrease	short to medium	PC-6, RS-3 in reverse	A-1, A-21
C-32	`landlords.controlled`	`compliance_cost` (board proceedings, conversion permit applications)	decrease	medium (after the board is dissolved)	OC-2 in reverse	A-1, A-3
C-7	`tenants.controlled`	`turnover`	increase	medium (mostly after the phase-in)	PC-7 in reverse	A-1, A-6
C-8	`landlords.controlled`	`maintenance_spending`	increase	short to medium	PC-4, RS-2 in reverse, TS-5	A-1, A-3
C-9	`landlords.controlled`	`reclassification` (new conversions to condominiums)	decrease	medium	RS-5, PC-5 in reverse	A-1, A-3
C-10	`landlords.controlled`	`quantity_supplied` of formerly covered rental units	increase	short to medium	PC-5 in reverse	A-1, A-3, A-15
C-11	`tenants.controlled`	`side_payments` received (payments from landlords to leave during the phase-in)	increase	short	RS-8, LW-5	A-1, A-13
C-12	`tenants.prospective`	`quantity_traded` (lettings of covered units that fall vacant in the announcement window)	decrease	short (announcement window only)	RS-8	A-11
C-13	`tenants.controlled.displaced`	`search_cost`	increase	medium	PC-3 in reverse, OC-2	A-1, A-6
C-14	`tenants.controlled`, `landlords.controlled`	`lobbying_effort`	ambiguous (see section 7)	medium	RS-7 (repeal clause)	A-14

C-10 has two quick sources: units that owners were holding off the market, and buildings in the conversion permit process whose owners withdraw (A-15). It does not include converted condominiums returning to rental; see C-23 and the metamorphic check.

C-11 and C-12 are both timing responses (RS-8), and neither has a counterpart in WE-001. Because a vacancy is decontrolled at once while a sitting tenant's rent is phased in, a landlord gains from a vacancy during the phase-in and may pay the tenant to leave. Before the effective date, a landlord with a vacant unit gains by leaving it empty until decontrol rather than letting it at the controlled rent.

C-13 applies only to tenants who move because of the rent increase. Sitting tenants who would have moved anyway now search in a market without a shortage, which is part of C-4.

C-4 to C-6 and C-30 reverse the margins WE-001 found open and used under the ordinance (its A-3, A-15, A-16): searching, selection by connections, key money and illegal sublets. They reverse only where those margins were in fact used, which is A-21. WE-001 does not sign `wait_time` (there is no waiting list), so the repeal does not either. C-28 is the other side of PC-3 in reverse: seekers who would have obtained a covered unit at the controlled rent now pay the market rent for it.

C-32: the board's proceedings end at the second anniversary (R3), and fewer owners apply for conversion permits (C-9), so landlords spend less on them. Phase-in disputes continue for two years, so the decrease starts at the medium horizon.

C-14: RS-7's repeal clause applies. Both classes lobbied in the counterfactual too: sitting tenants to keep the ordinance, owners to loosen or repeal it (WE-001 C-11). After the repeal, tenants lobby to restore it and owners to defend the repeal. Each class lobbied in both worlds, so the change in each class's effort is `ambiguous` without a premise. RS-7's premise, that each can organize at a cost below its stake, is A-14.

Not reported: reconversion of condominiums to rental. Converted units have been sold to separate owners, and reassembling a building costs more than the difference in return (TS-6; A-9, p = 0.9; its converse A-16, p = 0.1, is below the cut-off).

6. Second-order effects

ID	Actor class	Variable	Sign	Horizon	Rules	Market relation	Assumptions
C-15	`tenants.prospective` in exempt units	`price_substitute` (rent of exempt units)	decrease	medium (from the effective date)	PC-8 in reverse; CR-12b with an accounting identity (CR-12e)	covered and exempt units: `same_market_other_segment`, close	A-1, A-5
C-27	`tenants.prospective` in exempt units	`price_substitute` (rent of exempt units)	increase	short (announcement window)	RS-8, PC-8	as C-15	A-5, A-11
C-16	`developers`	`investment_target` (new rental construction)	decrease (see section 7)	medium	PC-8, LW-4 in reverse		A-5, A-8
C-17	`landlords.controlled`	`asset_value` (formerly covered buildings)	increase	short (from the announcement)	SB-4, TS-5		A-1, A-2, A-3
C-29	`buyers.covered_buildings`	`price_paid` (covered buildings)	increase	short to medium	SB-4, OC-7	sale of buildings	A-1, A-3
C-31	`tenants.prospective` (subtenants)	`disputes_settled_outside_courts`	decrease	short to medium	RS-10 in reverse	the sublet (tenant to subtenant)	A-1, A-21
C-18	`homebuyers`	`quantity_substitute` (condominiums from new conversions)	decrease	medium	RS-5 in reverse	rental and owner-occupied housing: `substitute`, partial	A-1, A-3
C-19	`tenants.controlled`	`job_changes`	increase	medium (mostly after the phase-in)	RS-11, PC-7 in reverse	controlled tenancy and jobs: `complement`, note `location`	A-1, A-12
C-20	`tenants.controlled`	`quality_per_unit`	increase	medium	PC-4 in reverse, TS-1, TS-5		A-1, A-3
C-21	`renters.neighbors`	`price_paid` in Eastfield and Northgate	ambiguous (see section 7)	medium	RS-4, PC-3	Riverton and neighbors: `across_border`, partial	A-6, A-7
C-22	`landlords.controlled`	`investment_target` (major renovation of formerly covered buildings)	increase	medium to long	PC-5 in reverse, TS-5		A-1, A-4
C-23	`landlords.controlled`	`capital_stock_target` (formerly covered rental stock)	increase	long	PC-5 in reverse, TS-1 (upward clause), TS-5, TS-6		A-1, A-4, A-9, A-18

C-15 and C-27: PC-8 in reverse holds only from the effective date. In the announcement window, landlords hold vacant covered units empty (C-12), so seekers who would have taken them turn to exempt units and exempt rents rise (C-27; RS-8 with PC-8). From the effective date two chains reach exempt rents. (−) Seekers who were rationed out of the covered segment can now rent there (PC-8 in reverse), and withheld units return to use (C-10). (+) Sitting tenants who will not pay the higher rent move, and some move to exempt units (A-6). The (+) chain is offset one for one: each sitting tenant who leaves frees a covered unit that a seeker takes, so households move into the covered segment, net, only as the number of covered units in use rises (C-10, C-23, C-24 are all increases). This is an accounting identity used as a dominance premise (CR-12e), and it is the same identity WE-001 uses (rationing changes who occupies the covered stock, not how many households it houses). So the combined sign is a decrease (section 7). PC-8's Unless clause does not apply: the repeal pushes no sellers into the exempt segment.

C-31: RS-10 in reverse. As illegal sublets fall (C-30), fewer subtenants hold arrangements with no recourse to the courts. No previously excluded sellers enter, so RS-10's quality-variability caveat does not arise.

C-17 applies SB-4, which covers a lasting change in returns in either direction: the ordinance lowered the expected return of a fixed stock of buildings, and the repeal raises it again, so building prices rise as soon as the repeal is expected. The gain goes to whoever owns the buildings at the announcement. Owners who sold during the ten years, at prices that already reflected the control, bore the loss and do not share in the gain. Expectations of re-imposition (TS-5) make the rise smaller but do not change its sign as long as owners give the repeal some chance of lasting (A-3). By OC-7's symmetry clause, the gain to owners who sell is the outlay of those who buy: later buyers pay the higher price (C-29), and so neither gain nor lose from the repeal as it stands.

C-22 and C-23 are Tier B in catalog 0.3 (Tier C in the 0.2 draft). PC-5 in reverse says investment returns when the ceiling goes; TS-5 adds that investment paying back only over ten or more years is smaller and later than the mirror of the ordinance's effect, and goes ahead only as far as owners expect decontrol to last that long (A-4, p = 0.6). C-23 also uses TS-6, under which the stock converted under the ordinance does not come back (A-9), and the upward clause of TS-1, under which rebuilding runs at the speed of new investment rather than as the mirror of wear-out (A-18). Its persistence against a Riverton that never had the ordinance is partial (metamorphic check, WE-001 C-9).

C-21 is at depth 3 and reaches a diffuse group of bystanders, so the chain stops there.

7. Combined effects

Actor class, variable, horizon	Chains (+)	Chains (−)	Dominance rule	Combined sign
`developers`, `investment_target` (new rental construction), medium	Repeal removes the risk that the new-construction exemption would be withdrawn while the ordinance stood. Rests on developers having feared that (negation of A-8, A-17, p = 0.4).	The rent premium that spillover demand gave exempt units shrinks (C-15), so new building pays less (LW-4). Rests on A-5 and A-8 (p = 0.6). A further chain under TS-5: if developers expect controls to return and to cover new buildings next time, expected returns fall further. Rests on A-19 (p = 0.35), so it is a conditional possibility, not reported; it would push the same way.	None	**decrease**. The + chain rests on an assumption below the 0.5 cut-off and is not reported. This is the mirror of WE-001's C-13 and rests on the same judgment about what developers expected.
`tenants.controlled`, `quality_adjusted_price`, medium	Rent rises (C-1).	Quality rises (C-20).	None	**ambiguous**. Same as in WE-001, with both chains reversed.
`tenants.prospective` in exempt units, `price_substitute`, medium	Sitting tenants who move out of covered units, some to exempt units (A-6).	Rationed-out seekers move into covered units (PC-8 in reverse); withheld units return (C-10).	Accounting identity (CR-12e): every sitting tenant who leaves frees a covered unit that a seeker takes, so net demand for exempt units falls by the rise in covered units in use (C-10, C-23, C-24).	**decrease** (C-15). In the announcement window, **increase** (C-27), a separate window.
`landlords.controlled`, `resources_available_other_uses`, short to medium	Higher rent on every covered unit at renewal and on vacancy (C-2); fewer relocation payments and permit costs (C-9, C-32).	Key money no longer received by the landlords who took it (C-6); payments to sitting tenants to leave (C-11).	A-2 with A-21: rents rise on every covered unit each month, while key money was confined to a minority of lettings (WE-001's premise that the cap was enforced against most landlords); payments to leave are made only where the landlord expects the vacancy to be worth more (CR-5). Empirical premise (CR-12e).	**increase** (C-34).
`renters.neighbors`, `price_paid`, medium	Displaced sitting tenants move to cheaper neighboring cities (C-13, A-6, A-7).	Riverton seekers who had been pushed out to the neighbors can now find units in Riverton (C-3, C-15).	None	**ambiguous**. WE-001's C-15 does not simply reverse: the repeal pushes some people out of Riverton as well as drawing others back.
`tenants.controlled`, `landlords.controlled`, `lobbying_effort`, medium	Sitting tenants now bear a concentrated loss that re-imposition would undo, and organize for it (RS-7, A-14). Owners who have capitalized the gain (C-17) defend the repeal.	In the counterfactual, tenants would have kept lobbying to defend the ordinance and owners to loosen it (WE-001 C-11); that effort ends.	None (RS-7, repeal clause)	**ambiguous** for each class: each lobbied in both worlds. The repeal changes what each lobbies for; the method cannot say whether either class's effort rises or falls.

Metamorphic check against WE-001

"Reverses" means the expected sign under repeal is the opposite of WE-001's. "Partly reverses" means the sign reverses for the flow (what happens from now on) but not for a stock that the ordinance already changed. **Persistence** (TS-6; `02_ontology.md`, 2.5) is how much of WE-001's effect remains after the repeal, compared with a Riverton that never had the ordinance: none, partial or full. It is "partial" or "full" where the effect ran through a stock change that costs something to reverse (conversions, deferred maintenance that became structural damage, displaced tenants), and also during the phase-in, while rents are still held below the path without the ordinance; "none" where a flow simply stops; "n/a" for transition effects with no counterpart on that path. It may be given per horizon (`02_ontology.md`, 2.5). The same values are recorded on each consequence in the consequence list.

WE-001 ID	Actor class, variable	Sign in WE-001	Expected sign under repeal (this example)	Reverses?	Persistence (TS-6)	Reason
C-1	`tenants.controlled`, `price_paid` (medium to long)	decrease	increase (C-1)	Reverses	partial (short: the 15% steps); none (from the second anniversary)	Phased by R1; vacancies go to market at once (C-28).

WE-001 ID	Actor class, variable	Sign in WE-001	Expected sign under repeal (this example)	Reverses?	Persistence (TS-6)	Reason
C-21	`tenants.controlled`, `price_paid` (first months after the effective date)	ambiguous	no counterpart	Does not reverse	n/a	A transition effect of the original's announcement (pre-emptive raises against the cap). The repeal sets no base date.
C-17	`tenants.controlled` renewing before the effective date, `price_paid`	increase (short)	no counterpart	Does not reverse	n/a	The repeal's announcement effect works through a different margin: landlords hold vacant units empty (C-12) instead of raising rents early.
C-27	`landlords.controlled`, `net_price_received`	decrease	increase (C-2)	Reverses	partial (short: the 15% steps); none (from the second anniversary)	The mirror of C-1.
C-2	`tenants.prospective`, `excess_demand`	increase	decrease (C-3)	Reverses	none	The shortage exists only at the controlled price.
C-3	`tenants.prospective`, `search_cost`	increase	decrease (C-4)	Reverses	none	For seekers. Displaced sitting tenants newly bear search costs (C-13), which has no counterpart in WE-001.
C-4	`tenants.prospective`, `allocation_by_non_price_criteria`	increase	decrease (C-5)	Reverses	none	Landlords still screen on credit, as in any rental market; what falls is selection made possible by the shortage.
C-7	`tenants.prospective`, `side_payments` (key money)	increase	decrease (C-6)	Reverses	none	Key money loses its reason once rents can rise. Payments now flow the other way, from landlords to sitting tenants (C-11), which has no counterpart in WE-001.
C-22	`tenants.controlled`, `informal_activity` (illegal sublets)	increase	decrease (C-30)	Reverses	none	The gap between the tenant's rent and the market rent closes over the phase-in.
C-23	subtenants, `disputes_settled_outside_courts`	increase	decrease (C-31)	Reverses	none	RS-10 in reverse, with the sublets.
C-5	`tenants.controlled`, `turnover`	decrease	increase (C-7)	Reverses, delayed	none	Lock-in lasts while a below-market rent remains, so mostly after the phase-in.
C-18	`tenants.controlled`, `job_changes`	decrease	increase (C-19)	Reverses, delayed	none	Mostly after the phase-in, when the remaining below-market rent ends.
C-6	`landlords.controlled`, `maintenance_spending`, `quality_per_unit`	decrease	increase (C-8 spending, C-20 quality)	Partly reverses	partial	Maintenance spending responds at once (A-3; TS-5, quick payback). Ten years of deferred maintenance have partly become structural damage (A-10, `reversal_cost`), which is repaired only as owners choose to invest, and repairs with long payback depend on owners trusting the repeal (TS-5, A-4). Quality does not jump back.
C-8	`landlords.controlled`, `reclassification`	increase	decrease (C-9)	Partly reverses	full for buildings already converted; none for the flow	New conversions fall and pending applications are withdrawn. Buildings already converted stay condominiums: reconversion is not worth its cost (TS-6; A-9; A-16 below the cut-off).
C-9	`landlords.controlled`, `quantity_supplied`	decrease	increase (C-10 short to medium; C-23 long)	Partly reverses	partial	Withheld units and pending conversions return quickly. Units lost to conversion do not return (TS-6, A-9). Rebuilding runs at the speed of new investment (TS-1 upward clause, A-18) and depends on credibility (TS-5, A-4), not as a mirror of the wear-out in WE-001.
C-24	`landlords.controlled`, `units_held_vacant` (medium)	increase	increase in the announcement window (C-12, units held for decontrol); then units return to use (C-10)	Reverses after a temporary rise	partial	Units held vacant pending conversion return to use from the effective date; units that need major repair return only as owners invest (A-18).
C-10	`tenants.controlled` in converted buildings, `turnover`	increase	no separate consequence; follows from C-9 (fewer conversions)	Partly reverses	full for tenants already displaced	Tenants already displaced by past conversions are not restored; their buildings stay converted (TS-6, A-9). Fewer tenants are displaced by conversion from now on.
C-30	`tenants.controlled` in converted buildings, `resources_available_other_uses` (relocation payment)	increase	no separate consequence; follows from C-9	Partly reverses	full for payments already made	Fewer conversions mean fewer relocation payments from now on; payments already received stay with the tenants.
C-25	`landlords.controlled`, `compliance_cost`	increase	decrease (C-32, after the board is dissolved)	Partly reverses	partial	Board proceedings end; the K3 permit process is not repealed, so owners who still convert still bear it.
C-11	landlords, tenants, developers, `lobbying_effort`	increase	ambiguous (C-14)	Does not reverse	n/a	RS-7, repeal clause: each class lobbied in both worlds. The repeal changes what each lobbies for.

WE-001 ID	Actor class, variable	Sign in WE-001	Expected sign under repeal (this example)	Reverses?	Persistence (TS-6)	Reason
C-12	`tenants.prospective`, `price_substitute` (exempt units)	increase	decrease from the effective date (C-15); increase in the announcement window (C-27)	Reverses after a temporary rise	partial	Against a Riverton that never had the ordinance, the stock lost to conversion keeps the whole market tighter (TS-6, A-9), so part of WE-001's effect remains.
C-13	`developers`, `investment_target` (new rental)	increase	decrease (C-16)	Reverses	partial	Rests on the same judgment about developers' expectations (A-8 here, A-9 in WE-001). Buildings built under the ordinance remain.
C-14	`homebuyers`, `quantity_substitute` (condominiums)	increase	decrease (C-18)	Partly reverses	partial	The flow of new conversions falls; the condominiums already converted stay on the owner-occupied market (TS-6, A-9).
C-15	`renters.neighbors`, `price_paid`	increase	ambiguous (C-21)	Does not reverse	none	Displaced sitting tenants move out while other seekers move back.
C-16	`tenants.prospective`, `landlords`, `quantity_traded` (covered rentals)	decrease	decrease in the announcement window (C-12); increase from the effective date (C-24)	Reverses after a temporary dip	partial	RS-8 holds vacant units off the market until decontrol. Lettings of units already converted do not come back (TS-6, A-9).
C-20	`landlords.controlled`, `asset_value`	decrease	increase (C-17)	Partly reverses	partial	It reverses for the asset, not for those who sold under the ordinance at a depressed price (SB-4), and converted or damaged buildings do not regain it (A-9, A-10).
C-28	`landlords.controlled`, `resources_available_other_uses`	decrease	increase (C-34)	Reverses	partial (short: the 15% steps); none (from the second anniversary)	Same dominance premise as WE-001 (rent on every unit against key money for a minority).
C-29	`tenants.controlled`, `resources_available_other_uses`	increase	decrease (C-33, those who stay)	Reverses	partial (short: the 15% steps); none (from the second anniversary)	The other side of C-34.
C-19	`taxpayers`, `resources_available_other_uses`	decrease	increase (C-25)	Reverses, delayed	none	The board runs for two more years (R3), and the saving reaches taxpayers only if the city does not redirect the board's budget (A-20).
C-26	`officials.enforcement`, `enforcement_cost`	increase	decrease (C-26)	Reverses, delayed	none	After the board is dissolved.
—	`tenants.controlled`, `quality_adjusted_price`	ambiguous	ambiguous	Reverses (stays ambiguous)	—	Both chains reverse.

Consequences with no counterpart in WE-001: C-11 (payments to leave), C-12 (vacant units held back, as a window effect), C-13 (search costs of displaced tenants), C-22 and C-23 (renovation and rebuilding), C-27 (exempt rents in the announcement window), C-28 (new tenants of covered units, the other side of PC-3), C-29 (later buyers of covered buildings).

8. Time structure

- **Announcement window (six months).** Building prices rise as soon as the repeal is expected (C-17). Landlords with vacancies hold them empty until decontrol (C-12), which pushes seekers into exempt units and raises exempt rents in the window (C-27); exempt rents fall below the counterfactual only from the effective date (C-15). Owners pause conversion applications (C-9, C-10).
- **Phase-in (two years).** Vacancies go to market rent at once; sitting tenants' rents rise in two steps of at most 15% (C-1). Because a vacancy is worth more to the landlord than a phased tenancy, payments to leave are concentrated here (C-11). Lock-in weakens but does not end, so most of the rise in turnover and job changes comes after the second anniversary, when R2 also ends (C-7, C-19). The board keeps running, so taxpayers save nothing yet (C-25), and landlords' board proceedings continue (C-32).
- **Flows reverse quickly; stocks do not.** The repeal changes what owners do from now on: maintenance spending, conversion applications, key money and lettings respond within months. What ten years of the ordinance left behind changes only as new flows accumulate. Deferred maintenance that has become structural damage is repaired only as owners choose to invest (A-10). Converted buildings stay converted (A-9). This is TS-6, and the main reason several WE-001 consequences only partly reverse; the persistence column of the metamorphic check records it.
- **Rebuilding is not the mirror of decline.** Under the ordinance, the covered stock shrank at the speed of wear-out and conversion (TS-1). Under the repeal, it grows at the speed of new investment, which is governed by planning, permits, finance and construction (TS-1, upward clause; A-18). C-23 is set to long for this reason.
- **Expectations of re-imposition.** Riverton has shown it will impose rent control, and sitting tenants now have a reason to campaign for it again (C-14). Repairs that pay back within a few years proceed if owners expect the repeal to last that long (A-3, $p = 0.8$). Major renovation and new construction, which pay back over ten or more years, depend on a weaker expectation (A-4, $p = 0.6$), and are smaller and later than the mirror of WE-001's effects (TS-5).
- **Who gains from the capital gain.** The rise in building values (C-17) goes to owners at the time of the announcement. Owners who sold during the ten years bore the capital loss and get nothing back. The persons who gain from the repeal are not the persons who lost from the ordinance.

9. Opportunity costs

ID	Actor class	Variable	Sign	What is given up	Rules
C-24	`tenants.prospective`, `landlords.controlled`	`quantity_traded` (lettings of formerly covered units)	increase	Nothing given up; exchanges are freed. Rentals at market rent that the cap prevented now happen, and both sides expect to gain from them.	OC-4 in reverse
C-25	`taxpayers`	`resources_available_other_uses`	increase (after the board is dissolved)	Nothing given up; the board's budget is freed for other uses, if the city does not redirect it to other spending (A-20).	OC-3, OC-1 in reverse
C-26	`officials.enforcement`	`enforcement_cost` (rent board)	decrease	Nothing given up; board staff and budget move to other work.	OC-3 in reverse
C-32	`landlords.controlled`	`compliance_cost`	decrease	Nothing given up; time spent on board proceedings and permit applications is freed.	OC-2 in reverse
C-33	`tenants.controlled` (those who stay)	`resources_available_other_uses`	decrease	The higher rent they keep paying, which they would otherwise have spent on other things.	CR-8b (matching C-34)
C-34	`landlords.controlled`	`resources_available_other_uses`	increase	The higher rent received, net of key money no longer taken and payments to leave (section 7).	CR-8b; CR-12b with A-2, A-21

C-33 and C-34 are the two sides of the rent transfer (CR-8b), the mirror of WE-001's C-28 and C-29. C-33 is signed only for sitting tenants who stay and keep paying the higher rent; those who move may spend less on housing, so no sign is given for them. Search time for displaced tenants is C-13. C-25 rests on OC-3's first funding case, as in WE-001 (the board was paid from a budget that rose with the task), and on A-20 for what the city does with the saving.

10. Assumptions

ID	Statement	Type	p	Evidence	If false
A-1	For most covered units, the controlled rent is now well below the rent of a comparable exempt unit.	binding	0.95	Rent board registry against rents of comparable exempt units; ten years of a 2% cap against the market trend.	Almost every consequence weakens or disappears.
A-2	Landlords raise rents toward the market level at renewal and on vacancy, as R1 allows.	other	0.85	Ownership structure of covered buildings; landlord behavior in exempt units.	C-1, C-2, C-17, C-28, C-33 and C-34 shrink; the other chains weaken.
A-3	Owners expect decontrol to last at least a few years, long enough for routine repairs to pay back.	expectations	0.8	Size of the ballot majority; council composition; how hard it would be to put a new ordinance on the ballot.	C-8, C-9, C-10, C-17, C-18 and C-20 weaken.
A-4	Owners expect decontrol to last ten to fifteen years, long enough for major renovation and new building to pay back.	expectations	0.6	As A-3, plus the strength of tenant organizing after the repeal (C-14).	C-22 and C-23 not reported.
A-5	Covered and exempt units are close substitutes.	substitutability	0.85	Similar locations and sizes; same tenant pool (WE-001 A-7).	C-15 and C-16 weaken.
A-6	Some sitting tenants will not pay the higher rent and will move.	mobility	0.85	Incomes of sitting tenants against market rents for their units.	C-7, C-13 and C-21 weaken.
A-7	Commuting from neighboring cities is practical for many renters.	mobility	0.7	Travel times of 15 to 25 minutes; transit links.	C-21 not reported.
A-8	Developers had expected the new-construction exemption to last, so the exempt-segment premium was feeding new building.	expectations	0.6	Same evidence as WE-001 A-9.	The sign of C-16 flips to increase.
A-9	Condominium conversions are costly to undo: converted units have been sold to separate owners.	reversal_cost	0.9	Condominium registry; number of owners per converted building.	Some converted buildings could return to rental; C-8, C-9 and C-14 in the metamorphic check would reverse more fully.
A-10	Some deferred maintenance has become structural damage that costs more to repair than ongoing maintenance would have.	reversal_cost	0.75	Inspection records; age of the covered stock.	Quality recovers faster (C-20).
A-11	Some covered units fall vacant in the announcement window, and landlords can keep them empty until decontrol at small cost.	expectations	0.7	Normal vacancy rate; rent forgone for a few months against the gain from decontrol.	C-12 and C-27 not reported.
A-12	Some job opportunities for sitting tenants would require moving or a much longer commute.	mobility	0.75	Size of the region; where jobs are (WE-001 A-12).	C-19 not reported.
A-13	Payments to leave are legal and some sitting tenants accept them.	other	0.7	The measure does not regulate them; practice in comparable markets.	C-11 not reported.
A-14	Sitting tenants and owners can organize at a cost below their stake.	other	0.75	The ballot campaigns on both sides.	C-14 not reported.
A-15	Some buildings are in the conversion permit process at the effective date, and their owners can withdraw.	other	0.7	Pending permit applications.	C-10 rests only on units held off the market.
A-16	Reconverting condominiums to rental is worth its cost for some buildings.	reversal_cost	0.1	Number of owners per building; cost of buying out owners.	Below cut-off; not reported.
A-17	Developers had feared that the new-construction exemption would be withdrawn (negation of A-8).	expectations	0.4	As A-8.	Below cut-off; the + chain for C-16 is not reported.
A-18	Restoring capacity in formerly covered buildings (major renovation, repair of structural damage, bringing units back into use) takes several years of permits, finance and construction.	durability	0.85	Permit times for major alterations; construction lead times; the age of the covered stock.	C-23 comes sooner; its horizon would be medium.

ID	Statement	Type	p	Evidence	If false
A-19	Developers give a meaningful probability to controls returning, and covering new buildings, within the payback period of a new rental building (ten to fifteen years).	expectations	0.35	The original ordinance exempted new construction; size of the repeal majority; strength of tenant organizing after the repeal (C-14).	Below cut-off; the further − chain for C-16 is not reported.
A-20	The city reduces taxes or other borrowing, or funds other spending it would otherwise have cut, by the amount of the board's budget, rather than leaving the saving unused or absorbed by administration elsewhere.	revenue_use	0.7	City budget process; whether the board's appropriation is a separate line.	C-25 not reported.
A-21	Under the ordinance, the search, selection, key-money and illegal-sublet margins were open and used (WE-001 A-3, A-15, A-16).	other	0.8	WE-001's evidence for those premises; rent board overcharge and sublet complaints.	C-4, C-5, C-6, C-30 and C-31 not reported for the margins that were not used; the C-34 dominance premise is unaffected.

11. Report

The policy. Ten years after Riverton froze and capped rents on existing rental units, the city repeals the ordinance. Units that fall vacant can be let at any rent from the effective date. Sitting tenants' rents may rise by at most 15% a year for two years, after which there is no limit; their protection against eviction ends at the same time. The rent board closes after two years. The condominium-conversion permit rules stay.

Visible consequences. Sitting tenants pay more than they would have under the ordinance, and their landlords receive more (C-1, C-2).

Invisible consequences.

- *People looking for a home in Riverton* find more formerly covered units on offer, with no shortage, less searching, less selection by connections and no key money (C-3 to C-6, C-10). Those who would have obtained a covered unit at the controlled rent now pay the market rent for it (C-28). From the effective date, rents in newer, exempt buildings are lower than they would have been (C-15). In the months before the repeal takes effect, some vacant units are held empty instead (C-12), and exempt rents are higher in those months (C-27). Illegal sublets, and subtenants' disputes settled outside the courts, decline (C-30, C-31).
- *Sitting tenants* move more often, and change jobs more often, once the phase-in ends (C-7, C-19). Some are paid by their landlords to leave early (C-11). Those who cannot pay the higher rent must search for other housing (C-13), in Riverton or in neighboring cities. Those who stay have less to spend on other things (C-33). Maintenance and quality rise (C-8, C-20), but not at once.
- *Owners of covered buildings* see building values rise as soon as the repeal is expected (C-17), convert fewer buildings to condominiums (C-9), and, to the extent they trust the repeal to last ten years or more, invest in major renovation (C-22). The owners who gain are those who own at the announcement, not necessarily those who bore the loss when the ordinance began; later buyers pay the higher price (C-29). Owners have more to spend on other things (C-34) and spend less on board proceedings and permits (C-32).
- *Developers* build less new rental housing than they would have, because the premium that the shortage gave new, exempt units shrinks (C-16).
- *Homebuyers* see fewer new conversions to condominiums (C-18).
- *Taxpayers* stop funding the rent board after two years, provided the city does not redirect its budget (C-25).

Ambiguous effects. Whether sitting tenants are better off overall, taking higher rent and higher quality together. Whether rents in neighboring cities rise or fall: some Riverton seekers move back while some displaced tenants move out (C-21). Whether tenants' or owners' lobbying effort rises or falls: both lobbied with the ordinance in force, and both lobby after the repeal, for different ends (C-14).

What does not reverse. Buildings converted to condominiums under the ordinance stay condominiums; tenants already displaced by those conversions are not restored; structural damage from deferred maintenance is repaired only over time; and the capital loss borne by owners who sold during the ten years is not returned to them. The formerly covered rental stock grows again (C-23), but at the speed of new investment, held back by the chance that controls return, and without the units already converted.

Key conditions. Almost everything rests on controlled rents being well below market rents at the time of repeal (A-1, p = 0.95). The fall in exempt rents from the effective date rests on an accounting identity: sitting tenants who leave free covered units that seekers take. Long-lived investment in the formerly covered stock rests on owners expecting the repeal to last (A-4, p = 0.6). The fall in new construction rests on developers having trusted the exemption (A-8, p = 0.6); if they had not, it would flip.

What this analysis cannot say. How large any of these effects will be; how far, or how fast, Riverton's rental market comes to resemble the one it would have had without the ordinance; whether gains to owners and seekers outweigh losses to sitting tenants; whether the repeal is desirable.

Consequence list (normal form)

ID	Actor class	Variable	Sign	Horizon	Visibility	Tier	Derivation	Assumptions	Persistence
C-1	tenants.controlled	price_paid	increase	short–medium	visible	B	PC-3 in reverse, TS-3	A-1, A-2	partial (short: the 15% steps); none (from the second anniversary)
C-2	landlords.controlled	net_price_received	increase	short–medium	visible	B	PC-3 in reverse	A-1, A-2	partial (short: the 15% steps); none (from the second anniversary)
C-3	tenants.prospective	excess_demand (formerly covered units)	decrease	short	invisible	B	PC-1 in reverse	A-1, A-2	none
C-4	tenants.prospective	search_cost	decrease	short	invisible	B	PC-2 in reverse	A-1, A-2, A-21	none
C-5	tenants.prospective	allocation_by_non_price_criteria	decrease	short	invisible	B	PC-2 in reverse	A-1, A-2, A-21	none
C-6	tenants.prospective	side_payments (key money, sublet premiums)	decrease	short–medium	invisible	B	PC-6, RS-3 in reverse	A-1, A-2, A-21	none
C-7	tenants.controlled	turnover	increase	medium	invisible	B	PC-7 in reverse	A-1, A-6	none

ID	Actor class	Variable	Sign	Horizon	Visibility	Tier	Derivation	Assumptions	Persistence
C-8	landlords.controlled	maintenance_spending	increase	short–medium	invisible	B	PC-4, RS-2 in reverse, TS-5	A-1, A-3	partial (A-10)
C-9	landlords.controlled	reclassification (new conversions)	decrease	medium	invisible	B	RS-5, PC-5 in reverse	A-1, A-3	full for buildings already converted (A-9); none for the flow
C-10	landlords.controlled	quantity_supplied (formerly covered units)	increase	short–medium	invisible	B	PC-5 in reverse	A-1, A-3, A-15	partial (A-9)
C-11	tenants.controlled	side_payments received (payments to leave)	increase	short	invisible	B	RS-8, LW-5	A-1, A-13	n/a (transition)
C-12	tenants.prospective	quantity_traded (vacant covered units, announcement window)	decrease	short	invisible	B	RS-8	A-11	n/a (transition)
C-13	tenants.controlled.displaced	search_cost	increase	medium	invisible	B	PC-3 in reverse, OC-2	A-1, A-6	n/a (no counterpart)
C-14	tenants.controlled, landlords.controlled	lobbying_effort	ambiguous	medium	invisible	B	RS-7 (repeal clause)	A-14	n/a
C-15	tenants.prospective	price_substitute (exempt units)	decrease	medium	invisible	B	PC-8 in reverse; CR-12b, CR-12e (accounting identity)	A-1, A-5	partial (A-9)
C-16	developers	investment_target (new rental)	decrease	medium	invisible	B	PC-8, LW-4 in reverse	A-5, A-8	partial (buildings built under the ordinance remain)
C-17	landlords.controlled	asset_value (formerly covered buildings)	increase	short	invisible	B	SB-4, TS-5	A-1, A-2, A-3	partial (owners who sold under the ordinance do not recover; A-9, A-10)
C-18	homebuyers	quantity_substitute (condominiums from new conversions)	decrease	medium	invisible	B	RS-5 in reverse	A-1, A-3	partial (A-9)
C-19	tenants.controlled	job_changes	increase	medium	invisible	B	RS-11, PC-7 in reverse	A-1, A-12	none
C-20	tenants.controlled	quality_per_unit	increase	medium	invisible	B	PC-4 in reverse, TS-1, TS-5	A-1, A-3	partial (A-10, A-18)
C-21	renters.neighbors	price_paid	ambiguous	medium	invisible	B	RS-4, PC-3	A-6, A-7	none
C-22	landlords.controlled	investment_target (major renovation, formerly covered buildings)	increase	medium–long	invisible	B	PC-5 in reverse, TS-5	A-1, A-4	partial (A-10, A-18)
C-23	landlords.controlled	capital_stock_target (formerly covered rental stock)	increase	long	invisible	B	PC-5 in reverse, TS-1, TS-5, TS-6	A-1, A-4, A-9, A-18	partial (A-9, A-18)
C-24	tenants.prospective, landlords.controlled	quantity_traded (lettings of formerly covered units)	increase	medium	invisible	B	OC-4 in reverse	A-1, A-2	partial (A-9)
C-25	taxpayers	resources_available_other_uses	increase	medium	invisible	B	OC-3, OC-1 in reverse	A-20	none
C-26	officials.enforcement	enforcement_cost (rent board)	decrease	medium	invisible	A	OC-3 in reverse	none	none
C-27	tenants.prospective	price_substitute (exempt units, announcement window)	increase	short	invisible	B	RS-8, PC-8	A-5, A-11	n/a (transition)
C-28	tenants.prospective	price_paid (covered units let on vacancy)	increase	short–medium	invisible	B	PC-3 in reverse	A-1, A-2	none
C-29	buyers.covered_buildings	price_paid (covered buildings)	increase	short–medium	invisible	B	SB-4, OC-7	A-1, A-3	none
C-30	tenants.controlled	informal_activity (illegal sublets)	decrease	short–medium	invisible	B	PC-6, RS-3 in reverse	A-1, A-21	none
C-31	tenants.prospective (subtenants)	disputes_settled_outside_courts	decrease	short–medium	invisible	B	RS-10 in reverse	A-1, A-21	none
C-32	landlords.controlled	compliance_cost	decrease	medium	invisible	B	OC-2 in reverse	A-1, A-3	partial (the K3 permit process is not repealed)
C-33	tenants.controlled (those who stay)	resources_available_other_uses	decrease	short–medium	invisible	B	CR-8b (matching C-34)	A-1, A-2	partial (short: the 15% steps); none (from the second anniversary)
C-34	landlords.controlled	resources_available_other_uses	increase	short–medium	invisible	B	CR-8b; CR-12b with A-2, A-21	A-1, A-2, A-21	partial (short: the 15% steps); none (from the second anniversary)
—	tenants.controlled	quality_adjusted_price	ambiguous	medium	invisible	B	C-1 vs C-20	A-1, A-3	none

Gaps found

- GAP-022-1 (closed in catalog 0.3: TS-5). Expectations of re-imposition. C-8, C-17, C-20, C-22 and C-23 now cite TS-5; C-22 raised from Tier C to Tier B on A-4 (p = 0.6); the developers' re-imposition chain in section 7 rests on A-19 (p = 0.35) and is listed as a conditional possibility.
- GAP-022-2 (closed in catalog 0.3: TS-6, assumption type `reversal_cost`, `persistence` field). Irreversible changes. A-9, A-10 and A-16 retyped as `reversal_cost`; persistence recorded in the metamorphic check; the rows for WE-001 C-6, C-8, C-9, C-10 and C-14 are no longer Tier C.

- GAP-022-3 (closed in catalog 0.3: TS-1, upward clause). Speed of rebuilding. The lag is stated as A-18 (`durability`); C-23 raised from Tier C to Tier B.
- GAP-022-4 (closed in catalog 0.3: `persistence` may be given per horizon (2.5)). Persistence has no horizon. The `persistence` field compares the repeal with the path on which the ordinance was never imposed, but does not say at what horizon. The effect of conversions on exempt rents (WE-001 C-12) is partial in the medium run and may fall to none in the long run, as new building replaces the converted stock; deferred maintenance (C-6) may be repaired within a decade. One value per consequence forces the analyst to pick a horizon silently. *Proposed fix:* in `02_ontology.md` 2.5, define `persistence` at the consequence's horizon, and allow one value per horizon where they differ (for example "partial (medium); none (long)"), with the `reversal_cost` assumption naming what would end the persistence.
- Catalog 0.5: corrected after the core-only comparison (see ../05_examples_core/WE-022_rent_freeze_repeal_core.md): exempt rents split by window (C-15 decrease from the effective date, resting on the occupancy identity; new C-27 increase in the announcement window); lobbying kept ambiguous under RS-7's repeal clause; added new tenants of covered units (C-28), later buyers of buildings (C-29, OC-7 symmetry), illegal sublets and subtenants' disputes (C-30, C-31, RS-10 in reverse), landlords' compliance cost (C-32) and both sides of the rent transfer (C-33, C-34); `wait_time` dropped from C-4 and the PC-2 margins tied to A-21; C-26 recorded as `enforcement_cost`; the cross-variable total-cost row removed (CR-12c); persistence recorded on every consequence, with the core version's values for rents during the phase-in; metamorphic check rebuilt against WE-001 as corrected for catalog 0.5.

WE-046 Congestion charge for a central zone, with a residents' discount and a bus earmark

Analyst: draft (Handbook 0.9 held-out test, corrected under 1.0) | Rule catalog version: 1.0 | Status: awaiting expert review

A stylized policy in a fictional city. Several real cities charge drivers to enter a central zone; this one is not modeled on any of them. It is a held-out test of catalog 0.9 on a case none of the 0.9 changes were drawn from. It probes CR-12's 0.9 rule for a Tier C chain against a Tier B chain (road congestion is a `shared_facility`, so every chain through travel speeds is Tier C under CR-9 0.7, and it opposes Tier B chains on drivers' costs, shops' sales, property values and wages), CR-12's 0.9 note on what the response argument gives (vehicle numbers, where faster roads draw some drivers back), RS-9's 0.9 clause on location complements (shops inside the zone and just outside it), RS-16 and its 0.9 clause on a cost attached to the substitute itself (ride-hail cars pay the charge too), CR-6's 0.9 clauses on splitting a shift between price and quantity and on a public supplier that adds capacity only after a lag (buses), CR-7c capitalization inside the zone and at its edge, TX-1 and TX-2 incidence where the "seller" of road space is the city, CR-8d and SB-7 on an earmark with no maintenance-of-effort clause, and OC-3's 0.9 clause on a fixed budget together with its right-baseline clause. The unclear provision is whether delivery vans that enter and leave within 30 minutes pay. The counterfactual throughout is **Calderford without the charge**: the same roads, buses and parking, with traffic as it would otherwise have been. Effects on air quality, noise, road safety and health are outside the scope of version 1. The probabilities are illustrative; the evidence column says what a real analysis would cite.

1. Policy record

Calderford Central Zone Charging Ordinance. Calderford is a city of about 900,000. Its central zone, the area inside the Inner Ring Road, covers about 7 km² and has about 58,000 residents, 14,500 cars registered to zone addresses, 260,000 jobs and about 4,200 shops, cafés and restaurants. Just outside the Inner Ring lies Ringside, a band of about 1,100 shops and offices with three large parking garages, five to ten minutes' walk from the zone's edge. Beyond Ringside are the city's outer districts and, beyond the city line, the suburban town of Merrowdale with a regional shopping centre.

On an average weekday, about 118,000 vehicles are driven or parked in the zone between 7:00 and 19:00: about 92,000 cars (about 9,500 of them residents'), 19,000 goods vehicles (about 7,000 of which make a single visit lasting under 30 minutes), 4,000 taxis and ride-hail cars, and 3,000 buses and coaches. Average speed in the zone in charging hours is about 12 km/h. Of commuters to zone jobs, about 62% come by bus or tram, 22% by car and 16% on foot or by bicycle. A typical car commute ends with about 4 km inside the zone (about 20 minutes at 12 km/h); the same commute by bus takes about 15 minutes longer door to door. There are about 31,000 off-street parking spaces in the zone, at an average of $21 a day and about 90% weekday occupancy; the three Ringside garages have about 7,500 spaces at $9 a day and about 88% occupancy (about 900 spare). Calderford Transit runs about 850 buses and two tram lines, with fare revenue of about $180 million a year and general-fund support of about $140 million; peak load on routes into the zone is about 85% of capacity, and new buses take 14 to 18 months from order to service. Through traffic can go round the zone on the Inner Ring Road, which is itself busy at peak. Ride-hail fares are set by the platforms; there is no fare regulation.

The city's model projects that with the charge about 88,000 vehicles a day would be in the zone in charging hours, and the average speed would rise to about 15 km/h (saving about 4 minutes on the typical 4 km of a car commute).

The ordinance was adopted and published on 4 January 2027; charging starts on 1 November 2027 (ten months later). No expiry; the council must review the charge level every three years.

Field	Value	Source sentence
Jurisdiction (and what lies just outside)	The zone inside the Inner Ring Road. Just outside: Ringside (shops, offices, garages), the Inner Ring Road itself, then the outer districts and Merrowdale.	§2: "The zone is the area bounded by, but not including, the Inner Ring Road."
Announced / enacted / effective / expires	Announced and enacted 4 January 2027; effective 1 November 2027; no expiry.	§1, §14.
Enforcement (who, penalties, resources)	The City Charging Office (CCO): cameras reading number plates at all 74 entry points and at 40 points inside the zone. Unpaid charge: penalty notice of 130(65 if paid within 14 days). Appeals against notices go to the existing Parking Adjudication Office (PAO).	§9 to §11.

Component	Instrument	Target	Bound	Parameters	Exemptions
K1 Daily charge	`per_unit_tax`	A vehicle-day in the zone	Registered keeper of each vehicle driven or parked in the zone on a weekday between 7:00 and 19:00	$15 per vehicle per day, payable by midnight the next day. One charge covers any number of entries that day.	Scheduled buses; emergency vehicles.
K2 Residents' discount	`per_unit_tax` (reduced rate)	Same	Registered keepers whose registered address is in the zone	90% discount: $1.50 a day. Residence is taken from the vehicle register, which keepers update by declaration.	—
K3 Earmark	`revenue_use`	Net revenue of K1 and K2	The city	"The net revenue of the charge, after the costs of the City Charging Office, shall be applied to bus services." No clause on the level of general-fund support for buses.	—

Component	Instrument	Target	Bound	Parameters	Exemptions
K4 Enforcement and administration	Enforcement	K1, K2	CCO, funded from gross revenue (K3: "after the costs of the City Charging Office"); PAO, whose budget the ordinance does not change	About $45 million a year to run; about $70 million to set up, advanced from city reserves and repaid from the first year's revenue.	—

Existing rules that stay in the counterfactual: Calderford Transit's fares and its general-fund support (set each year in the city budget); on-street parking rules and their enforcement in the zone (about $11 million a year); the PAO, which hears about 44,000 parking appeals a year, about 14,000 of them on zone tickets.

Unclear provision:

- **U1, short delivery visits.** §3(1): "The charge is payable for each day on which a vehicle is driven or parked in the zone during charging hours." §7: "A goods vehicle which enters and leaves the zone within thirty minutes in the course of a delivery shall be recorded as a short visit." The schedule of charges sets a single daily rate and nowhere refers to short visits. Reading U1-a: a short visit is recorded (the CCO's freight statistics) and pays the daily charge like any other day in the zone. Reading U1-b: a short visit is not a chargeable day; such vans pay nothing. The CCO has issued no guidance.

Readings get no probability; where a sign depends on U1, one sign per reading is given.

Figures. Static revenue (118,000 vehicles, 252 charging days): about $410 million from non-residents and $3.6 million from residents. Projected revenue (88,000 vehicles, of which about 8,000 residents'): about $303 million, less about $26 million under U1-b if the 7,000 short delivery visits do not pay. Net of the CCO's $45 million, about $230 million to $260 million a year is earmarked for buses, against $140 million of general-fund support now. For a typical car commuter, the charge is $15 a day and the time saved by faster roads is about 4 minutes each way inside the zone, about 8 minutes a day. For a multi-drop van covering about 40 km in the zone, the same speed change saves about 40 minutes a day. The PAO's workload: about 180,000 penalty notices a year are projected, about 5% appealed (about 9,000 appeals); fewer cars in the zone are projected to cut zone parking tickets by about a fifth (about 2,800 fewer appeals): a net rise of about 6,200 appeals a year, about 14%, with the same budget.

2. Classification

Component	Rothbard type	Binding? (assumption ID)	Rule families
K1	Binary: a payment to the state attached to the option of driving in the zone on a weekday (CR-1, operation 3, charged per day rather than per entry). Road space in the zone is supplied by the city at a money price of zero in the counterfactual. Congestion among its users is the `shared_facility` relation, not CR-6 rationing: the public-supply clause applies only where demand at the set price exceeds a fixed quantity (CR-6, 1.0). K1 adds a money price.	Yes (A-1).	TX-1, TX-2, TX-9, RS-1, RS-3, RS-4, RS-5, RS-7 (1.0), RS-8, RS-9 (0.9), RS-12, RS-16 (1.0), RS-22, CR-6 (0.9, 1.0), CR-7c, CR-12 (0.9, 1.0)
K2	Binary: the same operation at a tenth of the rate, for a class defined by registered address. Against the counterfactual it is still a burden, not a subsidy; it draws a line at the zone boundary.	Yes, for residents who drive in the zone (A-1).	RS-5, RS-11, CR-7c
K3	Binary: the use of the revenue; funds an existing public supplier whose budget is set each year.	Yes	SB-7, CR-8d, CR-9 (`shared_facility`: crowding in vehicles below capacity, CR-6 1.0), RS-13 (1.0), RS-16 Unless
K4	Binary: administration funded from the policy's own revenue (CCO) and from an unchanged budget (PAO).	Yes	OC-3 (own revenue; 0.9 fixed budget; right baseline)

The roads of the zone are a facility every driver, van and bus shares: one vehicle fewer speeds every other vehicle. That is a `shared_facility` relation (02_ontology.md, 2.8), not an exchange. Under CR-9's 0.7 clarification every chain through travel speeds is Tier C, pending Q20. Where such a chain opposes a Tier B chain on the same key, CR-12's 0.9 rule applies: the Tier B sign is reported, qualified "not counting effects through the roads", and the combined sign is reported separately at Tier C. The same holds for riders sharing a bus or tram below its capacity: crowding among them is the `shared_facility` relation (the vehicle), at Tier C, not CR-6 rationing (CR-6, 1.0; PC-19, 1.0).

3. Actor classes

Actor class	Role	How its options change
`drivers.nonresident` (`buyers.current` of road space)	Drivers of cars into the zone who do not live there: commuters, shoppers, visitors	A weekday in the zone costs $15 more. Split by response (CR-1, 0.7): `drivers.keep` (keep driving in), `drivers.switch` (switch to bus, tram, ride-hail or Ringside parking), `drivers.retime` (move trips to evenings or weekends), `drivers.through` (cross the zone without stopping; may go round it) and `drivers.stop` (stop coming).
`residents.zone`	Zone residents who drive in the zone on weekdays	Pay $1.50 a day; would pay $15 if they moved out and kept driving in.
`goods_vehicle_operators`	Carriers and own-account suppliers delivering into the zone	Pay $15 per van-day; under U1-b, not for visits under 30 minutes.
`businesses.zone`	Shops, cafés, restaurants and services in the zone	Customers who drove in face the charge; deliveries cost more; streets are less congested (Tier C).
`businesses.edge`	Shops and services in Ringside	Customers who park outside the zone and shoppers who stop short of it pass their doors.
`owners.zone_commercial`, `owners.zone_residential`, `owners.edge_commercial` (`factor_owners.specific`)	Owners of property in the zone and in Ringside at the time the charge became expected	Values capitalize the change in expected yields (CR-7c).
`owners.parking.zone`, `owners.parking.edge`	Owners of garages in the zone and in Ringside	Demand for parking in the zone falls (complement of driving in); demand in Ringside rises (substitute).
`bus_riders`	Existing riders of buses and trams into the zone	New riders join; service may expand from the earmark; buses run faster (Tier C).
`transit_operator`	Calderford Transit	Receives the earmark; general-fund support may fall.

Actor class	Role	How its options change
`ride_hail_drivers` (`sellers.substitutes`)	Ride-hail and taxi drivers	More riders; each car pays $15 a day in the zone.
`employers.zone`, `workers.zone`	Employers in the zone and their staff who commute by car	The cost of reaching a zone job by car rises.
`ring_road_users`	Drivers on the Inner Ring Road	Through traffic diverted round the zone joins them (Tier C).
`shoppers`	Buyers who drove into the zone to shop	May shop in Ringside or at Merrowdale instead.
`officials.enforcement`	The City Charging Office	Collects the charge, runs cameras, issues penalty notices.
`officials.adjudication`, `parking_appellants`	The Parking Adjudication Office and those who appeal parking tickets	New appeals against penalty notices; fewer zone parking tickets.
`taxpayers`, recipients of other city spending	Funders and beneficiaries of the general fund	General-fund support for buses may fall (SB-7).
`input_suppliers`	Sellers of cameras, payment systems and buses	New orders.
`transit_workers`	Bus drivers and depot staff	More service once new buses arrive.
`keepers.cloned_plates`	Registered keepers whose plates are cloned by evaders	Receive penalty notices for trips they did not make.
`buyers.prospective` (zone housing)	People who work in the zone, drive, and live outside it	Living in the zone now costs $1.50 a driving day against $15 from outside.
`officials.parking`	The city's zone parking enforcement	Fewer cars parked in the zone, so fewer fines.
`lobbying groups`	Zone retailers, delivery firms, Ringside residents, the transit union	Concentrated stakes in the charge level, exemptions and U1.

4. Direct effect

ID	Actor class	Variable	Sign	Horizon	Rules	Assumptions
C-1	`drivers.nonresident`	`price_paid` (money cost of a weekday in the zone by car)	increase	short (from 1 November 2027)	CR-1, TX-2	A-2
C-2	`residents.zone`	`price_paid` (money cost of a weekday driving in the zone, at the discount)	increase	short	CR-1, TX-2	A-2
C-3	`officials.enforcement`	`tax_revenue` (charge)	increase	short	TX-1	none
C-4	`drivers.nonresident`, `goods_vehicle_operators`	`quantity_traded` (vehicle-days in the zone, weekdays 7:00 to 19:00; not counting effects through the roads)	decrease	short to medium	CR-4c, TX-2, CR-12 (0.9)	A-1, A-2
C-5	`drivers.keep`, `goods_vehicle_operators`, `bus_riders`	`travel_time` (per km in the zone in charging hours)	decrease	short to medium	CR-4b, CR-9 (`shared_facility`; Tier C), CR-6 (1.0: not rationing)	A-1, A-2, A-3
C-6	`drivers.nonresident`, `residents.zone`, `goods_vehicle_operators`, `ride_hail_drivers`	`quantity_traded` (all charged vehicle-days in the zone, weekdays 7:00 to 19:00, all classes together)	decrease	short to medium	CR-9 (`shared_facility`; Tier C), CR-12e (response argument), CR-12 (0.9, 1.0: what the response argument gives and needs)	A-1, A-2, A-3, A-34, A-35

C-1 to C-4 are what the ordinance's supporters expect: drivers pay, the city collects, fewer vehicles enter. The residents' discount does not make residents better off against the counterfactual: they pay $1.50 a day where they paid nothing (C-2). K1 is a charge per day, not per entry, so it bears on whether a vehicle is in the zone on a given day; a driver already in the zone pays nothing more for further trips that day (CR-4, units).

Road space has no seller (TX-2). TX-2 gives `price_paid` up and/or `net_price_received` down, with the split decided by the two sides' alternatives (DM-1). Here the "seller" of road space is the city, which supplies a fixed quantity whatever the price: there is no seller whose net price can fall. So the charge is borne in the first instance by those who keep driving (C-1, C-2), and its further burden travels along the relations of the trips it prices: to businesses whose customers or deliveries it reaches (C-27, C-46) and, through CR-7b, to the immobile means in the zone (C-33). This is TX-2 read for a public supply at a zero price (Gaps found, c1).

Fewer vehicles, and faster roads (C-4 to C-6). Without the charge, each vehicle in the zone slows the others: the time cost of a trip rises with the number of vehicles. That is the `shared_facility` relation (CR-9, 0.7), not CR-6 rationing: CR-6's public-supply clause applies only where demand at the set price exceeds a fixed quantity, and then gives a group sign over the margins; congestion among users of a road is Tier C (CR-6, 1.0). The charge attaches $15 to the option of a weekday in the zone; those for whom the trip was worth less than $15 more than its other costs switch, retime or stop (CR-4b; A-1). That is C-4 at Tier B. With fewer vehicles, the rest move faster (C-5): one vehicle's use of the road changes the time cost to every other user, which is the `shared_facility` relation, so C-5 is Tier C. Faster roads draw some drivers back (a lower time cost on the option of driving in). That chain opposes C-4, and it runs over the shared facility, so CR-12's 0.9 rule applies: C-4 stands at Tier B, qualified "not counting effects through the roads", and the combined sign is recorded at Tier C (C-6). The combined sign is decided by CR-12e's response argument: the drivers drawn back respond to a speed gain that exists only because drivers left in response to the charge; if the number of vehicle-days returned to its old level, so would the speed, and every driver would face the old time cost plus the charge, so the number would fall again. The response cannot reverse its cause. CR-12's 1.0 note says what the argument needs. (1) The total it bounds: all charged vehicle-days in the zone in charging hours, every class together (non-residents' cars, residents' cars, vans, ride-hail and taxis), since the speed depends on all of them; so C-6 is recorded on that total, not on C-4's two classes alone. (2) The analogue of constant cost: at a given number of vehicles, the time cost is the same with and without the charge (same roads, signals and parking; A-34), so that the old number would bring back the old speed. (3)

Partial response: by CR-12's 0.9 note the argument alone gives "a decrease or no change"; A-1 makes the response partial (at the old time cost, some drivers value a weekday in the zone by less than $15 more than its other costs). (4) Second responses to the same change are separate chains: drivers switching to ride-hail add ride-hail car-days, but each carries the trips of more than one former private car-day (A-35), so the switch lowers the total; the extra buses bought from the earmark take road space and slow, not speed, other traffic. Both push the same way. The Tier C chain does not lower the tier of C-4, and C-6 is Tier C because its chain runs over the roads.

5. Responses

ID	Actor class	Variable	Sign	Horizon	Rules	Assumptions
C-7	`drivers.switch`	`quantity_substitute` (bus and tram trips into the zone)	increase	short to medium	RS-1, CR-4	A-5
C-8	`drivers.switch`	`quantity_substitute` (day parking in Ringside garages)	increase	short	RS-1, RS-16, CR-6 (0.9: splitting a shift)	A-5, A-6
C-9	`owners.parking.edge` (`sellers.substitutes`)	`price_substitute` (Ringside garage day rate)	increase	short to medium	RS-16, CR-6 (0.9: splitting a shift)	A-5, A-6
C-10	`drivers.switch`	`quantity_substitute` (ride-hail and taxi trips into the zone)	increase	short	RS-1	A-25
C-11	`ride_hail_drivers`	`price_substitute` (ride-hail fares for trips into the zone; not counting effects through the roads)	increase	short	RS-16 (1.0: per unit of quality; costs on the substitute), CR-4 (0.8), CR-6, CR-12a, CR-12 (0.9)	A-25, A-26
C-12	`ride_hail_drivers`	`number_of_sellers` (ride-hail and taxi cars working in the zone on a weekday)	ambiguous	short	RS-16 (1.0: a cost fixed per facility), CR-4 (0.8), CR-12b	A-25, A-26
C-13	`drivers.retime`	`quantity_traded` (car trips into the zone on weekdays before 7:00 and after 19:00)	increase	short to medium	RS-8, CR-10a, CR-4	A-17
C-14	`drivers.retime` (shopping and personal trips)	`quantity_traded` (car trips into the zone at weekends)	increase	short to medium	RS-8, CR-4	A-17
C-15	`drivers.through`	`travel_time` (through trips diverted round the zone)	increase	short	CR-4 (x: cross the zone and pay, y: go round), CR-3a	A-18, A-31
C-16	`ring_road_users`	`travel_time` (on the Inner Ring Road)	increase	short to medium	CR-9 (`shared_facility`; Tier C)	A-3, A-18
C-17	`drivers.nonresident`	`reclassification` (vehicles registered to zone addresses to claim the discount)	increase	short to medium	RS-5	A-11
C-18	`drivers.nonresident`, `goods_vehicle_operators`	`legal_breach` (unpaid charges; false, cloned or obscured plates)	increase	short	RS-3, CR-4 (x: pay, y: evade)	A-16
C-19	`lobbying groups`	`lobbying_effort` (on questions that exist only under the ordinance: the level at each review, a shoppers' exemption, U1, eligibility for the discount, the earmark)	increase	short to medium	RS-7 (1.0), CR-4 (1.0), CR-12 (0.9: activities that exist only under the policy)	A-15
C-20	`residents.zone`	`turnover` (moves out of the zone by residents who drive in it)	decrease	medium	RS-11, CR-3	A-22
C-21	`goods_vehicle_operators`	`reclassification` (routes reshaped into visits of under 30 minutes)	U1-a: no_change; U1-b: increase	short to medium	RS-5, CR-1	A-13
C-22	`goods_vehicle_operators`	`quantity_traded` (delivery visits into the zone lasting under 30 minutes)	U1-a: decrease; U1-b: increase	short to medium	CR-4, RS-5	A-1, A-13
C-23	`goods_vehicle_operators`	`compliance_cost` (records proving that a visit was a delivery under 30 minutes)	U1-a: no_change; U1-b: increase	short	CR-8c, OC-2	A-13
C-24	`goods_vehicle_operators`	`legal_breach` (false short-visit claims by vehicles not delivering)	U1-a: no_change; U1-b: increase	short	RS-3	A-13, A-16
C-25	`businesses.zone` (retailers whose customers mostly drive: furniture, hardware, bulky goods)	`number_of_sellers` (in the zone)	decrease	medium	RS-4, RS-6, CR-7a	A-7, A-20
C-26	`shoppers`	`cross_border_purchases` (purchases made outside the zone, in Ringside or Merrowdale, by former car shoppers)	increase	short to medium	RS-4, RS-9	A-7
C-61	`drivers.through`	`quantity_substitute` (through trips on the Inner Ring Road)	increase	short to medium	RS-1, CR-4 (x: cross the zone and pay, y: go round)	A-18
C-64	`goods_vehicle_operators`	`transaction_size` (drops per charged van-day in the zone)	increase	short to medium	CR-4c, CR-1 (0.9: operation per range)	A-12, A-32
C-65	`keepers.cloned_plates`	`compliance_cost` (contesting notices for trips they did not make)	increase	short to medium	CR-9 (`contract`: the plate register), CR-1, CR-8c	A-16
C-66	`residents.zone`	`quantity_demanded` (cars kept at zone addresses)	decrease	medium	CR-4b, CR-4c	A-2, A-33
C-67	`buyers.prospective` (zone housing)	`quantity_demanded` (zone housing, by people who work in the zone and drive; not counting effects through the roads)	increase	short (from 4 January 2027) to medium	CR-4 (x: live outside and pay $15, y: move in and pay $1.50), CR-10a, CR-12 (0.9)	A-2, A-27

Substitutes (C-7 to C-12). RS-1 gives the direction of each switch; CR-4 gives only that more buyers turn to the substitute, and a traded quantity or a price needs a premise on the substitute's supply (RS-1; RS-16; CR-6, 0.9). Buses and trams run to a fixed schedule at a set fare, with peak load at about 85% of capacity: the riders drawn from cars are carried in the space that is there, so the quantity rises (C-7). Below capacity this is not rationing (CR-6, 1.0); the added riders crowd those already aboard through the shared vehicle, a Tier C chain (C-38). Ringside's garages have about 900 spare spaces against the several thousand drivers the city model expects to park outside the zone: the quantity rises until the garages are full (C-8), and beyond that the shift goes to price (C-9). By CR-6's 0.9 clause, the split between price and quantity is signed only with a premise on supply: A-6 (garages cannot be added within a year). For ride-hail, RS-16's clauses on costs on the substitute (0.9) and on prices per unit of quality (1.0) apply: the policy attaches a cost to the substitute itself, but it is $15 per car-day in the zone, not a cost per trip sold. A cost fixed per facility and period bears on whether the car works in the zone that day, not on the price of each trip (CR-4, 0.8; RS-16, 1.0). On the fare per trip (C-11), more riders push it up where supply cannot expand at an unchanged cost at peak (A-26); the charge pushes the same way through fewer cars working in the zone, where some leave. CR-12a gives `increase`, and the demand chain alone suffices (CR-12d). On the number of ride-hail cars working in the zone (C-12), the chains oppose: more riders draw cars in (+), the charge per car-day keeps some out (−); no premise ranks them, so `ambiguous` (CR-12b). Through the roads, faster trips let each car carry more trips per hour, which adds supply of trips and opposes the fare rise: a Tier C chain, so C-11 is qualified "not counting effects through the roads" and the combined fare is `ambiguous` at Tier C (C-60; CR-12, 0.9).

Timing (C-13, C-14). A trip into the zone happens with and without the charge, at different times. Trips that can be moved go to the free windows: weekday evenings and early mornings (C-13) and weekends (C-14). Each window is signed against the counterfactual (02_ontology.md, 2.5); the weekday 7:00 to 19:00 window is C-4. These are flows (trips in each window), not stocks (CR-12, 0.9).

Through traffic and the ring road (C-15, C-16). A driver who crossed the zone without stopping now pays $15 to do so, or goes round on the Inner Ring Road, which takes longer (C-15; CR-4, CR-3a: the chosen alternative was ranked lower before). Diverted traffic slows the ring road for everyone on it (C-16): `shared_facility`, Tier C. Both push C-15 the same way, so no combination is needed. CR-3a alone gives only that the diverted drivers are worse off by their own ranking; that going round takes longer is a premise (A-31: the Inner Ring Road route is longer and busy at peak). The diverted trips themselves are recorded on the ring road (C-61; RS-1, CR-4; A-18).

The discount draws registrations (C-17). The residents' discount depends on the registered address, which keepers update by declaration. Non-residents who can name a zone address (a relative's flat, a business address) save $13.50 a day by doing so (RS-5; A-11). Penalties for false registration exist but checks are by audit.

Evasion (C-18, C-65). Foreign plates, cloned plates and obscured plates escape the cameras; some drivers risk the penalty (RS-3; A-16). The trade itself stays in a legal activity, so it is `legal_breach`. A cloned plate sends the notice to the keeper whose plate it copies: the plate register links them (a `contract` relation, CR-9), and the keeper spends time contesting notices for trips they did not make (C-65; CR-8c).

Lobbying (C-19). By CR-4's 1.0 note and RS-7 (1.0), lobbying is signed against what would have been contested on the counterfactual path. The questions C-19 names (the level at each three-yearly review, a shoppers' exemption, the reading of U1, eligibility for the discount, the earmark) exist only under the ordinance, so lobbying on them is an `increase` against none (CR-12, 0.9). Whether the general question of a charge would have stayed contested without the ordinance (there were hearings in 2026) is a premise the case does not supply, so lobbying for or against a charge as such is not signed.

Lock-in (C-20). A resident who drives in the zone and moves out would pay $15 instead of $1.50 for each weekday in the zone. Moving now costs that difference too (CR-3), so some residents who would have moved stay (RS-11; A-22).

Residents' cars and movers (C-66, C-67). The charge is due for each day a vehicle is "driven or parked" in the zone, so a resident's car parked at home on a weekday costs $1.50 a day where it cost nothing. Residents who value keeping a car at a zone address by less than that over their next-best (giving it up, keeping it outside the zone) give it up (C-66; CR-4b, CR-4c; A-33). In the other direction, a person who works in the zone and drives compares living outside and paying $15 a day with living in the zone and paying $1.50; against the counterfactual, where both cost nothing, the zone gains, and some move in (C-67; CR-4, CR-10a; A-27). Faster roads lower the time cost of driving in from outside, which opposes C-67 through the roads; 8 minutes a day is worth much less than the $13.50 difference for most such workers (A-4), so the combined Tier C sign is also `increase` (C-68; CR-12e, dominance).

Deliveries and U1 (C-21 to C-24). The charge is per day, so a van making twenty drops pays the same as one making one. Under U1-a, a supplier making a single short drop pays $15 for it; some such visits are given up, consolidated onto carriers' multi-drop rounds, or moved to before 7:00 (C-22, U1-a: decrease; A-1). Under U1-b, short visits are free, while a van that stays longer pays for the day; operators reshape long rounds into several in-and-out loops of under 30 minutes where the layout allows (C-21; A-13), so short visits rise against the counterfactual (C-22, U1-b: increase). Proving that a visit was a delivery and lasted under 30 minutes needs records (C-23), and other vehicles may claim the status falsely (C-24). Under U1-a none of this exists. Under both readings, a van that pays for the day pays nothing more for further drops that day: the first drop bears the charge and later ones none (CR-1, 0.9: operation per range), so carriers move drops onto days already charged and consolidate rounds (C-64; CR-4c; A-32). Faster roads let a van make more drops in a day too, which pushes the same way.

Relocation (C-25, C-26). Shoppers who drove in to buy bulky goods can buy them in Ringside or at Merrowdale, where they can park (RS-4). Retailers that depend on such customers move out of the zone as their leases end (C-25; RS-6, CR-7a; A-7). Where they reopen in Ringside, the activity is new to Ringside and lost to the zone (OC-5, 0.9: new to the place).

6. Second-order effects

ID	Actor class	Variable	Sign	Horizon	Rules	Market relation	Assumptions
C-27	`businesses.zone`	`quantity_traded` (sales to customers who drove into the zone; not counting effects through the roads)	decrease	short to medium	RS-9 (0.9: location complements), CR-9	purchases at zone shops: `complement` (note `location`) to being in the zone	A-1, A-7
C-28	`businesses.zone`	`quantity_traded` (all sales; not counting effects through the roads)	ambiguous	medium	RS-9 (0.9: new occupants), CR-12b	as C-27; buses: `complement` to reaching the zone	A-1, A-7, A-8, A-10
C-29	`businesses.edge`	`quantity_traded` (sales; not counting effects through the roads)	increase	short to medium	RS-9 (0.9), RS-4, CR-9	purchases in Ringside: `complement` (note `location`) to parking there	A-7

ID	Actor class	Variable	Sign	Horizon	Rules	Market relation	Assumptions
C-30	businesses.edge	quantity_traded (sales)	ambiguous	short to medium	RS-9, CR-9 (shared_facility; Tier C), CR-12 (0.9)	as C-29; Inner Ring Road: shared_facility	A-3, A-7, A-18
C-31	owners.edge_commercial	asset_value (Ringside shops and offices; not counting effects through the roads)	increase	short (from 4 January 2027)	CR-7c, CR-10c, SB-4	follows C-29	A-7, A-20
C-32	owners.edge_commercial	asset_value (Ringside shops and offices)	ambiguous	short	CR-7c, CR-9 (shared_facility; Tier C), CR-12 (0.9)	as C-30	A-3, A-7, A-18, A-20
C-33	owners.zone_commercial	asset_value (zone shops and offices; not counting effects through the roads)	ambiguous	short	CR-7b, CR-7c, CR-10c, CR-12b	follows C-28, C-46	A-7, A-8, A-10, A-20
C-34	owners.zone_residential	asset_value (zone housing; not counting effects through the roads)	ambiguous	short	CR-7c, CR-10c, CR-6 (0.9: splitting a shift), CR-12b	zone housing and outer housing: substitute	A-2, A-8, A-20, A-27, A-33
C-35	owners.parking.zone	price_complement (zone garage day rate)	decrease	short to medium	RS-12, CR-6 (0.9: splitting a shift)	zone parking: complement to driving in	A-1, A-21
C-36	owners.parking.zone	asset_value (zone garages)	decrease	short	CR-7c, CR-10c, SB-4	as C-35	A-1, A-20, A-21
C-37	owners.parking.edge	asset_value (Ringside garages)	increase	short	CR-7c, CR-10c, SB-4	follows C-9	A-6, A-20
C-38	bus_riders	quality_per_unit (crowding on routes into the zone, before new buses arrive)	decrease	short	CR-9 (shared_facility: the vehicle; Tier C), CR-6 (1.0: below capacity, not rationing), PC-19 (1.0)	buses and cars: substitute; riders: shared_facility	A-5, A-28
C-39	bus_riders	quality_per_unit (crowding, after new buses arrive)	ambiguous	medium	RS-16 (Unless), RS-13, SB-7, CR-9 (shared_facility; Tier C), CR-12b	as C-38	A-5, A-8
C-40	bus_riders	wait_time (headways)	decrease	medium	SB-7, CR-8d	earmark funds service	A-8
C-41	bus_riders	travel_time (bus journeys in the zone)	decrease	short to medium	CR-9 (shared_facility; Tier C)	roads: shared_facility	A-1, A-3
C-42	transit_operator	quantity_supplied (bus service-km)	increase	medium	SB-7, CR-8d	earmark	A-8
C-43	taxpayers, recipients of other city spending	resources_available_other_uses	at_least_one_increase	medium	SB-7, CR-8d, CR-8b	general fund	A-9
C-44	goods_vehicle_operators	full_cost_per_unit (a van-day in the zone: charge plus time; not counting effects through the roads)	U1-a: increase; U1-b: increase	short to medium	CR-1, CR-3, CR-12 (0.9)	—	A-2, A-12
C-45	goods_vehicle_operators	full_cost_per_unit (a van-day in the zone: charge plus time)	U1-a: ambiguous; U1-b: ambiguous	short to medium	CR-9 (shared_facility; Tier C), CR-12b, CR-12 (0.9)	roads: shared_facility	A-2, A-3, A-12
C-46	businesses.zone	price_paid (deliveries; not counting effects through the roads)	U1-a: increase; U1-b: increase	short to medium	RS-14, CR-7b, DM-1	deliveries input_to zone businesses	A-2, A-12, A-14
C-47	businesses.zone	tax_share_borne (charge on deliveries, against carriers)	U1-a: increase; U1-b: increase	medium	DM-1, CR-7b	as C-46	A-12, A-14
C-48	drivers.keep	full_cost_per_unit (a weekday in the zone by car: charge plus time; not counting effects through the roads)	increase	short to medium	CR-1, CR-3, CR-12 (0.9)	—	A-2
C-49	drivers.keep	full_cost_per_unit (a weekday in the zone by car: charge plus time)	increase	short to medium	CR-9 (shared_facility; Tier C), CR-12 (0.9), CR-12e (dominance, A-4)	roads: shared_facility	A-2, A-3, A-4
C-50	drivers.switch	travel_time (door to door, by bus or by parking in Ringside and walking; not counting effects through the roads)	increase	short to medium	CR-3a, CR-4	—	A-5, A-23
C-51	drivers.switch	travel_time (door to door)	increase	short to medium	CR-9 (shared_facility; Tier C), CR-12 (0.9), CR-12e (dominance, A-29)	roads: shared_facility	A-3, A-5, A-23, A-29
C-52	employers.zone, workers.zone	wage_rate (zone jobs reached by car) (+), employment (zone) (−); not counting effects through the roads	at_least_one_as_stated	medium	RS-22, CR-5c, CR-7a, CR-12 (0.9)	labor: workers choose between zone jobs and jobs elsewhere	A-19

ID	Actor class	Variable	Sign	Horizon	Rules	Market relation	Assumptions
C-53	`employers.zone`, `workers.zone`	`wage_rate` (zone jobs reached by car), `employment` (zone)	ambiguous	medium	RS-22, CR-9 (`shared_facility`; Tier C), CR-12b, CR-12 (0.9)	as C-52; roads: `shared_facility`	A-3, A-19
C-54	`input_suppliers`	`quantity_traded` (cameras and payment systems; later, buses)	increase	short to medium	CR-9, SB-7	`input_to` K4 and to bus service	A-8
C-60	`ride_hail_drivers`	`price_substitute` (ride-hail fares for trips into the zone)	ambiguous	short	CR-9 (`shared_facility`; Tier C), CR-12b, CR-12 (0.9)	as C-11; roads: `shared_facility`	A-3, A-25, A-26
C-62	`residents.zone`	`full_cost_per_unit` (a weekday driving or parking in the zone: charge plus time; not counting effects through the roads)	increase	short to medium	CR-1, CR-3, CR-12 (0.9)	—	A-2
C-63	`residents.zone`	`full_cost_per_unit` (a weekday driving in the zone: charge plus time)	ambiguous	short to medium	CR-9 (`shared_facility`; Tier C), CR-12b, CR-12 (0.9)	roads: `shared_facility`	A-2, A-3
C-68	`buyers.prospective` (zone housing)	`quantity_demanded` (zone housing, by people who work in the zone and drive)	increase	short to medium	CR-9 (`shared_facility`; Tier C), CR-12 (0.9), CR-12e (dominance, A-4)	zone housing and outer housing: `substitute`; roads: `shared_facility`	A-2, A-3, A-4, A-27
C-70	`transit_workers`	`employment` (bus operation)	increase	medium	CR-9, LW-10	labor `input_to` bus service	A-8, A-28
C-71	`officials.parking`	`non_tax_receipts` (zone parking fines)	decrease	short to medium	CR-9, CR-4b	parking tickets follow cars parked in the zone	A-1

Shops inside and outside the zone (RS-9, 0.9; C-25 to C-30). Purchases made while in the zone are tied to being there: a shopper who no longer drives in buys where she now parks or goes instead (RS-9, location complements). So sales to customers who drove in fall in the zone (C-27) and rise in Ringside (C-29), and some go to Merrowdale (C-26). RS-9's 0.9 clause adds: when a policy changes who occupies the location, record the new occupants' purchases as the opposing chain. Here the charge changes who is in the zone in two ways. Through the earmark, more frequent buses bring people who would not otherwise have come (A-8, A-10): a Tier B chain. Through the roads, quieter and faster streets attract some visitors who avoided the congestion (Tier C). With the bus chain at $p \geq 0.5$, the zone's total sales are `ambiguous` at Tier B (C-28; CR-12b; CR-12 0.8, kind 2). A premise that the loss of car shoppers dominates (A-30, $p = 0.4$) is below the cut-off. Counting the roads, the combined sign is still ambiguous (CR-12, kind 3), so it is not recorded again. For Ringside (C-29), the Tier B chain is a clean increase; the Tier C chain opposes it: traffic diverted round the zone slows the Inner Ring Road on which Ringside's customers arrive (C-16). Under CR-12's 0.9 rule C-29 stands at Tier B, qualified, and the combined sign at Tier C is `ambiguous` (C-30), since no premise compares the two.

Property values (CR-7c; C-31 to C-37). Each value moves when the charge becomes expected, on 4 January 2027 (CR-7c, CR-10c; A-20); owners then capture the gain or bear the loss, and later buyers pay the new price.

- *Ringside commercial (C-31, C-32).* Its sales rise at Tier B (C-29), so its value rises at Tier B (C-31). The ring road chain opposes it at Tier C, so the combined Tier C sign is `ambiguous` (C-32).
- *Zone commercial (C-33).* The yield of a zone shop falls with sales to car shoppers (C-27) and rises with new bus-borne visitors (C-28's (+) chain). The share of the charge that shoppers and deliveries pass back to zone businesses settles, as far as businesses can move, on the immobile means: the buildings (CR-7b). No premise ranks the chains: `ambiguous`.
- *Zone housing (C-34).* Three chains. Residents who drive or park in the zone pay $1.50 a day and their visitors and tradespeople pay 15(−); *someresidentsgiveuptheircarsorleave*(*C*−66,*A*−33).*Carcommuterstozonejobscomparelivinginthezone*(1.50 a day) with living outside and driving in ($15); against the counterfactual, where both cost nothing, the zone gains relative to outside, and some of them would move in (C-67, A-27) (+). More frequent buses (A-8) raise the value of every address near a route, in the zone and outside (+, for the zone). With the housing stock fixed in the short run, the demand shift goes to price (CR-6, 0.9). The chains oppose: `ambiguous`. Quieter streets (Tier C) do not change that (kind 3).
- *Parking (C-35 to C-37).* Parking in the zone is a complement of driving in (RS-12): demand falls, and with the stock fixed (A-21) the day rate falls (C-35, CR-6 0.9) and so does the garages' value (C-36). C-6 shows that the number of vehicle-days falls at Tier C too, so no opposing chain reaches C-35. Ringside garages gain (C-37).

Buses (C-38 to C-42, C-70). Buses run at a set fare with peak load at about 85% of capacity. Demand at the fare does not exceed the fixed quantity, so CR-6's public-supply clause (and with it the 0.9 clause on a public seller that adds capacity after a lag) does not apply: crowding among riders below capacity is the `shared_facility` relation (the vehicle), at Tier C (CR-6, 1.0; PC-19, 1.0). Riders drawn from cars take space in vehicles that existing riders share, and the fleet cannot grow for 14 to 18 months (A-28), so in the first year crowding worsens for existing riders (C-38, Tier C). A Tier B group (crowding, pass-ups, waiting) would need a premise that demand at the fare exceeds crush capacity on some routes, which the case does not supply. Once new buses arrive, RS-16's Unless clause applies: the policy adds supply to the substitute's market (RS-13), so crowding is `ambiguous` (C-39), at Tier C for the same reason. The added service draws drivers into bus operation (C-70; CR-9, LW-10; A-8, A-28). Headways shorten as service grows (C-40), on A-8. Faster roads speed the buses (C-41, Tier C); no Tier B chain opposes it. The new riders join existing ones; the riders themselves are better off by their own ranking only as far as the switch was their best remaining option (CR-3a: those who switched were made worse off, C-50).

The earmark (CR-8d, SB-7; C-42, C-43). The ordinance applies net revenue "to bus services" and says nothing about general-fund support, which is set each year. Money is fungible (CR-8d): the council can reduce the $140 million of general-fund support as the earmark arrives, so bus spending rises by less than the earmark, and the rest funds other spending or lower taxes (SB-7; A-9, $p = 0.7$). Service still rises if the reduction is less than the earmark (A-8, $p = 0.7$). The freed general funds reach a group: taxpayers, or recipients of other city spending (C-43, `at_least_one_increase`; CR-8, 0.7, payers from the general budget, in reverse). The CCO's costs come out of gross revenue first (K3), so they reduce what the earmark does (OC-3, own revenue; C-57).

Deliveries (C-44 to C-47). A van-day in the zone costs $15 more (C-44 at Tier B; under U1-b only for vans that stay over 30 minutes, A-12: some deliveries cannot be done in 30 minutes). Through the roads, a van covering 40 km in the zone saves about 40 minutes a day, worth more than

$15 at a driver's and van's hourly cost; a van making one short drop saves almost nothing. So the combined sign at Tier C is `ambiguous` under both readings (C-45): there is no dominance premise either way across the class. CR-12e's response argument does not decide C-45: the speed gain comes from cars leaving, which is a response to the charge on cars, not to the charge on vans; for vans it is a second change (CR-12e, 0.8). Zone businesses buy deliveries: carriers can serve customers anywhere in the city, while a zone business cannot move its premises cheaply (A-14), so carriers pass more of the charge on and zone businesses bear more of it than carriers do (C-46, C-47; DM-1, CR-7b). The Tier C chain reaches C-46 through the carriers' costs: the combined sign at Tier C is `ambiguous` (kind 3 through C-45) and is not recorded again.

Drivers who keep driving (C-48, C-49). A weekday in the zone costs $15 more in money (C-48, Tier B). Through the roads, the same day takes less time. For a typical car commute the saving is about 8 minutes a day, worth much less than $15 for most drivers (A-4, p = 0.8: arithmetic on the case's figures and a premise on the value drivers place on time). That premise dominates the Tier C chain, so the combined sign at Tier C is also `increase` (C-49; CR-12e, dominance). Some drivers who value their time highly gain on balance; A-4 is about the class. The response argument would not give C-49: it binds the number of vehicles as a whole (C-6), not each driver's cost (CR-12e, 0.8).

Residents (C-62, C-63). A resident's weekday driving or parking in the zone costs $1.50 more in money (C-62, Tier B). Through the roads, a resident who drives in the zone saves time as other drivers do. A-4 compares the time saved with $15, not with $1.50; for many residents 8 minutes a day may be worth more than $1.50. No premise compares the two, so the combined Tier C sign is `ambiguous` (C-63; CR-12b).

Drivers who switch (C-50, C-51). A driver who switches to the bus or to Ringside parking took an option ranked below driving before the charge (CR-3a), and for most such trips it takes longer (A-23: the bus commute is about 15 minutes longer). Faster roads shorten the bus trip by a few minutes (Tier C). A-29 (the gap exceeds the saving) dominates that chain, so C-51 at Tier C is `increase` too.

Zone jobs (C-52, C-53). A zone job reached by car now costs its holder $15 a day more to reach; workers compare zone jobs with jobs elsewhere (CR-7a), so the terms on which zone employers can hire car commuters move against them (CR-5c): they pay more, or employ fewer, as a mixed group (C-52; A-19). This is RS-22 (compensating differential from a non-wage term), promoted in 1.0, with the cost of reaching the job as the term; its group form is the one recorded. Through the roads, the remaining commute is faster by car and by bus, which lowers the cost of reaching zone jobs (Tier C). No premise compares the two for the class, so the combined Tier C sign is `ambiguous` (C-53).

Parking fines (C-71). Fewer cars parked in the zone mean fewer parking tickets, so the city's zone parking fine receipts fall (CR-9, CR-4b; A-1). The same fall reduces the PAO's parking appeals (C-58).

Stopping points. The suppliers of zone and Ringside shops, the spending of workers whose hours change, Merrowdale's own traffic and the suppliers of new buses beyond the first order: depth 3 to 4, diffuse; stopped. The drivers and buses drawn into bus service come from other uses, whose output falls (RS-13, 1.0; CR-8a); that chain is diffuse at depth 3 and stopped. Air quality, noise and road safety: out of scope.

7. Combined effects

Actor class, variable, horizon	Chains (+)	Chains (−)	Dominance rule	Combined sign
`drivers.nonresident`, `goods_vehicle_operators`, `quantity_traded` (vehicle-days in the zone, weekdays 7:00 to 19:00), short to medium	Faster roads draw some drivers back (Tier C).	The charge (CR-4b; A-1).	CR-12e, response argument on the total of charged vehicle-days: the return is a response to a speed gain that the leaving drivers caused and cannot restore the old number; A-34 (same time cost at a given number of vehicles), A-1 (partial response), A-35 (ride-hail switch lowers the total) (CR-12, 0.9, 1.0).	Tier B: **decrease** (C-4, the two classes). Tier C: **decrease** (C-6, all charged vehicle-days).
`drivers.keep`, `full_cost_per_unit`, short to medium	—	The charge (+ to cost); faster roads (− to cost, Tier C).	A-4 (arithmetic on the case's figures, with a premise on drivers' value of time).	Tier B: **increase** (C-48). Tier C: **increase** (C-49).
`goods_vehicle_operators`, `full_cost_per_unit`, short to medium	The charge (under U1-b, for vans staying over 30 minutes; A-12).	Faster roads (Tier C), large for multi-drop vans.	None: the saving exceeds $15 for some vans and is near zero for others. Response argument not applicable (the speed gain is a second change for vans).	Tier B: U1-a **increase**, U1-b **increase** (C-44). Tier C: **ambiguous** under both readings (C-45).
`drivers.switch`, `travel_time`, short to medium	Bus or walk is slower door to door (A-23).	Faster buses (Tier C).	A-29 (the gap far exceeds the saving).	Tier B: **increase** (C-50). Tier C: **increase** (C-51).
`businesses.zone`, `quantity_traded` (all sales), medium	New bus-borne visitors (A-8, A-10); visitors drawn by quieter streets (Tier C).	Car shoppers who buy elsewhere (C-27).	None (A-30, p = 0.4, below the cut-off).	Tier B: **ambiguous** (C-28). Tier C: ambiguous (kind 3), not recorded again.
`businesses.edge`, `quantity_traded`, short to medium	Shoppers who park in Ringside (C-29).	Slower Inner Ring Road (Tier C, C-16).	None	Tier B: **increase** (C-29). Tier C: **ambiguous** (C-30).
`owners.edge_commercial`, `asset_value`, short	Follows C-29.	Follows C-16 (Tier C).	None	Tier B: **increase** (C-31). Tier C: **ambiguous** (C-32).
`owners.zone_commercial`, `asset_value`, short	New bus-borne visitors.	Lost car shoppers; delivery costs borne by businesses (C-46, C-47).	None	**ambiguous** (C-33).
`owners.zone_residential`, `asset_value`, short	Car commuters moving in (A-27); more buses (A-8).	Residents' $1.50; visitors' and tradespeople's $15.	None	**ambiguous** (C-34).
`bus_riders`, `quality_per_unit` (crowding), short	—	New riders on a fixed fleet below capacity (A-28; `shared_facility`, Tier C).	None needed	Tier C: **decrease** (C-38).
`bus_riders`, `quality_per_unit` (crowding), medium	New buses from the earmark (A-8).	New riders.	None	Tier C: **ambiguous** (C-39).

Actor class, variable, horizon	Chains (+)	Chains (−)	Dominance rule	Combined sign
`ride_hail_drivers`, `price_substitute`, short	More riders (A-25, A-26); the charge per car-day, through fewer cars.	More trips per car-hour on faster roads (Tier C).	None	Tier B: **increase** (C-11). Tier C: **ambiguous** (C-60).
`ride_hail_drivers`, `number_of_sellers` (cars working in the zone), short	More riders.	The charge per car-day (a cost fixed per car and day, RS-16 1.0).	None	**ambiguous** (C-12).
`residents.zone`, `full_cost_per_unit`, short to medium	The $1.50 charge.	Faster roads (Tier C).	None: A-4 compares the time saved with $15, not $1.50.	Tier B: **increase** (C-62). Tier C: **ambiguous** (C-63).
`buyers.prospective` (zone housing), `quantity_demanded`, short to medium	$1.50 in the zone against $15 from outside (A-27).	Faster roads lower the time cost of driving in from outside (Tier C).	A-4 (the time saved is worth much less than the $13.50 difference).	Tier B: **increase** (C-67). Tier C: **increase** (C-68).
`employers.zone`, `workers.zone`, `wage_rate/employment`, medium	Faster commutes (Tier C) lower the cost of reaching zone jobs.	The charge raises it.	None	Tier B: **at_least_one_as_stated** (C-52). Tier C: **ambiguous** (C-53).
`officials.adjudication`, workload, short to medium	Appeals against penalty notices (about 9,000).	Fewer zone parking tickets (about 2,800 fewer appeals).	Arithmetic on the case's figures (A-24).	Net rise in work on an unchanged budget: **at_least_one_as_stated** (C-58).

The charge and the earmark are recorded on both sides: drivers pay (C-1, C-2, C-55), the city receives (C-3), the CCO's costs come out of the receipts (C-57), the set-up cost is advanced from reserves and repaid (C-69), and the net revenue funds buses (C-42) and, through fungibility, other spending or lower taxes (C-43).

Final check (`03_workflow.md`).

1. Uncombined chains: every key reached by a chain through the roads has its Tier B and Tier C records (C-4/C-6, C-11/C-60, C-29/C-30, C-31/C-32, C-44/C-45, C-48/C-49, C-50/C-51, C-52/C-53, C-62/C-63, C-67/C-68); same-direction Tier C chains (C-15, C-64) need no second record.
2. Group signs: zone jobs (C-52), the PAO (C-58) and the general-fund beneficiaries (C-43) are groups; bus crowding is not signed as a Tier B member of a CR-6 group (C-38).
3. Baseline: every sign is against Calderford without the charge; residents' $1.50 is an increase against $0.
4. Worse off is not poorer: `resources_available_other_uses` is signed only for those who keep paying (C-55) and for the reserves (C-69).
5. Timing shifts: trips in and outside charging hours are signed per window (C-4, C-13, C-14).
6. Both sides of every transfer: above; parking fines lost (C-71) against the PAO baseline (C-58).
7. Every actor class, every core rule: CR-7c for garages, homes and shop property; CR-6 only for the PAO (a fixed quantity of hearings at a zero price), not for roads or buses below capacity; enforcement costs and the payer's side are recorded.
8. Unstated premises: A-31 to A-35 added (through trips, consolidation, residents' cars, the response argument's premises).
9. Baseline of every pattern: TX-2, RS-1, RS-4, RS-5, RS-8, RS-9, RS-12, RS-16, RS-22, SB-4 and SB-7 are signed against no charge. RS-7 is re-signed against the counterfactual contest (C-19, restricted to questions that exist only under the ordinance). OC-3's right baseline is used (C-58). DM-1 and TX-9 compare by definition (shares between sides; a static projection) and are recorded on those variables. RS-20, EN-11 and EN-13 are not used.

8. Time structure

- **4 January to 31 October 2027: expectation.** Property and garage values move at once (C-31, C-33, C-34, C-36, C-37). Driver-dependent retailers whose leases end in 2027 begin to move to Ringside or close (C-25); delivery firms reorganize rounds, more so under U1-b (C-21). No rush to drive in the zone before November: a trip cannot be stored, and the charge applies per day from 1 November. Address re-registrations for the discount begin before charging starts (C-17), and zone workers who drive can move into the zone before November (C-67). The $70 million set-up cost is advanced from reserves in 2027 (C-69).
- **November 2027 to end 2028: first year.** Vehicle-days fall at once (C-4, C-6) and speeds rise (C-5). Drivers move to buses, Ringside garages, ride-hail and other hours (C-7 to C-14); these responses grow over months as routines change (TS-8). Buses are crowded (C-38) and Ringside garages fill and raise their rates (C-8, C-9). The PAO's queue lengthens (C-58). The reserves advanced for set-up are repaid from the first year's revenue (C-69).
- **2029 onward.** New buses arrive from the earmark (C-40, C-42), turning crowding from worse to ambiguous (C-39), and transit employment rises (C-70). General-fund support is set at each budget (A-9). Retail relocation continues as leases turn over (C-25, TS-1). The charge is fixed at $15 in money and reviewed every three years; as prices and incomes rise, it binds less unless raised (TS-3).

9. Opportunity costs

ID	Actor class	Variable	Sign	What is given up	Rules
C-55	`drivers.keep`, `residents.zone`, `goods_vehicle_operators` (those who keep paying)	`resources_available_other_uses` (money)	decrease	What the $15 (or $1.50) a day would otherwise have bought or saved. Only those who keep paying are counted (CR-3, note); those who switch are worse off by their own ranking (C-50) but may spend more elsewhere.	TX-1, OC-1, CR-3
C-56	`drivers.nonresident`, `residents.zone`, `goods_vehicle_operators`	`compliance_cost` (accounts, payments, penalty disputes)	increase	Time spent registering, paying and disputing.	OC-2, TX-5, CR-8c
C-57	`officials.enforcement` (CCO)	`enforcement_cost`	increase	About $45 million a year, and $70 million to set up, paid from gross revenue before the earmark: buses get that much less.	OC-3, CR-8c, CR-8d

ID	Actor class	Variable	Sign	What is given up	Rules
C-58	`officials.adjudication`, `parking_appellants`	`wait_time` (decisions on other parking appeals) (+), `enforcement_cost` (time per other appeal) (−)	at_least_one_as_stated	The PAO's other work: its budget is unchanged and its caseload rises about 14% net.	OC-3 (0.9: fixed budget; right baseline), CR-6 (public supply: a fixed quantity of hearings), CR-8a
C-59	`officials.enforcement`	`tax_revenue_vs_static_projection`	decrease	Revenue is about $303 million against a static $414 million.	TX-9
C-69	`taxpayers`, recipients of other city spending (city reserves)	`resources_available_other_uses`	decrease	$70 million of reserves held for about a year until repaid from the first year's revenue.	CR-8a, CR-12g

OC-3, three ways. The CCO is funded from the policy's own revenue (K3: net revenue goes to buses after the CCO's costs), so its cost reduces what the earmark buys (C-57; OC-3, own revenue; CR-8d), not general taxpayers' means. The PAO's budget is not changed (A-24), so OC-3's 0.9 fixed-budget clause applies: its added work comes from some other use of its time: slower decisions on other appeals, or less time per case, recorded as a group with `at_least_one_as_stated` (C-58). The right baseline is the PAO's work without the charge, not zero: fewer cars in the zone mean fewer zone parking tickets and appeals. Arithmetic on the case's figures (about 9,000 new appeals against about 2,800 fewer) gives the net direction. The PAO is a fixed quantity of hearings at a zero price, so here, unlike the roads and buses, CR-6's public-supply clause does apply, and it gives the margins as a group. Zone parking enforcement ($11 million a year) also has fewer cars to watch; that is a separate budget, and its staff are not cut by the ordinance, so it is not signed.

Prevented trips. Road use in the zone is not an exchange with a seller, so OC-4 does not apply to it as such. Those who stop driving in are worse off by their own ranking (CR-3a); where a shopper who stops coming does not buy the same goods elsewhere, the purchases are prevented exchanges (CR-5a), but whether she buys elsewhere is C-26's question and no separate sign is given.

10. Assumptions

ID	Statement	Type	p	Evidence	If false
A-1	Some drivers who now enter the zone on weekdays value the day there by less than $15 more than its other costs, at today's speeds (the charge binds, and the response is partial).	binding	0.95	Trip surveys: many trips are discretionary or could use other modes; the city model projects about 88,000 vehicles against 118,000.	Nothing in C-4 to C-6 follows; C-6's response argument gives only "decrease or no change"; C-71 is not reported.
A-2	The charge is enforced: cameras read most plates and penalties are collected.	enforcement	0.85	74 entry and 40 internal camera points; penalty $130.	C-1, C-2 and the consequences that follow shrink; C-18 grows.
A-3	Fewer vehicles in the zone raise speeds for those that remain.	other	0.85	City traffic model (12 to 15 km/h); experience of road closures and holidays.	Every Tier C consequence that runs over the roads (C-5, C-6, C-16, C-30, C-32, C-41, C-45, C-49, C-51, C-53, C-60, C-63, C-68) fails.
A-4	For most car trips into the zone, the time saved by faster roads is worth much less than $15.	other	0.8	About 8 minutes a day on a typical commute; wage levels in Calderford.	C-49 and C-68 become `ambiguous`.
A-5	Some drivers switch to bus, tram or Ringside parking.	substitutability	0.9	62% of commuters already use transit; peak load about 85% of capacity; routes from most districts; Ringside is 5 to 10 minutes' walk.	C-7, C-8, C-38, C-50, C-51 are not reported.
A-6	Ringside's garages, about 88% full, cannot add spaces within a year.	durability	0.75	About 900 spare spaces; planning and construction times.	C-9 and C-37 shrink; the shift goes to quantity.
A-7	Some shoppers who drove into the zone buy instead in Ringside or Merrowdale, where they can park.	other	0.7	Shopper surveys; Ringside's and Merrowdale's ranges.	C-25 to C-33 lose their location chains.
A-8	General-fund support for buses is cut by less than the earmark, so bus service rises.	revenue_use	0.7	Earmark about $230 to $260 million against $140 million of support; council statements; past budgets.	C-28's (+) chain, C-39, C-40, C-42, C-70 and C-54's bus part are not reported.
A-9	The council reduces general-fund support for buses below what it would otherwise have been (no maintenance-of-effort clause).	revenue_use	0.7	No clause in K3; budget pressures.	C-43 is not reported; the earmark adds its full amount to buses.
A-10	More frequent buses bring to zone shops some visitors who would not otherwise have come.	other	0.55	Effect of earlier service changes on footfall.	C-28 becomes `decrease` at Tier B.
A-11	Some non-residents can register a vehicle to a zone address without detection.	enforcement	0.6	Registration by declaration; audits rare.	C-17 is not reported.
A-12	Some deliveries cannot be completed within 30 minutes (installations, large drops, multi-drop rounds).	other	0.85	Freight survey: 12,000 of 19,000 goods vehicles stay longer.	C-44, C-46 and C-47 under U1-b are not reported; C-64 is not reported.
A-13	Under U1-b some rounds can be reshaped into visits under 30 minutes.	other	0.7	Many zone drops lie near the edge; Ringside depots.	C-21 to C-24 under U1-b are not reported; C-22 under U1-b is `no_change`.
A-14	Carriers have better alternatives (customers elsewhere) than zone businesses have (premises elsewhere).	substitutability	0.65	Carriers serve the whole region; zone businesses' leases and locations.	C-47 is not reported; C-46 rests on RS-14 alone.
A-15	Zone retailers, delivery firms, Ringside residents and the transit union can organize at a cost below their stakes.	other	0.85	Existing associations; hearings in 2026.	C-19 is not reported.
A-16	Some drivers evade (foreign, cloned or obscured plates).	enforcement	0.6	Experience of camera enforcement elsewhere.	C-18, C-24 and C-65 are not reported.
A-17	Some weekday trips can move to before 7:00, after 19:00 or to weekends.	other	0.7	Shopping hours; shift work.	C-13 and C-14 are not reported.
A-18	Some through traffic diverts to the Inner Ring Road.	mobility	0.85	Origin-destination survey: about a tenth of vehicle-days are through trips.	C-15, C-16, C-30, C-32 and C-61 are not reported.

ID	Statement	Type	p	Evidence	If false
A-19	Some zone workers commute by car and cannot switch cheaply, and zone employers compete with employers elsewhere for them.	other	0.55	22% of commuters drive; shift workers and those far from transit.	C-52 and C-53 are not reported.
A-20	Buyers of property expect the charge to last.	expectations	0.75	No expiry; three-yearly review of the level only.	C-31, C-33, C-34, C-36, C-37 shrink.
A-21	Zone garages cannot be converted to other uses within a year.	durability	0.8	Structure and planning rules.	C-35 and C-36 shrink; the shift goes to quantity.
A-22	Some residents who drive in the zone would have moved out of it within a few years.	mobility	0.6	Household turnover in the zone.	C-20 is not reported.
A-23	For most who switch, the trip by bus or by Ringside parking and walking takes longer than by car.	other	0.8	Bus commute about 15 minutes longer door to door.	C-50 and C-51 are not reported.
A-24	The PAO's budget is not raised for the new appeals, and its net caseload rises (about 9,000 new appeals against about 2,800 fewer).	revenue_use	0.8	The ordinance gives the PAO no funds; projected notice and appeal rates.	C-58 is replaced by OC-3's rising-budget case.
A-25	Some drivers switch to ride-hail or taxis.	substitutability	0.6	Current ride-hail use; parking costs in the zone.	C-10 to C-12 are not reported.
A-26	Ride-hail supply into the zone at peak cannot expand at an unchanged cost at once.	durability	0.6	Peak surge pricing now.	C-11 rests on the charge alone, through fewer cars; C-12 stays `ambiguous` (more cars come in at an unchanged cost, against the charge).
A-27	Some people who work in the zone and drive would move into it to pay $1.50 rather than $15.	mobility	0.55	Car commuters' household moves.	C-67 and C-68 are not reported; C-34 loses one (+) chain; still ambiguous with A-8.
A-28	Calderford Transit cannot add buses for at least a year (14 to 18 months to order).	durability	0.8	Order books; fleet size; peak load 85%.	C-38 becomes ambiguous; C-70 comes earlier.
A-29	The gap in door-to-door time between car and bus far exceeds what faster roads save the bus (15 minutes against a few).	other	0.75	Timetables; the city model.	C-51 becomes `ambiguous`.
A-30	The loss of car shoppers dominates the gain from new visitors in zone sales.	other	0.4	No direct evidence; depends on the bus service actually added.	Below the cut-off: C-28 stays `ambiguous`.
A-31	For most diverted through trips, going round on the Inner Ring Road takes longer than crossing the zone did.	other	0.75	The ring road is longer than the cross-zone route and busy at peak; origin-destination survey.	C-15 is not reported; diverted drivers are worse off by their own ranking only.
A-32	Some carriers can consolidate zone deliveries into fewer van-days (more drops per charged day).	other	0.7	One charge covers any number of entries in a day; carriers already plan rounds; Ringside depots.	C-64 is not reported.
A-33	Some zone residents value keeping a car at a zone address by less than $1.50 a weekday more than their next-best (giving it up, keeping it outside the zone, moving out).	substitutability	0.6	14,500 cars registered at zone addresses against about 9,500 resident cars driven in the zone on a weekday, so many are little used; car-sharing and transit in the zone.	C-66 is not reported; C-34 loses part of a (−) chain.
A-34	At a given number of vehicles in the zone, travel time is the same with and without the charge: the roads, signals and parking are unchanged, and the extra buses from the earmark add to congestion rather than relieve it.	other	0.8	The ordinance changes no road layout; the city model.	C-6's response argument fails: the old number need not bring back the old speed, and C-6 is `ambiguous`.
A-35	Each ride-hail car-day added carries the trips of more than one former private car-day, so switching to ride-hail lowers the total of vehicle-days.	other	0.8	Ride-hail cars make many zone trips a day; the city model.	C-6 needs a premise on the size of the ride-hail switch; ride-hail car-days are a second, opposing response.

11. Report

The policy. From 1 November 2027, every vehicle driven or parked in Calderford's central zone on a weekday between 7:00 and 19:00 pays $15 for the day; vehicles registered to zone addresses pay $1.50. Scheduled buses are exempt. The net revenue, after the charging office's costs, goes to bus services. It is unclear whether delivery vans that enter and leave within 30 minutes pay.

Visible consequences. Drivers pay (C-1, C-2) and the city collects (C-3). Fewer vehicles are in the zone in charging hours (C-4, and C-6 counting the roads), and traffic there moves faster (C-5, Tier C).

Invisible consequences.

- *Drivers.* Those who keep driving pay more per day even counting the time they save (C-48, C-49), and have less to spend (C-55). Those who switch to buses, Ringside parking or ride-hail take longer door to door (C-50, C-51). Some drive in the evenings and at weekends instead (C-13, C-14); through drivers go round on the ring road, which slows it (C-15, C-16). Through traffic moves to the Inner Ring Road (C-61). Some register cars at zone addresses (C-17) or evade (C-18), and keepers whose plates are cloned spend time contesting notices (C-65). Zone residents who drive move less (C-20); they pay more per day in money (C-62), though counting faster roads their full cost may rise or fall (C-63); some give up cars kept at zone addresses (C-66). People who work in the zone and drive move into it (C-67, C-68).
- *Substitutes.* Bus and tram trips rise (C-7) and buses are more crowded until new buses arrive (C-38, Tier C); after that, crowding may go either way (C-39), while headways shorten (C-40) and buses run faster (C-41, Tier C). Ringside's garages fill and charge more (C-8, C-9). Ride-hail trips and fares rise (C-10, C-11), though counting faster roads the fare may rise or fall (C-60); the number of ride-hail cars working in the zone may rise or fall (C-12).
- *Shops and property.* Zone shops lose sales to customers who drove in (C-27); whether their total sales fall depends on how many new visitors the added buses bring (C-28). Driver-dependent retailers leave the zone (C-25), and shoppers buy in Ringside or Merrowdale (C-26). Ringside's shops sell more (C-29) and their property rises in value (C-31), unless the busier ring road outweighs it (C-30, C-32, Tier C). Zone commercial and residential values may rise or fall (C-33, C-34). Zone garages lose value and charge less (C-35, C-36); Ringside garages gain (C-37). These values moved in January 2027, when the charge became expected.
- *Deliveries.* Van operators pay more per day (C-44), though counting faster roads, multi-drop rounds may gain (C-45). Zone businesses pay more for deliveries and bear more of the charge on them than carriers do (C-46, C-47).

- *Deliveries, more.* Carriers put more drops into each charged van-day (C-64).
- *Jobs.* Zone employers pay more or employ fewer car commuters (C-52); counting faster commutes, the sign is open (C-53). More bus staff are employed once new buses arrive (C-70).
- *The budget.* The earmark raises bus service only as far as the council does not cut its own support (C-42); what it cuts funds other spending or lower taxes (C-43). The charging office's costs come out of the earmark (C-57), and its set-up cost ties up city reserves for a year (C-69). The adjudication office's other work slows (C-58). Zone parking fine receipts fall (C-71). Revenue falls short of a static projection (C-59).

Per reading of U1. If short delivery visits pay (U1-a), single short drops are given up or consolidated (C-22). If they do not (U1-b), delivery rounds are reshaped into short loops (C-21, C-22), with records to prove them (C-23) and false claims (C-24). Under both readings, delivery costs and their burden on zone businesses rise (C-44, C-46, C-47).

Ambiguous effects. Zone shops' total sales and zone property values (C-28, C-33, C-34); the number of ride-hail cars in the zone (C-12); bus crowding after new buses (C-39, Tier C); counting the roads, Ringside's sales and values (C-30, C-32), van operators' cost (C-45), zone employers' wages and jobs (C-53), ride-hail fares (C-60) and residents' full cost (C-63).

Key conditions. That the charge binds and is enforced (A-1, A-2); that fewer vehicles mean faster roads (A-3, p = 0.85), on which every Tier C claim through the roads rests; bus crowding is a Tier C claim through the shared vehicle; that the time saved is small against $15 for car trips (A-4, p = 0.8); that the council does not cut bus support by the full earmark (A-8, p = 0.7).

What this analysis cannot say. How large any effect is; whether drivers as a whole gain or lose counting the time saved; whether the city as a whole is better off; air quality, noise, safety and health (out of scope for version 1); whether the policy is desirable.

Consequence list (normal form)

ID	Actor class	Variable	Sign	Horizon	Visibility	Tier	Derivation	Assumptions
C-1	drivers.nonresident	price_paid (weekday in the zone by car)	increase	short	visible	B	CR-1, TX-2	A-2
C-2	residents.zone	price_paid (weekday in the zone, discounted)	increase	short	visible	B	CR-1, TX-2	A-2
C-3	officials.enforcement	tax_revenue (charge)	increase	short	visible	A	TX-1	none
C-4	drivers.nonresident, goods_vehicle_operators	quantity_traded (vehicle-days in the zone; not counting effects through the roads)	decrease	short–medium	visible	B	CR-4c, TX-2, CR-12	A-1, A-2
C-5	drivers.keep, goods_vehicle_operators, bus_riders	travel_time (per km in the zone)	decrease	short–medium	visible	C	CR-4b, CR-9 (shared_facility), CR-6 (1.0)	A-1, A-2, A-3
C-6	drivers.nonresident, residents.zone, goods_vehicle_operators, ride_hail_drivers	quantity_traded (all charged vehicle-days in the zone)	decrease	short–medium	visible	C	CR-9 (shared_facility), CR-12e, CR-12 (0.9, 1.0)	A-1, A-2, A-3, A-34, A-35
C-7	drivers.switch	quantity_substitute (bus and tram trips)	increase	short–medium	invisible	B	RS-1, CR-4	A-5
C-8	drivers.switch	quantity_substitute (Ringside garage parking)	increase	short	invisible	B	RS-1, RS-16, CR-6	A-5, A-6
C-9	owners.parking.edge	price_substitute (Ringside day rate)	increase	short–medium	invisible	B	RS-16, CR-6	A-5, A-6
C-10	drivers.switch	quantity_substitute (ride-hail trips)	increase	short	invisible	B	RS-1	A-25
C-11	ride_hail_drivers	price_substitute (ride-hail fares into the zone; not counting effects through the roads)	increase	short	invisible	B	RS-16, CR-4, CR-6, CR-12a, CR-12	A-25, A-26
C-12	ride_hail_drivers	number_of_sellers (ride-hail cars working in the zone)	ambiguous	short	invisible	B	RS-16, CR-4, CR-12b	A-25, A-26
C-13	drivers.retime	quantity_traded (weekday trips before 7:00 and after 19:00)	increase	short–medium	invisible	B	RS-8, CR-10a, CR-4	A-17
C-14	drivers.retime	quantity_traded (weekend trips)	increase	short–medium	invisible	B	RS-8, CR-4	A-17
C-15	drivers.through	travel_time (diverted through trips)	increase	short	invisible	B	CR-4, CR-3a	A-18, A-31
C-16	ring_road_users	travel_time (Inner Ring Road)	increase	short–medium	invisible	C	CR-9 (shared_facility)	A-3, A-18
C-17	drivers.nonresident	reclassification (registrations to zone addresses)	increase	short–medium	invisible	B	RS-5	A-11
C-18	drivers.nonresident, goods_vehicle_operators	legal_breach (unpaid charges, false plates)	increase	short	invisible	B	RS-3, CR-4	A-16
C-19	lobbying groups	lobbying_effort (questions that exist only under the ordinance)	increase	short–medium	invisible	B	RS-7, CR-4, CR-12	A-15
C-20	residents.zone	turnover (moves out of the zone)	decrease	medium	invisible	B	RS-11, CR-3	A-22
C-21	goods_vehicle_operators	reclassification (rounds reshaped into short visits)	U1-a: no_change; U1-b: increase	short–medium	invisible	B	RS-5, CR-1	A-13
C-22	goods_vehicle_operators	quantity_traded (delivery visits under 30 minutes)	U1-a: decrease; U1-b: increase	short–medium	invisible	B	CR-4, RS-5	A-1, A-13
C-23	goods_vehicle_operators	compliance_cost (short-visit records)	U1-a: no_change; U1-b: increase	short	invisible	B	CR-8c, OC-2	A-13

ID	Actor class	Variable	Sign	Horizon	Visibility	Tier	Derivation	Assumptions
C-24	goods_vehicle_operators	legal_breach (false short-visit claims)	U1-a: no_change; U1-b: increase	short	invisible	B	RS-3	A-13, A-16
C-25	businesses.zone (driver-dependent retailers)	number_of_sellers (in the zone)	decrease	medium	invisible	B	RS-4, RS-6, CR-7a	A-7, A-20
C-26	shoppers	cross_border_purchases (outside the zone)	increase	short–medium	invisible	B	RS-4, RS-9	A-7
C-27	businesses.zone	quantity_traded (sales to car customers; not counting effects through the roads)	decrease	short–medium	invisible	B	RS-9, CR-9	A-1, A-7
C-28	businesses.zone	quantity_traded (all sales; not counting effects through the roads)	ambiguous	medium	invisible	B	RS-9, CR-12b	A-1, A-7, A-8, A-10
C-29	businesses.edge	quantity_traded (Ringside sales; not counting effects through the roads)	increase	short–medium	invisible	B	RS-9, RS-4, CR-9	A-7
C-30	businesses.edge	quantity_traded (Ringside sales)	ambiguous	short–medium	invisible	C	RS-9, CR-9 (shared_facility), CR-12	A-3, A-7, A-18
C-31	owners.edge_commercial	asset_value (not counting effects through the roads)	increase	short	invisible	B	CR-7c, CR-10c, SB-4	A-7, A-20
C-32	owners.edge_commercial	asset_value	ambiguous	short	invisible	C	CR-7c, CR-9 (shared_facility), CR-12	A-3, A-7, A-18, A-20
C-33	owners.zone_commercial	asset_value (not counting effects through the roads)	ambiguous	short	invisible	B	CR-7b, CR-7c, CR-10c, CR-12b	A-7, A-8, A-10, A-20
C-34	owners.zone_residential	asset_value (not counting effects through the roads)	ambiguous	short	invisible	B	CR-7c, CR-10c, CR-6, CR-12b	A-2, A-8, A-20, A-27, A-33
C-35	owners.parking.zone	price_complement (zone day rate)	decrease	short–medium	invisible	B	RS-12, CR-6	A-1, A-21
C-36	owners.parking.zone	asset_value (zone garages)	decrease	short	invisible	B	CR-7c, CR-10c, SB-4	A-1, A-20, A-21
C-37	owners.parking.edge	asset_value (Ringside garages)	increase	short	invisible	B	CR-7c, CR-10c, SB-4	A-6, A-20
C-38	bus_riders	quality_per_unit (crowding, before new buses)	decrease	short	invisible	C	CR-9 (shared_facility), CR-6 (1.0), PC-19	A-5, A-28
C-39	bus_riders	quality_per_unit (crowding, after new buses)	ambiguous	medium	invisible	C	RS-16, RS-13, SB-7, CR-9 (shared_facility), CR-12b	A-5, A-8
C-40	bus_riders	wait_time (headways)	decrease	medium	invisible	B	SB-7, CR-8d	A-8
C-41	bus_riders	travel_time (bus journeys in the zone)	decrease	short–medium	invisible	C	CR-9 (shared_facility)	A-1, A-3
C-42	transit_operator	quantity_supplied (bus service-km)	increase	medium	visible	B	SB-7, CR-8d	A-8
C-43	taxpayers, recipients of other city spending	resources_available_other_uses	at_least_one_increase	medium	invisible	B	SB-7, CR-8d, CR-8b	A-9
C-44	goods_vehicle_operators	full_cost_per_unit (van-day; not counting effects through the roads)	U1-a: increase; U1-b: increase	short–medium	invisible	B	CR-1, CR-3, CR-12	A-2, A-12
C-45	goods_vehicle_operators	full_cost_per_unit (van-day)	U1-a: ambiguous; U1-b: ambiguous	short–medium	invisible	C	CR-9 (shared_facility), CR-12b, CR-12	A-2, A-3, A-12
C-46	businesses.zone	price_paid (deliveries; not counting effects through the roads)	U1-a: increase; U1-b: increase	short–medium	invisible	B	RS-14, CR-7b, DM-1	A-2, A-12, A-14
C-47	businesses.zone	tax_share_borne (charge on deliveries, against carriers)	U1-a: increase; U1-b: increase	medium	invisible	B	DM-1, CR-7b	A-12, A-14
C-48	drivers.keep	full_cost_per_unit (weekday by car; not counting effects through the roads)	increase	short–medium	invisible	B	CR-1, CR-3, CR-12	A-2
C-49	drivers.keep	full_cost_per_unit (weekday by car)	increase	short–medium	invisible	C	CR-9 (shared_facility), CR-12, CR-12e	A-2, A-3, A-4
C-50	drivers.switch	travel_time (door to door; not counting effects through the roads)	increase	short–medium	invisible	B	CR-3a, CR-4	A-5, A-23
C-51	drivers.switch	travel_time (door to door)	increase	short–medium	invisible	C	CR-9 (shared_facility), CR-12, CR-12e	A-3, A-5, A-23, A-29
C-52	employers.zone, workers.zone	wage_rate (+), employment (−) (not counting effects through the roads)	at_least_one_as_stated	medium	invisible	B	RS-22, CR-5c, CR-7a, CR-12	A-19
C-53	employers.zone, workers.zone	wage_rate, employment	ambiguous	medium	invisible	C	RS-22, CR-9 (shared_facility), CR-12b, CR-12	A-3, A-19
C-54	input_suppliers	quantity_traded (cameras, payment systems, buses)	increase	short–medium	invisible	B	CR-9, SB-7	A-8
C-55	drivers.keep, residents.zone, goods_vehicle_operators	resources_available_other_uses (money)	decrease	short–long	invisible	B	TX-1, OC-1, CR-3	A-2

ID	Actor class	Variable	Sign	Horizon	Visibility	Tier	Derivation	Assumptions
C-56	drivers.nonresident, residents.zone, goods_vehicle_operators	compliance_cost (accounts, payments, disputes)	increase	short	invisible	A	OC-2, TX-5, CR-8c	none
C-57	officials.enforcement (CCO)	enforcement_cost	increase	short–long	invisible	A	OC-3, CR-8c, CR-8d	none
C-58	officials.adjudication, parking_appellants	wait_time (other appeals) (+), enforcement_cost (time per other appeal) (−)	at_least_one_as_stated	short–medium	invisible	B	OC-3, CR-6, CR-8a	A-24
C-59	officials.enforcement	tax_revenue_vs_static_projection	decrease	short	invisible	B	TX-9	A-1
C-60	ride_hail_drivers	price_substitute (ride-hail fares into the zone)	ambiguous	short	invisible	C	CR-9 (shared_facility), CR-12b, CR-12	A-3, A-25, A-26
C-61	drivers.through	quantity_substitute (Inner Ring Road through trips)	increase	short–medium	invisible	B	RS-1, CR-4	A-18
C-62	residents.zone	full_cost_per_unit (weekday in the zone; not counting effects through the roads)	increase	short–medium	invisible	B	CR-1, CR-3, CR-12	A-2
C-63	residents.zone	full_cost_per_unit (weekday in the zone)	ambiguous	short–medium	invisible	C	CR-9 (shared_facility), CR-12b, CR-12	A-2, A-3
C-64	goods_vehicle_operators	transaction_size (drops per charged van-day)	increase	short–medium	invisible	B	CR-4c, CR-1	A-12, A-32
C-65	keepers.cloned_plates	compliance_cost (contesting notices for others' trips)	increase	short–medium	invisible	B	CR-9 (contract), CR-1, CR-8c	A-16
C-66	residents.zone	quantity_demanded (cars kept at zone addresses)	decrease	medium	invisible	B	CR-4b, CR-4c	A-2, A-33
C-67	buyers.prospective (zone housing)	quantity_demanded (zone housing by car-commuting zone workers; not counting effects through the roads)	increase	short–medium	invisible	B	CR-4, CR-10a, CR-12	A-2, A-27
C-68	buyers.prospective (zone housing)	quantity_demanded (zone housing by car-commuting zone workers)	increase	short–medium	invisible	C	CR-9 (shared_facility), CR-12, CR-12e	A-2, A-3, A-4, A-27
C-69	taxpayers, recipients of other city spending (city reserves)	resources_available_other_uses	decrease	short (2027 to first-year repayment)	invisible	A	CR-8a, CR-12g	none
C-70	transit_workers	employment (bus operation)	increase	medium	invisible	B	CR-9, LW-10	A-8, A-28
C-71	officials.parking	non_tax_receipts (zone parking fines)	decrease	short–medium	invisible	B	CR-9, CR-4b	A-1

Corrections after the core-only comparison

Each conclusion of the comparison (`05_examples_core/WE-046_congestion_charge_core.md`) whose cause is P, or E against this version, each valid core-only claim this version omitted, and each correction in its Findings was checked against the core rules and catalog 1.0 before it was applied.

Changed.

- C-15: the premise that going round takes longer is now stated (A-31); CR-3a alone gives only a loss by the drivers' own ranking (comparison row C-15, P).
- C-38: Tier B to Tier C. At about 85% peak load, demand at the fare does not exceed the fixed quantity, so CR-6's public-supply clause and its 0.9 lag clause do not apply; crowding among riders below capacity is the `shared_facility` relation (CR-6, 1.0; PC-19, 1.0). PC-19 was cited for a single member of its own group (comparison row C-38 and Findings). A Tier B group would need a premise that demand exceeds crush capacity, which the case does not give.
- C-39: Tier B to Tier C, for the same reason (both chains act on crowding in shared vehicles).
- C-5, C-7 and the text of sections 2, 4 and 6: road congestion and bus crowding are no longer described as CR-6 rationing (CR-6, 1.0, scope of the public-supply clause). Signs unchanged.
- C-6: re-based on CR-12's 1.0 note on what the response argument needs. The total it bounds is now named (all charged vehicle-days, every class), the analogue of the constant-cost premise is stated (A-34), and the second response through ride-hail is handled by A-35. Sign and tier unchanged; the actor classes widen to the named total.
- C-11: qualified "not counting effects through the roads", with its Tier C companion C-60 (comparison row C-11, E: the Tier C record was missing). Re-derived under RS-16 1.0: the $15 is a cost per car-day, not per trip, so it raises the fare only through fewer cars working in the zone; sign unchanged.
- C-12: re-signed under RS-16 1.0 (only a cost per unit sold opposes sellers' net price; a cost fixed per facility bears on the number of facilities). Its variable changes from drivers' net price per trip to the number of ride-hail cars working in the zone, `ambiguous`.
- C-19: restricted to questions that exist only under the ordinance and re-signed against the counterfactual contest (CR-4, 1.0; RS-7, 1.0); lobbying on a charge as such needs a premise the case lacks and is not signed.
- C-52, C-53: cite RS-22 (promoted in 1.0) instead of the candidate pattern.
- C-58: CR-6 added to the derivation: the PAO's hearings are a fixed quantity at a zero price, the one place in this case where the public-supply clause applies (Findings).
- C-34: its chains now cite C-66 and C-67.

Added (core-only consequences this version omitted, each derivable from the core rules): C-60 (ride-hail fare at Tier C, `ambiguous`), C-61 (through trips on the Inner Ring Road), C-62 and C-63 (residents' full cost: Tier B `increase` not counting the roads, Tier C `ambiguous`), C-64 (drops per charged van-day, A-32), C-65 (keepers of cloned plates contest notices), C-66 (fewer cars kept at zone addresses: "driven or parked", A-33), C-67 and C-68 (zone housing demand from car-commuting zone workers, Tier B and Tier C), C-69 (city reserves tied up by the set-up advance), C-70 (transit employment after new buses), C-71 (zone parking fine receipts fall). New assumptions A-31 to A-35.

Removed. None.

Comparison conclusions not applied.

- Comparison row C-27 (J, zone sales to former car customers as a separate record): a recording choice that catalog 1.0 does not settle; kept.
- Comparison row C-3 (Tier A against Tier B for the revenue): no cause recorded and TX-1 gives Tier A given collection; kept.
- Comparison rows C-35, C-52 and C-58 (S, E against the core version): the comparison found this version's signs valid; only citations were updated (C-52, C-58).
- RS-21 was considered for C-64 and not cited: its precondition is a cap or right counted in physical units, and K1 is a charge, not a cap; CR-4c gives the move directly.

Gaps found

(a) Calculus gaps (a step no core rule covers). None new. Every chain through travel speeds runs over the `shared_facility` relation (the zone's roads, the Inner Ring Road): that is the open GAP-026-1 (also raised in WE-032 for congestion), covered by CR-9's 0.7 clarification at Tier C pending Q20.

(b) Unclear core rules.

1. *CR-12 0.9, Tier C against Tier B, where both give the same sign.* For vehicle-days (C-4, C-6), drivers' cost (C-48, C-49) and switchers' time (C-50, C-51), the Tier C combined sign equals the Tier B sign. The rule says "record two consequences" without saying whether the second is needed when it adds no information. Recorded both here, because the Tier C one rests on extra premises (A-3, A-4, A-29) and a reader needs to see that the Tier B sign survives them.
2. *CR-12e's response argument across a shared facility.* The argument applies to vehicle-days (C-6): the drivers drawn back are responding to a speed gain caused by drivers leaving. It does not apply to vans (C-45): for them the speed gain is caused by cars' response, a second change (CR-12e, 0.8). Nor to each driver's full cost (C-49): it binds the class's number, not each member's cost. The 0.8 and 0.9 notes cover both limits, but only when read together; a sentence saying "the argument signs the quantity of the responding class; costs to members or to other classes need their own premise" would help.
3. *CR-6 0.9, public seller adding capacity after a lag.* Worked, but it needed CR-12g to keep the two horizons apart (C-38 short, C-39 medium). The clause could say so. (Superseded under 1.0: at 85% peak load the clause does not apply, and C-38 and C-39 are Tier C `shared_facility` chains; see Corrections.)
4. *CR-1, operation by registered address.* The residents' discount is a reduced rate for a class defined by a declaration the actor controls. CR-1 0.7 says to split a class by response; here the class boundary is itself something actors choose (C-17). RS-5 covers the response, but CR-1 does not say that a class defined by a declared status is open to entry.

(c) Missing or wrong patterns.

1. *TX-2 (and DM-1) where the seller of the taxed good is the state.* TX-2 gives `price_paid` up and/or `net_price_received` down. Road space has no seller whose net price can fall: it is a public supply at a zero money price, rationed by time (CR-6, 0.7). Applied as written, TX-2's "and/or" leaves the seller's side empty and says nothing about where the rest of the burden goes. It was read as: buyers bear the charge first, and the burden passes along the trips' other relations (customers' purchases, deliveries) to immobile means in the zone (CR-7b). No sign had to be overridden, but a clause for a charge on the use of a public facility would help.
2. *RS-16 0.9, costs on the substitute.* Worked as written for ride-hail (C-11, C-12): both chains raise the fare, and they oppose on drivers' net price. The clause speaks of "the substitute's net price to its sellers"; the buyers' price is the other half and needed CR-12a. (Resolved in 1.0: RS-16's clause on prices per unit of quality says a cost fixed per facility bears on the number of facilities; C-12 is re-signed accordingly.)
3. *RS-9 0.9, location complements and new occupants.* Applied as written, but "the new occupants" came through two different relations: buses funded by the earmark (Tier B, a second component) and quieter streets (Tier C, the shared facility). The clause does not say that the new occupants' chain may run through another component or another tier; both were needed here (C-28, C-30).
4. *SB-7 needs two premises, not one.* SB-7 states that spending on the purpose "rises by less than the earmarked amount", with one premise (general funding can be cut). Whether it still rises at all needs a second premise, that the cut is less than the earmark (A-8). Without it, C-42 could not be signed. SB-7 should say so.
5. No pattern gave a sign that had to be overridden.

(d) Vocabulary.

1. No variable for vehicle-days in a zone or use of a road; recorded as `quantity_traded` with a qualifier, though no exchange takes place.
2. `full_cost_per_unit` is defined for a buyer's wait, search and side payments; it was used for the money charge plus travel time (C-48, C-49), which fits the definition's spirit but not its examples.
3. `wait_time` serves both for bus headways (C-40) and for adjudication queues (C-58); no variable for an agency's other work slowed by a fixed budget.
4. No actor class for users of a shared facility as such; `ring_road_users` was coined.

(e) Test of 0.9.

- *CR-12 0.9, Tier C against Tier B.* Worked as written on nine keys: three where a dominance premise or the response argument made the Tier C sign equal the Tier B sign (C-6, C-49, C-51), four where the Tier C sign is `ambiguous` (C-30, C-32, C-45, C-53), and two where the Tier B sign was already ambiguous and kind 3 made a second record unnecessary (C-28, C-34). The Tier C chain never lowered a Tier B tier. Needed interpretation only on whether to record an identical Tier C sign ((b) 1).
- *CR-12 0.9, what the response argument gives.* Worked as written for vehicle-days (C-6): the argument gave "decrease or no change", and A-1 (some drivers at the margin at the old speed) supplied the partial-response premise. Correctly did not apply to vans or to each driver's cost ((b) 2).
- *RS-9 0.9, location complements.* Worked as written for zone and Ringside shops (C-27, C-29), and its "new occupants" clause turned the zone's total into `ambiguous` (C-28). Needed interpretation on new occupants arriving through another component and another tier ((c) 3).
- *RS-16 (0.8 split premise; 0.9 costs on the substitute; Unless with RS-13).* Worked as written for Ringside garages (C-9, with A-6), for ride-hail (C-11, C-12), and for buses once new buses arrive (C-39, Unless clause).
- *CR-6 0.9, splitting a shift into price and quantity.* Worked as written: Ringside garages (quantity until full, then price, on A-6), zone garages (price, on A-21), zone housing (price, stock fixed). Its public-seller-with-a-lag clause worked for buses, with CR-12g ((b) 3).
- *CR-7c (inside and at the edge).* Worked as written: values move at announcement for Ringside commercial (C-31), zone garages (C-36) and Ringside garages (C-37); zone commercial and residential values are `ambiguous` (C-33, C-34). The Tier C rule carried through to capitalization (C-32).

- *TX-1, TX-2 incidence.* TX-1 worked as written (C-3, C-55). TX-2 needed interpretation: road space has no seller ((c) 1). DM-1 worked for deliveries between carriers and zone businesses (C-47, on A-14).
- *CR-8d, SB-7 (earmark fungibility).* Worked, with a second premise SB-7 does not name ((c) 4): C-42 on A-8, C-43 on A-9, recorded as a group of payers in reverse (CR-8, 0.7).
- *OC-3 0.9, fixed budget, and right baseline.* Worked as written: the PAO's added work is a group with `at_least_one_as_stated` (C-58), and the right baseline (fewer zone parking appeals) was netted by arithmetic on the case's figures (A-24). The CCO's cost fell under the own-revenue case (C-57).
- *RS-8 0.9 (more timing chains).* No opposing timing chain arose; per-window signs for evening and weekend trips (C-13, C-14) followed 02_ontology.md, 2.5, signed on flows (CR-12 0.9).
- *OC-5 0.9 (new to the place).* Worked as written for retailers moving from the zone to Ringside (C-25).
- *Candidates from 06_gaps.md, 6.2e.* "Compensating differential from a non-wage term" (WE-032) was needed for zone jobs reached by car (C-52): CR-5c and CR-7a, with the cost of reaching a job in place of a non-wage term. The use is against the no-charge counterfactual (a cost the charge adds to taking a zone job), so it is valid; this is its second use. Its name should be widened to "a change in the cost of taking a job at a place". "A claim payable inside a boundary draws items from outside" (WE-027) resembles the residents' discount drawing registrations (C-17), but RS-5 covered it; not counted as a use.
- Re-run note: the core-only comparison contradicted two verdicts above: CR-6 0.9's public-seller-with-a-lag clause did not apply to the buses, which run below capacity (C-38, C-39 are Tier C under CR-6 1.0), and CR-12 0.9 was not applied to ride-hail fares, whose Tier C record (C-60) was missing.